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A Mathematical Modeling Approach from Nonlinear Dynamics to Complex Systems

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A Mathematical Modeling Approach from Nonlinear Dynamics to Complex Systems

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Preface

Nonlinear systems are the ones that can be modeled by nonlinear algebraic or nonlinear differential equations. In regard to its physical behavior, the relations between changes in the outputs, and in the inputs, are not proportional to each other. A nonlinear system presents a chaotic dynamic, if its dynamics are, on average, exponentially sensitive to changes in its initial condition.

The study of nonlinear systems is very relevant because the nature and all its phenomena are intrinsically nonlinear. The understanding of nonlinear systems is a continuous challenge and requires an interdisciplinary approach that brings together methodologies and tools from different research areas, and also inspires the development of new approaches based on the theory of dynamical systems.

Nowadays, from the technological point of view, solutions are being conceived that exploit the intrinsic and unique characteristics of these nonlinear systems to allow the development of technological systems with unprecedented level of efficiency.

As a result, the study of complex systems is becoming prominent these days. Although there isn't a prevalent definition for complex systems, we can consider a complex system as the one that presents a collective behavior out of the interaction of their constitutive parts, so that this collective behavior cannot be expected if just the properties of their parts are considered. It means that out of this interaction appears the so-called emergent property, so that "The whole is more than the sum of its parts."[1]

As so, a system is considered to be a complex one according to its dynamical behavior. Usually, complex behavior can be seen in an ensemble or a combination of several subsystems, and in a spatial distributed system as, for example, in which even a low-dimensional system can present dynamical characteristics to be considered as a complex one.

[1] Aristotle.

The emergent property that characterizes a complex system may appear as a consequence of several mechanisms, as nonlinear interactions, self-adaptation, self-organization, feedback loops, among others.

São José dos Campos, Brazil Elbert E. N. Macau

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Chapter 1
From Nonlinear Dynamics to Complex Systems: Introduction



Elbert E. N. Macau

Nonlinear dynamics is about systems whose dynamics is ruled by nonlinear algebraic or nonlinear differential equations. In regard to their physical behavior, the relationships between changes in their inputs and the resultant behavior in their outputs are not proportional to one another. This behavior characterizes them as nonlinear systems. A nonlinear system may present chaotic dynamics if its dynamics is on average exponentially sensitive to changes in its initial condition [1]. In this case, although generated by a deterministic system, a chaotic trajectory appears to be complicated and even resembles having random behavior.

The study of nonlinear systems is very relevant because the nature and all its phenomena are intrinsically nonlinear. The understanding of nonlinear systems is a continuous challenge and requires an interdisciplinary approach that brings together methodologies and tools from different research areas, and also uses new specific approaches mainly based on the theory of dynamical systems [2, 3].

Nowadays, from the technological point of view, solutions that exploit the intrinsic and unique characteristics of these nonlinear systems are being conceived so as to allow the development of technological systems with an unprecedented level of efficiency.

Nonlinear dynamics also plays a key role in complex systems. Although there isn't a prevalent definition for a complex system, we can consider it to be the one in which its dynamics appear out of the interaction of *its* constitutive subsystems or "parts," so that these dynamics cannot be expected if just the dynamics of its

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constitutive subsystems or "parts" *are* taken in consideration [4]. It means that out of the nonlinear interactions the so-called "emergent property" appears, so that "The whole is more than the sum of its parts."[1]

The nonlinear interactions that make up a complex system may appear as a consequence of several mechanisms, as nonlinear interactions, self-adaptation, self-organization, feedback loops, among others [5]. The resulting dynamical behavior is neither completely orderly and predictable nor completely random and unpredictable, and its time evolution may reveal sets of different patterns, over several scales, that may keep changing over the time. This behavior *is* prevalent in systems with many degrees of freedom, but may also occur in low dimensional systems.

Of the characteristics involved, mathematical models that properly capture the essential behavior of both nonlinear and complex systems are essential in allowing us to better understand the key properties that rule these systems. This actually is a very active research area in which new approaches or modified methodologies originating from other areas and properly modified are proposed and frequently appear in scientific literature.

The main goals of this book are to present the reader with some of these recent approaches of modeling and analyzing nonlinear and complex systems.

Complex behavior frequently occurs in high dimensional systems and in space extended systems. The latter is described by partial differential equations (PDE). Conservation law plays a prominent role in these equations. For example, they can be used to investigate the existence, uniqueness, and stability of solutions of PDEs. Bruzón et al. [6] study the generalized Gardner equation by using Lie symmetries. The multipliers method is used to obtain results in conservation laws. Similarity variables and similarity solutions are obtained. The authors also show how to construct some travelling wave solutions by using a simplified equation derived from the generalized Gardner equation.

The theory of symmetry reduction, based on the Lie classical method, and conservation law construction were also used by Gandarias et al. [7] to analyze the generalized Fisher–Kolmogorov–Petrovskii–Piskunov equation. This is a reaction–diffusion equation used to model a variety of problems, ranging from population genetics to neurobiology and pattern formation. The authors point out that local symmetries are useful in finding out whether a PDE can be linearized by an invert contact transformation. This occurs, if and only if it has an infinite-parameter Lie group of contact transformation satisfying specific criteria. By using a symmetry generator, the authors were able to find exact solutions of physical and chemical interest for the underlying PDE.

Low dimensional non-integrable Hamiltonian systems present very complex dynamics. Their phase space presents chaotic seas, invariant tori, and stability islands that are intertwined among them. This scenario is considerably accentuated if a time-dependent perturbation is added to the system. Livorati [8] analyzes a system similar to that being investigated, which is a driven stadium-like billiard

[1] Aristotle.

modeled by a four-dimensional nonlinear mapping. The author seeks to understand the behavior of a so-called tunable orbit and its consequences regarding the dynamics of the underlying system whose boundary is time dependent. He points out the existence of a critical resonance velocity, which separates the ensemble energy, where high initial velocities produce Fermi acceleration, while low initial velocities do not. He claims that tunable orbits, acting in partnership with stickiness orbits are responsible for the separation of the ensembles and for the steady state regime for long time series.

Intermittency and transient chaos are prevalent phenomena in chaotic systems. In the former, for a closed chaotic system, if a hole is present or appears, a trajectory can escape from a trapping region and wander for some time, after which it returns to the trapping region. In the latter, the trajectory eventually settles with a different attractor. However, transport barriers can prevent trajectories from moving among regions of the phase space. Chimanski et al. [9] investigate transport barriers and intermittency in dynamical systems. First, a symplectic map is analyzed with regard to variations of a control parameter. For same values, a robust torus is present, which implies a barrier. The effect of this barrier *on* the system dynamics is studied. Subsequently, the authors analyze a Rayleigh–Bérnard convection in which by varying the magnitude of the Rayleigh number, an interior crisis occurs, which leads to an intermittency between quasiperiodic and hyper-chaotic dynamics.

A specific dissipative version of the Fermi–Ulam model consists of a particle, whose movement is confined between two rigid and infinitely heavy walls. One of the walls is fixed, while the other moves periodically in time. The collisions are assumed to be inelastic. Leonel and Marques [10] investigate this system showing that a transient chaos appears after a boundary crisis. They show that the attractor that existed prior to the boundary crisis is destroyed and replaced by a chaotic transient. For this phenomenon, the authors calculate statistical properties that allow them to show that the relaxation time associated with the survival probability of a moving particle surviving in the region of the phase space where the chaotic attractor likely exists before the crisis decays exponentially.

Nonlinear dynamical and complex systems can also be used as building blocks for technological applications. By exploiting the nonlinearities, systems with unprecedented levels of efficiency can be built. Eisencraft et al. [11] present applications of chaotic signal and systems in Telecommunication Engineering and Signal Processing. Four research projects which are under development are described: a message authentication method; a spectral characterization of chaotic systems; a proposed filter based on a chaotic system; a communication system that presents a equalization and switching scheme between chaos-based and conventional modulations. These examples properly highlight the potentialities of nonlinear dynamical and complex systems to be exploited in high efficient technological applications.

One of the most stunning effects regarding nonlinear dynamical and complex systems is the synchronization of some species of fireflies. This phenomenon involves hundreds of thousands of fireflies. In the beginning, their lights flash in an uncorrelated manner with one another. After some time, they flash simultaneously,

exhibiting synchronization. This is a remarkable example of a self-organized phenomenon in which the coupling between the active elements originates as a result of the dynamics of the neighboring elements. The control mechanism that rules this phenomenon is completely decentralized. Ramírez-Ávila et al. [12] review models that were created to explain how and why this phenomenon takes place. Furthermore, they discuss how the mechanisms that rule the firefly synchronization phenomenon can be applied to technological issues, such as in communication networks.

In a flocking, a large number of birds flight together, are capable of avoiding collisions, and this set of birds can turn sharply and suddenly, keeping the relative position with one another. Algar et al. [13] present a story about the mathematics involved in flocking at the level of local behavior and global results. They explore the origin of global coherent motion in flocks builds from simple interacting agents with only local coherence. To do this they motivate the standard Lagrangian models and show that it is in these standard models that the interesting global behavior emerges—including self-organization, emergence, and coherent collective motion. The chapter outlines the standard approach and discusses some key models. This is done with two key stages in mind: Grouping and flocking. Note that these two concepts are usually treated quite separately in the literature. That is, they explain how do local rules drive coagulation of particles, and then how do these groups generate coordinated collective motion.

Nowadays, complex networks are considered to be one of the most promising and prominent approaches for modeling complex systems. They have been used in a variety of different areas, with remarkably successful results. Donner et al. [14] summarize recent developments of this approach to studying flow systems. Some illustrative examples are provided in which correlated based functional network approaches are used to infer statistical relationships, and Lagrangian flow networks provide a very efficient toll to identify relevant structures in the phase space of complex systems.

Modeling a complex system by using a complex network implies in to build the network that represents the relevant interactions among the constitutive parts of the underlying system. After that, the obtained complex network is analyzed by using well-defined quantifiers. They allow for a proper characterization of a complex network. One of them is the so-called *centrality*, which brings relevant information about the organization of the complex network in regard to the relevance of the nodes for the underlying dynamical process that has been modeled. In [15] Rodrigues presents a review about this quantifier and shows the several variants of it that can be defined to measure and pinpoint specific properties of the network. After that, they present how centrality can be applied to understand key mechanisms that rule the underlying modeled system. Finally, he discusses some perspectives and challenges in regard to conceive a definition for centrality measures in the context of multilayer and temporal networks.

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Chapter 2
An Overview of the Generalized Gardner Equation: Symmetry Groups and Conservation Laws



M. S. Bruzón, M. L. Gandarias, and R. de la Rosa

2.1 Introduction

Many nonlinear phenomena are described by differential equations, especially by partial differential equations (PDEs). Symmetry group analysis of a differential equation appears as a powerful method to analyse PDEs [24, 32]. Among its many applications, we highlight the fact that they allow us to obtain exact solutions of a PDE, directly or by using similarity solutions [8, 19]; classify invariant equations; reduce the number of independent variables and construct conservation laws [5, 12, 36].

Given a PDE with dependent variable u and independent variables (t, x), a local conservation law is a space-time divergence expression which vanishes on all solutions $u(t, x)$ of the PDE. Although this concept has its origin in physics, conservation laws play an important role in the study of differential equations. For instance, the existence of a large number of conservation laws of a PDE is a strong indicator of its integrability. In addition, they can be used in numerical methods and mathematical analysis to investigate the existence, uniqueness and stability of solutions of PDEs.

There are different methods to construct conservation laws. It is necessary to distinguish the variational problems from the non-variational problems. For non-variational problems there is a theorem on conservation laws proved by Ibragimov [21] which makes use of the concept of self-adjoint equations and its generalizations [17, 22, 23]. Ibragimov's method has been applied to find conservation laws in several papers [8, 9, 16, 34, 36–39]. Furthermore, in [35], by considering the property of nonlinear self-adjointness, a criterion for linearization of PDEs was proved.

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However, this method for the translation generators yields trivial conservation laws and does not produce all admitted conservation laws. On the other hand, all the low-order conservation laws can be derived by using the multiplier method given by Anco and Bluman [2, 5–7, 15].

An equivalence transformation is a non-degenerate transformation acting on the dependent and independent variables which maps a PDE into another equation of the same family. The main advantage of the equivalence transformations is that instead of considering individual equations they permit an analysis for complete equivalent classes. Thus, equivalence transformations appear as a powerful method to study PDEs with variable coefficients.

The symmetry group of a partial differential equation is defined as the largest group of transformations acting on independent and dependent variables which transforms solutions of the equation into other solutions. Probably the most famous method used to obtain symmetries is the Lie classical method. The basic idea of Lie method is that, when a differential equation is invariant under a Lie group of transformations, there is a transformation which reduces the number of independent variables, in the case of a PDE; either the order of the equation, in the case of an ordinary differential equation (ODE).

A very successful method to obtain exact travelling wave solutions of numerous nonlinear PDEs is the method of simplest equation, especially its version called modified method of simplest equation [27, 28, 30]. Method of simplest equation is based on a procedure analogous to the first step of the test for the Painlevè property [29].

In this paper, we consider a generalized Gardner equation given by

$$u_t + A(t) f(u)\, u_x + C(t)\, f^2(u)\, u_x + B(t)\, u_{xxx} + Q(t)\, F(u) = 0, \tag{2.1}$$

and the particular case $f(u) = u^n$,

$$u_t + A(t) u^n\, u_x + C(t)\, u^{2n}\, u_x + B(t)\, u_{xxx} + Q(t)\, F(u) = 0, \tag{2.2}$$

where n is a positive constant, $A(t)$, $B(t)$, $C(t)$ and $Q(t)$ are arbitrary smooth functions of t, $f(u)$ and $F(u)$ are arbitrary smooth functions of u.

Equation (2.2) with $n = 1/2$, $A(t) = \alpha$, $C(t) = \beta$, $B(t) = \delta$ and $Q(t) = 0$ is

$$u_t + \left(\alpha u^{\frac{1}{2}} + \beta u\right) u_x + \delta u_{xxx} = 0, \qquad \alpha, \beta \neq 0,$$

where α and β are constants which represent the convection and the dispersion coefficients, respectively. This equation is known as the Schamel-Korteweg-de Vries equation with constant-coefficients. This equation describes many phenomena in plasma physics and it plays an important role in the study of the effect of electron trapping on the nonlinear interaction of ion-acoustic waves by including a quasi-potential. In [1], a wide class of exact solutions of this equation with time-dependent coefficients is found. The authors showed that the equation admits travelling wave solutions and they travel with time-dependent speed along the characteristic curves.

In [14] we considered the generalized variable-coefficient Gardner equation (2.2) with $n = 1$ and $F(u) = u$

$$u_t + A(t)u\,u_x + C(t)\,u^2 u_x + B(t)\,u_{xxx} + Q(t)\,u = 0. \tag{2.3}$$

For Eq. (2.3), classical symmetries were obtained. Moreover, we determined the subclasses of equation (2.3) which are nonlinearly self-adjoint, as well as the multipliers admitted by (2.3). We derived conservation laws by using both methods: the method given in [21] and the direct method of the multipliers [6, 7]. We showed that some of these conservation laws yield conserved integrals with physical meaning, such as mass and energy. Finally, as an example of another application of the conserved vectors, we applied the double reduction method to get exact solutions of the Gardner equation (2.3) from solutions of a second-order ordinary differential equation.

In [15] we studied the generalized variable-coefficient Gardner equation involving arbitrary functions and nonlinear terms of any order which is given by

$$u_t + A(t)u^n\,u_x + C(t)\,u^{2n} u_x + B(t)\,u_{xxx} + Q(t)\,u = 0. \tag{2.4}$$

This equation is Eq. (2.2) with $F(u) = u$. We obtained the equivalence group of equation (2.4) which is infinite-dimensional. Equivalence group allows us to consider a transformation which leads us to a subclass of equation (2.4) with a smaller number of arbitrary functions

$$u_t + A(t)u^n\,u_x + u^{2n} u_x + u_{xxx} + Q(t)\,u = 0. \tag{2.5}$$

Lie symmetries of Eq. (2.5) have been obtained. Furthermore, we have proved that Eq. (2.5) is nonlinearly self-adjoint. Nontrivial conservation laws have been obtained by using the theorem given by Ibragimov [21] and the direct method of the multipliers of Anco and Bluman [6, 7].

In [41] adding the term $E(t)u_x$ to Eq. (2.4), with $E(t)$ being an arbitrary smooth function of t, the authors found exact non-travelling solutions, which include soliton solutions, combined soliton solutions, triangular periodic solutions, Jacobi elliptic function solutions and combined Jacobi elliptic function solutions. In [25] Johnpillai and Kalique obtained the optimal system of one-dimensional subalgebras of the Lie symmetry algebras when $n = 1$, $A(t) = 1$ and $C(t) = 0$. In [26] the authors also have constructed conservation laws for the same equation for some special forms of the functions $B(t)$ and $Q(t)$. Lie symmetries of Eq. (2.4) were obtained by Molati and Ramollo [31], when $n = 1$ and $Q(t) = 0$. In [18] Hong and Lu obtained exact solutions using the general mapping deformation method of a generalization of Eq. (2.4) which includes a term $E(t)u_x$ and a forcing term $R(t)$, where $E(t)$ and $R(t)$ are arbitrary smooth functions of t. Bruzón et al. obtained a symmetry classification and conservation laws for Eq. (2.4) with $Q = 0$, $n = 1$ and $Q = 0$, $n = \frac{1}{2}$ in [11], and for $C \neq 0$, $n = 1$ and $C = 0$ in [10].

In [13] we studied the classical and nonclassical symmetries admitted by the generalized constant-coefficient Gardner equation

$$u_t + a\,u^n\,u_x + b\,u^{2n}\,u_x + c\,u_{xxx} + d\,u_x + e\,u + f = 0. \tag{2.6}$$

We established a symmetry classification of Eq. (2.6) in terms of the arbitrary constants a, b, c, d, e, f and n. We proved that the nonclassical method applied to this equation leads to new symmetries. Furthermore, we obtained similarity variables and similarity solutions of equation (2.4) from the optimal system of subalgebras. Taking into account the elements of the optimal system of subalgebras we reduced equation (2.6) to an ordinary differential equation. Finally, we constructed some travelling wave solutions by using the modified simplest equation method.

This work is organized as follows. In Sect. 2.2, we construct some nontrivial conservation laws for Eq. (2.1) by using the direct method of the multipliers [6, 7]. In Sect. 2.3, we obtain the continuous equivalence transformations of class (2.2) in order to reduce the number of arbitrary functions. In Sect. 2.4, we obtain the Lie symmetries of the reduced equation obtained by using equivalence transformations. From reduced equation we derive exact travelling waves solutions by using the simplest equation method given by Kudryashov and Loguinova [27, 30]. The conclusions are presented in Sect. 2.6.

2.2 Conservation Laws

Conservation laws appear in many of physical, chemical and mechanical processes, such laws enable us solve problems in which certain physical properties do not change over time within an isolated physical system. In this section we will construct conservation laws of Eq. (2.1)

$$u_t + A(t)\,f(u)u_x + C(t)\,f(u)^2u_x + u_{xxx} + Q(t)F(u) = 0.$$

A conservation law of Eq. (2.1) is a space-time divergence such that

$$D_t T(x,t,u,u_x,u_t,\ldots) + D_x X(x,t,u,u_x,u_t,\ldots) = 0, \tag{2.7}$$

on all solutions $u(x,t)$ of Eq. (2.1). Here, T represents the conserved density and X the associated flux [5], and D_x, D_t denote the total derivative operators with respect to x and t, respectively.

Each conservation law (2.7) has an equivalent characteristic form in which has been eliminated u_t and differential consequences from T and X by using Eq. (2.1). We eliminate u_t, u_{tx}, ... to obtain

$$\widehat{T} = T\,|_{u_t=\Delta} = T - \Phi,$$
$$\widehat{X} = X\,|_{u_t=\Delta} = X - \Psi,$$

where $\Delta = -A(t)f(u)u_x - C(t)f(u)^2u_x - u_{xxx} - Q(t)F(u)$, so that

$$\left(D_t\widehat{T}(x,t,u,u_x,u_{xx},\ldots) + D_x\widehat{X}(x,t,u,u_x,u_{xx},\ldots)\right)|_{u_t=\Delta} = 0,$$

and where

$$\begin{aligned} D_t\,|_{u_t=\Delta} &= \partial_t + \Delta\partial_u + D_x(\Delta)\partial_{u_x} + \cdots \\ D_x\,|_{u_t=\Delta} &= \partial_x + u_x\partial_u + u_{xx}\partial_{u_x} + \cdots = D_x \end{aligned}$$

is verified on all solutions of equation (2.1). In particular, moving off of solutions, we have the identity

$$\begin{aligned} D_t = D_t\,|_{u_t=\Delta} &+(u_t + A(t)f(u)u_x + C(t)f(u)^2u_x + u_{xxx} + Q(t)F(u))\partial_u \\ &+D_x(u_t + A(t)f(u)u_x + C(t)f(u)^2u_x + u_{xxx} + Q(t)F(u))\partial_{u_x} + \cdots \end{aligned}$$

These expressions yield us to the characteristic form of the conservation law (2.7)

$$\begin{aligned} &D_t\widehat{T}(x,t,u,u_x,u_{xx},\ldots) + D_x\left(\widehat{X}(x,t,u,u_x,u_{xx},\ldots) + \widehat{\Psi}(x,t,u,u_x,u_t,\ldots)\right) \\ &= (u_t + A(t)f(u)u_x + C(t)f(u)^2u_x + u_{xxx} + Q(t)F(u))\Lambda(x,t,u,u_x,u_{xx},\ldots), \end{aligned} \tag{2.8}$$

where

$$\begin{aligned} \widehat{\Psi}(x,t,u,u_x,u_t,\ldots) = &E_{u_x}(\widehat{T})(u_t + A(t)f(u)u_x + C(t)f(u)^2u_x + u_{xxx} \\ &+Q(t)F(u)) \\ &+E_{u_{xx}}(\widehat{T})D_x(u_t + A(t)f(u)u_x + C(t)f(u)^2u_x + u_{xxx} \\ &+Q(t)F(u)) \\ &+\cdots \end{aligned}$$

is a trivial flux [5], and the function

$$\Lambda = E_u(\widehat{T}),$$

is a multiplier, where $E_u = \partial_u - D_x\partial_{u_x} + D_x^2\partial_{u_{xx}} - \cdots$, denotes the (spatial) Euler operator with respect to u.

A function $\Lambda(x,t,u,u_x,u_{xx},\ldots)$ is called multiplier if it verifies that $(u_t + A(t)f(u)u_x + C(t)f(u)^2u_x + u_{xxx} + Q(t)F(u))\Lambda$ is a divergence expression for all functions $u(x,t)$, not just solutions of equation (2.1).

We point out that the multipliers $\Lambda(x,t,u,u_x,u_{xx},\ldots)$ derived by using the multipliers of Anco and Bluman method proposed in [6] correspond to the substitutions of the condition of nonlinear self-adjointness [21, 23].

Equation (2.1) is said to be *nonlinearly self-adjoint* if the equation obtained from its adjoint equation by the substitution $v = h(x,t,u,u_x,\ldots)$, with a certain function $h(x,t,u,u_x,\ldots)$ such that $h(x,t,u,u_x,\ldots) \neq 0$ is identical to the original equation (2.1), i.e:

$$\Delta^*_{|v=h} = \Lambda\Delta,$$

and Λ is the multiplier.

The adjoint equation to Eq. (2.1) has the form

$$\Delta^* \equiv \frac{\delta(v\,\Delta)}{\delta u} = 0\,, \tag{2.9}$$

where

$$\frac{\delta}{\delta u} = \frac{\partial}{\partial u} - D_t\left(\frac{\partial}{\partial u_t}\right) - D_x\left(\frac{\partial}{\partial u_x}\right) + D_x^2\left(\frac{\partial}{\partial u_{xx}}\right) - D_x^3\left(\frac{\partial}{\partial u_{xxx}}\right) - \cdots\,, \tag{2.10}$$

denotes the variational derivatives (the Euler-Lagrange operator) and v is a new dependent variable, and D_t, D_x are the total differentiations.

In [21] (see also [20]) a general theorem on conservation laws for arbitrary differential equations which do not require the existence of Lagrangians has been proved. This new theorem is based on the concept of adjoint equations for nonlinear equations. In [3, 4] Anco showed that Ibragimov's theorem is a special case of a standard conservation law formula using symmetries and adjoint-symmetries, and this formula is not always able to produce all conservation laws.

In this paper, in order to obtain conservation laws, we use (2.8) from which is deduced that all nontrivial conserved densities in the form (2.7) arise from multipliers Λ of Eq. (2.1), where Λ depends only on x, t, u and x derivatives of u. Multipliers are determined from divergence condition which is given by

$$\frac{\delta}{\delta u}\Big(\big(u_t + A(t)f(u)u_x + C(t)f(u)^2u_x + u_{xxx} + Q(t)F(u)\big)\Lambda\Big) = 0. \tag{2.11}$$

This yields a multiplier determining equation that can be split in u_t, u_{xxx} and their derivatives. Thus, we obtain an overdetermined system of equations for Λ and the arbitrary functions that Eq. (2.1) involves which consist of

$$D_x^3\Lambda + D_x\big((Af + Cf^2)\Lambda\big) + D_t\Lambda - \big(Af'u_x + 2Cff'u_x + QF'\big)\Lambda = 0,$$

and

$$\Lambda_u = E_u(\Lambda), \qquad \Lambda_{u_x} = -E_u^{(1)}(\Lambda), \qquad \Lambda_{u_{xx}} = -E_u^{(2)}(\Lambda), \ldots$$

which are verified for all solutions $u(x,t)$ of Eq. (2.1).

Given a multiplier Λ, we can obtain the conserved density using a standard method [40]

$$T = \int_0^1 d\lambda \; u\Lambda(x, t, \lambda u, \lambda u_x, \lambda u_{xx}, \ldots).$$

We have determined the 0th-order multipliers, $\Lambda(t, x, u)$, which Eq. (2.1) admits. For $f(u)$ arbitrary function we obtain

$$\Lambda = F_1(t)u + F_2(t, x) \tag{2.12}$$

where $F_1(t)$, $F_2(t, x)$ and $F(u)$ must satisfy the equation

$$F_1 QF + QF_u\,(F_1 u + F_2) - CF_{2_x} f^2 - AF_{2_x} f - F_{1_t} u - F_{2_t} = 0 \tag{2.13}$$

Particular solutions of equation (2.13) lead to the following cases

Case 1. For $F(u) = \frac{1/2\,k_2 u^2 + k_1 u}{k_2 u + k_1}$ and $Q(t) = 1$

$$\Lambda = \exp(t)(k_1 u + k_2).$$

The conserved density obtained from this multiplier is

$$T = 1/2\,\mathrm{e}^t u\,(k_1 u + 2\,k_2)\,.$$

The flux obtained from this multiplier is

$$X = \int \mathrm{e}^t f\,(u)\,(C\,(t)\,f\,(u) + A\,(t))\,(k_1 u + k_2)\;\mathrm{d}u$$
$$+1/2\left(\left(2\,k_1 u + 2\,k2\right)u_{xx} - k_1 {u_x}^2\right)\mathrm{e}^t.$$

Case 2. For $F(u) = u$,

$$\Lambda = k_1 e^{2H} u + k_2 e^H.$$

The conserved density obtained from this multiplier is

$$T = 1/2\,k_1 e^{2H} u^2 + k_2 e^H u.$$

The flux obtained from this multiplier is

$$X = k_1 e^{2H} u u_{xx} - 1/2\,k_1 e^{2H} {u_x}^2 + k_2 e^H u_{xx}$$
$$+ \int f\,(u)\,e^H\,(C\,(t)\,f\,(u) + A\,(t))\left(k_1 e^H u + k_2\right)\mathrm{d}u.$$

Case 2.1 In the case that $F(u) = u$, $f(u) = u$, $A(t) = 1$, $C(t) = 0$ or $F(u) = u$, $f(u) = \sqrt{u}$, $A(t) = 0$, $C(t) = 1$ an additional multiplier is obtained

$$\Lambda = e^H x - e^{2H} L u.$$

In this case, we obtain the following conserved vector

$$T = \frac{1}{2}\left(2e^H xu - e^{2H} L u^2\right),$$

$$X = \frac{1}{6}\left(e^H x\left(6u_{xx} + 3u^2\right) - e^{2H} L\left(2u^3 - 3u_x^2 + 6uu_{xx}\right) - 6e^H u_x\right).$$

In the above cases, k_1 and k_2 are arbitrary constants, $H(t) = \int Q(t)dt$ and $L(t) = \int e^{-\int Q(t)dt} dt$.

2.3 Equivalence Transformations

In this section we determine the group of equivalence transformations of class (2.2). By using equivalence transformations class (2.2) can be reduced to a subclass with a smaller number of arbitrary functions. An equivalence transformation of class (2.2) is a nondegenerate point transformation, (t, x, u) to $(\tilde{t}, \tilde{x}, \tilde{u})$ in the augmented space (t, x, u, A, B, C, F, Q) which maps any equation of class (2.2) into an equation of the same class, where the arbitrary functions, $\tilde{A}(\tilde{t})$, $\tilde{B}(\tilde{t})$, $\tilde{C}(\tilde{t})$, $\tilde{F}(\tilde{u})$ and $\tilde{Q}(\tilde{t})$ are not necessarily the same that the original ones. Applying Lie's infinitesimal criterion [33] we determine the equivalence transformations of class (2.2). We must require the invariance of class (2.2) and the invariance of the auxiliary system

$$A_x = A_u = B_x = B_u = C_x = C_u = F_t = F_x = Q_x = Q_u. \tag{2.14}$$

We consider the one-parameter group of equivalence transformations in

$$(t, x, u, A, B, C, F, Q)$$

given by

$$\begin{aligned}
\tilde{t} &= t + \varepsilon\, \tau(t, x, u) + O(\varepsilon^2),\\
\tilde{x} &= x + \varepsilon\, \xi(t, x, u) + O(\varepsilon^2),\\
\tilde{u} &= u + \varepsilon\, \eta(t, x, u) + O(\varepsilon^2),\\
\tilde{A} &= A + \varepsilon\, \omega^1(t, x, u, A, B, C, F, Q) + O(\varepsilon^2),\\
\tilde{B} &= B + \varepsilon\, \omega^2(t, x, u, A, B, C, F, Q) + O(\varepsilon^2),\\
\tilde{C} &= C + \varepsilon\, \omega^3(t, x, u, A, B, C, F, Q) + O(\varepsilon^2),\\
\tilde{F} &= F + \varepsilon\, \omega^4(t, x, u, A, B, C, F, Q) + O(\varepsilon^2),\\
\tilde{Q} &= Q + \varepsilon\, \omega^5(t, x, u, A, B, C, F, Q) + O(\varepsilon^2),
\end{aligned} \tag{2.15}$$

where ε is the group parameter. In this case, the vector field takes the following form

$$Y = \tau\partial_t + \xi\partial_x + \eta\partial_u + \omega^1\partial_A + \omega^2\partial_B + \omega^3\partial_C + \omega^4\partial_F + \omega^5\partial_Q. \tag{2.16}$$

The invariance of the system given by Eqs. (2.2) and (2.14) under the one-parameter group of equivalence transformations (2.15) with infinitesimal generator (2.16) leads to a system of determining equations. After having solved the determining system, omitting tedious calculations, we obtain the associated equivalence algebra of class (2.2) which is infinite-dimensional and it is spanned by

$$Y_1 = -x\partial_x - 3t\partial_t + \tfrac{1}{n}u\partial_u + A\partial_A + 3F\partial_F,$$

$$Y_2 = -\tfrac{1}{n}u\partial_u + A\partial_A + 2C\partial_C,$$

$$Y_3 = x\partial_x + \tfrac{1}{2n}u\partial_u + \tfrac{1}{2}A\partial_A + 3B\partial_B,$$

$$Y_4 = \partial_X,$$

$$Y_\alpha = \alpha\partial_t - \alpha_t A\partial_A - \alpha_t B\partial_B - \alpha_t C\partial_C - \alpha_t Q\partial_Q,$$

where $\alpha = \alpha(t)$ is an arbitrary smooth function. We show the finite form of these equivalence transformations in the following theorem:

Theorem 1 *The equivalence group of class (2.2) consists of the transformations*

$$\tilde{t} = e^{-3\varepsilon_1}\lambda(t), \quad \tilde{x} = (x+\varepsilon_4)e^{\varepsilon_3-\varepsilon_1}, \quad \tilde{u} = e^{\frac{1}{n}\left(\varepsilon_1-\varepsilon_2+\frac{\varepsilon_3}{2}\right)}u,$$

$$\tilde{A} = \frac{e^{\varepsilon_1+\varepsilon_2+\frac{\varepsilon_3}{2}}}{\lambda_t}A, \quad \tilde{B} = \frac{e^{3\varepsilon_3}}{\lambda_t}B, \quad \tilde{C} = \frac{e^{2\varepsilon_2}}{\lambda_t}C, \quad \tilde{F} = e^{-3\varepsilon_1-\frac{1}{n}\left(\varepsilon_1-\varepsilon_2+\frac{\varepsilon_3}{2}\right)}F,$$

$$\tilde{Q} = \frac{1}{\lambda_t}Q$$

where ε_1, ε_2, ε_3 and ε_4 are arbitrary constants, $\lambda = \lambda(t)$ is an arbitrary smooth function verifying $\lambda_t \neq 0$.

From Theorem 1 we transform Eq. (2.2) into an equation in which the highest order linear term has been set to a nonzero constant value, i.e. we choose the gauging $\tilde{B}(\tilde{t}) = 1$

$$\tilde{u}_{\tilde{t}} + \tilde{A}(\tilde{t})\,\tilde{u}^n\,\tilde{u}_{\tilde{x}} + \tilde{C}(\tilde{t})\,\tilde{u}^{2n}\,\tilde{u}_{\tilde{x}} + \tilde{u}_{\tilde{x}\tilde{x}\tilde{x}} + \tilde{Q}(\tilde{t})\,\tilde{F}(\tilde{u}) = 0,$$

and perform it using the transformation

$$\tilde{t} = e^{3(\varepsilon_3-\varepsilon_1)}\int B(t)dt, \quad \tilde{x} = (x+\varepsilon_4)e^{\varepsilon_3-\varepsilon_1}, \quad \tilde{u} = e^{\frac{1}{n}\left(\varepsilon_1-\varepsilon_2+\frac{\varepsilon_3}{2}\right)}u, \tag{2.17}$$

where the new arbitrary functions are given by $\tilde{A}(\tilde{t}) = e^{\varepsilon_1+\varepsilon_2-\frac{5}{2}\varepsilon_3}\dfrac{A(t)}{B(t)}$, $\tilde{C}(\tilde{t}) = e^{2\varepsilon_2-3\varepsilon_3}\dfrac{C(t)}{B(t)}$ and $\tilde{Q}(\tilde{t}) = e^{-3\varepsilon_3}\dfrac{Q(t)}{B(t)}$. This allows us to restrict, without loss of generality, our study to the class

$$u_t + A(t)\,u^n\,u_x + C(t)\,u^{2n}\,u_x + u_{xxx} + Q(t)\,F(u) = 0. \tag{2.18}$$

2.4 Lie Symmetries

To apply the Lie classical method to Eq. (2.18) we consider the one-parameter Lie group of infinitesimal transformations in (t, x, u) given by

$$t^* = t + \varepsilon\tau(t, x, u) + O(\varepsilon^2), \tag{2.19}$$

$$x^* = x + \varepsilon\xi(t, x, u) + O(\varepsilon^2), \tag{2.20}$$

$$u^* = u + \varepsilon\eta(t, x, u) + O(\varepsilon^2), \tag{2.21}$$

where ε is the group parameter. We require that this transformation leaves invariant the set of solutions of equation (2.18). This yields to an overdetermined, linear system of equations for the infinitesimals $\tau(t, x, u)$, $\xi(t, x, u)$ and $\eta(t, x, u)$. A general element of the symmetry algebra of (2.18) has the form

$$\mathbf{v} = \tau(t, x, u)\partial_t + \xi(t, x, u)\partial_x + \eta(t, x, u)\partial_u. \tag{2.22}$$

Having determined the infinitesimals, the symmetry variables are found by solving the characteristic equation which is equivalent to solving the invariant surface condition

$$\eta(t, x, u) - \tau(t, x, u)u_t - \xi(t, x, u)u_x = 0. \tag{2.23}$$

The set of solutions of equation (2.4) is invariant under the transformation (2.19)–(2.21) provided that

$$\mathrm{pr}^{(3)}\mathbf{v}(\Delta) = 0 \quad \text{when} \quad \Delta = 0,$$

where $\Delta = u_t + A(t)u^n u_x + u^{2n}u_x + u_{xxx} + Q(t)u$, and $\mathrm{pr}^{(3)}\mathbf{v}$ is the third prolongation of the vector field (2.22) which is given by

$$\mathrm{pr}^{(3)}\mathbf{v} = \mathbf{v} + \zeta^t\partial u_t + \zeta^x\partial u_x + \zeta^{xxx}\partial u_{xxx}, \tag{2.24}$$

where

$$\zeta^J(t, x, u^{(3)}) = D_J(\eta - \tau u_t - \xi u_x) + \tau u_{Jt} + \xi u_{Jx},$$

with $J = (j_1, \ldots, j_k)$, $1 \leq j_k \leq 2$, $1 \leq k \leq 3$, and $u^{(3)}$ denote the sets of partial derivatives up to third order [32]. Applying (2.24) we obtain a set of determining equations for the infinitesimals. Simplifying this system we get $\tau = \tau(t)$, $\xi = \frac{k_1}{3}x + \omega$ and $\eta = (\delta + \frac{k_1}{3})u + \beta$, where $\tau(t)$, $\beta(t)$, $\delta(t)$ and $\omega(t)$ must satisfy the following conditions:

$$\begin{aligned} &3C\beta_x u^{2n} + 3A\beta_x u^n + 3\delta_t u + 3Qh_u\delta u + Q\tau_t h_u u + \tau_{tt} u \\ &\quad -3Qh\delta + 3\beta_{xxx} + 3\beta_t + 3Qh_u\beta + 2Q\tau_t h + 3Q_t\tau h = 0, \\ &\quad -\tau_{tt} ux + 6Cn\delta u^{2n+1} + 2C\tau_t n u^{2n+1} + 2C\tau_t u^{2n+1} + 3C_t\tau u^{2n+1} \\ &\quad +3An\delta u^{n+1} + A\tau_t n u^{n+1} + 2A\tau_t u^{n+1} + 3A_t\tau u^{n+1} \\ &\quad +6Cn\beta u^{2n} + 3An\beta u^n - 3\omega_t u = 0. \end{aligned} \tag{2.25}$$

In order to find Lie symmetries of Eq. (2.18) it is necessary to distinguish according to powers of u which appears in (2.25). This leads to three different cases: $n \neq 1, \frac{1}{2}$, $n = 1$, and $n = \frac{1}{2}$.

2.4.1 Case 1: $n \neq 1, \frac{1}{2}$

- If $F(u) = u^k$, $k \neq 1$

$$\tau = 3k_1 t + k_3, \qquad \xi = k_1 x + k_2, \qquad \eta = -\frac{(3\,A_t\,k_1\,t + A_t\,k_2 + 2\,A\,k_1)\,u}{A\,n},$$

where $A = A(t)$, $C(t)$ and $Q = Q(t)$ must satisfy the following conditions:

$$3AC_t k_1 t - 6A_t C k_1 t + 3AC_t k_2 - 6A_t C k_2 - 2ACk_1 = 0, \tag{2.26}$$

$$AA_{tt} k_1 t - (A_t)^2 k_1 t + AA_{tt} k_2 - (A_t)^2 k_2 + AA_t k_1 = 0, \tag{2.27}$$

$$\begin{aligned} &-3A_t Qkk_1 t + 3A_t Qk_1 t - 3A_t Qkk_2 + 3A_t Qk_2 \\ &3AQ_t k_1 nt - 2AQkk_1 + 2AQk_1 + 3AQk_1 n + 3AQ_t k_2 n = 0. \end{aligned} \tag{2.28}$$

- If $F(u) = u$ and $C = 1$, the infinitesimals are given by

$$\tau = k_2 t + k_3, \qquad \xi = \frac{k_2}{3}x + k_1, \qquad \eta = -\frac{k_2}{3n}u,$$

where $A = A(t)$ and $Q = Q(t)$ must satisfy the following conditions:

$$(k_2 t + k_3)A_t + \frac{k_2}{3}A = 0, \tag{2.29}$$

$$(k_2 t + k_3)Q_t + k_2 Q = 0. \tag{2.30}$$

- If $F(u) = u$, $A = 0$, $C = c$, $Q = q$

$$\tau = 3k_3t + k_1, \qquad \xi = k_3x + k_2, \qquad \eta = -k_3\frac{u}{n}.$$

- If $F(u) = u$, $A = a$, $C = 0$, $Q = q$

$$\tau = 3k_3t + k_1, \qquad \xi = k_3x + k_2, \qquad \eta = -k_3\frac{2u}{n}.$$

2.4.2 Case 2: $n = 1$

- If $F(u) = u^2$,
  - For $A = b_0\exp(nt)$, $C = c_0\exp(2nt)$, and $Q = q_0\exp(nt)$, we obtain

$$\xi = 0, \quad \tau = 1, \quad \eta = -nu \tag{2.31}$$

  - For $A = a_0(3k_1t+k_2)^{-2/3}$, $C = c_0(3k_1t+k_2)^{-2/3}$ and $Q = q_0(3k_1t+k_2)^{-1}$,

$$\xi = k_1x + k_2, \quad \tau = 3k_1t + k_3, \quad \eta = 0 \tag{2.32}$$

- If $F(u) = u$ and $C = 1$, this case was already considered in [14], in that paper the authors obtained Lie symmetries of Eq. (2.4). We show the symmetries of the reduced equation (2.18), which can be deduced from [14] by using a transformation,

$$\tau = k_2t + k_3, \qquad \xi = \frac{k_2}{3}x + \beta, \qquad \eta = \gamma - \frac{k_2}{3}u,$$

  where $A = A(t)$, $Q = Q(t)$, $\beta = \beta(t)$ and $\gamma = \gamma(t)$ must satisfy the following conditions:

$$(k_2t + k_3)A_t + \frac{k_2}{3}A + 2\gamma = 0, \tag{2.33}$$

$$(k_2t + k_3)Q_t + k_2Q = 0, \tag{2.34}$$

$$\gamma A - \beta_t = 0, \tag{2.35}$$

$$\gamma Q + \gamma_t = 0. \tag{2.36}$$

- If $F(u) = u$, $C = 0$, $A = k$, $Q = 1$

$$\tau = k_3, \qquad \xi = -kk_1\exp(-t) + k_2, \qquad \eta = k_1\exp(-t).$$

- If $F(u) = \text{const.}$, $C = 0$, $A = k$, $Q = 1$

$$\tau = 3k_1 t + k_4, \quad \xi = \left(x + 2dt - \frac{5}{2}kft^2\right)k_1 + k_2 kt + k_3, \quad \eta = -k_1\left(5ft + 2u\right) + k_2.$$

- If $F(u) = 0$, $A(t) = a$, $C(t) = c$ and $Q(t) = d$

$$\tau = 6k_1 ct + k_3, \quad \xi = \left(2cx - a^2 t\right)k_1 + k_2, \quad \eta = -k_1\left(2cu + a\right).$$

2.4.3 Case 3: $n = \frac{1}{2}$

- If $F(u) = u^k$, $k \neq 1, 2$, for $A = a_0(3k_1 t + k_2)^{-2/3}$, $C = c_0(3k_1 t + k_2)^{-2/3}$ and $Q = q_0(3k_1 t + k_2)^{-1}$,

$$\xi = k_1 x + k_2, \quad \tau = 3k_1 t + k_3, \quad \eta = 0 \tag{2.37}$$

- If $F(u) = u$, $A(t) = 0$ and $C = 1$, we get the following symmetries

$$\tau = \tau(t), \qquad \xi = \frac{\tau_t}{3}x + \beta, \qquad \eta = \frac{\tau_{tt}}{3}x - \frac{2\tau_t}{3}u + \beta_t,$$

where $Q = Q(t)$, $\beta = \beta(t)$ and $\tau = \tau(t)$ must satisfy the following conditions:

$$\left(Q_{tt} + 4QQ_t\right)\tau + 2\left(Q_t + 2Q^2\right)\tau_t = 0, \tag{2.38}$$

$$\beta_t Q + \beta_{tt} = 0. \tag{2.39}$$

Moreover, we can solve Eq. (2.38) using the technique of separating variables, obtaining

$$\tau = \frac{k_4}{\sqrt{2Q^2 + Q_t}},$$

and where $Q(t)$ and $\beta(t)$ satisfy (2.39).

- If $F(u) = u$, $C = k$, $A = 0$, $Q = 1$

$$\tau = k_3, \qquad \xi = -kk_1 \exp(-t) + k_2, \qquad \eta = k_1 \exp(-t).$$

- If $F(u) = \text{const.}$, $C = k$, $A = 0$, $Q = 1$

$$\tau = 3k_1 t + k_4, \quad \xi = \left(x + 2dt - \frac{5}{2}kft^2\right)k_1 + k_2 kt + k_3, \quad \eta = -k_1\left(5ft + 2u\right) + k_2.$$

In the above cases, $k_1, k_2, k_3, k_4, k, a_0, b_0, c_0, d_0, a, b, c$ and d are arbitrary constants.

2.5 Reduced Equations: Some Travelling Wave Solutions

In this section we consider Eq. (2.18) with $A(t) = a$, $C(t) = b$, $Q(t) = 1$ and $F(u) = eu + f$

$$u_t + a\,u^n\,u_x + b\,u^{2n}\,u_x + c\,u_{xxx} + e\,u + f = 0, \tag{2.40}$$

where a, b, e, f y n are arbitrary constants.

For Eq. (2.40), the following Lie symmetries were obtained in [13]. If a, b, e, f and n are arbitrary, we get

$$\mathbf{v}_1 = \partial_x, \qquad \mathbf{v}_2 = \partial_t. \tag{2.41}$$

In the following cases, we obtain additional symmetries.

Case 1 $e \neq 0$

1.1. If $n = \dfrac{1}{2}, a = 0, b = k$ or $n = 1, a = k, b = 0$,

$$\mathbf{v}_3 = -\frac{k}{e}\exp(-e\,t)\partial_x + \exp(-e\,t)\partial_u. \tag{2.42}$$

Case 2 $e = 0$

2.1. If $n = \dfrac{1}{2}, a = 0, b = k$ or $n = 1, a = k, b = 0$,

$$\mathbf{v}_4 = \left(x - \frac{5}{2}kft^2\right)\partial_x + 3t\partial_t - (5ft + 2u)\,\partial_u, \tag{2.43}$$

$$\mathbf{v}_5 = kt\partial_x + \partial_u. \tag{2.44}$$

2.2. If $n \neq \dfrac{1}{2}, 1, a = 0, f = 0$,

$$\mathbf{v}_6 = x\partial_x + 3t\partial_t - \frac{u}{n}\partial_u. \tag{2.45}$$

2.3. If $n \neq 1, b = 0, f = 0$,

$$\mathbf{v}_7 = x\partial_x + 3t\partial_t - \frac{2u}{n}\partial_u. \tag{2.46}$$

2.4. If $n = 1, f = 0$,

$$\mathbf{v}_8 = \left(2bx - a^2t\right)\partial_x + 6bt\partial_t - (2bu + a)\,\partial_u. \tag{2.47}$$

In the previous cases, $k \neq 0$ represents an arbitrary constant.

Table 2.1 Similarity solutions and similarity variables of Eq. (2.4)

Subcase	Optimal system of subalgebras	Similarity variables	Similarity solutions
Arbitrary	$\langle \lambda \mathbf{v}_1 + \mu \mathbf{v}_2 \rangle$	$z = \mu x - \lambda t$	$u = h(z)$
1.1. a)	$\langle \lambda \mathbf{v}_1 + \mu \mathbf{v}_2 \rangle$	$z = \mu x - \lambda t$	$u = h(z)$
1.1. b)	$\langle \lambda \mathbf{v}_1 + \mu \mathbf{v}_3 \rangle$	$z = t$	$u = h(z) + \dfrac{\mu \exp(-et)x}{\lambda - \frac{k\mu}{e}\exp(-et)}$
2.1. a)	$\langle \lambda \mathbf{v}_1 + \mu \mathbf{v}_2 \rangle$	$z = \mu x - \lambda t$	$u = h(z)$
2.1. b)	$\langle \lambda \mathbf{v}_1 + \mu \mathbf{v}_5 \rangle$	$z = t$	$u = \dfrac{h(z) + \mu x}{\lambda + \mu k t}$
2.1. c)	$\langle \mathbf{v}_4 \rangle$	$z = x t^{-\frac{1}{3}} + \dfrac{f k t^{\frac{5}{3}}}{2}$	$u = t^{-\frac{2}{3}} h(z) - f t$
2.2. a)	$\langle \lambda \mathbf{v}_1 + \mu \mathbf{v}_2 \rangle$	$z = \mu x - \lambda t$	$u = h(z)$
2.2. b)	$\langle \mathbf{v}_6 \rangle$	$z = x t^{-\frac{1}{3}}$	$u = t^{-\frac{1}{3n}} h(z)$
2.3. a)	$\langle \lambda \mathbf{v}_1 + \mu \mathbf{v}_2 \rangle$	$z = \mu x - \lambda t$	$u = h(z)$
2.3. b)	$\langle \mathbf{v}_7 \rangle$	$z = x t^{-\frac{1}{3}}$	$u = t^{-\frac{2}{3n}} h(z)$
2.4. a)	$\langle \lambda \mathbf{v}_1 + \mu \mathbf{v}_2 \rangle$	$z = \mu x - \lambda t$	$u = h(z)$
2.4. b)	$\langle \mathbf{v}_8 \rangle$	$z = x t^{-\frac{1}{3}} + \dfrac{a^2 t^{\frac{2}{3}}}{4b}$	$u = t^{-\frac{1}{3}} h(z) - \dfrac{a}{2b}$

Suppose that $\mathscr{A}$ is an r-dimensional Lie algebra, and $\mathbf{v}_1, \mathbf{v}_2, \ldots \mathbf{v}_r$, form a basis for $\mathscr{A}$. If we consider that two subalgebras are related by a transformation of the group of symmetries, invariant solutions calculated from them will be related by the same transformation. We construct the optimal system of subalgebras to obtain those invariant solutions that cannot be derived from others [32]. By using the optimal system, we calculate the similarity variables and the similarity solutions. Thus, we allow us to transform Eq. (2.40) into an ODE, solving this characteristic system,

$$\frac{dx}{\xi} = \frac{dt}{\tau} = \frac{du}{\eta}. \tag{2.48}$$

In Table 2.1, we show the elements of the optimal system for each case in the symmetry classification along with their corresponding similarity variables and similarity solutions. Furthermore, in Table 2.2 we present the corresponding reduced ODEs.

We get the following generator

$$\lambda V_1 + \mu V_2 = \lambda \partial_x + \mu \partial_x. \tag{2.49}$$

We substitute (2.49) into the invariant surface condition

$$\eta(x,t,u) - \xi(x,t,u)\frac{\partial u}{\partial x} - \tau(x,t,u)\frac{\partial u}{\partial t} = 0, \tag{2.50}$$

Table 2.2 Reduced equations

Subcase	ODEs
Arbitrary	$\mu^3 h''' + b\mu h^{2n} h' + a\mu h^n h' - \lambda h' + eh + f = 0$
1.1. a)	$\mu^3 h''' + k\mu h h' - \lambda h' + eh + f = 0$
1.1. b)	$\lambda\, e\, \exp(e\, z)\,(h' + f) - \mu\,(e^2\, x + k\, h' - e\, h\, k + f\, k) = 0$
2.1. a)	$\mu^3 h''' + k\mu h h' - \lambda h' + f = 0$
2.1. b)	$h' + (\lambda + \mu\, k\, z)\, f = 0$
2.1. c)	$3\, h''' + 3\, k\, h\, h' - h'\, z - 2\, h = 0$
2.2. a)	$\mu^3 h''' + b\mu h^{2n} h' - \lambda h' = 0$
2.2. b)	$3\, n\, h''' + 3\, k\, n\, h^{2n}\, h' - n\, h'\, z - h = 0$
2.3. a)	$\mu^3 h''' + \mu a h^n h' - \lambda h' = 0$
2.3. b)	$3\, n\, h''' + 3\, a\, n\, h^n\, h' - n\, h'\, z - 2\, h = 0$
2.4. a)	$\mu^3 h''' + b\mu h^2 h' + a\mu h h' - \lambda h' = 0$
2.4. b)	$3\, h''' + 3\, b\, h^2\, h' - h'\, z - h = 0$

and we obtain the similarity variable and the similarity solution

$$z = \mu x - \lambda t, \qquad u(x,t) = h(z). \tag{2.51}$$

Substituting (2.51) into (2.40) we obtain

$$\mu^3 h''' + b\mu h^{2n} h' + a\mu h^n h' - \lambda h' + eh + f = 0. \tag{2.52}$$

In order to obtain travelling wave solutions, we apply the simplest method to Eq. (2.52) with $n = 1$

$$\mu^3 h''' + b\mu h^2 h' + a\mu h h' - \lambda h' + eh + f = 0. \tag{2.53}$$

The simplest method applied to a PDE is the following: we consider a nonlinear PDE for $u(x,t)$ is in the form

$$\Delta(u, u_t, u_x, u_{xx}, \ldots) = 0,$$

where Δ is polynomial in $u(x,t)$ and its partial derivatives, in which the highest order derivatives and nonlinear terms are involved. In order to obtain exact solitary wave solutions of the equation, we have to pursue the following fundamental steps [27, 30]:

Step 1 We consider the travelling wave variable

$$u(x,t) = h(z) = h(\mu x + \lambda t), \tag{2.54}$$

where λ represents the speed of the travelling wave. The wave variable (2.54) carries Eq. (2.18) into the following ordinary differential equation

$$\Delta'(h, \alpha h', h', h'', \ldots) = 0. \tag{2.55}$$

Step 2 To seek the travelling wave solution of equation (2.53), we assume that (2.55) has a solution in the following form

$$h(z) = k_0 + k_1 Y + \cdots + k_N Y^N + \kappa_1 \left(\frac{Y'}{Y}\right) + \cdots + \kappa_N \left(\frac{Y'}{Y}\right)^N, \tag{2.56}$$

where k_j $(j = 0, 1, \ldots, N)$ and κ_j $(j = 1, 2, \ldots, N)$ are unknown constants to be calculated, and $Y(z)$ is the general solution of the Riccati equation:

$$Y'(z) + Y^2(z) - \alpha Y(z) - \beta = 0, \tag{2.57}$$

with α and β constants which must be determined.

Step 3 The positive integer N in (2.56) can be determined by taking into consideration the homogeneous balance between the highest order linear terms and the nonlinear terms of highest order occurring in (2.55).

Step 4 Inserting (2.56) and the derivatives h', h'', ..., into (2.55) we get a polynomial in $Y(z)$ and its derivatives. Equating the coefficients of the different powers of the function $Y(z)$ and equating them to zero, we obtain an overdetermined system of equations which must be solved to find k_j, κ_j and λ. This completes the determination of the solution of the ODE.

We will make use of this method to construct travelling wave solutions to the PDE (2.53). Taking the homogeneous balance between the highest order derivative h''' and the nonlinear term of highest order h^2h' we obtain $N = 2$. Therefore, the solution of (2.53) takes the following form

$$h = a_0 + a_1 Y, \tag{2.58}$$

where a_0, a_1, a_2 are constant to be determined and $Y(z)$ is the general solution of the Riccati equation:

$$Y'(z) + Y^2(z) - \alpha Y(z) - \beta = 0, \tag{2.59}$$

with α and β constants which must be determined. Inserting (2.58) and the derivatives h', h'', ..., into (2.53) we get a polynomial in $Y(z)$ and its derivatives. Requiring the vanishing of the coefficients of the different powers of the function $Y(z)$, we obtain an overdetermined system of equations

$$\begin{aligned}
a_0 e + f &= 0,\\
a a_0 a_1 \alpha\mu + a_0^2 a_1 b\alpha\mu + a_1 e + a_1\alpha^3\mu^3 - a_1\alpha\lambda &= 0,\\
-a a_0 a_1 \mu + a a_1^2 \alpha\mu - a_0^2 a_1 b\mu + 2a_0 a_1^2 b\alpha\mu - 7a_1\alpha^2\mu^3 + a_1\lambda &= 0,\\
-a a_1^2 \mu - 2a_0 a_1^2 b\mu + a_1^3 b\alpha\mu + 12a_1\alpha\mu^3 &= 0,\\
-a_1\mu\left(a_1^2 b + 6\mu^2\right) &= 0,
\end{aligned} \tag{2.60}$$

Solving (2.60) we obtain that this system is satisfied when $f = e = 0$ and

$$a_0 = \frac{a a_1^2 - 6 a_1 \alpha \mu^2}{12 \mu^2}. \tag{2.61}$$

Equation (2.59) with $\beta = 0$ is the Bernoulli equation, so we can obtain the corresponding solution h of the ODE (2.53) in terms of this equation. As a result, the solution of the Bernoulli equation is

$$Y(z) = \alpha \left(\frac{Y_1 + Y_2}{1 + Y_1 + Y_2} \right), \tag{2.62}$$

where $Y_1(z) = \sinh(\alpha(z + \delta))$, $Y_2(z) = \cosh(\alpha(z + \delta))$ and δ is an arbitrary constant. Substituting (2.62) into (2.58) we obtain the following solution

$$h(z) = \frac{1}{12} a_1 \left(\frac{a a_1}{\mu^2} + 6\alpha \tanh\left(\frac{1}{2} \alpha (c + z) \right) \right).$$

By transformation (2.51) we can obtain a solution of equation (2.40) with $e = f = 0$ and $n = 1$.

2.6 Conclusions

In this paper we have considered a generalized variable-coefficient Gardner equation. We have used the method of the multipliers, proved by Anco and Bluman, in order to obtain new results in conservation laws. Equivalence transformations are employed to improve the study of equation (2.2). It has been shown that Eq. (2.4) admits an infinite dimensional equivalence transformations group. This allows us to reduce equation (2.4) to a subclass (2.18) with fewer number of arbitrary functions. Classical symmetries of Eq. (2.18) have been obtained involving the different arbitrary functions which can be used to determine similarity and exact solutions. Symmetries obtained in this work generalize those already obtained by other authors in equations which belong to the family of equations (2.2), such as KdV equation and other Gardner equations with time-dependent coefficients. Furthermore, the arbitrary constants that appear in the functions allow us to suit physical phenomena which can be modeled by the considered equations. Finally, we have constructed some travelling wave solutions by using the modified simplest equation method.

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Chapter 3
On Symmetries and Conservation Laws for a Generalized Fisher–Kolmogorov–Petrovsky–Piskunov Equation



María Luz Gandarias, María de los Santos Bruzón, and María Rosa

3.1 Introduction

In recent years, many physicists and mathematicians have paid much attention to the Fisher–Kolmogorov–Petrovsky–Piskunov (FKPP) equations due to their importance in mathematical physics. The theory of reaction–diffusion has been developed since the 1930s. It was widely used for the description and explanation of various physical, chemical, biological phenomena and growth of solid tumors. Reaction–diffusion equations such as Fisher equation appear in a variety of problems ranging from population genetics to neurobiology and pattern formation. The theory of reaction–diffusion waves began in the 1930s with the works by Fisher [14] and Kolmogorov, Petrovsky, and Piskunov [24] on propagation of a dominant gene and by Zeldovich, Frank-Kamenetskii in combustion theory [41]. They introduced the scalar reaction–diffusion equation

$$u_t = u_{xx} + f(u), \tag{3.1}$$

they defined travelling wave solutions and they studied their existence, stability, and speed of propagation. These works were followed by many others where reaction–diffusion waves were studied in connection with various problems in chemistry, physics, biology, and medicine. This Eq. (3.1) is known as KPP equation, which is one of the simplest examples of a nonlinear reaction–diffusion equation.

For $f(u) = u^2(1-u)$, the KPP equation (3.1) is reduced to the well-known Huxley equation and it has been studied by Hodgkin and Huxley and Kolmogorov. Another important equation of this class is, for $f(u) = u(1-u^2)$, the Fitzhugh–Nagumo equation which arises in the study of nerve cells [15, 30].

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The classic simplest case of a nonlinear reaction–diffusion is the Fisher–Kolmogorov equation. This equation was introduced by Fisher in 1937 [14] to model the advance of advantageous gene through a geographic region. It is given by

$$u_t = D\,u_{xx} + \rho\left(1 - \frac{u}{K}\right) \tag{3.2}$$

where $u(x,t)$ represents the gene frequency at location x and time t, D is the diffusion coefficient, K is the carrying capacity that is the saturation value beyond which the population cannot grow anymore, and ρ is the proliferation. D, K, and ρ are positive constants.

The equations analyzed in this chapter are particular cases of a generalized FKPP equation with three variable coefficients, where g is the diffusion coefficient depending on the variable u, being x and t the independent variables, $f(u)$ an arbitrary function, and $c(x)$ an arbitrary function depending on the space variable x. Let $u(x,t)$ denote the density of tumor cells.

$$u_t = f(u) + \frac{1}{c(x)}\,(c(x)g(u)u_x)_x \tag{3.3}$$

For special cases Eq. (3.3) has been studied by other authors.

The transient heat conduction equation with a heat source term following a power law in a rectangular, cylindrical coordinate system has been considered by Moitsheki in [28] by using Lie classical symmetries. In [7] Bokhari et al. have considered the following particular case of (3.3) where $g(u) = u$ and $f(u) = u(1-u)$

$$u_t = u(1-u) + \frac{1}{x}\,[xuu_x]_x\,, \tag{3.4}$$

for which the authors derived an exact solution in terms of the Bessel functions by using Lie classical reductions.

In [8] Bokhari et al. have considered Eq. (3.3) but only when $g(u)$ is a linear function $g(u) = \alpha_1\,u + \alpha_2$ and $c(x) = x$. They state that a classification of (3.3) can only be achieved when g is linear in u. When f and g follow a power law they give a stationary solution. Over the last two decades a lot of attention has been paid to using Lie point symmetry methods to exploit the invariance of the generalized equation

$$u_t = (A(u)u_x)_x + B(u)u_x + C(u). \tag{3.5}$$

A complete Lie symmetry classification for the nonlinear heat equation (3.5) with $B = C = 0$ was described by Ovsiannikov in [32]. In [9], the authors constructed conservation laws for Eq. (3.5) associated with its Lie symmetries obtained in [10, 11].

Equation (3.5) with $B(u) = 0$ becomes the so-called density-dependent equation

$$u_t = f(u) + (g(u)u_x)_x\,, \tag{3.6}$$

which has been considered by J.D. Murray in [29]. Conservation laws for Eq. (3.6) were obtained in [19] using the property of nonlinear self-adjointness and the general theorem of conservation laws [23].

The aim of this chapter is to give a review of the generalized Fisher equation: symmetry groups and conservation laws, which have been derived in the last years by using Lie classical method, a conservation law theorem, and the multipliers method. These results have been published in several journals.

It is well known that the Lie symmetries, originally advocated by the Norwegian mathematician Sophus Lie in the beginning of the nineteenth century [27], have been widely applied to investigate nonlinear differential equations (including multi-component systems of partial differential equations (PDEs) and ordinary differential equations (ODEs)), notably, for constructing their exact and explicit solutions. It is known that conservation laws play a significant role in the solution process of an equation or a system of differential equations and its applications to PDEs [12, 16, 17, 19, 22, 23, 35, 38, 39]. In [2], Anco and Bluman gave a general algorithmic method to find all conservations laws for evolution equations like Eq. (3.3). Many recent papers using this method have been published [1, 18, 20, 21, 33].

Local symmetries admitted by a nonlinear PDE are also useful to discover whether or not the equation can be linearized by an invertible mapping and construct an explicit linearization when one exists. A nonlinear scalar PDE is linearizable by an invertible contact (point) transformation if and only if it admits an infinite-parameter Lie group of contact transformations satisfying specific criteria [3, 5, 6]. An obvious limitation of group-theoretic methods based on local symmetries, in their utility for particular PDEs, is that many of these equations do not have local symmetries. It turns out that PDEs can admit nonlocal symmetries whose infinitesimal generators depend on integrals of the dependent variables in some specific manner. It also happens that if a nonlinear scalar PDE does not admit an infinite-parameter Lie group of contact transformations, it is not linearizable by an invertible contact transformation. However, most of the interesting linearizations involve non-invertible transformations; such linearizations can be found by embedding given nonlinear PDEs in auxiliary systems of PDEs [5]. Krasilshchik and Vinogradov [25] gave criteria which must be satisfied by nonlocal symmetries of a PDE when realized as local symmetries of a system of PDEs which covers the given PDE.

In [5, 6], Bluman introduced a method to find a new class of symmetries for a PDE. By writing a given PDE, denoted by $R\{x, t, u\}$ in a conserved form, a related system denoted by $S\{x, t, u, v\}$ with potentials as additional dependent variables is obtained. Any Lie group of point transformations admitted by $S\{x, t, u, v\}$ induces a symmetry for $R\{x, t, u\}$; when at least one of the generators of the group depends explicitly on the potential, then the corresponding symmetry is neither a point nor a Lie–Bäcklund symmetry. These symmetries of $R\{x, t, u\}$ are called potential symmetries. The nature of potential symmetries allows one to extend the uses of point symmetries to such nonlocal symmetries. In particular:

(i) Invariant solutions of $S\{x, t, u, v\}$ yield solutions of $R\{x, t, u\}$ which are not invariant solutions for any local symmetry admitted by $R\{x, t, u\}$.

(ii) If $R\{x,t,u\}$ admits a potential symmetry leading to the linearization of $S\{x,t,u,v\}$, then $R\{x,t,u\}$ is linearized by a non-invertible mapping.

In [36], we have constructed conservation laws for Eq. (3.3). These conservation laws were derived by using a conservation theorem due to Ibragimov, as well as the multiplier method of Anco and Bluman [2].

In [37], we consider a particular case of Eq. (3.3) when $c(x) = k_1 e^{rx}$, $g(u) = k_2 u^{-2}$, and $f(u) = k_3 u$ with k_1, k_2, k_3, and r arbitrary constants

$$u_t = k_3 u + \frac{1}{k_1 e^{rx}} \left(k_1 e^{rx} k_2 u^{-2} u_x \right)_x . \tag{3.7}$$

By using the symmetry generators previously derived in [36], we perform the optimal system of subalgebras, the similarity reductions, and we find exact solutions of physical and chemical interest. For Eq. (3.7) it happens that some of the associated conserved systems admit symmetries that yield to potential symmetries. These symmetries are realized as local symmetries of a related auxiliary system, and lead to the construction of corresponding invariant solutions, as well as to the linearization of the equation by non-invertible mappings.

3.2 Lie Symmetries, Optimal Systems, and Reductions

In this section we perform a Lie group classification for Eq. (3.3)

$$u_t = f(u) + \frac{1}{c(x)} (c(x) g(u) u_x)_x$$

in the cases: (i) $c(x) = 1$, (ii) $c(x) = x$, and (iii) $c(x)$ follows a power law or an exponential law with $g(u) \neq 0$ and $c_x \neq 0$. That is, we classify all Lie symmetries depending on the forms of the arbitrary elements (functions $f(u)$ and $g(u)$). Let us consider a one-parameter Lie group of infinitesimal transformations in (x,t,u) given by

$$\begin{aligned} x^* &= x + \varepsilon \xi(x,t,u) + \mathscr{O}(\varepsilon^2), \\ t^* &= t + \varepsilon \tau(x,t,u) + \mathscr{O}(\varepsilon^2), \\ u^* &= u + \varepsilon \phi(x,t,u) + \mathscr{O}(\varepsilon^2), \end{aligned} \tag{3.8}$$

where ε is the group parameter. Then one requires that this transformation leaves invariant the set of solutions of Eq. (3.3). This yields to the overdetermined, linear system of eleven equations for the infinitesimals $\xi(x,t,u)$, $\tau(x,t,u)$, and $\phi(x,t,u)$. The associated Lie algebra of infinitesimal symmetries is the set of vector fields of the form

$$\mathbf{v} = \xi \frac{\partial}{\partial x} + \tau \frac{\partial}{\partial t} + \phi \frac{\partial}{\partial u}. \tag{3.9}$$

Having determined the infinitesimals, the symmetry variables are found by solving the invariant surface condition

$$\Phi \equiv \xi \frac{\partial u}{\partial x} + \tau \frac{\partial u}{\partial t} - \phi = 0. \tag{3.10}$$

After solving the determining equations for each case, we can distinguish different cases in which the symmetries are admitted by Eq. (3.3) for functional forms of $c(x)$, $f(u)$, and $g(u)$ as well as the corresponding generators are given below:

Case (i) For $c(x) = 1$
For $f(u)$ and $g(u)$ arbitrary, *the symmetries that are admitted* by (3.3) are

$$\mathbf{v}_1 = \frac{\partial}{\partial x}, \quad \mathbf{v}_2 = \frac{\partial}{\partial t},$$

For some special choices of the functions $f(u)$ and $g(u)$, $f \neq constant$ $g \neq constant$, it can be extended in the cases listed below.

(1) If $f(u) = u^m$ and $g(u) = u^n$ with $n \neq 0$ and $m \neq 0$, then besides $\mathbf{v}_1$ and $\mathbf{v}_2$ *the symmetry that is admitted* by (3.3) is

$$\mathbf{v}_3 = (n - m + 1)x\frac{\partial}{\partial x} + 2(1 - m)t\frac{\partial}{\partial t} + 2u\frac{\partial}{\partial u}.$$

(2) If $f(u) = c_2\, u^{n+1} - \dfrac{c_1\, u}{n}$ and $g(u) = u^n$ with $n \neq 0$, then besides $\mathbf{v}_1$ and $\mathbf{v}_2$ *the symmetry that is admitted* by (3.3) is

$$\mathbf{v}_4 = e^{c_1 t}\frac{\partial}{\partial t} - \frac{c_1 e^{c_1 t} u}{n}\frac{\partial}{\partial u}.$$

(3) If $f(u) = u^{-\frac{1}{3}}$ and $g(u) = u^{-\frac{4}{3}}$, then besides $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ *the symmetries that are admitted* by (3.3) are

$$\mathbf{v}_5 = e^{\frac{2x}{\sqrt{3}}}\frac{\partial}{\partial x} - \sqrt{3}e^{\frac{2x}{\sqrt{3}}}u\frac{\partial}{\partial u}, \quad \mathbf{v}_6 = e^{-\frac{2x}{\sqrt{3}}}\frac{\partial}{\partial x} + \sqrt{3}e^{-\frac{2x}{\sqrt{3}}}u\frac{\partial}{\partial u}.$$

(4) If $f(u) = e^{nu}$ and $g(u) = e^{mu}$ with $n \neq 0$ and $m \neq 0$, then besides $\mathbf{v}_1$ and $\mathbf{v}_2$ *the symmetry that is admitted* by (3.3) is

$$\mathbf{v}_7 = (m - n)x\frac{\partial}{\partial x} - 2nt\frac{\partial}{\partial t} + 2\frac{\partial}{\partial u}.$$

These results were published in [19] where we found the subclasses of Eq. (3.3) which are nonlinear self-adjoint.

Table 3.1 Functions and generators

i	f_i	g_i	$\mathbf{v}_k$
1	u^m	u^n	$\mathbf{v}_1, \mathbf{v}_2 = (n-m+1)x\partial_x + 2(1-m)t\partial_t + 2u\partial_u$
2	$\frac{u^{n+1}}{n+1}$	u^n	$\mathbf{v}_1, \mathbf{v}_3 = nt\partial_t - u\partial_u$
3	$c_2u^{n+1} - \frac{c_1u}{n}$	u^n	$\mathbf{v}_1, \mathbf{v}_4 = ne^{c_1t}\partial_t - c_1e^{c_1t}u\partial_u$
4	$-\frac{c_1u}{n}$	u^n	$\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4$
5	c_1u	u^{-1}	$\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4, \mathbf{v}_5 = (x\ \log(x) - x)\ \partial_x - 2\,u\ \log(x)\partial_u$
6	$c_2e^{nu} - \frac{c_1}{n}$	de^{nu}	$\mathbf{v}_1, \mathbf{v}_6 = ne^{c_1t}\partial_t - c_1e^{c_1t}\partial_u$
7	$\frac{-c_1}{n}$	de^{nu}	$\mathbf{v}_1, \mathbf{v}_6, \mathbf{v}_7 = nx\partial_x + 2\partial_u$
8	c_2e^{nu}	de^{nu}	$\mathbf{v}_1, \mathbf{v}_8 = nt\partial_x - \partial_u$

Table 3.2 Optimal systems

i	
1	$\mathbf{v}_1,\ \mathbf{v}_2$
2	$\mathbf{v}_1,\ \mathbf{v}_3$
3	$\mathbf{v}_1,\ \mathbf{v}_4$
4	$\mathbf{v}_1 + \alpha\mathbf{v}_2,\ a\mathbf{v}_2 + b\mathbf{v}_4,\ \mathbf{v}_2,\ \mathbf{v}_4$
5	$\mathbf{v}_1 + \mathbf{v}_2,\ a\mathbf{v}_2 + b\mathbf{v}_4,\ \mathbf{v}_1 + a\mathbf{v}_5,\ \mathbf{v}_4 + b\mathbf{v}_5,\ \mathbf{v}_5$
6	$\mathbf{v}_1,\ \mathbf{v}_6$
7	$\mathbf{v}_1 + \alpha\mathbf{v}_7,\ a\mathbf{v}_6 + b\mathbf{v}_7,\ \mathbf{v}_6,\ \mathbf{v}_7$
8	$\mathbf{v}_1,\ \mathbf{v}_8$

Case (ii) For $c(x) = x$

In ([34]), for this case, Eq. (3.3) is studied from the point of view of the theory of symmetry reductions in partial differential equations. A group classification is obtained. All the reductions are derived from the optimal system of subalgebras. Some of the reduced equations admit Lie symmetries which yield to further reductions.

The results obtained were as follows:

For $f(u)$ and $g(u)$ arbitrary, *the only symmetry that is admitted* by (3.3) is $\mathbf{v}_1 = \partial_t$. The functional forms of f and g which have extra symmetries are (Table 3.1). In order to construct the one-dimensional optimal system, following Olver in [31], we construct the corresponding generators of the optimal system of subalgebras (Table 3.2).

In Tables 3.3 and 3.4 respectively, similarity solutions and reductions of Eq. (3.3) to ODEs are obtained using the generators of the optimal system.

For $f(u)$ and $g(u)$ arbitrary functions the similarity variable and similarity solution are $z = x$, $u = h(z)$. Substituting them into (3.3) we obtain the reduced ODE, where $f = f(h)$ and $g = g(h)$ are arbitrary functions:

$$\frac{g\,h_x}{x} + g\,h_{x\,x} + f = 0.$$

Table 3.3 Similarity solutions

i	j	$\mathbf{w}_j$	z_j	u_j
1	1	$\mathbf{v}_2$	$t^{\frac{n-m+1}{2(m-1)}}\, x$	$\frac{h(z)}{t^{\frac{1}{m-1}}}$
2	2	$\mathbf{v}_3$	x	$h\,(z)\, t^{\frac{-1}{n}}$
3	3	$\mathbf{v}_4$	x	$h(z)\, e^{-\frac{c_1 t}{n}}$
4	4	$\mathbf{v}_2$	t	$h\,(z)\, x^{\frac{2}{n}}$
4	5	$\mathbf{v}_1 + \alpha \mathbf{v}_2$	$e^{-\alpha\, n\, t}\, x$	$h\,(z)\, e^{2\,\alpha\, t}$
4	6	$\mathbf{v}_2 + b\mathbf{v}_4$	$e^{\frac{e^{-c_1 t}}{b c_1}}\, x$	$h(z)\, e^{-\frac{2 e^{-c_1 t}}{b c_1 n} - \frac{c_1 t}{n}}$
5	7	$\mathbf{v}_5$	t	$\frac{h}{x^2\,(\log x - 1)^2}$
5	8	$\mathbf{v}_1 + a\mathbf{v}_5$	$\frac{\log(\log x - 1) - a\, t}{a}$	$\frac{h(z)}{(x\, \log x - x)^2}$
5	9	$\mathbf{v}_4 + b\mathbf{v}_5$	$\frac{e^{-c_1 t}}{c_1} - \frac{\log(\log x - 1)}{b}$	$h(z)\, e^{-2\, e^{\frac{b e^{-c_1 t}}{c_1} - b\, z} - \frac{2 b e^{-c_1 t}}{c_1} + c_1 t}$
6	10	$\mathbf{v}_6$	x	$h\,(z) - \frac{c_1 t}{n}$
7	11	$\mathbf{v}_1 + \alpha \mathbf{v}_7$	$e^{-\alpha\, n\, t}\, x$	$h\,(z) + \frac{2\, \log x}{n}$
7	12	$a\mathbf{v}_6 + b\mathbf{v}_7$	$e^{\frac{b e^{-c_1 t}}{c_1}}\, x$	$h\,(z) - \frac{2 b\, e^{-c_1 t}}{c_1 n} - \frac{c_1 t}{n}$
8	13	$\mathbf{v}_8$	x	$h\,(z) - \frac{\log t}{n}$

Table 3.4 Reduced ODEs

i	j	ODE_j
1	1	$-\frac{h_z\,(-n+m-1)\, z}{2 h^n\,(m-1)} - \frac{h_z}{z} - \frac{(h_z)^2\, n}{h} - \frac{h^{1-n}}{m-1} - h^{m-n} - h_{z\,z} = 0$
2	2	$\frac{h_z}{z} + \frac{h}{n+1} + \frac{(h_z)^2\, n}{h} + \frac{h^{1-n}}{n} + h_{z\,z} = 0$
3	3	$\frac{h'}{z} + \frac{(h')^2\, n}{h} + h'' + c_2\, h = 0$
4	4	$-\frac{4\, h^{n+1}\,(n+1)}{n^2} + \frac{c_1\, h}{n} + h_z = 0$
4	5	$\frac{a\, h_z\, n\, z}{h^n} + \frac{h_z}{z} - \frac{h^{1-n}\,(2\, a\, n + c_1)}{n} + \frac{(h_z)^2\, n}{h} + h_{z\,z} = 0$
4	6	$\frac{h_z\, z}{b\, h^n} + \frac{h_z}{z} + \frac{(h_z)^2\, n}{h} - \frac{2 h^{1-n}}{b\, n} + h_{z\,z} = 0$
5	7	$h_z - c_1\, h - 2 = 0$
5	8	$h\, h_{zz} - (h_z)^2 + \left(a^2\, h^2 - a\, h\right)\, h_z + a^2\, c_1\, h^3 + 2\, a^2\, h^2 = 0$
5	9	$h\, h_{zz}\, e^{2\, b\, z} - (h_z)^2\, e^{2\, b\, z} + b\, h\, h_z\, e^{2\, b\, z} + e^2\, b^2\, h^2\, h_z - 2\, e^2\, b^3\, h^3 = 0$
6	10	$d\, (h_z)^2\, n\, z + d\, h_{zz}\, z + c_2\, z + d\, h_z = 0$
7	11	$e^{h\, n}\, \left(\left(d\, (h_z)^2\, n^2 + d\, h_{zz}\, n\right)\, z^2 + 5\, d\, h_z\, n\, z + 4\, d\right) - c_1 = 0$
7	12	$e^{h\, n}\, \left(\left(d\, (h_z)^2\, n^2 + d\, h_{zz}\, n\right)\, z + d\, h_z\, n\right) - 2\, b\, z = 0$
8	13	$e^{h\, n}\, \left(\left(d\, (h_z)^2\, n^2 + (d\, h_{zz} + c_2)\, n\right)\, z + d\, h_z\, n\right) + z = 0$

3.2.1 Some Further Reductions and Exact Solutions

In this section we give the exact solutions for some of the reduced equations in the case (ii), by some method of integration or reducing the order again using Lie classical method.

3.2.1.1 ODE_1

Among our reduced equations we have one equation related to this type of solution. For the reduced ODE_1 with

$$z = t^{\frac{n-m+1}{2(m-1)}} x,\ u = \frac{h}{t^{\frac{1}{m-1}}}.$$

In fact if we choose the parameter $n = m - 3$ the similarity variable has the form

$$u = \frac{1}{t^{\frac{1}{m-1}}} h\left(\frac{x}{t^{\frac{1}{m-1}}}\right)$$

For $m > 1$ it is clear that $u(x, t) \to \delta(x)$ as $t \to 0$ and the similarity solution is a source solution.

3.2.1.2 ODE_3

$$\frac{h'}{z} + \frac{(h')^2\, n}{h} + h'' + c_2\, h = 0. \tag{3.11}$$

Equation (3.11) admits the symmetry generator $\mathbf{w} = \dfrac{\partial}{\partial h}$. Taking into account the invariants of its first prolongation and using the new variables

$$\zeta = z,\ w = \frac{h'}{h}, \tag{3.12}$$

we obtain that Eq. (3.11) can be reduced to the first order Riccati equation

$$\frac{w}{z} + w_z + n\, w^2 + w^2 + c_2 = 0. \tag{3.13}$$

In order to solve Eq. (3.12) we substitute $w = \dfrac{y'}{(n+1)\, y}$ to get

$$\frac{y_z}{z} + y_{z\,z} + c_2\ (n+1)\ y = 0, \tag{3.14}$$

which is a Bessel equation whose general solution is

$$y = k_1 J_0(kz) + k_2 Y_0(kz).$$

Consequently

$$h = (k_1 J_0(kz) + k_2 Y_0(kz))^{1/n+1},$$

where $k = \sqrt{c_2(n+1)}$. Finally using the substitutions given in Table 3.3 we obtain an exact solution of the generalized Fisher equation in cylindrical coordinates, given by

$$u = e^{-\frac{c_1 t}{n}} (k_1 J_0(kx) + k_2 Y_0(kx))^{1/n+1}.$$

This solution for the particular case $n = 1$ has been derived in [7].

3.2.1.3 ODE$_4$

The reduced equation

$$-\frac{4\,h^{n+1}\,(n+1)}{n^2} + \frac{c_1\,h}{n} + h_z = 0, \tag{3.15}$$

can be integrated by quadratures. The solution is

$$h\,(z) = \left(\frac{4 + 4\,n + \mathrm{e}^{c_1\,z} k_1\,n c_1}{n c_1}\right)^{\frac{-1}{n}}, \tag{3.16}$$

where $k_1 \in R$. Using the substitutions given in Table 3.3 we obtain an exact solution of (3.3) for $c(x) = x$, given by

$$u\,(x,t) = x^{2/n} \left(\frac{4 + 4\,n + \mathrm{e}^{c_1\,t} k_1\,n c_1}{n c_1}\right)^{\frac{-1}{n}}. \tag{3.17}$$

3.2.1.4 ODE$_5$

$$\frac{h_z}{z} + \frac{(h_z)^2\,n}{h} + h_{z\,z} = 0 \tag{3.18}$$

For the reduced Eq. (3.18), we realize that the ODE admits the group corresponding to the generator

$$\mathbf{w} = z\frac{\partial}{\partial z} + h\frac{\partial}{\partial h}. \tag{3.19}$$

Taking into account the invariants of its first prolongation and using the new variables

$$w(\zeta) = h_z,\ \zeta = \frac{h}{z}, \tag{3.20}$$

we obtain that Eq. (3.18) can be reduced to the first order ODE

$$(w-\zeta)w_\zeta + w + \frac{nw^2}{\zeta} = 0, \tag{3.21}$$

whose implicit solution is

$$\zeta + \frac{w\,(t)}{n} - k_1\,(w\,(\zeta))^{\frac{1}{n+1}} = 0, \tag{3.22}$$

where $k_1 \in R$.

Setting $h = y^{1/(n+1)}$ into (3.18), we get

$$\frac{y_z}{z} + y_{z\,z} = 0,$$

whose solution is

$$y = k_1\,\log z + k_2,$$

$$h = (k_1\,\log z + k_2)^{(\frac{1}{n+1})}.$$

From Table 3.3 we get

$$u = e^{-\frac{c_1 t}{n}}(k_1\,\log x + k_2)^{(\frac{1}{n+1})}.$$

3.2.1.5 ODE$_8$

Simplest Equation Method

The reduced equation ODE$_7$

$$h\,h_{zz} - (h_z)^2 + \left(a^2\,h^2 - a\,h\right)\,h_z + a^2\,c_1\,h^3 + 2\,a^2\,h^2 = 0 \tag{3.23}$$

is an ODE that admits the symmetry generator

$$\mathbf{w} = \frac{\partial}{\partial h}, \tag{3.24}$$

which does not yield any symmetry reduction. However Eq. (3.23) is an autonomous equation and we can apply the modified method of the simplest equation which has been developed by Kudryashov [26]. The Kudryashov version of the method contains a procedure analogous to the first step of the test for the Painleve property. Frequently this method is applied for travelling wave solutions.

We apply the modified Kudryashov method [40] to Eq. (3.23), taking the Riccati equation as the simplest equation, we suppose that the solutions can be expressed by a polynomial in y in the form

$$h = \sum_{i=0}^{n} a_i \, (y)^i \, , \tag{3.25}$$

where $y = y(z)$ satisfies the Riccati equation

$$y_z - k_2 y^2 - k_1 y - k_0 = 0, \tag{3.26}$$

$a_i, i = 0, \ldots, n, k_i \; i = 1, \ldots, 2$ are constants to be determined later.

Considering the homogeneous balance between the highest derivative term and the highest power law term, we get $n = 1$ we can write (3.25) as

$$h = a_0 + a_1 y, \qquad a_1 \neq 0. \tag{3.27}$$

By substituting (3.27) with

$$y_z = k_2 y^2 + k_1 y + k_0,$$

$$y_{zz} = 2k_2 y + k_1,$$

into (3.23), we get setting equal zero the coefficients of the different powers of y the following conditions:

$$\begin{aligned}
&{a_1}^2 k_2 \left(k_2 + a^2 a_1\right) = 0,\\
&2\,a_0 {k_2}^2 + a_1 k_1 k_2 + 2\,a^2 a_0 a_1 k_2 - a\,a_1 k_2 + a^2 {a_1}^2 k_1 + a^2 {a_1}^2 c_1 = 0,\\
&3\,a_0 k_1 k_2 + a^2 {a_0}^2 k_2 - a\,a_0 k_2 + 2\,a^2 a_0 a_1 k_1 - a\,a_1 k_1 + a^2 {a_1}^2 k_0\\
&\quad +3\,a^2 a_0 a_1 c_1 + 2\,a^2 a_1 = 0,\\
&2\,a_0 k_0 k_2 + a_0 {k_1}^2 - a_1 k_0 k_1 + a^2 {a_0}^2 k_1 - a\,a_0 k_1 + 2\,a^2 a_0 a_1 k_0 - a\,a_1 k_0\\
&\quad +3\,a^2 {a_0}^2 c_1 + 4\,a^2 a_0 = 0,\\
&a_0 a_1 k_0 k_1 - {a_1}^2 {k_0}^2 + a^2 {a_0}^2 a_1 k_0 - a\,a_0 a_1 k_0 + a^2 {a_0}^3 c_1 + 2\,a^2 {a_0}^2 = 0.
\end{aligned} \tag{3.28}$$

By solving this system we get

$$a = -c_1, \quad k_2 = -c_1^2\, a_1, \quad k_0 = -\frac{a_0\, c_1\ (a_0\, c_1 + 2)}{a_1}, \quad k_1 = -2\, a_0\, c_1^2 - 2\, c_1. \tag{3.29}$$

The Riccati equation (3.26) becomes

$$y_z + a_1\, c_1^2\, y^2 - \left(-2\, a_0\, c_1^2 - 2\, c\right)\, y + \frac{a_0\, c_1\ (a_0\, c_1 + 2)}{a_1},$$

whose solution is

$$y = -\frac{(a_0\, c_1 + 2)\ e^{-2\, c_1\, z - 2\, k\, c_1} - a_0}{a_1\, c_1\, e^{-2\, c_1\, z - 2\, k\, c_1} - a_1}$$

and

$$h = \frac{2}{e^{2\, c_1\, z + 2\, k\, c_1} - c_1},$$

where k is the integrating constant. Finally using the substitutions given in Table 3.3 we obtain an exact solution of the generalized Fisher equation in cylindrical coordinates, given by

$$u = \frac{2}{(x\ \log x - x)^2\ \left(e^{2\, c_1\, k - 2\, (\log(\log x - 1) + c_1\, t)} - c_1\right)}.$$

For $c_1 < 0, x > 0$ it is a bounded solution.

3.2.1.6 $\mathbf{ODE_{10}}$

The reduced equation

$$d\ (h_z)^2\ n\, z + d\, h_{zz}\, z + c_2\, z + d\, h_z = 0. \tag{3.30}$$

is an ODE that admits the symmetry generator

$$\mathbf{w} = \frac{\partial}{\partial h}, \tag{3.31}$$

which does not yield any reduction. However in Eq. (3.30) the terms in h are missing allowing the substitution $h' = w$ and becoming the Riccati equation:

$$w_z + n w^2 + \frac{c}{d} + \frac{w}{z} = 0.$$

The solution to Eq. (3.30) can be given in terms of Bessel functions as:

$$h(z) = \frac{\ln\left(d\,n\,(\mathrm{J}_0(p)k_1 - k_2\,\mathrm{Y}_0(p))^2\,{c_2}^{-1}z^{-2}\,(\mathrm{J}_1(p)\mathrm{Y}_0(p) - \mathrm{Y}_1(p)\mathrm{J}_0(p))^{-2}\right)}{2n}, \tag{3.32}$$

being $p = \sqrt{\frac{c_2\,n}{d}}$, $k_1,\ k_2 \in R$.

Using the corresponding substitution given in Table 3.3 we obtain that an exact solution of (3.3) is

$$u = h(x) - \frac{c_1 t}{n}$$

where $h(x)$ is given by (3.32).

3.2.1.7 $\mathbf{ODE_{11}}$

The reduced equation

$$e^{h\,n}\left(\left(d\ (h_z)^2\ n^2 + d\,h_{zz}\,n\right)z^2 + 5\,d\,h_z\,n\,z + 4\,d\right) - c_1 = 0, \tag{3.33}$$

can be integrated by quadrature. The solution is

$$h(z) = \ln\left(\frac{c_1\,z^2 + 4\,k_1\,d\,n - 4\,k_2\,d\,n\ln(z)}{4dz^2}\right)n^{-1}, \tag{3.34}$$

where $k_1,\ k_2 \in R$.

3.2.1.8 $\mathbf{ODE_{13}}$

$$e^{h\,n}\left(\left(d\ (h_z)^2\ n^2 + (d\,h_{zz} + c_2)\ n\right)z + d\,h_z\,n\right) + z = 0. \tag{3.35}$$

The solution to Eq. (3.35) can be given in term of Bessel functions as

$$h(z) = \frac{\ln(H)^{-1}}{n}, \tag{3.36}$$

being $H = -\frac{\mathrm{J}_0(p)z\mathrm{Y}_1(p)+\mathrm{J}_0(p)k_1\,d\,n\frac{p}{z}-z\mathrm{J}_1(p)\mathrm{Y}_0(p)-k_2\,d\,n\frac{p}{z}\mathrm{Y}_0(p)}{zc_2n(-\mathrm{J}_1(p)\mathrm{Y}_0(p)+\mathrm{Y}_1(p)\mathrm{J}_0(p))}$, $p = \sqrt{\frac{c_2\,n}{d}}$ and $k_1,\ k_2 \in R$.

Table 3.5 Functions c_i, g_i, f_i with $i = 1 \cdots 9$ and generators $\mathbf{v}_k$ with $k = 1 \cdots 9$

i	c_i	g_i	f_i	$\mathbf{v}_k$
1	Arbitrary	Arbitrary	Arbitrary	$\mathbf{v}_1 = \partial_t$
2	$k_1 x^r$	$k_2 u^q$	$k_3 u^p$	$\mathbf{v}_1, \mathbf{v}_2 = (q+1-p)x\partial_x + 2(1-p)t\partial_t + 2u\partial_u$
3	$k_1 x^r$	$k_2 u^q$	$k_3 u$	$\mathbf{v}_1, \mathbf{v}_2, = qx\partial_x + 2u\partial_u, \mathbf{v}_3 = e^{-k_3 q t}\partial_t + k_3 e^{-k_3 q t} u\partial_u$
4	$k_1 e^{rx}$	$k_2 u^q$	$k_3 u^p$	$\mathbf{v}_1, \mathbf{v}_4 = \partial_x$
5	$k_1 e^{rx}$	$\frac{k_2}{u^2}$	$k_3 u$	$\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5 = \dfrac{e^{-rx}}{r}\partial_x + e^{-rx} u\partial_u$
6	$k_1 e^{rx}$	k_2	$k_3 u^p$	$\mathbf{v}_1, \mathbf{v}_4, \mathbf{v}_6 = \dfrac{(-x + k_2 rt)\,(p-1)}{2}\partial_x - (p-1)t\partial_t + u\partial_u$
7	$k_1 e^{rx}$	$k_2 e^{qu}$	$k_3 e^{pu}$	$\mathbf{v}_1, \mathbf{v}_4\ (p \neq q)$
8	$k_1 e^{rx}$	$k_2 e^{qu}$	$\frac{2k_2 e^{qu} r^2}{9q}$	$\mathbf{v}_1, \mathbf{v}_4, \mathbf{v}_7, = \frac{-3qe^{\frac{-rx}{3}}}{2r}\partial_x + e^{\frac{-rx}{3}}\partial_u,$ $\mathbf{v}_8 = -qt\partial_t + \partial_u$
9	$k_1 e^{rx}$	k_2	$k_3 e^{pu}$	$\mathbf{v}_1, \mathbf{v}_4, \mathbf{v}_9 = \dfrac{p\,(-x + k_2 rt)}{2}\partial_x - pt\partial_t + \partial_u$

Finally using the substitutions given in Table 3.3 we obtain an exact solution of the generalized Fisher equation in cylindrical coordinates, given by

$$u(x,t) = h(x) - \frac{logt}{n}$$

where $h(x)$ is given by (3.36).

Case (iii) For $c(x)$ following a power law or an exponential law, the functional forms of $c(x)$, $f(u)$ and $g(u)$ as well as the corresponding generators are given in Table 3.5.

These results were published in [36] where conservation laws were found using two different approaches.

3.2.2 *Conservation Laws for Eq. (3.3)*

In [2] Anco and Bluman gave a general treatment of a direct conservation law method for partial differential equations expressed in a standard Cauchy–Kovalevskaya form, in particular for evolution equations

$$u_t = G(x, u, u_x, u_{xx}, \ldots, u_{nx}).$$

The nontrivial conservation laws are characterized by a multiplier λ with no dependence on u_t satisfying

$$\hat{E}[u]\,(\Lambda u_t - \Lambda G(x, u, u_x, u_{xx}, \ldots, u_{nx})) = 0.$$

Here

$$\hat{E}[u] := \frac{\partial}{\partial u} - D_t \frac{\partial}{\partial u_t} - D_x \frac{\partial}{\partial u_x} + D_x^2 \frac{\partial}{\partial u_{xx}} + \ldots.$$

The conserved current must satisfy

$$\Lambda = \hat{E}[u]\Phi^t$$

and the flux Φ^x is given by [13]

$$\Phi^x = -D_x^{-1}(\Lambda G) - \frac{\partial \Phi^t}{\partial u_x} G + G D_x \left(\frac{\partial \Phi^t}{\partial u_{xx}} \right) + \ldots.$$

The conservation law will be written

$$D_t(\Phi^t) + D_x(\Phi^x) = 0.$$

For Eq. (3.3) we get the following multipliers and the corresponding conserved densities and fluxes:

1. For $f(u) = u,\ g(u) = u^q,\ c(x) = e^x$.

$$\begin{cases} \Lambda = e^{x-t}, & \Lambda = e^{-t}, \\ \phi^t = e^{x-t}u, & \phi^t = -e^{-t}u, \\ \phi^x = -e^{x-t}u^q u_x, & \phi^x = -e^{-t}\, u^q\, u_x - \frac{e^{-t}\, u^{q+1}}{q+1}. \end{cases}$$

2. For $f(u) = u,\ g(u) = u^q,\ c(x) = x^r$.

$$\begin{cases} \Lambda = x^r e^{-t}, & \Lambda = x e^{-t}, \\ \phi^t = x^r e^{-t}u, & \phi^t = x e^{-t}u, \\ \phi^x = -e^{-t}\, u^q\, u_x\, x^r, & \phi^x = -e^{-t} u^q\, u_x x - \frac{(r-1)\, e^{-t} u^{q+1}}{q+1}. \end{cases}$$

3. For $f(u) = au^p,\ g(u) = pu^{(p-1)},\ c(x) = e^{rx}$, we get that Λ must satisfy

$$\Lambda_{xx} - r\Lambda_x + a\Lambda = 0, \tag{3.37}$$

the solutions with $\Delta = r^2 - 4a$ are

$$\Lambda = \begin{cases} k_1 x e^{rx/2} + k_2 e^{rx/2} & \Delta = 0 \\ k_1\, e^{\frac{(\sqrt{\Delta}+r)\,x}{2}} + k_2\, e^{\frac{(r-\sqrt{\Delta})\,x}{2}} & \Delta > 0 \\ e^{\frac{rx}{2}} \left(k_1\, \sin\left(\frac{\sqrt{-\Delta}\,x}{2}\right) + k_2\, \cos\left(\frac{\sqrt{-\Delta}\,x}{2}\right)\right) & \Delta < 0 \end{cases}$$

- For $\Delta = 0$ we obtain the following:

$$\begin{cases} \Lambda = x e^{\frac{rx}{2}}, \\ \phi^t = x e^{\frac{x}{2}} u, \\ \phi^x = -p\, u^{p-1}\, u_x\, x\, e^{\frac{rx}{2}} - \dfrac{2\,a\,u^p\,x\,e^{\frac{rx}{2}}}{r} + \dfrac{4\,a\,u^p\,e^{\frac{rx}{2}}}{r^2} \end{cases}.$$

and

$$\begin{cases} \Lambda = e^{\frac{rx}{2}}, \\ \phi^t = u\, e^{\frac{rx}{2}}, \\ \phi^x = -p\, u^{p-1}\, u_x\, e^{\frac{rx}{2}} - \dfrac{2\,a\,u^p\,e^{\frac{rx}{2}}}{r}. \end{cases}$$

- For $\Delta > 0$ we obtain the following multipliers and the corresponding conserved densities and fluxes:

$$\begin{cases} \Lambda = e^{\frac{\left(\sqrt{\Delta}+r\right)x}{2}}, \\ \phi^t = e^{\frac{\left(\sqrt{\Delta}+r\right)x}{2}} u, \\ \phi^x = \dfrac{\left(\sqrt{\Delta}-r\right)\,u^p\,e^{\frac{\sqrt{\Delta}x}{2}+\frac{rx}{2}}}{2\,a} - \dfrac{p\,u^{p-1}\,u_x\,e^{\frac{\sqrt{\Delta}x}{2}+\frac{rx}{2}}}{a} \end{cases}$$

and

$$\begin{cases} \Lambda = e^{\frac{\left(r-\sqrt{\Delta}\right)x}{2}}, \\ \phi^t = u\, e^{\frac{\left(r-\sqrt{\Delta}\right)x}{2}}, \\ \phi^x = -p\, u^{p-1}\, u_x\, e^{\frac{rx}{2}-\frac{\sqrt{\Delta}x}{2}} - \frac{\sqrt{\Delta}\,u^p\,e^{\frac{rx}{2}-\frac{\sqrt{\Delta}x}{2}}}{2} - \frac{r\,u^p\,e^{\frac{rx}{2}-\frac{\sqrt{\Delta}x}{2}}}{2}. \end{cases}$$

- For $\Delta < 0$ we obtain the following multipliers and the corresponding conserved densities and fluxes:

$$\begin{cases} \Lambda = e^{\frac{rx}{2}}\, \sin\left(\dfrac{\sqrt{-\Delta}\,x}{2}\right), \\ \phi^t = u\, e^{\frac{rx}{2}}\, \sin\left(\dfrac{\sqrt{-\Delta}\,x}{2}\right), \\ \phi^x = -\dfrac{r\,u^p\,e^{\frac{rx}{2}}\,\sin\left(\frac{\sqrt{-\Delta}\,x}{2}\right)}{2} + \dfrac{\sqrt{-\Delta}\,u^p\,e^{\frac{rx}{2}}\,\cos\left(\frac{\sqrt{-\Delta}\,x}{2}\right)}{2} \\ \qquad -p\,u^{p-1}\,u_x\,e^{\frac{rx}{2}}\,\sin\left(\dfrac{\sqrt{-\Delta}\,x}{2}\right) \end{cases}$$

and

$$\begin{cases} \Lambda = e^{\frac{r\,x}{2}}\,\cos\left(\dfrac{\sqrt{-\Delta}\,x}{2}\right), \\ \phi^t = u\,e^{\frac{r\,x}{2}}\,\cos\left(\dfrac{\sqrt{-\Delta}\,x}{2}\right), \\ \phi^x = -\dfrac{\sqrt{-\Delta}\,u^p\,e^{\frac{r\,x}{2}}\,\sin\left(\frac{\sqrt{-\Delta}\,x}{2}\right)}{2} - \dfrac{r\,u^p\,e^{\frac{r\,x}{2}}\,\cos\left(\frac{\sqrt{-\Delta}\,x}{2}\right)}{2} \\ \qquad\quad -p\,u^{p-1}\,u_x\,e^{\frac{r\,x}{2}}\,\cos\left(\dfrac{\sqrt{-\Delta}\,x}{2}\right). \end{cases}$$

4. For $f(u) = ae^{pu}$, $g(u) = pe^{pu}$, $c(x) = e^{rx}$, we get that Λ must satisfy condition (3.37) with $\Delta = r^2 - 4a$ and we obtain the following:

- For $\Delta = 0$:

$$\begin{cases} \Lambda &= xe^{\frac{rx}{2}}, \\ \phi^t &= xe^{\frac{x}{2}}u, \\ \phi^x &= -p\,u_x\,x\,e^{\frac{r\,x}{2}+p\,u} - \dfrac{r\,x\,e^{\frac{r\,x}{2}+p\,u}}{2} + e^{\frac{r\,x}{2}+p\,u} \end{cases}$$

and

$$\begin{cases} \Lambda &= e^{\frac{rx}{2}}, \\ \phi^t &= u\,e^{\frac{r\,x}{2}}, \\ \phi^x &= -p\,u_x\,e^{\frac{r\,x}{2}+p\,u} - \dfrac{r\,e^{\frac{r\,x}{2}+p\,u}}{2}. \end{cases}$$

- For $\Delta > 0$:

$$\begin{cases} \Lambda = e^{\frac{\left(r+\sqrt{\Delta}\right)x}{2}}, \\ \phi^t = e^{\frac{\left(\sqrt{\Delta}+r\right)x}{2}}u, \\ \phi^x = -pu_x e^{\frac{rx}{2}+\frac{\sqrt{\Delta}x}{2}+pu} - \dfrac{re^{\frac{rx}{2}+\frac{\sqrt{\Delta}x}{2}+pu}}{2} + \dfrac{\sqrt{\Delta}e^{\frac{rx}{2}+\frac{\sqrt{\Delta}x}{2}+pu}}{2} \end{cases}$$

and

$$\begin{cases} \Lambda = e^{\frac{\left(r-\sqrt{\Delta}\right)x}{2}}, \\ \phi^t = e^{\frac{\left(r-\sqrt{\Delta}\right)x}{2}}u, \\ \phi^x = -pu_x e^{\frac{rx}{2}-\frac{\sqrt{\Delta}x}{2}+pu} - \dfrac{re^{\frac{rx}{2}-\frac{\sqrt{\Delta}x}{2}+pu}}{2} - \dfrac{\sqrt{\Delta}e^{\frac{rx}{2}-\frac{\sqrt{\Delta}x}{2}+pu}}{2}. \end{cases}$$

- For $\Delta < 0$:

$$\begin{cases} \Lambda = e^{\frac{r\,x}{2}}\,\sin\left(\dfrac{\sqrt{-\Delta}\,x}{2}\right), \\ \phi^t = u\,e^{\frac{r\,x}{2}}\,\sin\left(\dfrac{\sqrt{-\Delta}\,x}{2}\right), \\ \phi^x = -\dfrac{r\,e^{\frac{r\,x}{2}+p\,u}\,\sin\left(\frac{\sqrt{-\Delta}\,x}{2}\right)}{2} + \dfrac{\sqrt{-\Delta}\,e^{\frac{r\,x}{2}+p\,u}\,\cos\left(\frac{\sqrt{-\Delta}\,x}{2}\right)}{2} \\ \qquad -p\,u_x\,e^{\frac{r\,x}{2}+p\,u}\,\sin\left(\dfrac{\sqrt{-\Delta}\,x}{2}\right) \end{cases}$$

and

$$\begin{cases} \Lambda = e^{\frac{r\,x}{2}}\,\cos\left(\dfrac{\sqrt{-\Delta}\,x}{2}\right), \\ \phi^t = u\,e^{\frac{r\,x}{2}}\,\cos\left(\dfrac{\sqrt{-\Delta}\,x}{2}\right), \\ \phi^x = -\dfrac{r\,e^{\frac{r\,x}{2}+p\,u}\,\cos\left(\frac{\sqrt{-\Delta}\,x}{2}\right)}{2} - \dfrac{\sqrt{-\Delta}\,e^{\frac{r\,x}{2}+p\,u}\,\sin\left(\frac{\sqrt{-\Delta}\,x}{2}\right)}{2} \\ \qquad -p\,u_x\,e^{\frac{r\,x}{2}+p\,u}\,\cos\left(\dfrac{\sqrt{-\Delta}\,x}{2}\right). \end{cases},$$

5. For $f(u) = au^p$, $g(u) = pu^p$, $c(x) = x^r$.

$$\begin{aligned} \Lambda &= \mathrm{J}\left(\tfrac{|r-1|}{2}, \sqrt{a}\,x\right) x^{\frac{r}{2}+\frac{1}{2}}, \\ \phi^t &= \mathrm{J}\left(\tfrac{|r-1|}{2}, \sqrt{a}\,x\right) x^{\frac{r}{2}+\frac{1}{2}}\,u, \\ \phi^x &= \frac{1}{2}\sqrt{a}\,u^p\,x^{\frac{r}{2}+\frac{1}{2}}\,B_7 - \frac{1}{2}\sqrt{a}\,u^p\,x^{\frac{r}{2}+\frac{1}{2}}\,B_6 - p\,u^{p-1}\,u_x\,x^{\frac{r}{2}+\frac{1}{2}}B_4 \\ &\quad -\tfrac{r}{2}u^p\,x^{\frac{r}{2}-\frac{1}{2}}B_4 + \tfrac{1}{2}u^p x^{\frac{r}{2}-\frac{1}{2}}B_4 \\ &\quad +\frac{1}{4}\int x^{\frac{r+1}{2}}\,u^p\,a\,(-B_5 - 2B_4 - B_3)dx + \frac{1}{4}\int x^{\frac{r-3}{2}}u^p B_4\left(r^2 - 2r + 1\right)dx \\ &\quad +\frac{1}{4}\int \sqrt{a}\,u^p\,x^{\frac{r}{2}-1}\,(2\,B_1 - 2\,J_2)\,dx, \end{aligned}$$

with

$$\begin{aligned} &B_1 = J\left(\tfrac{|r-1|+2}{2}, \sqrt{a}\,x\right),\ B_2 = J\left(\tfrac{|r-1|-2}{2}, \sqrt{a}\,x\right),\ B_3 = J\left(\tfrac{|r-1|+4}{2}, \sqrt{a}\,x\right), \\ &B_4 = J\left(\tfrac{|r-1|}{2}, \sqrt{a}\,x\right), \\ &B_5{=}J\left(\tfrac{|r-1|-4}{2}, \sqrt{a}\,x\right),\quad B_6{=}J\left(\tfrac{|r-1|}{2}+1, \sqrt{a}\,x\right),\ B_7{=}J\left(\tfrac{|r-1|}{2}-1, \sqrt{a}\,x\right). \end{aligned}$$

$$
\begin{aligned}
\Lambda &= \mathrm{Y}\left(\tfrac{|r-1|}{2},\ \sqrt{a}x\right)x^{\frac{1}{2}+\frac{1}{2}r},\\
\phi^t &= \mathrm{Y}\left(\tfrac{|r-1|}{2},\ \sqrt{a}x\right)x^{\frac{1}{2}+\frac{1}{2}r}u,\\
\phi^x &= \frac{1}{2}\sqrt{a}\,u^p\,x^{\frac{r}{2}+\frac{1}{2}}\,B_7 - \frac{1}{2}\sqrt{a}\,u^p\,x^{\frac{r}{2}+\frac{1}{2}}\,B_6 - p\,u^{p-1}u_x\,x^{\frac{r}{2}+\frac{1}{2}}B_4\\
&\quad -\tfrac{r}{2}u^p\,x^{\frac{r}{2}-\frac{1}{2}}B_4 + \tfrac{1}{2}u^p x^{\frac{r}{2}-\frac{1}{2}}B_4\\
&\quad +\frac{1}{4}\int x^{\frac{r+1}{2}}u^p a\,(-B_5-2B_4-B_3)\,dx + \frac{1}{4}\int x^{\frac{r-3}{2}}\,u^p\,B_4\left(r^2-2r+1\right)dx\\
&\quad +\frac{1}{4}\int \sqrt{a}\,u^p\,x^{\frac{r}{2}-1}\ (2B_1-2B_2)\,dx
\end{aligned}
$$

with

$$
\begin{aligned}
&B_1 = Y\left(\tfrac{|r-1|+2}{2},\sqrt{a}\,x\right),\ B_2 = Y\left(\tfrac{|r-1|-2}{2},\sqrt{a}\,x\right),B_3 = Y\left(\tfrac{|r-1|+4}{2},\sqrt{a}\,x\right),\\
&B_4 = Y\left(\tfrac{|r-1|}{2},\sqrt{a}\,x\right),\\
&B_5{=}Y\left(\tfrac{|r-1|-4}{2},\sqrt{a}\,x\right),\quad B_6{=}Y\left(\tfrac{|r-1|}{2}+1,\sqrt{a}\,x\right),\ B_7{=}Y\left(\tfrac{|r-1|}{2}-1,\sqrt{a}\,x\right).
\end{aligned}
$$

3.3 Classical Potential Symmetries

In [4], Bluman introduced a method to find a new class of symmetries for a PDE. Suppose a given scalar PDE of second order

$$F(x,t,u,u_x,u_t,u_{xx},u_{xt},u_{tt}) = 0, \tag{3.38}$$

where the subscripts denote the partial derivatives of u, can be written as a conservation law

$$\frac{D}{Dt}f(x,t,u,u_x,u_t) - \frac{D}{Dx}g(x,t,u,u_x,u_t) = 0, \tag{3.39}$$

for some functions f and g of the indicated arguments. Here $\frac{D}{Dx}$ and $\frac{D}{Dt}$ are total derivative operators defined by

$$
\begin{aligned}
\frac{D}{Dx} &= \frac{\partial}{\partial x} + u_x\frac{\partial}{\partial u} + u_{xx}\frac{\partial}{\partial u_x} + u_{xt}\frac{\partial}{\partial u_t} + \ldots,\\
\frac{D}{Dt} &= \frac{\partial}{\partial t} + u_t\frac{\partial}{\partial u} + u_{xt}\frac{\partial}{\partial u_x} + u_{tt}\frac{\partial}{\partial u_t} + \ldots.
\end{aligned}
$$

Through the conservation law (3.39) one can introduce an auxiliary potential variable v and form an auxiliary potential system

$$\begin{aligned} v_x &= f(x,t,u,u_x,u_t),\\ v_t &= g(x,t,u,u_x,u_t). \end{aligned} \tag{3.40}$$

Any Lie group of point transformations

$$\mathbf{X_S} = \xi(x,t,u,v)\frac{\partial}{\partial x} + \tau(x,t,u,v)\frac{\partial}{\partial t} + \phi(x,t,u,v)\frac{\partial}{\partial u} + \psi(x,t,u,v)\frac{\partial}{\partial v}, \tag{3.41}$$

admitted by (3.40) yields a nonlocal symmetry *potential symmetry* of the given PDE (3.39) if and only if the following condition is satisfied

$$\xi_v^2 + \tau_v^2 + \phi_v^2 \neq 0. \tag{3.42}$$

3.3.1 Classical Potential Symmetries for a Particular Case of Eq. (3.3)

We consider the particular case of Eq. (3.3) where $c(x) = k_1 e^{rx}$, $g(u) = k_2 u^{-2}$, and $f(u) = k_3 u$ being k_1, k_2, k_3, and r arbitrary constants. That is

$$u_t = k_3 u + \frac{1}{k_1 e^{rx}}\left(k_1 e^{rx} k_2 u^{-2} u_x\right)_x. \tag{3.43}$$

In order to find potential symmetries of Eq. (3.43) we write the equation in a conserved form with

$$\begin{cases} \phi^t = -e^{-t}u,\\ \phi^x = -e^{-t}\,u^{-2}\,u_x + e^{-t}\,u^{-1}, \end{cases} \tag{3.44}$$

and the associated auxiliary system is given by

$$\begin{cases} v_x = e^{-t}u,\\ v_t = -e^{-t}\,u^{-2}\,u_x + e^{-t}\,u^{-1}, \end{cases} \tag{3.45}$$

A Lie point symmetry characterized by an infinitesimal transformation of the form

$$\begin{aligned} x^* &= x + \varepsilon\xi(x,t,u,v) + \mathcal{O}(\varepsilon^2),\\ t^* &= t + \varepsilon\tau(x,t,u,v) + \mathcal{O}(\varepsilon^2),\\ u^* &= u + \varepsilon\phi(x,t,u,v) + \mathcal{O}(\varepsilon^2),\\ v^* &= v + \varepsilon\phi(x,t,u,v) + \mathcal{O}(\varepsilon^2) \end{aligned} \tag{3.46}$$

is admitted by system (3.45). In the following, we present the point symmetries of (3.45) which were derived in [36] and we show which of these symmetries induces potential symmetries of Eq. (3.43). These symmetries are such that the condition (3.42) is satisfied. If the above relation does not hold, then the point symmetries of (3.45) project into point symmetries of (3.43). System (3.45) admits Lie symmetries if and only if

$$pr^{(1)}X(v_x - e^{-t}u) = 0,$$
$$pr^{(1)}X(v_t + e^{-t}u^{-2}u_x - e^{-t}u^{-1}) = 0,$$

where $pr^{(1)}X$ is the first extended generator of $\mathbf{X}_S$. We get

$$\begin{aligned}
\eta &= -g\,e^{-x} - \frac{k_5\,v^2}{2} + \frac{k_2\,v}{2} - \frac{k_5\,e^{-2t}}{2} + \frac{k_6}{2} + k_1,\\
\tau &= \frac{k_4\,e^{2t}}{2} - \frac{k_5\,e^{-2t}}{2} + k_6,\\
\phi &= -g\,u\,e^{-x} + g_v\,u^2\,e^{-x-t} + k_5\,e^{-t}\,u^2\,v - \frac{k_2\,e^{-t}\,u^2}{2} + \frac{k_4\,e^{2t}\,u}{2} + \frac{k_5\,e^{-2t}\,u}{2},\\
\psi &= k_5\,e^{-2t}\,v - k_6\,v - \tfrac{k_2\,e^{-2t}}{2} + k_3,
\end{aligned}$$

where $g = g(t, v)$ must satisfy

$$g_t\,e^{2t} + g_{vv} = 0. \tag{3.47}$$

The generators corresponding to k_2, k_5, and $g(t, v)$ correspond to potential symmetries, these are

$$\mathbf{v_1} = v\frac{\partial}{\partial x} - e^{-t}u^2\frac{\partial}{\partial u} - e^{-t}\frac{\partial}{\partial v},$$

$$\mathbf{v_2} = -e^{-2t}\frac{\partial}{\partial x} - e^{-2t}\frac{\partial}{\partial t} + e^{-t}(2u^2v + u)\frac{\partial}{\partial u} + e^{-2t}v\frac{\partial}{\partial v},$$

$$\mathbf{v}_\infty = -ge^{-x}\frac{\partial}{\partial x} - \left(g(t, v)ue^{-x} - \frac{\partial g(t, v)}{\partial v}u^2e^{-x-t}\right)\frac{\partial}{\partial u}.$$

The nonlinear Eq. (3.7) does not admit an infinite-parameter Lie group of contact transformations; however, its associated auxiliary system (3.6) admits an infinite-parameter Lie group of point transformations with infinitesimal generator $\mathbf{v}_\infty$, where $g(t, v)$ is an arbitrary function satisfying Eq. (3.47). One can obtain the invertible mapping

$$z_1 = v \qquad z_2 = e^{-t}$$
$$w_1 = e^x \qquad w_2 = \tfrac{-e^x}{u}$$

which transforms any solution $(w_1(z_1, z_2),\ w_2(z_1, z_2))$ of the linear system

$$\frac{\partial w_1}{\partial z_1} = -\frac{w_2}{z_2}$$
$$\frac{\partial w_2}{\partial z_1} = \frac{\partial w_1}{\partial z_2}$$

to a solution $(u(x, t), v(x, t))$ of the nonlinear system (3.6) and hence to a solution $u(x, t)$ of Eq. (3.43).

3.4 Conclusions

In different papers we have considered the generalized Fisher equation (3.3) and some particular cases from the point of view of the theory of symmetry reductions in partial differential equations. We have obtained the optimal system of one-dimensional subalgebras, the corresponding reductions to ODEs, and some exact solutions. Conservation laws for this equation were constructed by using two different methods. The potential system, in a particular case, has been achieved from the complete list of the conservation laws. We have constructed nonlocal symmetries (potential symmetries) which were realized as local symmetries of a related auxiliary system of differential equations. By using these potential symmetries we have also linearized Eq. (3.43) by an explicit non-invertible mapping.

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Chapter 4
Tunable Orbits Influence in a Driven Stadium-Like Billiard



André L. P. Livorati

4.1 Introduction

Hamiltonian dynamical systems are typically non-integrable and non-ergodic [1–4]. Their phase space presents mixed properties where chaotic seas, invariant tori, and stability islands are intertwined with themselves, which generates a huge variety of nonlinear phenomena, specially if a time-dependent perturbation is added in the system [3–7]. Considering dissipative and non-dissipative dynamics, the study of the chaotic properties of those systems can be found in many areas of research as fluids [8], plasmas [9, 10], nanotubes [11], complex networks [12], optics [13–15], acoustic [16], among others.

Particularly, billiard-like systems are often simple modeling and produce very interesting nonlinear phenomena [17]. Their dynamics are basically described by a particle moving in straight lines inside a closed domain, where at the collision point, the trajectory obeys Snell's law, i.e., the angle of reflection is equal to the angle of incidence. The notion of billiards is known since Birkhoff [18] in 1927, but were Sinai [19], Bunimovich [20, 21] and Gallavotti [22] who gave the mathematical support to the field. Nowadays, one may found billiards applications in microwaves, [23–25], quantum dots [26–29], synchronization [30, 31], soft wall effects [32, 33], astronomy [34], laser dynamics [35], and many others.

Bunimovich [36] introduced a billiard with the shape of a football stadium in 1974. The boundary of the billiard consists of two semicircular arcs connected with two parallel straight lines. An important property of the stadium billiard is a transition in the dynamics from mixed phase space to ergodic. Such transition is

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marked by the geometric control parameters, known as defocusing mechanism [17]. The mechanism basically consists of a narrow parallel beam of rays (trajectories), that after being focused due a reflection with the focusing boundary, may pass through a focusing point and become divergent, provided that a free path between two consecutive reflections from the boundary is long enough. The mechanism of defocusing works under the condition that divergence prevails over convergence [37].

A stadium-like billiard, where the focusing component can be either parabolic [37–39], elliptic [40], or just a circumference segment [38, 39], was introduced for the first time in 1999, by Loskutov, Ryabov, and Akinshin [41]. Together they developed the LRA conjecture[42], which claims that, if a billiard in its static boundary dynamics, has any chaotic component in the phase space, the introduction of a time perturbation in the boundary is enough condition for the dynamics to exhibit unlimited diffusion in the velocity of the particle, also known as Fermi acceleration (FA) [43]. However, the elliptical billiard is not included in the LRA conjecture. It is in fact integrable for the static boundary case presenting a separatrix curve separating two types of dynamical regimes: libration from rotation. When time dependence is introduced in the boundary, the separatrix curve turns into a stochastic layer, which allows a change in the dynamics behavior. In these cases, an orbit can suffer successive crossings from rotation to libration regions, in which were called tunable mechanism [44, 45]. This successive tunable crossings leads to an unlimited diffusion in energy [46].

Although the driven stadium-like billiard problem has been studied for many years [38, 39, 42, 47–49], concerning different aspects and applications, the implications of the nonlinear perturbation require an extensive and complex analysis where some chaotic properties are not yet well fully understood. In this chapter we seek to aim and understand the behavior of tunable orbits and their consequences to the dynamics of the stadium-like billiard with a time dependence of its boundary. We point out the existence of a critical resonance velocity, which separates the ensemble energy, where high initial velocities produce FA and low initial velocities do not experience the unlimited energy growth. We claim that tunable orbits, acting in partnership with stickiness orbits (basically described here as orbits that get trapped around a stability island for a finite time) are responsible for the separation of the ensembles and a steady state for long time series. These results were confirmed by statistical analysis of initial orbits, chosen initially inside stability islands, whose final behavior for long time series is a constant plateau, related to a different stability island in the phase space. Also, we made use of Lyapunov exponents and transport analysis to characterize such tunnelling behavior. Such results are original for the stadium-like billiard, and open new doors for the study of tunable orbits in similar dynamics systems with time-dependence perturbation.

The chapter was organized as follows: in Sect. 4.2 we described the dynamics of the stadium billiard and set up the scenario for the resonance and the ensemble separation. Section 4.3 is devoted to describe the numerical methods used to characterize the statistics for both ensembles, and its separation, giving emphasis

in the tunable orbits and the stickiness influence. Finally in Sect. 4.4 we draw some final remarks and conclusions of this work.

4.2 The Stadium-Like Billiard, the Mapping and Chaotic Properties

This section is devoted to present all the details needed to study the dynamics of an ensemble of non-interacting particles suffering perfectly elastic collisions with a stadium-like shape domain, as shown in Fig. 4.1. The focusing boundaries (radial ones) are time dependent according to a periodic function of the time, as $R(t) = R_0 + r\sin(wt)$, where $R_0 \gg r$. The velocity of the boundary is then obtained by $\dot{R}(t) = B(t) = B_0\cos(wt)$, with $B_0 = rw$ and r is the amplitude of oscillation of the moving boundary while w is the frequency of oscillation. The original radius can be obtained by simple geometric considerations as $R_0 = (a^2 + 4b^2)/(8b)$, and φ, which is the angle between the axial symmetry line and the border of the focusing component is given as $\varphi = \arcsin(a/2R_0)$. The geometric control parameters are: a which is the width of the circle arc, b indicates the depth of the curvature, and l is the strength of the parallel lines, they are displayed in Fig. 4.1b.

In our approach, the dynamics of the model is described in the static wall approximation. The approach considers that the boundary is assumed to be fixed, but at the instant of the impacts, the particles suffer an exchange of energy and momentum with the wall as if the boundary were moving. This procedure, although is an immediate simplification of the model, keeps the essence of the problem, therefore retaining the nonlinearities involved in the system and, at the same time brings the huge advantage of avoid to find solutions of transcendental equations that

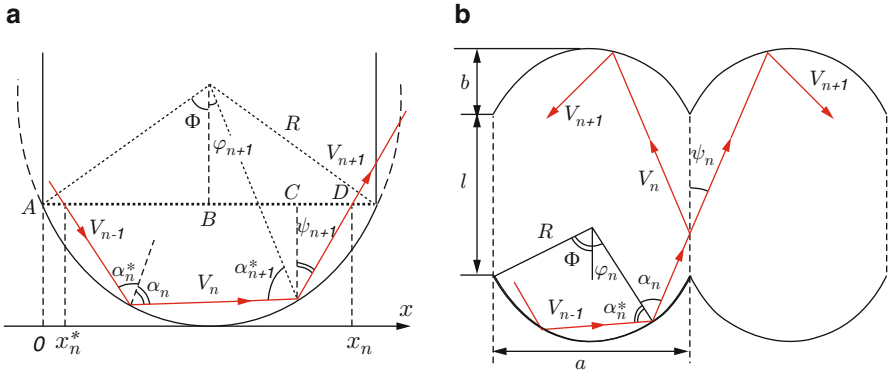


Fig. 4.1 An illustration of the stadium-like billiard with its variables and parameters. In (**a**) we have a schematics of the successive collisions dynamics, and in (**b**) a schematics of the indirect collisions dynamics

are very much time consuming in the statistical. Moreover, such an approach has been shown itself useful in many other dynamical systems [50–53].

The evolution of the dynamics is done considering the variables $(\alpha_n, \varphi_n, t_n, V_n)$. Here, α and φ are considered the polar angles of the dynamics, where α_n is the angle between the trajectory of the particle and the normal line at the collision point, φ_n is the angle between the normal line at the collision point and the vertical line in the symmetry axis. We assume that t_n is the time at the instant of the impact and V_n is the velocity of the particle, while the index n denotes the nth collision of the particle with the focusing boundary.

As an initial condition, we assume that at the initial time $t = t_n = 0$, the particle belongs to one of the focusing boundaries and the velocity vector directs towards to the billiard table. Just as a notation, all variables indexed with (*) are measured immediately before the collision. Also, in order to investigate more possible trajectories and orbits along the billiard domain, let us consider two distinct situations: (1) successive collisions with the same focusing component, displayed in Fig. 4.1a, and (2) indirect collisions, where after suffering a collision with a focusing boundary, the next collision of the particle is with the opposite one, where the particle can in principle collide many times with the parallel borders. In order to attenuate this last particular orbit, we made use of the unfolding method [17, 37], since the billiard has an axial symmetry, we can mirror successive stadium shapes, as one can see in Fig. 4.1b, until the particle reach the opposite focusing component. Figure 4.1b gives an illustrative example of one mirrored collision using the unfolding method.

For the occurrence of a successive collision, it is necessary that $\mid \varphi_{n+1} \mid \leq \varphi$. Therefore, according to Fig. 4.1a and the specular reflection law, we have that

$$\begin{cases} \alpha^*_{n+1} = \alpha_n \\ \varphi_{n+1} = \varphi_n + \pi - 2\alpha_n \pmod{2\pi} \\ t_{n+1} = t_n + \frac{2R\cos(\alpha_n)}{V_n} \end{cases}, \tag{4.1}$$

However, if we have $|\varphi_{n_1}| > \varphi$, then we have indirect collision, and the particle collides with the opposite focusing component. In this case, because of the unfolding method, two auxiliary variables are introduced, ψ, which is the angle between the vertical line at the collision point and the particle trajectory, and x_n, representing the projection over the horizontal axis. An illustrative schematic of the indirect collisions is displayed in Fig. 4.1b, and we may obtain the following expressions

$$\begin{cases} \psi_n = \alpha_n - \varphi_n \\ x_n = \frac{R}{\cos(\psi_n)}[\sin(\alpha_n) + \sin(\varphi - \psi_n)] \\ x_{n+1} = x_n + l\tan(\psi_n) \\ \alpha^*_{n+1} = \arcsin[\sin(\psi_n + \varphi) - x_{n+1}\cos(\psi_n)/R] \\ \varphi_{n+1} = \psi_n - \alpha^*_{n+1} \\ t_{n+1} = t_n + \frac{R[\cos(\varphi_n)+\cos(\varphi_{n+1})-2\cos(\varphi)]+l}{V_n\cos(\psi_n)} \end{cases}. \tag{4.2}$$

For both kinds of collisions (successive and indirect), the expression for the final velocity $\mathbf{V}_{n+1}$ is obtained by decomposing it into two separate components, which

are

$$\begin{cases} \mathbf{v}_n \cdot \mathbf{T}_n = v_n \sin(\alpha_n^*) \\ \mathbf{v}_n \cdot \mathbf{N}_n = -v_n \cos(\alpha_n^*) \end{cases}, \tag{4.3}$$

where $\mathbf{T}_n$ and $\mathbf{N}_n$ are the respective tangent and normal unit vectors at the impact point in the nth collision. Since the collision occurs in a non-inertial referential frame, we have to make change of referential frames from inertial to non-inertial. Then, the reflection law is given by

$$\begin{cases} \mathbf{V}'_{n+1} \cdot \mathbf{N}_{n+1} = -\gamma \mathbf{V}'_n \cdot \mathbf{N}_{n+1} \\ \mathbf{V}'_{n+1} \cdot \mathbf{T}_{n+1} = \zeta \mathbf{V}'_n \cdot \mathbf{T}_{n+1} \end{cases}, \tag{4.4}$$

where $\gamma \in [0, 1]$ and $\zeta \in [0, 1]$ are the respective restitution coefficients for the normal and the tangent components, and the superscript $'$ denotes the non-inertial referential frame. Since in this paper, the non-dissipative case dynamics is considered we have $\gamma = \zeta = 1$. A careful discussion of a dissipative version can be checked in Ref.[47].

Moving back to the inertial referential frame, the components of the velocity of the particle after collision are given by

$$\begin{cases} \mathbf{V}_{n+1} \cdot \mathbf{N}_{n+1} = -\gamma \mathbf{V}_n \cdot \mathbf{N}_{n+1} + (1+\gamma)\mathbf{B}(t_{n+1}) \cdot \mathbf{N}_{n+1} \\ \mathbf{V}_{n+1} \cdot \mathbf{T}_{n+1} = \zeta \mathbf{V}_n \cdot \mathbf{T}_{n+1} + (1-\zeta)\mathbf{B}(t_{n+1}) \cdot \mathbf{T}_{n+1} \end{cases} \tag{4.5}$$

Finally, the expression of the velocity of the particle after any kind of collision is given by

$$\mid \mathbf{V}_{n+1} \mid = \sqrt{(\mathbf{V}_{n+1} \cdot \mathbf{T}_{n+1})^2 + (\mathbf{V}_{n+1} \cdot \mathbf{N}_{n+1})^2}\,. \tag{4.6}$$

Also, considering the specular reflection law, the tangent and normal components of the velocity can be used to obtain the angle α recurrence expression after any kind of collision, which furnishes us

$$\alpha_n = \arcsin\left(\frac{\mid \mathbf{V}_n \mid}{\mid \mathbf{V}_{n+1} \mid} \sin(\alpha_n^*)\right)\,. \tag{4.7}$$

After that, the dynamics of the ensemble of non-interacting particles suffering collisions inside the stadium-like shape domain is then evolved by considering simultaneously Eqs. (4.1), (4.2), (4.6), and (4.7).

Figure 4.2 displays the behavior of the phase space (α, φ) for two different combinations of the geometric control parameters, for 100 initial conditions for $V_0 = 20$ and $B_0 = 0.01$. One can see that the dynamics has mix properties, with chaotic seas and stability islands of different periods. In particular, for Fig. 4.2b, the combination of the parameters a, b, and l, furnishes us a quasi-integrable

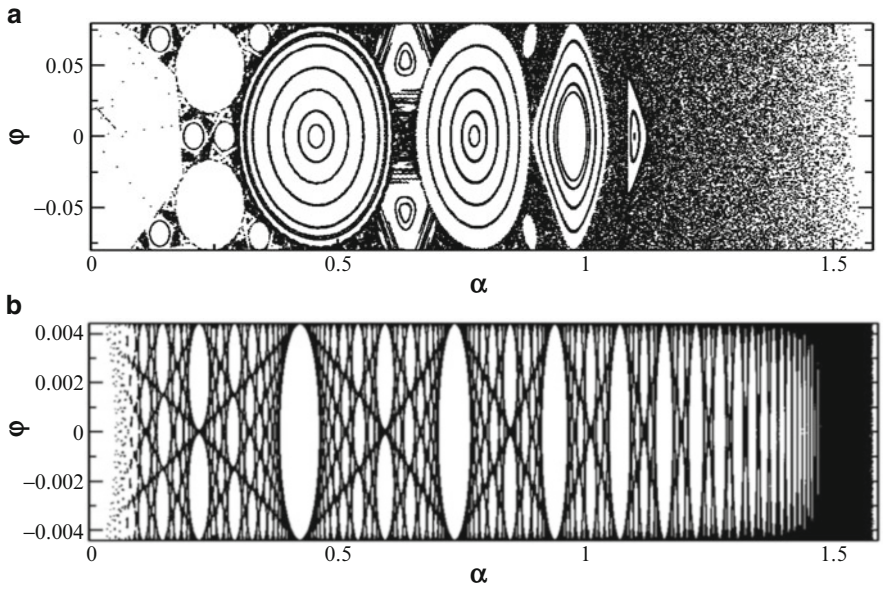


Fig. 4.2 Phase space $(\alpha\varphi)$ for the stadium-like billiard for 100 initial conditions for an initial velocity $V_0 = 20$ and $B_0 = 0.01$. In (**a**) we have $a = 0.5$, $b = 0.01$, and $l = 1$, and in (**b**) we have $a = 0.9$, $b = 0.001$, and $l = 2$

system, since the curvature b is basically three orders of length less than the other parameters. That is the reason we observe a huge amount of stability islands.

Also, the chaotic region for Fig. 4.2b is in a range where the polar angle α is near its maximum value $(\pi/2)$, indicating that the trajectory of a particle, in principle, could collide several times with the parallel lines, before collides again with a focusing component. In this case, the phase of the moving boundary would be uncorrelated between two collisions with the focusing boundary, which would give the status of pseudo-random to the orbit, originating then the chaotic sea for that region. A similar phenomena can be observed in the Fermi-Ulam model [54], which is considered a one-dimensional billiard.

4.3 Resonance, Ensemble Separation, and Tunable Orbits

In this section we will discuss the critical phenomena that are observed in the stadium-like billiard and their consequence to the dynamics. We start revisiting the resonance between the external perturbation and the rotation orbits around period-1 fixed points, and its consequences for separation of the ensemble of velocities. Then, we will focus on the lower ensemble of velocity, and made a numerical investigation of the behavior of the tunable orbits.

4.3.1 Resonance

Let us start this investigation considering the linearization of the unperturbed mapping [38, 39, 48], given as

$$\begin{cases} \xi_{n+1} = \xi_n + \frac{l}{a}\tan(\psi_n) \mod(1) \\ \psi_{n+1} = \psi_n + \frac{8bl}{a^2}(2\xi_{n+1} - 1) \mod(\pi/2) \end{cases} . \tag{4.8}$$

Considering the above expressions and according to the action-angle variables, it is possible to calculate the rotation number around a period-1 orbit as $\sigma = \arccos\left(1 - \frac{8bl}{a^2\cos^2(\psi_n^*)}\right)$, where $\psi_n^* = \arctan(\frac{ma}{l})$ is the fixed point [37–39], with $m \geq 1$ is the number of mirrored stadiums that the particle can go through in a trajectory, considering the unfolding method.

Now, one may consider a trajectory, where an orbit moves around some period-1 fixed point in the unperturbed version of the billiard. The time spent between two successive collisions, i.e., the time between collisions with the two focusing components is $\tau \approx \frac{l}{\cos(\psi_n^*)V_n}$ [38, 39]. Thus, one can obtain the rotation period of this orbit around the fixed point $T_{rot} = \frac{2\pi\tau}{\sigma}$.

One can check in Refs.[37, 49] that according to the convenience, either using angular coordinates α_n and φ_n (as shown in Fig. 4.2) or the auxiliary variables ψ and ξ, we may obtain an extremely similar phase space for the stadium-like billiard. Here, we set $\xi = 0.5 + R_0\sin(\varphi_{n+1})/a$, and it is related to the projection along the horizontal axis, and is usually normalized at mod(1). Also, the fixed points ψ^* are deeply connected with α_n because of the axial symmetry of the billiard. There is a sequence of fixed points corresponding to orbits always within the stadium at $\varphi_n = 0$ (period-1), that intersects different multiples of the parameter a in the horizontal direction, according to the unfolding method [17, 37].

Considering now, the case where a time perturbation is introduced in the focusing boundaries, one can obtain an external perturbation period as $T_{ext} = \frac{2\pi}{w}$. We have a resonance between the oscillation of the moving boundaries and the rotation orbits around the fixed points, if $T_{rot} = T_{ext}$. Finally, considering this equality among periods and after some rearrangement in the expressions, we can obtain a resonance velocity as

$$V_r = \frac{wl}{\cos(\psi_n^*)\arccos\left(1 - \frac{8bl}{a^2\cos^2(\psi_n^*)}\right)} . \tag{4.9}$$

Indeed, the resonance obtained in Eq. (4.9) only holds when the defocusing mechanism is not active in the dynamics, which restricts the range of the control parameters a, b, and l [37]. Also, depending the fixed point ψ^*, the critical resonance velocity value may vary. For the control parameters displayed in Fig. 4.2a, b the critical velocity is $V_r \approx 1.2$ and $V_r \approx 5.0$, respectively.

4.3.2 Ensemble Separation

The criticality described in the previously subsection is responsible for a mechanism of an ensemble separation in the dynamics. The separation occurs according to the initial velocity of a particle inside the stadium-like billiard. Let us set up the dynamical scenario.

For initial velocity higher than the resonance one, if the initial condition is chosen in the chaotic sea, we will see a gain in the velocity as time evolves. This gain, leads the dynamics to an unlimited diffusion, that is a characteristic of the FA phenomenon. Otherwise, if one chooses an initial condition, inside a stability island, the orbit will remain there, in a regular behavior.

On the other hand, considering now an initial velocity lower than the critical resonance one, the velocity of the particle may decrease, whatever in the phase space the initial condition is chosen, in the chaotic sea or inside a regular region. Such decrease is due to tunable orbits, when the resonance is active in the dynamics. Since, with the introduction of the time perturbation, the invariant curves that surround the fixed points become porous, and they behave in a similar way to a stochastic layer [44–46], as a consequence, the whole phase space becomes accessible for the particle. So, an orbit initially in the chaotic sea may penetrate in the vicinity of a stability island, and the opposite may occur as well. Such behavior will be investigated in the next section.

Figure 4.3 shows the ensemble separation considering the root mean square velocity (V_{RMS}) for the same combination of control parameters displayed in Fig. 4.2b, where the critical resonance velocity is around 5, according to Eq. (4.9). The V_{RMS} was evaluated according

$$V_{RMS} = \sqrt{\overline{V^2}} = \sqrt{\frac{1}{M}\sum_{i=1}^{M}\frac{1}{n}\sum_{j=1}^{n} {V_{i,j}}^2}\,, \tag{4.10}$$

where M is the ensemble of 1000 initial conditions, and n is the number of collisions (iterations). The average is taken along the orbit and along the ensemble of initial conditions. Also, the initial conditions were always chosen in the chaotic sea, uniformly distributed along $\varphi \in [-\varphi, +\varphi]$, and with a constant $\alpha = 1.55$.

One can see that the V_{RMS} experiences the ensemble separation according to their initial velocity. If $V_0 > V_r$, the curves stay in a constant plateau for short times, and then they bend towards a growth regime, in a typical behavior of FA (growth as $\sqrt{(n)}$). On the other hand, if $V_0 < V_r$, the V_{RMS} stays constant for all time or suffer a decrease of their velocity. Such ensemble separation was already observed for the combination of control parameters of Fig. 4.2a, (see Refs.[38, 39, 48, 49]).

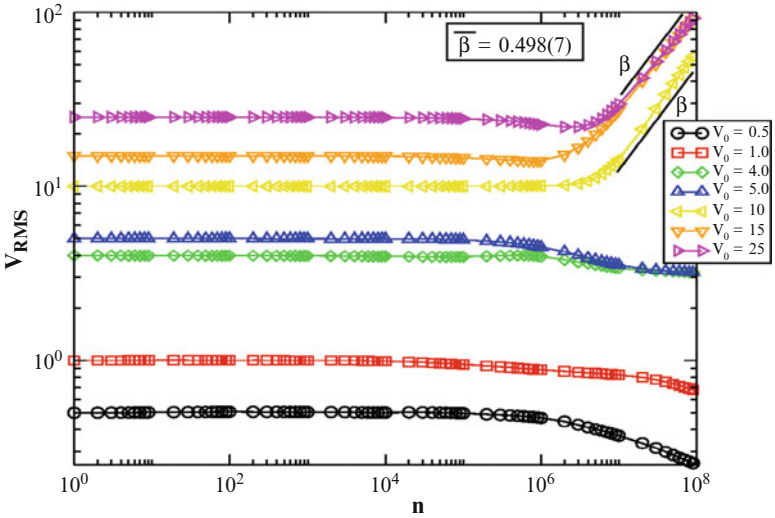


Fig. 4.3 Behavior of the V_{RMS} curves as function of n. There is an ensemble separation at the critical resonance velocity. If $V_0 > V_r$, the curves experience unlimited diffusion, with a growth regime as n^{β}, where $\beta \approx 0.5$; and, if $V_0 < V_r$ the V_{RMS} curves decrease and stay in a constant plateau. The control parameters used were $a = 0.9$, $b = 0.001$, and $l = 2$, which furnish us $V_r \approx 5.0$

4.3.3 Tunable Orbits

In this subsection we are particularly interested in the investigation of the lower velocity ensemble, which is the one that is influenced by the critical resonance. Usually, in a mixed phase space dynamical system, we have basically two behaviors, (1) chaotic motion and (2) regular. These two dynamical regimes are independent from each other, and depend on the initial conditions, also, they often depend on the control parameters [1–3].

However, with the introduction of a time perturbation in the dynamics, this notorious behavior might not be observed anymore in some particular cases, like the elliptical billiard [44–46]. Here, since the billiard boundary is vibrating, orbits are allowed to change their behavior from rotator to librator, and vice versa. The mechanism lies on the separatrix curve, which separates these two regimes in the static boundary case. This separatrix curve becomes a stochastic layer, and would allow this change in the dynamics, when time perturbation is introduced.

In the stadium-like billiard, we claim that the same mechanism applies, when the resonance is active. In this way, the invariant curves that surround the fixed points might become porous, and behave in a very similar way to a stochastic layer (just like the elliptical billiard). So, this mechanism allows a change between chaotic and regular orbits, and as a consequence, the whole phase space becomes accessible for

the particle. Also, stickiness may act together with the resonance, where orbits that get trapped around stability islands have more chances to change its behavior [49].

In order to clarify the behavior of these tunable orbits, let us follow the evolution a single initial condition, chosen inside a stability island, when the initial velocity is lower than the critical resonance one. Figure 4.4 shows such dynamical evolution, for the same combination of the control parameter a, b, and l displayed in Fig. 4.2a. Since the resonance phenomenon is based on period-1 fixed points, we decide to choose an initial condition inside the stability island located in $\psi* = \alpha* = \arctan(ma/l)$, with $m = 1$, which is the first period-1 regular region from left to right in Fig. 4.2a.

Analyzing the evolution of the coordinate φ_n, as function of the number of collisions n, one can see in Fig. 4.4a, for an initial velocity $V_0 = 0.7$, that after about 10^4 iterations, the orbit leaves the region of regularity, then it wanders through the chaotic sea for another 10^4 iteration, until goes inside another period-1 regular region (the fixed point with $m = 3$), and stays there for about another 10^4 iterations. The orbit then, leaves this region and keeps again wandering in the chaotic sea, where it passes through a stickiness region around the island with $m = 4$, and finally it reaches the period-1 regular structure with $m = 2$, where it is doomed to stay for really long times (as far we can tell, up to 10^{10} collisions). The same dynamical

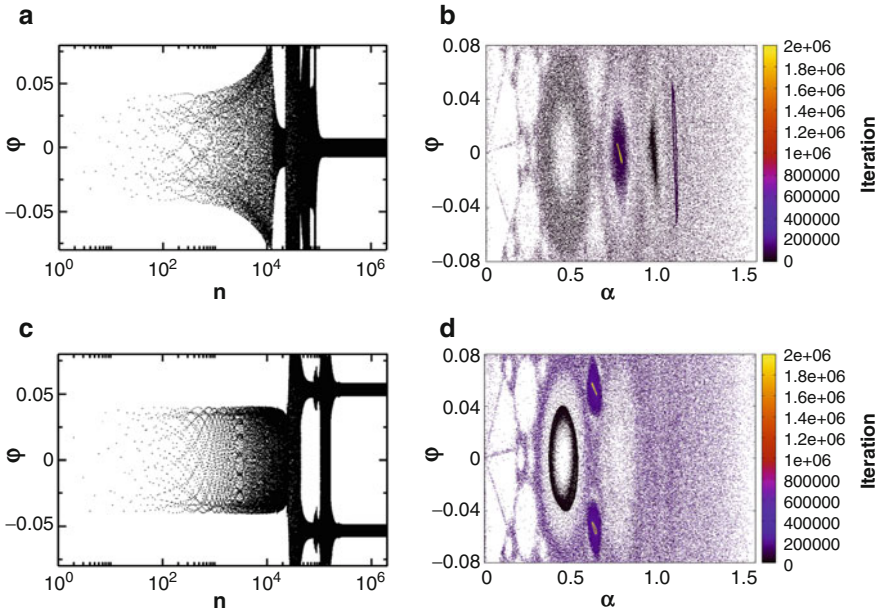


Fig. 4.4 Dynamical evolution of a tunable orbit setup initially inside a stability island for $a = 0.5$, $b = 0.01$, $l = 1.0$, and $B_0 = 0.01$. In (**a**) and (**b**) $V_0 = 0.7$, and in (**c**) and (**d**) $V_0 = 0.5$. The orbit can tunable several times, wander in the chaotic see, and come back tunnelling, before reaching a final plateau. In (**b**) and (**d**) the linear color gradient indicates the iteration number

behavior is seen in Fig. 4.4b, where the phase space (α, φ) is displayed, according to linear a color gradient representing the iteration number. Here, black denotes the first 20,000 collisions, violet (dark gray) are intermediate collisions, and orange and yellow (light gray) represent long time iterations (up to 2×10^6).

To compare with, Fig. 4.4c, d shows the same dynamical evolution of Fig. 4.4a, b, but now the initial velocity is lower as $V_0 = 0.5$. We started with the same initial condition of the previously case, in the same regular structure. Basically, the same dynamical behavior can be observed, as the orbit leaves the stability island, wanders for a while in the chaotic sea, and then the orbit "converged to a period-2" island and remained there, in a regular behavior for long times (again, as far as we can tell, up to 10^{10} collisions). In Fig. 4.4d we display this tunable behavior according to the phase space variables, where the iteration number is set as the same linear color gradient of Fig. 4.4b.

Now, it allows us to do the opposite. Let us start with an initial condition in the chaotic sea, and see how it evolves. Figure 4.5 displays the evolution of a single orbit chosen in the chaotic sea, for the same geometric control parameters of Fig. 4.2b, with an initial velocity $v_0 = 1.0$. One can see in Fig. 4.5a that after 10^5 collisions in the chaotic sea, the orbit evolves to stable behaviors. It stays trapped in a period-3 orbit, then it comes back to the chaotic sea. After that, it follows successive trappings, like a period-9 and a period-7, always alternating with chaotic behavior.

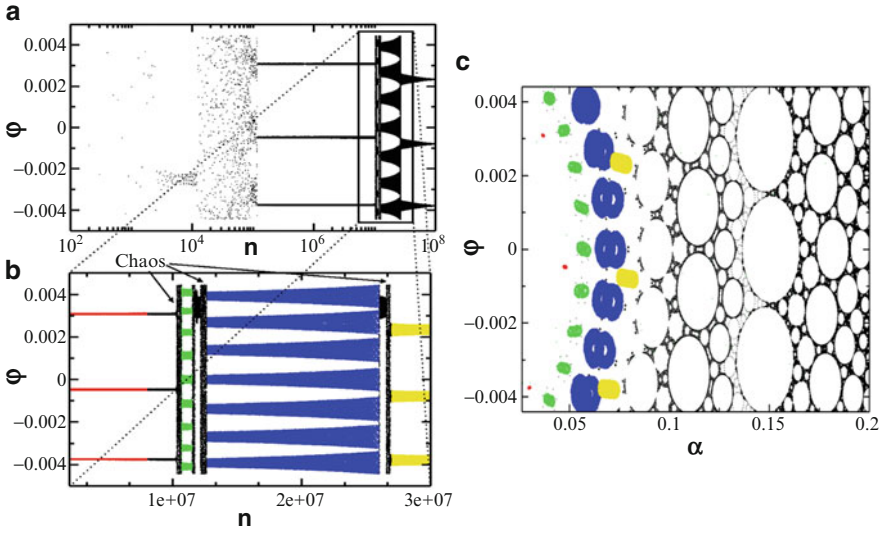


Fig. 4.5 Dynamical evolution of a tunable orbit setup initially in the chaotic sea for $a = 0.9$, $b = 0.001$, $l = 2.0$, $B_0 = 0.01$, and $V_0 = 1.0$. In (**a**), we have the whole evolution until 10^8 collisions, in (**b**) there is a zoom window in a successive trapping/tunnelling behavior for long times, and finally in (**c**) this successive trapping/tunnelling is display in the phase space, indicating the preferential islands in the low region of the α angle

Finally, it ends its dynamics in a period-3 orbit. Figure 4.5b, c displays a zoom-in in the trapping behavior as n evolves, and the phase space trapping, respectively. Also, the trappings occur with lower values of the α angle, which indicates that the orbit got itself in a regime of multiple successive collisions.

Also, one could argue that these trappings could be related to stickiness phenomenon. Indeed some of them truly are stickiness influence. However, one can take a careful look in the period-7 trapping in Fig. 4.5b, which is depicted around 1.3×10^7 and 2.6×10^7 collision. During the trapping, the φ angle is increasing its value as n evolves, indicating that the orbits are spiralling out of the stable region. Such behavior is due to tunable orbits. This behavior is the same one observed in Fig. 4.4a.

4.3.4 Lyapunov Exponents

In order to give support to the tunable orbits, let us calculate the Lyapunov exponent for those orbits and see how it behaves. The Lyapunov exponent is an important tool that has been widely used to quantify the average expansion or contraction rate, for a small volume of initial conditions [4]. If the Lyapunov exponent is positive, the trajectory shows sensibility on the initial conditions, and the orbit is said to be chaotic. However, a non-positive Lyapunov exponent indicates regularity and the dynamics can be in principle periodic or quasi-periodic [4].

The basic principle of the obtainance of the Lyapunov exponents is to verify if two different, by very close orbits, diverge exponentially in average from each other, as time evolves. So, we evaluate an original orbit and its neighbor orbit (satellite), separated by an infinitesimal distance δx_n, which is in average, limited or can linearly grow as time evolves, if the system is not chaotic. On the other hand, for chaotic systems, this distance grows exponentially with time, according to

$$\delta x_n \propto \delta x_n(0) e^{\lambda n} \,, \tag{4.11}$$

where λ is the local exponential expansion rate and the zero index refers to the initial distance between the orbits. Thus, we may define a parameter that is able to characterize how unstable a given orbit is, such as

$$\lambda = \lim_{n\to\infty} \lim_{\delta x_n(0)\to 0} \frac{1}{n} \ln\left(\frac{|\delta x_n|}{|\delta x_n(0)|}\right) . \tag{4.12}$$

Here, λ is the quantity responsible for the estimation of the average exponential rate of separation, between the original orbit and its satellite one, known as Lyapunov exponent [55]. Also, to consider an infinity time in Eq. (4.12) is not possible, so we truncate the time evolution until a finite time, said to be a large one. This technique gives us the Finite Time Lyapunov Exponent (FTLE) [56, 57].

Considering the above description, we shall give an initial original orbit $(\alpha_0, \varphi_0, V_0, t_0)$ and its satellite one $(\alpha_0{}', \varphi_0{}', V_0{}', t_0{}')$. Then, we evolve both orbits according to the mapping dynamics, and after a defined finite time (ℓ), we measure the distance between them as

$$\delta_1 = \sqrt{(V_1 - V_1{}')^2 + (t_1 - t_1{}')^2 + (\alpha_1 - \alpha_1{}')^2 + (\varphi_1 - \varphi_1{}')^2} \, . \tag{4.13}$$

The distance between the orbits δ_1 is then used to rescale the satellite orbit to the same direction of the original orbit. And then the procedure is restarted. The "new" rescaled values for the satellite orbit are given by

$$\begin{cases} \alpha_0{}' = \alpha_1 + [\delta_0(\alpha_1{}' - \alpha_1)]/\delta_1 \\ \varphi_0{}' = \varphi_1 + [\delta_0(\varphi_1{}' - \varphi_1)]/\delta_1 \\ V_0{}' = V_1 + [\delta_0(V_1{}' - V_1)]/\delta_1 \\ t_0{}' = t_1 + [\delta_0(t_1{}' - t_1)]/\delta_1 \end{cases} , \tag{4.14}$$

where δ_0 is the initial difference between the orbits set as $\delta_0 = 10^{-10}$. The FTLE is then computed considering n successive ℓ time intervals, until the finite time predetermined is reached, according to the following expression

$$\lambda_\ell = \frac{1}{n\ell} \sum_{j=1}^{n} \ln\left(\frac{\delta_j}{\delta_0}\right) \, . \tag{4.15}$$

Figure 4.6 displays the behavior of an orbit initially set up in the chaotic sea for two different combinations of control parameters, with initial velocities lower than the resonant one, and their respective Lyapunov exponent, set up by Eq. (4.15). In Fig. 4.6a–c, we have $a = 0.5$, $l = 1$ and $b = 0.01$, and $V_0 = 0.5$. We first show the phase space (α, φ) in Fig. 4.6a and the evolution of the φ angle as the dynamics evolves according n. Then, we display a comparison with the Lyapunov exponent in Fig. 4.6c, with step-size $\ell = 100$. One can see that the behavior of the Lyapunov exponent follows the same line of the behavior of $\varphi \times n$. While the orbit is in the chaotic sea, the Lyapunov exponent is positive, and when it passes through stickiness regime, the Lyapunov exponent goes near zero, indicating a regular behavior. At the end, when the orbit is doomed to a stable condition, the Lyapunov exponent, stays in a constant plateau very near the zero value, confirming thus the regular behavior for long times.

The same applies to Fig. 4.6d–f, when the combination of control parameters is $a = 0.9$, $l = 2$ and $b = 0.001$, and $V_0 = 3$. In Fig. 4.6d, we have the phase space (α, φ), with an initial condition setup in the chaotic sea. And in Fig. 4.6e, f we have a comparison of the angle φ and the respective Lyapunov exponent λ_{100} as function of the number of collisions n. One can see some "bursts" in the Lyapunov exponent of Fig. 4.6f. These are related to the transition of the tunable orbits, from chaos to stability, and vice versa. Also, for long times, the Lyapunov goes to zero in average, indicating a stable behavior.

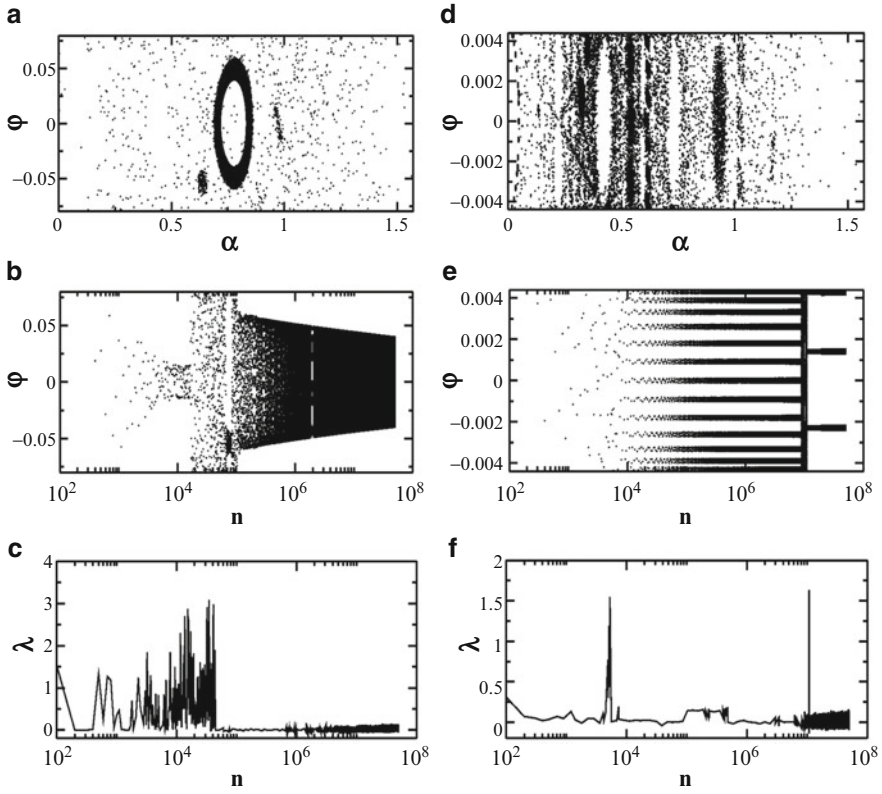


Fig. 4.6 An illustration of the stadium-like billiard with its variables and parameters. In (**a**) we have a schematics of the successive collisions dynamics, and in (**b**) a schematics of the indirect collisions dynamics

4.4 Final Remarks

To summarize, we investigated the dynamics of a particle confined inside a domain with a shape of a stadium-like billiard, where the focusing components are periodically time-dependent. A four-dimensional nonlinear mapping was obtained considering distinct kinds of collisions. Also, a critical resonance was setup between the rotation period-1 orbits and the external perturbation of the system. The resonance plays the role of a watershed in the ensemble of velocities, where for initial velocities higher than the critical one, we observe an unlimited diffusion of the velocity of the particle. On the other hand, for initial velocities lower than the critical one, we observe convergence to steady state plateaus for long times.

The critical resonance also allows a tunable behavior of orbits near the vicinity of stability islands. Here, the invariant curves that delimited these islands shapes start to behave like stochastic layers. So a chaotic orbit could access the interior of

an island and vice versa. Such peculiar dynamics allows the whole phase space to become accessible for the particle. The tunable behavior of the orbits was characterized via transport analysis during the dynamical evolution and through the finite time Lyapunov exponent. Also in a close future, it would be interesting to investigate if this mechanism that allows the orbit to tunnelling could be observed in other billiards with time-dependent boundaries.

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Chapter 5
Intermittency and Transport Barriers in Fluids and Plasmas



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5.1 Introduction

Intermittency is one of the most commonly observed phenomena in chaotic systems. It refers to the random alternation between phases of different behavior and, usually, is seen in time series as an interplay between laminar and bursty, or between periodic and chaotic phases. Several works have reported on the onset of intermittency in diverse systems, such as economic cycles [9], nonlinear wave–wave interactions in space plasmas [29], solar dynamo [38], and accretion disk simulations [28]. The basic mechanism behind intermittency is the coupling between nonattracting chaotic sets (chaotic saddles), which are sets responsible for transient chaos [22]. Thus, a

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trajectory or orbit in the vicinity of a chaotic saddle experiences chaotic motion for a finite time, before it is repelled along its unstable foliation toward some other set. If the second set is another chaotic saddle, it will eventually repel the orbit, which can go back to the first chaotic saddle or move toward another region of the phase space and so forth. The nonlinear coupling between chaotic saddles is done by a set of unstable periodic orbits that have components in both saddles. Intermittency is usually observed in the neighborhood of a global bifurcation, such as an interior or attractor-merging crisis or an embedded saddle-node bifurcation. In these kinds of bifurcations, a stable set collides with an unstable set (e.g., an unstable periodic orbit or a chaotic saddle), opening a "hole" or *leak* in the stable set, that causes its destabilization, consequently permitting the escape (*leakage*) of trajectories from the previously stable region of the phase space. If there is some reinjection mechanism capable of leading the trajectory back to the original region, the system will experience intermittency. Otherwise, the trajectory never returns and only a transient behavior is observed.

Leaking systems can occur in both dissipative and conservative systems. For instance, in chaotic scattering, a hole in the domain leads the system to behave chaotically for a finite time, before the trajectory hits the hole and escapes the domain of interest [2]. Here, two examples of leaking systems are studied. In the first case, a simple Hamiltonian map is introduced to study transport of magnetic field lines in fusion plasmas. In a tokamak, one wishes to prevent the radial transport (leaking) of magnetic field lines toward the device walls, since the plasma-wall interaction could impair the plasma confinement. Thus, the presence of transport barriers in the plasma is crucial in the context of fusion. We report on the effects of a transport barrier known as robust torus in our map. In the second example, crisis-induced intermittency is studied in direct numerical simulations of three-dimensional Rayleigh-Bénard convection. In a plane layer heated from below, a viscous fluid undergoes changes from periodic to chaotic and hyperchaotic regimes as the Rayleigh number is increased. The attractors are destroyed in crises that lead to the leaking of trajectories away from the trapping regions, as described above.

5.2 Transport Barriers in Nonintegrable Dynamical Systems

Through a known Hamiltonian derivation, a non-twist map with a transport barrier called robust torus (RT) is presented [27], in which the RT is introduced in a systematic way [43]. The motivation to create this map was the relevance of transport barriers to plasma confinement in tokamaks [16] and also because the map could be a natural extension of usual maps [36] applied to theoretically study transport in tokamaks. In another work [26] adapted the map with a single robust torus and adjusted the parameters according to the experimental ones used in the tokamak TCABR. We have verified that the RT regularizes the orbits close to its

position in the phase space and blocks the radial transport inside the tokamak plasma. The main idea to create a RT is to introduce an appropriate polynomial pre-factor multiplying the resonant perturbations which act on the system in such a way that their effects vanish on particular positions of the phase space even if the control parameter is turned on. This vanishing will occur in the values of the roots of the pre-factor. The RT in fact corresponds to an invariant curve which remains intact under the action of any generic perturbation.

In this section, two variations of the map are used: one non-twist and another twist, both of them with or without a robust torus. The non-twist maps present isochronous resonances, which can reconnect themselves and generate the well-known shearless barriers [44].

We compare the obtained topologies for the non-twist and twist maps, with and without RT, and we identify the effect of the RT on the dynamical transport. We create a RT near a secondary resonant mode and we show that the robustness of the RT is transferred to its neighborhood.

5.2.1 Symplectic Maps

We propose an equilibrium Hamiltonian, H_0, with toroidal symmetry and we define the toroidal angle, φ, as the canonical time t. This non-perturbed Hamiltonian corresponds to an integrable system and it is described by an analytical solution of the nonlinear Grad-Shafranov equation. H_0 is given in terms of the action J, corresponding to the toroidal normalized flux, and ϑ the poloidal angle canonically associated to J. The perturbations are generated by a helical electric current applied from thin ergodic magnetic limiters (EML) distributed along the poloidal direction in toroidal sections of the tokamak. The perturbing Hamiltonian, H_1, is a function of J, ϑ, and t, and it is represented by a Fourier expansion of delta-kicks due to the EML rings. The perturbations create a region with chaotic magnetic field lines at the plasma edge.

We expand H_0 around a magnetic surface with action J_0 and frequency $\Omega_0 = n/m$ where Ω_0 is equal to the inverse of the safety factor q and n and m are integers. In the Fourier perturbation expansion, we keep only the two main resonances in such a way the Hamiltonian has only the dominant resonant modes with poloidal wave numbers m and $(m + 1)$,

$$H(\Delta J, \vartheta, t) = H_0(\Delta J) + P(\Delta J)[\beta \cos(m\vartheta) + \eta \cos((m+1)\vartheta + nt)] \sum_{k=-\infty}^{+\infty} \delta\left(t - \frac{2\pi}{N_r}k\right) \tag{5.1}$$

where $\Delta J = J - J_0$ and $N_r = 4$ is the quantity of EML rings equally spaced around the tokamak.

We also consider two equilibrium configurations for the frequency profiles $\Omega_0 \equiv dH_0/d\Delta J$ given by:

$$H_0(\Delta J) = \frac{1}{2}\Delta J^2 \tag{5.2}$$

for the twist case and

$$H_0(\Delta J) = \frac{1}{2}\Delta J^2 - \frac{\alpha}{3}\Delta J^3 \tag{5.3}$$

for the non-twist case. The parameters β and η in Eq. (5.1) represent the perturbing electric current applied in EML rings and the parameter α in Eq. (5.3) is determined by the safety factor profile defined by H_0. The pre-factor $P(\Delta J)$ in Eq. (5.1) is a polynomial function and it allows us to introduce the RT.

In order to investigate the characteristics of the phase space concerning transport, due to the presence of a RT, we consider the Hamiltonian of Eq. (5.1) with the non-perturbed Hamiltonian, H_0, of Eqs. (5.2) and (5.3), twist and non-twist cases, respectively. We also choose two configurations for the pre-factor, one with $P(\Delta J) = 1$ for the usual Hamiltonian without RT and another one with $P(\Delta J) = (\Delta J - a)$ introducing one RT in the position $\Delta J = a$. However, instead of working with the Hamiltonian formulation we will take the corresponding nonlinear map associated to the Hamiltonian of Eq. (5.1) that was derived in detail by us in [27]. Thus, the non-twist map with two resonant modes and without RT is

$$\begin{aligned}
\Delta J_{k+1} &= \Delta J_k + \beta m \sin(m\vartheta_k) + \eta(m+1)\sin[(m+1)\vartheta_k + nt_{k+1}] \\
\vartheta_{k+1} &= \vartheta_k + \frac{2\pi}{N_r}(\Delta J_{k+1} - \alpha \Delta J_{k+1}^2) \\
t_{k+1} &= = t_k + \frac{2\pi}{N_r m}
\end{aligned} \tag{5.4}$$

The non-twist map with two resonant modes and with one RT is

$$\begin{aligned}
\Delta J_{k+1} &= \Delta J_k + (\Delta J_{k+1} - a)\{\beta m \sin(m\vartheta_k) + \eta(m+1)\sin[(m+1)\vartheta_k + nt_{k+1}]\} \\
\vartheta_{k+1} &= \vartheta_k + \frac{2\pi}{N_r}(\Delta J_{k+1} - \alpha \Delta J_{k+1}^2) + \beta\cos(m\vartheta_k) + \eta\cos[(m+1)\vartheta_k + nt_{k+1}] \\
t_{k+1} &= t_k + \frac{2\pi}{N_r m}
\end{aligned} \tag{5.5}$$

5.2.2 Numerical Results

In all numerical calculations we choose $m = 3$ and $n = 1$ introducing the resonant modes (1:3) and (1:4). The constant α that appears in Eq. (5.3) is adjusted for both maps, in the twist case $\alpha = 0$ and in the non-twist case $\alpha = 160.15$. We will split

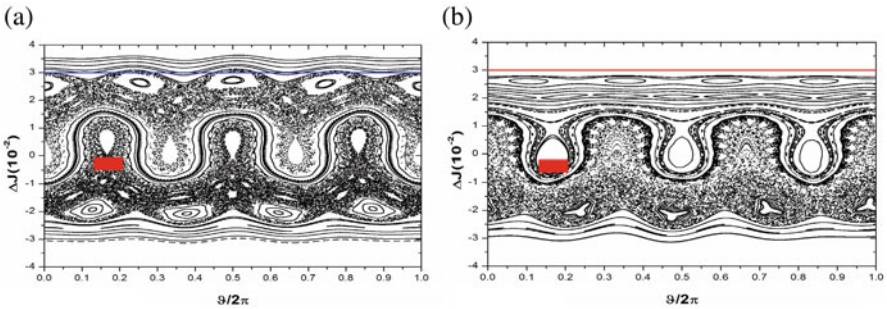


Fig. 5.1 Poincaré sections for the resonant modes (1:3) and (1:4) for $\alpha = 160.15$: **(a)** non-twist map, Eq. (5.4), without RT with $\beta = -1.3 \times 10^{-4}$ and $\eta = 1.15 \times 10^{-4}$, the blue line at $\Delta J = 3$ is only to guide the eyes, it is not a RT; **(b)** non-twist map, Eq. (5.5), with RT (in red) at $\Delta J = 3$, with $\beta = -4.0 \times 10^{-3}$ and $\eta = 9.2 \times 10^{-3}$

our study in four analyses: (i) Case non-twist without RT from Eq. (5.4); (ii) Case non-twist with one RT from Eq. (5.5); (iii) Case twist without RT from Eq. (5.4); (iv) Case twist with one RT from Eq. (5.5).

For the non-twist equilibrium, there are isochronous resonances in the phase space. Figure 5.1a corresponds to the case (i), where two isochronous resonances (1:3) have already reconnected and they are dimerized and separated by shearless curves [6]. Such invariant curves encircle both sets of surviving islands (1:3) and they exist only in non-twist maps. In plasma confinement approaches, shearless curves trap the magnetic field lines for a long time hampering the radial diffusion in such a way that an internal transport barrier (ITB) takes place due to a strong stickiness effect [41].

Figure 5.1b corresponds to the case (ii), the RT is indicated in blue color at the position $\Delta J = 3 = a$, and the alterations introduced in the dynamics are noticeable. We still have the two (1:3) and (1:4) island chains, but now an interesting topological rearrangement occurred, the chaotic sea that was close to the upper island chain (1:4), in Fig. 5.1a, had been suppressed due to the presence of the RT. The motion near the RT is kept integrable while on the other side of the shearless curve there is still a significant destruction of the invariant curves around the low resonance (1:4), showing a visible chaotic sea.

Figure 5.2a corresponds to the case (iii) and we see the dominant island chain (1:3) and the main secondary resonance (1:4) embedded in a chaotic sea with several other secondary resonances. In this configuration, the barriers are in fact the island structures, which will be destroyed as the parameters are varied. On the other hand, in Fig. 5.2b, which corresponds to the case (iv), we can see the stabilizing alterations introduced by the RT, which is indicated in red color in $\Delta J = a = 3$. Note that, for the parameters used to plot Fig. 5.2b, the chaotic sea near the RT, around the island (1:3), is suppressed and the neighborhood around $\Delta J = 3$ is more stable than the one of Fig. 5.2a, evidencing the local effect of the RT.

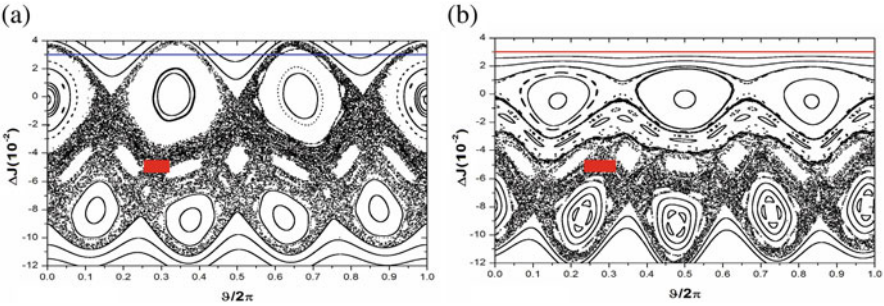


Fig. 5.2 Poincaré sections for the resonant modes (1:3) and (1:4) for $\alpha = 0$: **(a)** twist map, Eq. (5.4), without RT with $\beta = -6.0 \times 10^{-4}$ and $\eta = 2.99 \times 10^{-4}$, the blue line at $\Delta J = 3$ is only to guide the eyes, it is not a RT; **(b)** twist map, Eq. (5.5), with RT (in red) at $\Delta J = 3$, with $\beta = -7.5 \times 10^{-3}$ and $\eta = 8.2 \times 10^{-3}$

We should emphasize that different Hamiltonians (maps) govern the system without RT (Figs. 5.1a and 5.2a) and with RT (Figs. 5.1b and 5.2b) and hence, the scale of the parameters is different. In addition, the ranges of β are different because the non-twist system has one term that does not appear in the twist system.

For the non-twist cases, shearless curves play the role of partial barriers and they interfere differently on the transport depending on the presence of the RT. When the system does not have a RT, case (i), the trajectories reach the reference line after the destruction of the shearless curve. However, this kind of curve has a discontinuous behavior, it disappears for some values of β but it exists for many other intercalated values of β. So, for the case (i) without RT, the fluctuations seen in Fig. 5.2a show exactly the sensibility of the shearless curve for different values of the perturbation, while for the case (ii) with a RT, in Fig. 5.2b the trajectories reach the reference line, which is in fact the RT, only after the destruction of the resonance mode (1:4). Hence, in this case, the destruction or existence of the shearless curves is not a sufficient condition for the considered orbits to reach the reference position. For the twist case, the small red boxes in Figs. 5.1 and 5.2 are above the resonance (1:4), thus we are going to investigate the effect of the RT in the neighborhood of the resonance mode (1:3).

Figure 5.3c and d corresponds to the cases (iii) and (iv), respectively, and we observe a similar behavior in the transport rates but with a small nuance. In the case without RT, Fig. 5.3c, as the structures are being destroyed the transport increases but the trajectories still find some regions with a soft stickiness. In the case with RT, Fig. 5.3d, as the resonance structures are destroyed the transport increases abruptly and we observe a soft stickiness only around the RT.

The non-liner maps that we use here may describe the equilibrium magnetic field lines perturbed by resonances created by rings of ergodic magnetic limiters. In previous works [1,14], robust torus had proved to be, theoretically, an efficient transport barrier that prevented the leaking of magnetic field lines toward the tokamaks wall, avoiding plasma—wall interactions.

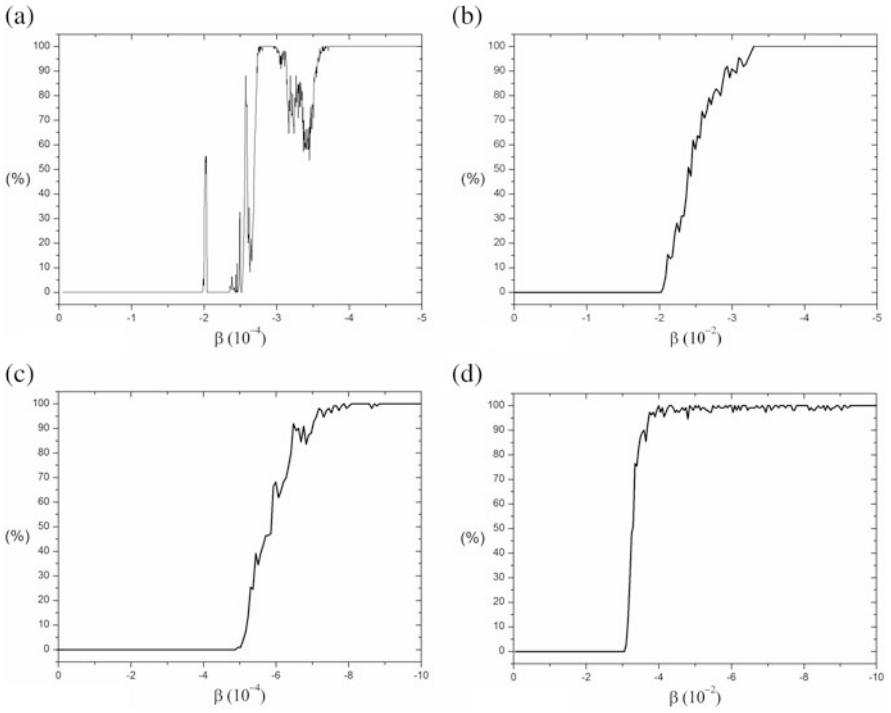


Fig. 5.3 Percentage of initial conditions that arrived $\triangle J = 3$ up to 2500 iterations: (**a**) Non-twist map without RT, Eq. (5.4); (**b**) Non-twist map with RT, Eq. (5.5); (**c**) Twist map without RT, Eq. (5.4); (**d**) Twist map with RT, Eq. (5.5)

We showed here the behavior of two different transport barriers, one formed by the reconnection process of isochronous resonances, the meandering curves, and another formed by the vanishing of the perturbation in a particular region, through the introduction of a robust torus (RT). In our approach the RT is stronger than the meandering barriers due to the stabilizing effect it induces in its neighborhood. We checked the behavior of dynamical transport from our maps for different resonant modes with rotation number close to the resonant mode (1:3) used here, and we have observed that the qualitative behavior of the transport is similar with the plots showed in Fig. 5.3. The RT plays a remarkable influence on the dynamical transport; this is the reason why this kind of robust barrier is relevant for Hamiltonian plasma approaches.

5.3 Hyperchaos and Intermittency in 3D Rayleigh-Bénard Convection

Convective instability is one of the most common mechanisms of fluid motion in the Universe, acting in a wide variety of systems from plasma motion in astrophysical objects to simple boiling processes of daily human tasks. In natural convection the density of the particles in a fluid is affected by a gradient of temperature, when a warmer (lighter) bottom layer tends to rise and the cooler top layer tends to sink in a motion induced by buoyancy forces [18]. This fluid motion plays an important role in many branches of physics, for instance, in the Earth's climate [10, 21] and in the granulation patterns (fluid cells) on the Sun's surface [19, 30]. The solar magnetic field is produced by convective flows (in the lowest levels of its convective zone) [32] and in the case of Earth's magnetic field a convective motion of a conducting fluid is the key mechanism for magnetic field generation [13, 39]. Experiments on thermal convective systems started in the beginning of the last century by Henri Bénard. He observed space periodic hexagonal patterns in a thin horizontal fluid layer with a free upper surface [3]. Using a linear stability analysis, Lord Rayleigh provided a first theoretical explanation basis to the Bénard's experiments [23]. Since then, a Newtonian fluid flow heated from bellow in a plane layer is known as Rayleigh-Bénard (RB) convection and much theoretical and experimental effort has been employed in studies of pattern formation and nonlinear effects in thermal convective systems [8, 31, 40, 42, 45].

In this work we follow the works by Podvigina [34, 35], where several attractors and bifurcations of the convective system are identified but no details on the chaotic attractors are shown, neither are their Lyapunov exponents computed. We classify all attractors according to the number of positive Lyapunov exponents and investigate the transient chaos and intermittency present before a transition to hyperchaos.

5.3.1 Rayleigh-Bénard Convection and Numerical Methods

We model RB convection under the Boussinesq approximation in a three-dimensional Euclidean system where vertical and horizontal boundaries are periodic and stress free, respectively. The temperatures of the top and bottom boundaries are held constant (see Fig. 5.4). The fluid flow in the presence of the gravitational force is described by the Navier-Stokes (5.6), continuity (5.7), and heat conduction (5.8) equations [7]. RB convection is characterized by two dimensionless parameters, the Prandtl number $P = \frac{\nu}{\kappa}$ representing the physical properties of the fluid and the Rayleigh number $Ra = \frac{\alpha g \delta T d^3}{\nu \kappa}$ giving the magnitude of the buoyancy force. The constants ν, κ, and α are, respectively, the kinematic viscosity, thermal diffusivity, and thermal expansion coefficients of the fluid.

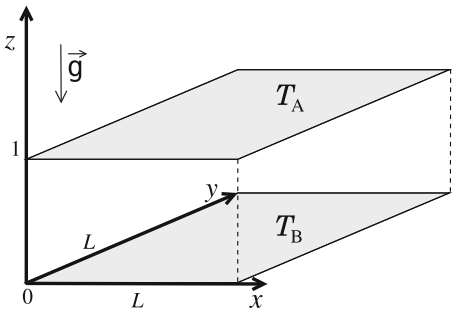


Fig. 5.4 Computational domain: fluid is confined in a square periodic cell $\mathscr{D} = [0, L]^2 \times [0, 1]$. Temperatures of upper T_A and bottom T_B planes remain fixed

$$\frac{\partial \mathbf{v}}{\partial t} = P\nabla^2\mathbf{v} + \mathbf{v} \times (\nabla \times \mathbf{v}) + PRa\theta\mathbf{e}_z - \nabla p, \tag{5.6}$$

$$\nabla \cdot \mathbf{v} = 0, \tag{5.7}$$

$$\frac{\partial \theta}{\partial t} = \nabla^2\theta - (\mathbf{v} \cdot \nabla)\theta + v_z, \tag{5.8}$$

where $\theta(\mathbf{x}, t) = T(\mathbf{x}, t) - (T_B + (T_A - T_B)z)$ represents the temperature profile. We solve Eqs. (5.6)–(5.8) numerically by applying the pseudospectral method [5]. The equations are integrated in time using the ETDRK3 (third-order exponential time differencing method) [12] with constant time step.

5.3.1.1 Lyapunov Exponents

The three largest Lyapunov exponents ($\lambda_1 \geq \lambda_2 \geq \lambda_3$) were computed to classify all convective attractors found in this work. Positive values correspond to exponential divergence in time of nearby initial conditions. A chaotic attractor has at least one positive Lyapunov exponent, $\lambda_1 > 0$, whereas hyperchaotic regimes have two or more positive exponents (e.g. $\lambda_1 > 0$ and $\lambda_2 > 0$). Periodic solutions have one vanishing exponent and the total number of null Lyapunov exponents in quasiperiodic sets correspond to the total number of their incommensurate frequencies [1, 20]. We apply the algorithm described in [17] to compute the three largest Lyapunov exponents. The exponents are obtained averaging, in time, finite time local exponential divergences of amplitudes of perturbations. The set of perturbative states is constructed using the Gramm-Schmidt orthonormalization process.

5.3.2 Numerical Results

Attractors of Rayleigh-Bénard convection are studied for $Ra \in [1720, 2500]$. All other parameters were fixed, with the domain size given by $L \times L \times 1$, with $L = 4$

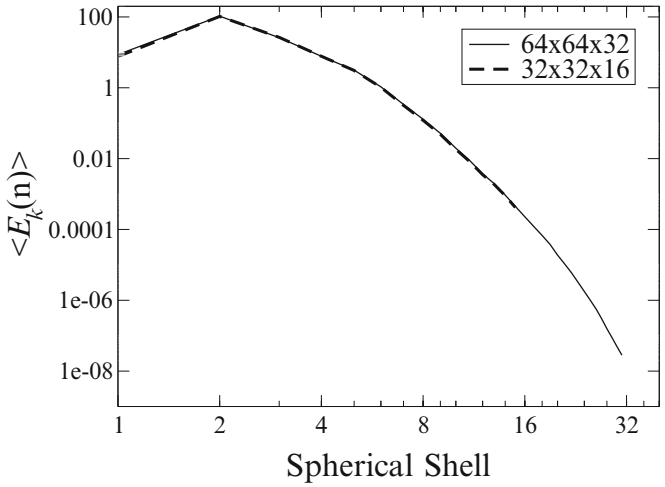


Fig. 5.5 Time-averaged kinetic energy spectra for the hyperchaotic attractor at $Ra = 2500$. Results for numerical spatial resolution of $32 \times 32 \times 16$ (solid line) and $64 \times 64 \times 32$ (dashed line) Fourier harmonics are plotted

and $P = 0.3$. In this work, we use the following vector notation for the Fourier coefficients and wave vectors:

$$\hat{\mathbf{v}}_{\mathbf{k}} = (\hat{v}_{\mathbf{k}}^{x}(t), \hat{v}_{\mathbf{k}}^{y}(t), \hat{v}_{\mathbf{k}}^{z}(t)), \ \mathbf{k} = (k_x, k_y, k_z).$$

Two spatial numerical resolution of $32 \times 32 \times 16$ and $64 \times 64 \times 32$ Fourier harmonics were employed as a test of the numerical convergence of the system solution. The kinetic energy spectra of the smaller spatial resolution decreases at least by five orders of magnitude for all regimes studied in this work, e.g., see Fig. 5.5 for a hyperchaotic attractor at $Ra = 2500$. A higher spatial resolution leads to a significant increase of computational time with no significant changes in the solutions. We also performed several tests with different time steps and all results remained unchanged. Thus, we applied $32 \times 32 \times 16$ Fourier harmonics with $dt = 10^{-3}$ as the constant time step.

The schematic bifurcation diagram in Fig. 5.6 summarizes all convective attractors found in this work. Dotted lines represent regions where the attractor becomes unstable but a transient resembling its behavior is still observed. The inset shows the range of time-averaged kinetic energy $< E_k >$ for each stable branch.

Figure 5.7 shows the three largest Lyapunov exponents for attractors in the bifurcation diagram of Fig. 5.6. In what follows we present the regimes found and briefly discuss the global and local bifurcations involved.

For $1720 \leq Ra \leq 1790$, a periodic attractor (PA) is observed, with $\lambda_1 = 0$ and $\lambda_{2,3} < 0$. Near $Ra = 1790$, PA is substituted by a quasiperiodic attractor (QPA1), with $\lambda_{1,2} = 0$ and $\lambda_3 < 0$. PA becomes QPA1 in a Hopf bifurcation giving rise to

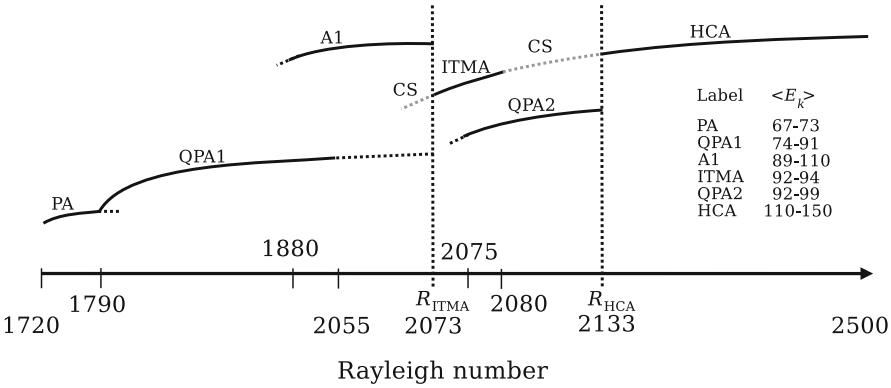


Fig. 5.6 Bifurcation diagram and the time averaged kinetic energy (inset) of the regimes found for $Ra \in [1720, 2500]$

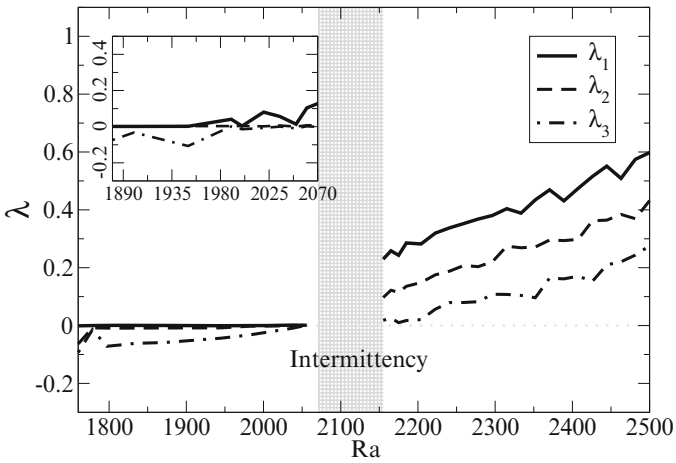


Fig. 5.7 The three largest Lyapunov exponents for attractors in the bifurcation diagram of Fig. 5.6. The upper inset shows the exponents for the A1 family. Attractors with at least two positive Lyapunov exponents are hyperchaotic

one drifting incommensurate frequency in the laboratory frame (this corresponds to the vanishing second Lyapunov exponent, see Fig. 5.7). Kinetic energy time series of PA and QPA1 are shown in Fig. 5.8a and b, respectively. QPA1 loses stability in a boundary crisis at $Ra \approx 2055$, leading to a quasiperiodic transient behavior (dotted line after QPA1 in Fig. 5.6), before the trajectories leak toward attractor A1.

For some range of Ra the QPA1 coexists with the A1 family of attractors. As Ra is increased, A1 bifurcates unto a set of quasiperiodic and chaotic attractors, where the latter are characterized by one positive Lyapunov exponent, as seen in the inset of Fig. 5.7. The first regime of A1 is a quasiperiodic attractor with two vanishing Lyapunov exponents. Figure 5.9a shows the kinetic energy time series for

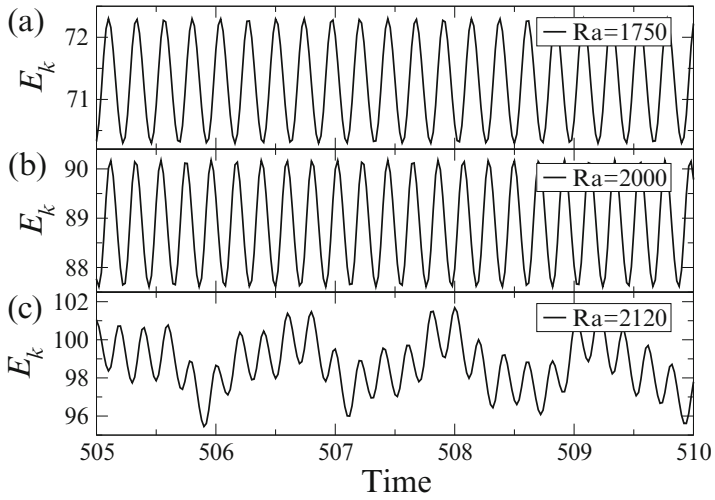


Fig. 5.8 Kinetic energy time series for $Ra = 1750$ (**a**), $Ra = 2000$ (**b**) and $Ra = 2120$ (**c**)

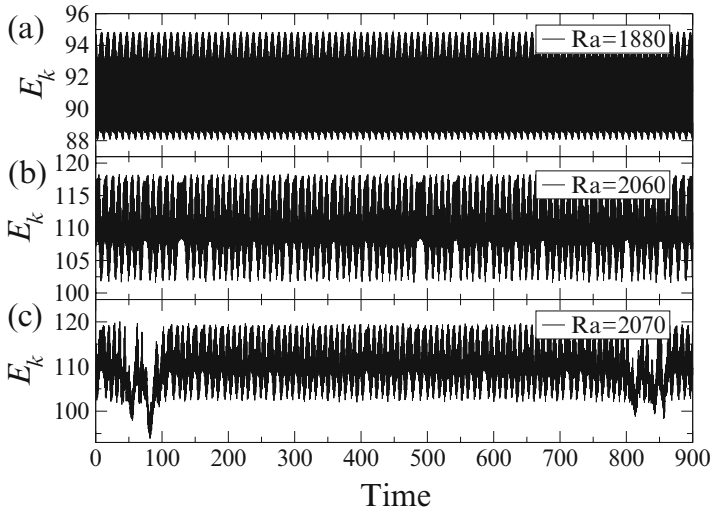


Fig. 5.9 Kinetic energy time series for three attractors of A1 family. $Ra = 1880$ (**a**), $Ra = 2060$ (**b**), and $Ra = 2070$ (**c**)

a quasiperiodic attractor of A1 family at $Ra = 1880$. A complete characterization of the A1 family is out of the scope of this paper but we stress that some of the A1 chaotic attractors present irregular energy bursts, as can be seen in the kinetic time series for $Ra = 2060$ in Fig. 5.9b (e.g., $100 \leq t \leq 150$) and $Ra = 2070$ Fig. 5.9c (e.g., $800 \leq t \leq 850$). The last stage of A1, $Ra = 2070$, is a chaotic attractor that is destroyed in an interior crises, when the intermittent attractor ITMA takes place.

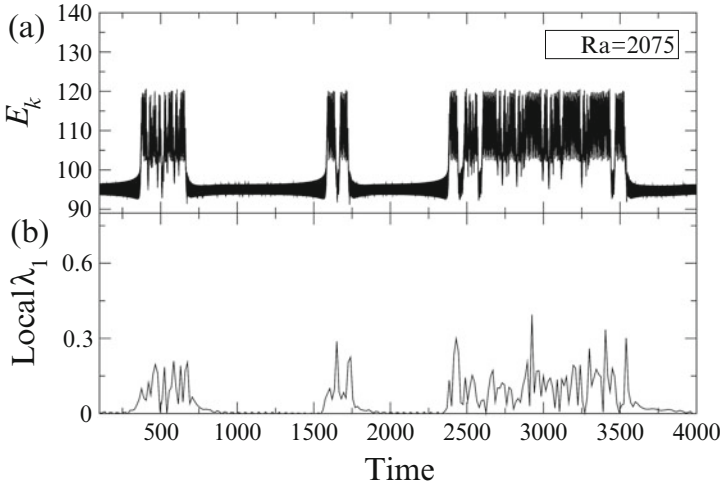


Fig. 5.10 Kinetic energy time series for ITMA $Ra = 2075$ and the local exponential divergence. The convergence of the Lyapunov exponents is compromised by the intermittent behavior of the fluid flow

The kinetic energy time series for $Ra = 2075$, shown in Fig. 5.10a, reveals the "on-off" intermittent nature of ITMA. We found this attractor to be composed by random switches between two main phases—a quasiperiodic phase (on—related to the destabilized QPA1), and a chaotic one (off—related to the destabilized A1). The ITMA creation portrays an interior crisis scenario, where a collision involving the destabilized QPA1 and attractor A1 takes place [37]. ITMA emerges at $Ra = 2073$ and is destroyed in its own boundary crisis at $Ra \approx 2080$, leaving behind a chaotic saddle CS represented by dotted gray lines in Fig. 5.6. Figure 5.10b shows the local Lyapunov exponents for $Ra = 2075$. Since the fluid flow switches intermittently between laminar and chaotic motions, seen as phases with positive and zero values of the largest local Lyapunov exponent, the statistical convergence of the Lyapunov exponents may not be reached in a reasonable computational time. Thus, this region of intermittency is represented by the gray area in Fig. 5.7.

For $2075 \leq Ra \leq 2130$ a new quasiperiodic attractor QPA2 appears, with three incommensurate frequencies $\lambda_{1,2,3} = 0$, as can be seen in Fig. 5.7. The time series of the kinetic energy for QPA2 at $Ra = 2120$ is presented in Fig. 5.8c. The similarity of the averaged kinetic energy (see inset of Fig. 5.6) and projections of phase space (not shown here) when compared to the QPA1 regime indicate a connection between both attractors, which was left to be studied in future works. The destruction of QPA2 occurs in a global bifurcation detailed bellow.

The hyperchaotic attractor HCA emerges at $Ra_{\mathrm{HCA}} = 2133$ in a second interior crisis. This global bifurcation involves QPA2 and the chaotic saddle CS, resulting in an enlarged set. The HCA appears with at least two positive Lyapunov exponents. The second positive exponent emerges during the CS regime between $2080 \leq Ra \leq 2133$.

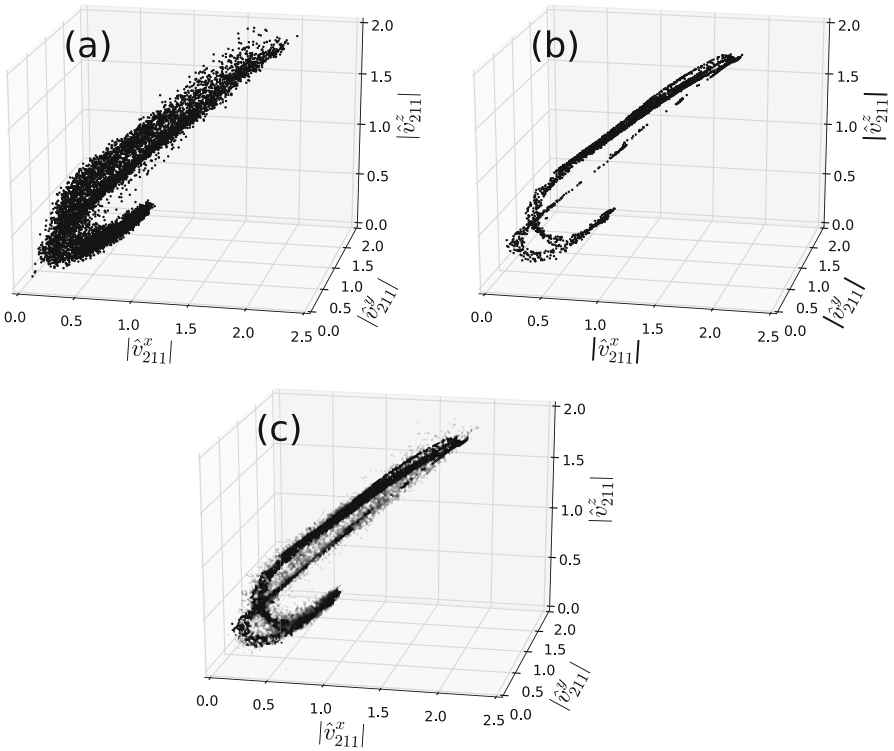


Fig. 5.11 Poincaré section of HCA attractor $Ra = 2160$ (**a**), QPA2 attractor $Ra = 2120$ (**b**), chaotic saddle (gray dots) and QPA2 (black dots) (**c**)

In order to visualize regimes involved in the second interior crisis we set up a Poincaré map defined by $|v^z_{2,1,2}| = 0.25$, with $\frac{d|v^z_{2,1,2}|}{dt} > 0$. The Fourier harmonics $|v^y_{2,1,1}|$, $|v^x_{2,1,1}|$, $|v^z_{2,1,1}|$ are plotted whenever a trajectory meets the conditions given by the Poincaré map. Figure 5.11a shows the Poincaré section of HCA for $Ra = 2160$ and Fig. 5.11b shows QPA2 for $Ra = 2120$. Both HCA and QPA2 have approximately the same shape, but the points in HCA are more spread out, as a result of the aggregation of SCS. Figure 5.11c shows both QPA2 (black points) and CS (gray points) extracted from long transients time series just before the creation of HCA. The union of both sets resembles the hyperchaotic attractor, confirming that CS becomes coupled to QPA2 in an interior crisis scenario.

5.3.2.1 Hyperchaotic Intermittency

The hyperchaotic attractor HCA also displays an intermittent behavior for values of the parameter close to R_{HCA}. The destabilized QPA2 becomes imbibed in it and is responsible for the laminar (quasiperiodic) phases of the kinetic time series, see for instance, Fig. 5.12a. In order to study this intermittency, we define $\Delta\tau$ as the

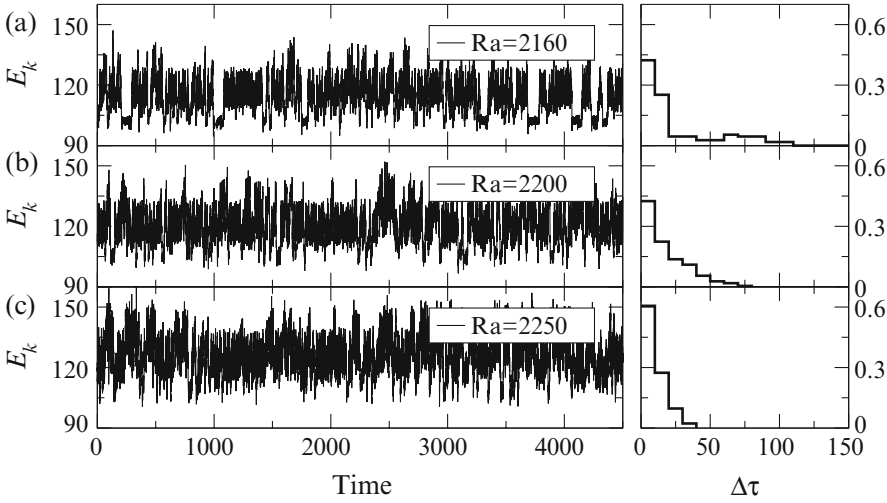


Fig. 5.12 Kinetic energy time series (left panels) and histogram of the duration of laminar phases in hyperchaotic attractors (right panels) for $Ra = 2160$ (**a**), $Ra = 2200$ (**b**), and $Ra = 2250$ (**c**)

time spent by the trajectory in the laminar phase. We computed $\Delta\tau$ for different increasing Rayleigh numbers $Ra > Ra_{\mathrm{HCA}}$. Figure 5.12 presents three kinetic time series and a histogram of $\Delta\tau$ for three different values of the Rayleigh number in the hyperchaotic regime. For $Ra = 2160$, Fig. 5.12a shows that the trajectory spends a large time near the destabilized QPA2 phase space before escaping to a chaotic motion. Values of $\Delta\tau$ are up to $\Delta\tau \approx 100$ time units. As we increase Ra the laminar phases become shorter, as can be seen in Fig. 5.12b for $Ra = 2200$. If we keep increasing Ra, e.g., Fig. 5.12c for $Ra = 2250$, a higher frequency of switches is identified. In this case, the trajectory spends most of the time in the chaotic rather than the quasiperiodic phase. For larger values of Ra the laminar phases cannot be distinguished and the intermittency is hidden in a fully hyperchaotic attractor.

The decrease of laminar time phases is a known phenomenon in intermittent regimes. As the system moves away from the critical bifurcation point, one of the phases of the time series tends to shorten and eventually vanishes [15]. In our case this suppression of the laminar phases follows a linear scaling dependence with the control parameter, with a slope of -1.17 (see Fig. 5.13). For values of $Ra > 2250$ the quasiperiodic phases can barely be seen and the chaotic phases dominate.

5.4 Conclusions

We have shown two examples of the importance of studying transport between regions of the phase space in chaotic systems. In the symplectic maps, the behavior of two different transport barriers was described, one formed by the reconnection

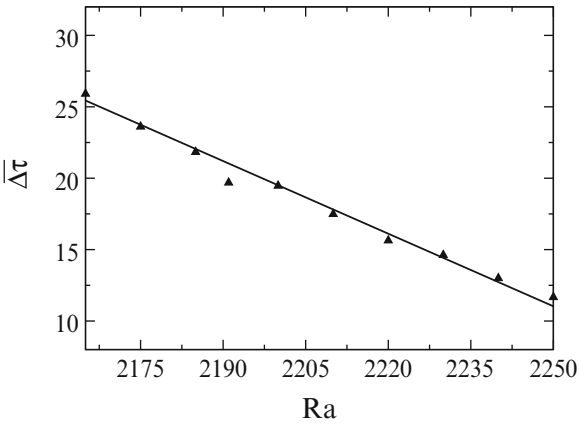


Fig. 5.13 Mean duration of the laminar phases for HCA as a function of *Ra*

process of isochronous resonances, the shearless curve, and another formed by the vanishing of the perturbation in a particular region through the introduction of a robust torus (RT). In our approach the RT is stronger than the shearless curve due to the stabilizing effect it induces in its neighborhood. The RT plays a remarkable influence on the dynamical transport; this is the reason why this kind of robust barrier is relevant for Hamiltonian plasma approaches. In the Rayleigh-Bénard convection simulation, a series of global bifurcations lead to diverse types of transient and intermittent behaviors, with boundary crises resulting in chaotic transients and interior crises resulting in intermittent switchings. The most striking result is the leaking of trajectories between a quasiperiodic region and a hyperchaotic region of the phase space after an interior crisis.

Note that the first study on transition to chaos in Rayleigh-Bénard convection in the absence of magnetic fields was conducted by Edward Lorenz [24], who adopted a reduced model that was later shown to exhibit dynamical phenomena such as crises and Pomeau-Manneville intermittency (see, e.g., Alligood et al. [1]). The *classical* Lorenz model has only three coupled ordinary differential equations and, therefore, is unable to produce hyperchaos due to its low-dimensional phase space. A subsequent generalization that included a horizontal background magnetic field was proposed by Macek and Strumik [25], who reported the presence of hyperchaos in the system. By conducting a direct numerical simulation of the 3D MHD equations, we confirm the existence of hyperchaos in magnetized RB convection.

The main bifurcations found in the present work were also studied in experiments with different setups in the past, e.g., route to convective chaos through quasiperiodicity and period doubling bifurcations of periodic states are reported in [14] and intermittency route to chaos in [4]. Our results have similarities with quasiperiodic route to stationing patterns in Rayleigh-Bénard convection [11], we also note similarities of our results compared to two-dimensional RB convection reported by Paul et al. [33], where traveling waves, multistability, quasiperiodic phase-locking route to chaos, as well as an attractor-merging crisis and detection of hyperchaos are discussed.

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Chapter 6
An Investigation of the Chaotic Transient for a Boundary Crisis in the Fermi-Ulam Model



Edson D. Leonel and Murilo F. Marques

6.1 Introduction

Dynamical systems are often tuned by control parameters [1, 2]. The variation of them may lead to interesting properties. In conservative systems transitions such as integrability to nonintegrability [3] are observed as the parameters are varied, transport of particles can be investigated [4] as invariant tori are broken due to a parameter change and many other different consequences emerge [5, 6]. In dissipative systems, no matter if described by ordinary differential equations [7] or by mappings [8, 9], bifurcations are often observed. They can be classified into two different types [10]: (1) local and (2) global bifurcations. The local bifurcations [11] are identified and characterized only by the investigation of local dynamics, mostly by the analysis of fixed points and their stability. The set of nonlinear equations describing the system is linearized near the fixed point allowing to only local inferences. The local knowledge of a given phenomena cannot be extended to all the phase space due to the linear stability analysis. Birth of attractors such as fixed points may lead to pitchfork bifurcations,[1] saddle node bifurcation with one fixed point stable and other unstable, transcritical bifurcation where there is a coexistence of fixed points but at the bifurcation they exchange stability, period doubling bifurcations, tangent bifurcations, Hopf bifurcations among others. In a global bifurcation [11], the knowledge of the stability of a fixed point is not enough to

[1]Depending on the type of the fixed points born at the bifurcations as well as their stability, the pitchfork bifurcation can be classified as supercritical or subcritical.

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foresee a bifurcation. Generally the change in the phase space due to the occurrence of a global bifurcation is catastrophic. It leads then to a major modification in the form of the phase space. Such types of bifurcations are commonly called in the literature as crisis [12, 13]. Three well-known crises [14] in the scientific community are: (i) interior crisis; (ii) merging attractor crisis; (iii) boundary crisis. In case (i) one can observe in the system a sudden increase in the size of the chaotic attractor due to a collision of a periodic orbit with the chaotic attractor. For the case (ii) two or more attractors collide simultaneously among each other and with a periodic orbit; as a result, the attractors become merged. Finally in case (iii) there is a crossing of an unstable with a stable manifold of a saddle fixed point leading to a sudden destruction of the chaotic attractor. If the manifolds are produced by the same saddle fixed point there is homoclinic crossing when a heteroclinic crossing is produced by intersections of manifolds generated from different saddle fixed points.

In this chapter we address a specific system to describe some properties for the chaotic transient observed in a boundary crisis. The model under consideration is a dissipative version of the Fermi-Ulam model [15]. The system consists of a classical particle which is confined to move between two rigid and infinitely heavy walls. One of them is fixed while the other moves periodically in time. Collisions of the particle with moving wall may lead to velocity changes in the particle. For elastic collisions [16], this was a frustrated model idealized by Fermi as an attempt to describe the acceleration of cosmic particles reaching the earth. The cosmic particles were supposed to be affected by the moving magnetic fields from the extra-galactic bodies therefore furnishing energy to the moving particle. The oscillating wall was introduced with this objective, to mimic the action of the magnetic fields. The fixed wall is an effective re-injection mechanism directing the moving particle for a further collision with the periodically varying wall. It is known that if the motion of the moving wall is not smooth enough, then the particle may experience no bound for the velocity. On the other hand, for a sinusoidally motion of the wall, the phase space of the model is composed by coexistence of periodic islands, invariant spanning curves, and chaotic seas. The invariant spanning curves work as barriers [17] not letting the particle to cross therefore confining the energy gain for a bouncing particle. The existence of them as a superior limit for energy of a chaotic orbit warrants local chaos.[2] Scaling properties give an overall description of the chaotic sea [17] proving it to be scaling invariant with respect to the control parameter, namely, the amplitude of the oscillation wall [18]. The determinant of the Jacobian matrix leads to measure preservation hence attractors are not observed. The scenario changes significantly when a fractional loss of energy upon collision is considered [19]. This is made by assuming the particle experiences an inelastic

[2]By local chaos we want to say the chaos observed is confined by invariant spanning curves. In a situation of global chaos the particle can diffuse unlimitedly in the velocity axis therefore leading to a phenomena called as Fermi acceleration. A prototype system exhibiting such phenomena is an analogous 1-D model called as a bouncer. The only difference from the Fermi-Ulam model is the returning mechanism for a further collision. In the Fermi-Ulam it is provided by a fixed wall whereas in the bouncer it is due to the gravitational field only.

collision each time it interacts with the walls. The determinant of the Jacobian matrix for the dissipative case is less than the unit. According to the Liouville's theorem [20], the dissipation leads to contraction of the phase space, hence yielding the system to exhibit attractors that may be periodic [21] or chaotic [22]. Our main goal in this chapter is then to investigate some properties for the chaotic transient for a boundary crisis observed due to the crossing of a stable and unstable manifold obtained from the same saddle fixed point.

We chose a specific combination of control parameters that lead the system to exhibit two types of attractors before the crisis: (i) an attracting fixed point[3] and (ii) a chaotic attractor. The chaotic attractor was confirmed by means of a positive Lyapunov exponent [23]. As soon as the control parameter is varied and the crossing is observed, the chaotic attractor is destroyed. It is then replaced by a chaotic transient [12]. Given an initial condition inside of the region of the phase space where the attractor likely existed prior to the crisis, the dynamics of a particle will run such region for a while until the particle finds the appropriate scape path and heads towards the attracting fixed point being captured by it and reaching the steady state at a sufficiently long time. Once the convergence to the steady state started there is no way to escape from there without an external action such as application of stochastic perturbation, a type of control with noise or other intervention on the system. We want to investigate the behavior of the survival probability a particle has to survive inside a region of the phase space where a likely chaotic attractor existed prior the crisis. As we shall see, the survival probability will be described by an exponential function whose relaxation time is given by a power law [22] measured in a parameter given the distance of the boundary crisis. Our interest is in the investigation of such a relaxation time, particularly the power law describing it. We will do it using two different procedures. The first one is by using extensive numerical simulations whereas the second considers the eigenvalues of the saddle point at the parameters leading to the boundary crisis. Investigation of transient in a gearbox model was also made in Refs. [24, 25].

The organization of this chapter is as follows. Section 6.2 describes the model constructing the equations that describe the dynamics of the system. It gives also a short presentation of the properties observed for the conservative case showing the phase space, organization of the stability islands, illustration of the first invariant spanning curve as well as the chaotic sea. In the presence of dissipation, the chaotic sea is replaced, depending on the combination of control parameters and initial conditions, by a chaotic attractor, which is also illustrated. In Sect. 6.3 we discuss the boundary crisis observed in a dissipative Fermi-Ulam model. The stable and unstable manifolds for a saddle point are plotted before the crisis. The inverse of the mapping is obtained and is used to construct the stable manifolds. The basin of attraction of either the chaotic attractor and attracting fixed point are plotted together with the chaotic attractor. Remarkably, the stable manifolds give the border of the two basin of attraction. A discussion of the relaxation time given the behavior of

[3]Indeed it is an asymptotically stable focus.

the survival probability for a particle to survive in the region of the phase space where a chaotic attractor likely existed is given. Numerical simulations determine the relaxation time and a connection with the results well settled [12] in the literature are made. Conclusions and final remarks are presented in Sect. 6.4 while a short Appendix at the end of the chapter gives the expressions for the eigenvalues of the saddle point.

6.2 The Model and the Map

We describe in this section the model we consider in the chapter. Our numerical simulations illustrating the nondissipative as well as the dissipative case will be made here too.

The system we consider is composed of a classical particle which is confined to move inside of two rigid and infinitely heavy walls. By choice, one of them is fixed at $x = l$ and the other one is periodically moving at $x(t) = \tilde{\varepsilon}\cos(\omega t)$, where $\tilde{\varepsilon}$ is the amplitude of oscillation, ω is the frequency of oscillation, l is the distance of the fixed wall from the equilibrium position of the moving wall, and t is the time. Figure 6.1 shows a schematic description of the model.

Because there are no forces acting in the particle along the flight it moves with a constant velocity between the collisions. The only modification in the velocity happens at the instant of the collision. We assume the particle experiences inelastic collisions with the walls. The restitution coefficient for collisions with the fixed wall is denoted as $\alpha \in [0, 1]$. The limit of $\alpha = 0$ terminates the dynamics since a single collision leads the particle to reach the state of rest therefore staying glued in the static wall. Collisions with the moving wall are characterized with a restitution coefficient $\beta \in [0, 1]$. The limit of $\beta = 0$ leads the particle to be re-injected in the dynamics with the maximum wall's velocity. This is a typical case of a dynamical

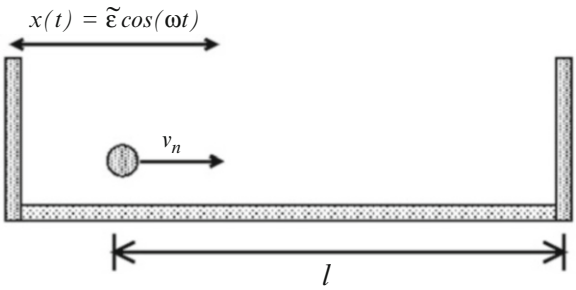


Fig. 6.1 Plot of a schematic description of the model. The motion of the oscillating wall is described by $x(t) = \tilde{\varepsilon}\cos(\omega t)$. The fixed wall is placed at $x = l$ measured from the equilibrium position of the moving wall. The restitution coefficients for collisions with fixed and moving walls are, respectively, $\alpha \in [0, 1]$ and $\beta \in [0, 1]$

regime called as locking [26]. The case of $\alpha = \beta = 1$ gives the nondissipative dynamics [27] whereas the phase space shows features typical of nonintegrable Hamiltonian systems.

The traditional way of describing the dynamics of the particle is the use of a mapping therefore updating the velocity of the particle after an impact with the moving wall. We see there is a large number of control parameters and that not all of them are relevant for the complete characterization of the model. We then define a set of dimensionless and hence more convenient variables as $V = v/(\omega l)$, $\varepsilon = \tilde{\varepsilon}/l$ and measure the time in terms of the number of oscillations that the moving wall completes in between the collisions hence measuring phases $\phi = \omega t$. The mapping is an application $T(V_n, \phi_n) = (V_{n+1}, \phi_{n+1})$ where V and ϕ denote both the dimensionless velocity of the particle and phase of the moving wall,

$$T : \begin{cases} V_{n+1} = V_n^* - (1+\beta)\varepsilon \sin(\phi_{n+1}) \\ \phi_{n+1} = [\phi_n + \Delta T_n] \mod 2\pi \end{cases}, \tag{6.1}$$

where the expressions for V_n^* and ΔT_n are determined by the kind of collision happened: (i) direct collisions[4] or (ii) indirect collisions.[5] For case (i), the expressions are $V_n^* = -\beta V_n$ and $\Delta T_n = \phi_c$ where $\phi_c \in (0, 2\pi]$ is the smallest solution of the equation $G(\phi_c) = 0$ with $G(\phi_c) = \varepsilon\cos(\phi_n + \phi_c) - \varepsilon\cos(\phi_n) - V_n\phi_c$. The numerical procedure used to solve $G(\phi_c) = 0$ was the bisection method with an accuracy of 10^{-14}. The determinant of the Jacobian matrix for this case is given by

$$\det(J) = \beta^2 \left[\frac{V_n + \varepsilon \sin(\phi_n)}{V_{n+1} + \varepsilon \sin(\phi_{n+1})} \right]. \tag{6.2}$$

For direct collisions the mapping is conservative only for the case of $\beta = 1$ where it preserves the following measure $d\mu = [V + \varepsilon \sin(\phi)]dVd\phi$.

For indirect collisions the expressions assume the following forms: $V_n^* = \beta\alpha V_n$ and $\Delta T_n = \phi_r + \phi_l + \phi_c$ where the auxiliary terms are $\phi_r = (1 - \varepsilon\cos(\phi_n))/V_n$, corresponding the time the particle spends traveling to the right-hand side until collides with the fixed wall and $\phi_l = (1-\varepsilon)/(\alpha V_n)$ is the time the particle spends moving backwards until reaches the collision zone at $x = \varepsilon$ while $\phi_c \in [0, 2\pi]$ is obtained from the smallest solution of $F(\phi_c) = 0$ where $F(\phi_c) = \varepsilon\cos(\phi_n + \phi_r + \phi_l + \phi_c) - \varepsilon + \alpha V_n\phi_c$. Again here the same numerical method was applied for the solution of $F(\phi_c) = 0$ with the same accuracy as before. For this case the determinant of the Jacobian matrix is written as

[4]Direct collisions are defined as the collisions the particle has with the moving wall without leaving the collision zone, a region defined by $x \in [-\varepsilon, +\varepsilon]$.

[5]In an indirect collision a particle leaves the collision zone, collides with the fixed wall, and is rebounded back for a further collision with the moving wall.

$$\det(J) = \alpha^2\beta^2\left[\frac{V_n + \varepsilon\sin(\phi_n)}{V_{n+1} + \varepsilon\sin(\phi_{n+1})}\right]. \tag{6.3}$$

The measure $d\mu$ is preserved only when $\beta = \alpha = 1$.

The term $-(1+\beta)\varepsilon\sin(\phi_{n+1})$ appearing in first equation of mapping (6.1) was obtained performing calculations in the referential frame of the moving wall. It is assumed to be instantaneously at rest at the moment of the impact. In such case the law of conservation of momentum is applied.

In what follows, we discuss two distinct regimes in separate sections. The first is the nondissipative case where the second one is the dissipative case leading to the boundary crisis for specific combinations of control parameters.

6.2.1 Nondissipative Case

When the two restitution coefficients assume $\alpha = \beta = 1$, the collisions of the particle with either walls are elastic and no fractional loss of energy is observed. The phase space is mixed with the existence of stability islands surrounded by a chaotic sea which is limited by a set of invariant spanning curves [6, 17]. These curves, in particular the first one, work as barriers not letting a flux of particle through them. Figure 6.2 illustrates well a typical phase space of the model for the nondissipative case. The parameter used is $\varepsilon = 10^{-2}$. The first invariant spanning curve is highlighted in the figure and is identified as *fisc*. It works as a barrier not letting any flux of particle through it.

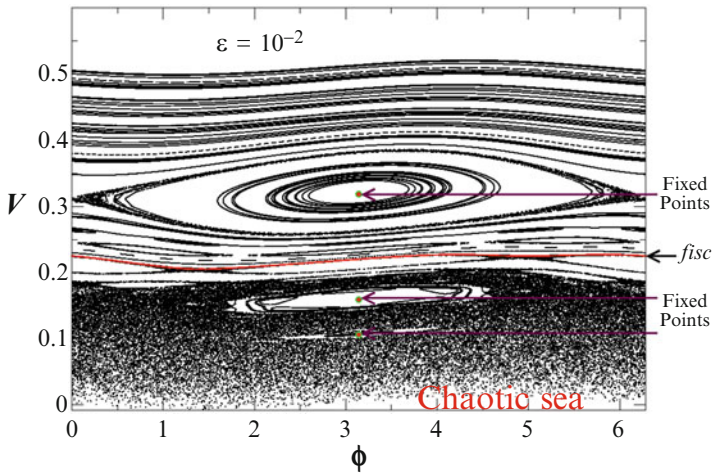


Fig. 6.2 Plot of the phase space for mapping (6.1) considering $\varepsilon = 10^{-2}$. The invariant spanning curve, represented by *fisc*, is shown in red (gray) color. Periodic islands are present as well as a chaotic sea

The periodic fixed points classified as elliptic are identified in the figure centering the periodic islands. For the parameter $\varepsilon = 10^{-2}$ there are only three elliptic fixed points, namely $V_i^* = 1/(i\pi)$ for $i = 1, 2, 3$ and $\phi = \pi$. For the regime of low velocity there is a region dominated by a chaotic sea. Some relevant information [17] of the chaotic sea are (1) Its amplitude depends on the position of the first invariant spanning curve. A first order approximation furnishes the position of the curve as $V_{fisc} \cong 2\sqrt{\varepsilon/0.9716}$, therefore scaling with $\varepsilon^{1/2}$. Starting with a low initial velocity, typically $V_0 < \varepsilon$, the average velocity grows to start with, its diffusion is given by a slope 1/2 in n, i.e. $\overline{V}(n) \cong \sqrt{\varepsilon^2 n}$. The changeover from the growth to a regime of constant velocity is marked by $n_x \propto \frac{1}{\varepsilon}$. A scaling law [17] describing the invariance of the chaotic sea with respect to either the parameter ε or n is given by $z = \frac{\tilde{\alpha}}{\tilde{\beta}} - 2$, where $\tilde{\alpha}$ gives the saturation of the velocity, $\tilde{\beta}$ marks the regime of growth and z gives the exponent of crossover. For the Fermi-Ulam model [17] the critical exponents are $\tilde{\beta} = 1/2, \tilde{\alpha} = 1/2$, and $z = -1$. The Lyapunov exponent [23] was measured for the chaotic sea. Indeed the Lyapunov exponents are defined as

$$\lambda_j = \lim_{n\to\infty} \frac{1}{n} \ln |\Lambda_j|, \quad j = 1, 2, \tag{6.4}$$

where Λ_j are the eigenvalues of the matrix $M = \Pi_{i=1}^{n} J_i(V, \phi)$ where J_i is the Jacobian matrix of the mapping (6.1) evaluated along the orbit. Figure 6.3 shows a

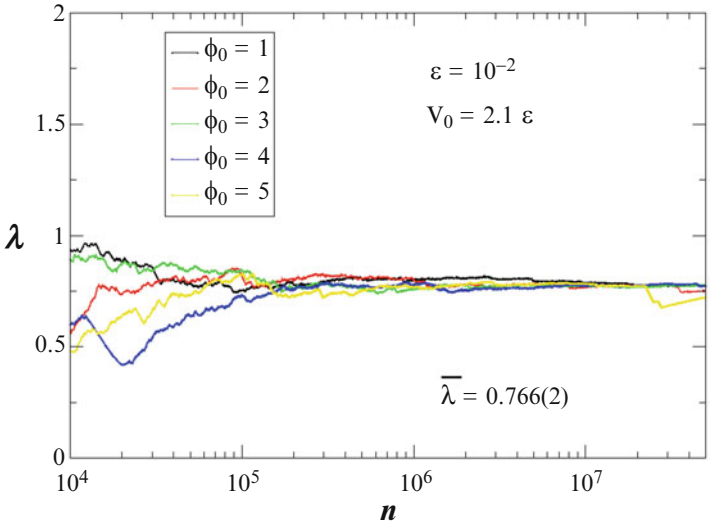


Fig. 6.3 Plot of five different Lyapunov exponent series obtained for the parameter $\varepsilon = 10^{-2}$, initial velocity $V_0 = 2.1\varepsilon$ and different initial phases as $\phi_{01} = 1$, $\phi_{02} = 2$, $\phi_{03} = 3$, $\phi_{04} = 4$, $\phi_{05} = 5$. An average of the asymptotic behavior gives us $\overline{\lambda} = 0.766(2)$. The positive finding for the Lyapunov exponent confirms chaos in the phase space. Horizontal axis is plotted logarithmic only for visual purposes. Due to the fluctuations the initial $n = 10^4$ were omitted from the plot

plot of five different series generated by different initial conditions along the chaotic sea for the parameter $\varepsilon = 10^{-2}$. The initial velocity was fixed as $V_0 = 2.1\varepsilon$ while the initial phases chosen were $\phi_{01} = 1$, $\phi_{02} = 2$, $\phi_{03} = 3$, $\phi_{04} = 4$, $\phi_{05} = 5$. One can see from the Lyapunov exponent time series there is a relative large fluctuation at short n when the series converge for large n. The initial 10^4 points of the series were not plotted in the figure since the fluctuations obtained for short time are large for the scale of the figure. Horizontal axis was plotted logarithmic for visual purposes only. From the five series we obtained an average positive Lyapunov exponent as $\overline{\lambda} = 0.766(2)$, therefore confirming the existence of chaos in the phase space. We must stress here the chaos observed is nondissipative since the phase space preserves a measure $d\mu$. In sequence we discuss the consequences of the dissipation in the collisions.

6.2.2 *Dissipative Case*

When the collisions are considered inelastic, the structure of the phase space changes significantly [19]. The mixed properties are not observed anymore. Depending on the combination of control parameters as well as initial conditions, attractors—either periodic or chaotic—are observed in the phase space. Figure 6.4 shows a plot of a chaotic attractor and an attracting fixed point for the parameters $\varepsilon = 0.04$, $\beta = 1$, and[6] $\alpha = 0.93624$. The chaotic attractor is marked by the dark region on the bottom part of the plot and is limited from below by the velocity of the moving wall, plotted as a continuous line (red). Attracting fixed point is highlighted as a bullet when saddle point is plotted as a star. The Lyapunov exponent measured for the chaotic attractor was $\overline{\lambda} = 1.7743(5)$ as reported in Ref. [19].

In the dissipative case, each attractor has now its own basin of attraction [10]. The basin corresponds to a set of points in the phase space whose dynamics heads towards the attractor for sufficiently long time. It is known that a boundary crisis may be investigated by analyzing the behavior of the basin of attraction of the chaotic attractor. Indeed a boundary crisis is produced [14] by a collision of the border of the basin of attraction of the chaotic attractor with the chaotic attractor itself. Such a collision implies in a sudden destruction of the chaotic attractor and also of its basin of attraction. The collision creates a homoclinic orbit produced by successive crossings of the stable and unstable manifolds generated by the same saddle fixed point and that the basin of attraction of the attracting fixed point becomes porous and suitable to be visited by the dynamics of the chaotic attractor. However once such visit happens the dynamics is then completely dominated by the convergence to the steady state. The chaotic attractor is destroyed and is therefore replaced by a chaotic transient [10].

[6]This set of control parameters was chosen just before the crisis.

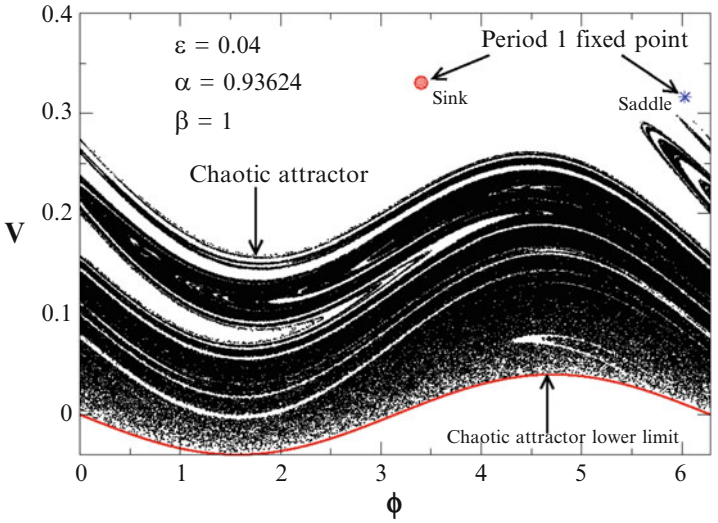


Fig. 6.4 Plot of a chaotic attractor and attracting fixed point for the control parameters $\varepsilon = 0.04$, $\beta = 1$ and $\alpha = 0.93624$. The lower limit of the chaotic attractor corresponds to the velocity of the moving wall (red curve). Sink and saddle are identified in the figure. The positive Lyapunov exponent measured for the chaotic attractor was $\overline{\lambda} = 1.7743(5)$, as obtained in Ref. [19]

One way to investigate the boundary crisis is by constructing the stable and corresponding unstable manifolds. A stable manifold consists of points along the eigenvector of the stable direction of the saddle point associated with the eigenvalue smaller than the unity whose dynamics heads directly towards the saddle point. The usual way to obtain the stable manifolds is to iterate the mapping inversely, i.e., instead of moving forward from the dynamics, the procedure moves backwards. For this end a specific set of initial conditions must be placed along the eigenvectors and the inverse mapping is updated. On the other hand, the unstable manifold consists of points along the eigenvector of the associated eigenvalue larger than the unity whose dynamics moves apart from the saddle point. The knowledge of the eigenvectors and initial conditions along them allow a direct iteration of the mapping to produce such curves.

6.3 Boundary Crisis Characterization

In this section we investigate the boundary crisis properly. To start with we obtain the fixed points that include also the saddle point from where the manifolds responsible for generating a homoclinic orbit are emanating. The fixed points [28] are obtained from the condition $V_{n+1} = V_n = V^*$ and $\phi_{n+1} = \phi_n = \phi + 2m\pi$ where $m = 1, 2, 3, \ldots$ is any integer number corresponding to the number of oscillations

that the moving wall completed between collisions. Matching these two conditions in the mapping (6.1) we obtain the coordinates of the fixed points are

$$V = \left[\frac{1+\beta}{\beta\alpha-1}\right]\varepsilon\,\sin(\phi), \tag{6.5}$$

$$\phi_{\pm} = \begin{cases} \arccos\left[\frac{\varepsilon+\tilde{\gamma}\sqrt{\varepsilon^2+\tilde{\gamma}^2-1}}{\varepsilon^2+\tilde{\gamma}^2}\right], \\ 2\pi - \arccos\left[\frac{\varepsilon-\tilde{\gamma}\sqrt{\varepsilon^2+\tilde{\gamma}^2-1}}{\varepsilon^2+\tilde{\gamma}^2}\right] \end{cases},$$

where the auxiliary term $\tilde{\gamma}$ assumes the following expression

$$\tilde{\gamma} = \frac{2\varepsilon\alpha m\pi}{\alpha+1}\left[\frac{1+\beta}{\beta\alpha-1}\right], \tag{6.6}$$

for any $m = 1, 2, 3, \ldots$. The classification of the fixed points are then P_{sink} and P_{saddle} with

$$P_{sink} = \left(\left[\frac{1+\beta}{\beta\alpha-1}\right]\varepsilon\,\sin(\phi), \phi = \arccos\left[\frac{\varepsilon+\tilde{\gamma}\sqrt{\varepsilon^2+\tilde{\gamma}^2-1}}{\varepsilon^2+\tilde{\gamma}^2}\right]\right) \tag{6.7}$$

$$P_{saddle} = \left(\left[\frac{1+\beta}{\beta\alpha-1}\right]\varepsilon\,\sin(\phi), \phi = 2\pi - \arccos\left[\frac{\varepsilon-\tilde{\gamma}\sqrt{\varepsilon^2+\tilde{\gamma}^2-1}}{\varepsilon^2+\tilde{\gamma}^2}\right]\right). \tag{6.8}$$

The construction of the unstable manifolds is simple. They are generated by the evolution of the dynamics produced by the mapping (6.1) with an appropriate set of initial conditions. Indeed the initial conditions must lie along the eigenvectors of the corresponding unstable eigenvalue. There are two branches of the unstable manifolds of the saddle point, as we see from Fig. 6.5. One of them, in black, leaves the saddle point and converges towards the attracting fixed point spiraling around it counterclockwise reducing gradually its amplitude of motion as the time goes one. For long enough time such branch converges to the steady state and gets trapped there forever. The second branch of the unstable manifold, in red color, moves away from saddle point in the downwards direction converging then to the chaotic attractor. The construction of the stable manifolds requests an explicit form for the inverse of mapping (6.1). Therefore instead of iterate $T(V_n, \phi_n) = (V_{n+1}, \phi_{n+1})$ we must obtain the inverse of T such that given a set of pair (V, ϕ) at instant $n+1$ we obtain the new pair at instant n. The expression for T^{-1} is given by

$$T^{-1}: \begin{cases} V_n = \frac{V_{n+1}+(1+\beta)\varepsilon\,\sin(\phi_{n+1})}{\alpha} \\ h(\phi_n) = \phi_n + \frac{1-\varepsilon\cos(\phi_n)}{V_n} + \frac{1-\varepsilon\cos(\phi_{n+1})}{\alpha V_n} - \phi_{n+1} \end{cases}, \tag{6.9}$$

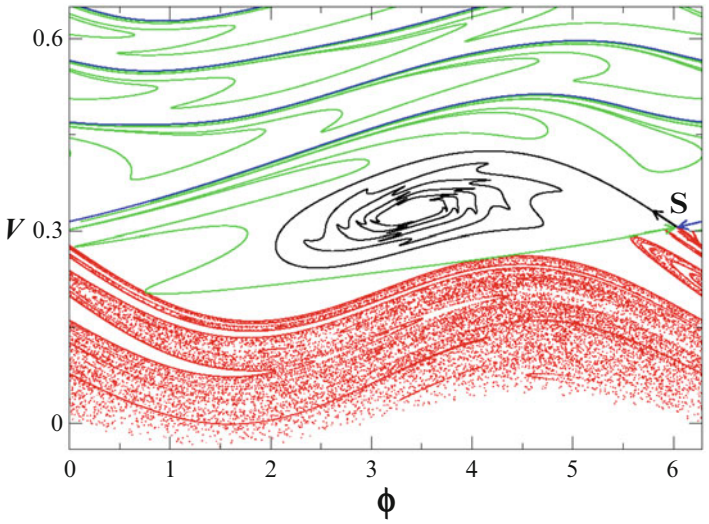


Fig. 6.5 Plot of the manifolds for the saddle point for the control parameters $\varepsilon = 0.04$, $\beta = 1$, and $\alpha = 0.93624$. Black denotes one branch of the unstable manifold leaving the saddle point and converging towards the attracting fixed point. Red represents the other branch of the unstable manifold converging to the chaotic attractor itself. The stable branches, plotted as green and blue draw the corresponding border of influence for the basin of attraction of the attracting fixed point and the chaotic attractor

where ϕ_n is obtained by the numerical solution of $h(\phi_n) = 0$. The numerical procedure used to solve $h(\phi_n) = 0$ was the well-known Newton's method with an accuracy of 10^{-14}. Because the stable manifolds always grow in velocity and since the velocity never gets smaller than ε, the condition of direct collisions[7] does not need to be considered in the stable manifold construction therefore leading us to take into account only the case of indirect collisions.

Let us now discuss the properties observed for the basin of attraction of the chaotic attractor at a boundary crisis. As soon as the control parameter α is varied, say increased, the distance between the attractor and its basin of attraction decreases until a limit they touch each other at a parameter α_c, leading to the occurrence of a crisis. The encounter of the chaotic attractor with its basin of attraction happens at the same instant that an unstable orbit touches the chaotic attractor producing a homoclinic tangency. After the event the chaotic attractor no longer exists anymore but is rather replaced by a chaotic transient. If an initial condition is given in the region of the phase space where the chaotic attractor likely existed prior to the crisis, the dynamics will follow it for a while, as a type of ghost version of the chaotic attractor, until it finds the appropriate route of escape and makes an incursion towards the direction of the fixed point therefore getting trapped there for further time. The region where the chaotic attractor existed before the crisis is now part of

[7]Such condition is given by solution of $G(\phi_c)$ in mapping (6.1).

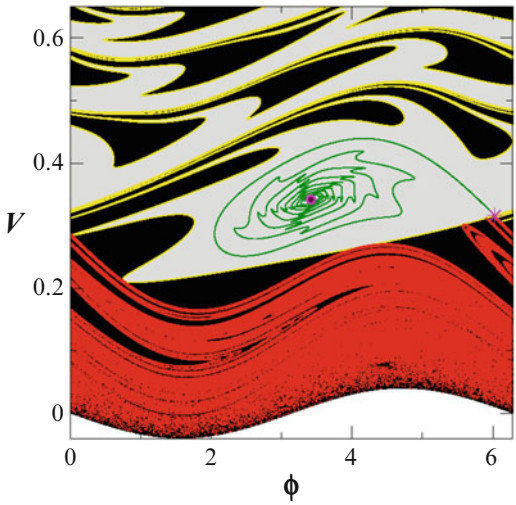


Fig. 6.6 Plot of the basin of attraction of the chaotic attractor (black) and the sink (gray). The boundary of the two basin of attraction is delimited by the stable manifolds, as painted in yellow. Attracting fixed point is plotted as a bullet while saddle is plotted as a star. Chaotic attractor is shown in red while the branch of the unstable manifold converging to the fixed point is shown in green. The control parameters used were $\beta = 1$, $\alpha = 0.93624$, and $\varepsilon = 0.04$

the basin of attraction of the attracting fixed point. Figure 6.6 shows a plot of the basin of attraction of the chaotic attractor as well as the basin of attraction of the attracting fixed point. Black identifies the basin of attraction of the chaotic attractor whereas gray gives the basin of attraction for the attracting fixed point. The chaotic attractor is painted as red in the figure. The attracting fixed point (bullet) and saddle (star) points are also shown in the figure. The control parameters used for construct Fig. 6.6 were $\varepsilon = 0.04$, $\beta = 1$, and $\alpha = 0.93624$, immediately before the boundary crisis. We notice the border separating the two basin of attractions are constructed by the stable manifolds, as shown in yellow. Figure 6.7 shows a plot of V vs. ϕ for the parameters $\varepsilon = 0.04$, $\beta = 1$, and $\alpha = 0.9375$, immediately after the crisis, exhibiting a chaotic like dynamics in black for short time. We see the particle moves along the previous region where the chaotic attractor likely existed until escape the region and heads towards the attracting fixed point spiraling around it. There is no return once the dynamics dives into the attracting fixed point. It spirals around the fixed point reducing gradually the amplitude of oscillation until eventually reaches the fixed point at a sufficiently long time.

The crossing of the unstable and stable manifolds is a clear evidence that the boundary crisis has happened. Figure 6.8 shows the stable and unstable manifolds for the saddle point constructed after the boundary crisis. The parameters used to construct the figure were $\varepsilon = 0.04$, $\beta = 1$, and $\alpha = 0.9375$.

The time measured in the number of collisions a particle stays wondering in a pseudo-chaotic motion after the crisis depends on the initial condition. The probability an initial condition is given in the chaotic region prior to the crisis to survive such region without escaping it and being trapped permanently by the attracting fixed point is described by an expression of the type

$$P(n) \cong e^{-\frac{n}{\tau}}, \tag{6.10}$$

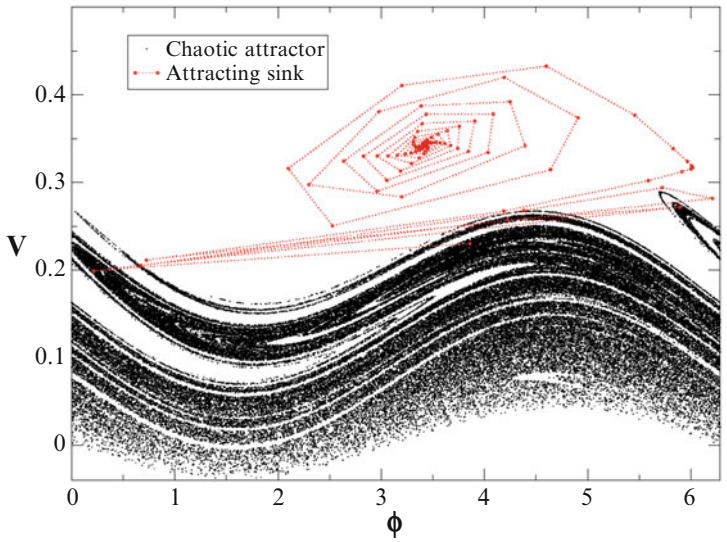


Fig. 6.7 Plot of V vs. ϕ for the parameters $\beta = 1$, $\varepsilon = 0.04$, and $\alpha = 0.9375$, immediately after the boundary crisis. Black points indicate the chaotic like dynamics (transiently) while the red small bullets show the convergence to the attracting fixed point. The dotted line is used only as a guide to the eye

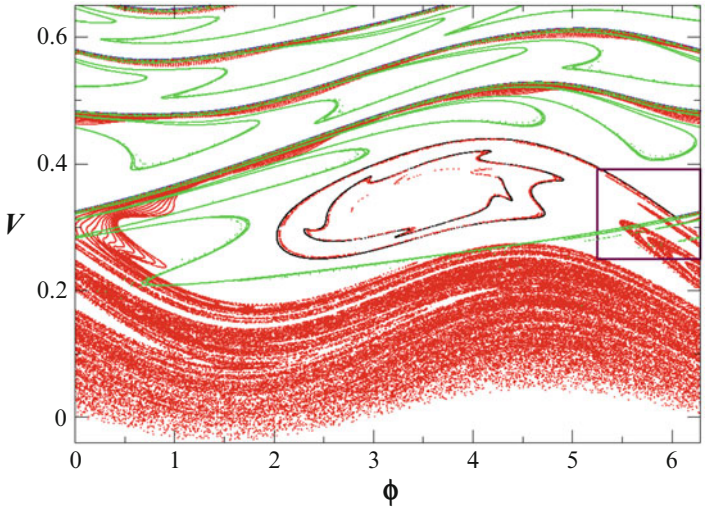


Fig. 6.8 Plot of the manifolds for the saddle point for the control parameters $\varepsilon = 0.04$, $\beta = 1$, and $\alpha = 0.9375$. Black denotes one branch of the unstable manifold leaving the saddle point and converging to the attracting point. Red represents the other branch of the unstable manifold passing near the region of the phase space where the chaotic attractor likely existed before the crisis. The stable branches, plotted as green and blue draw are also evident. The box highlights the many crossings of the manifolds confirming the boundary crisis

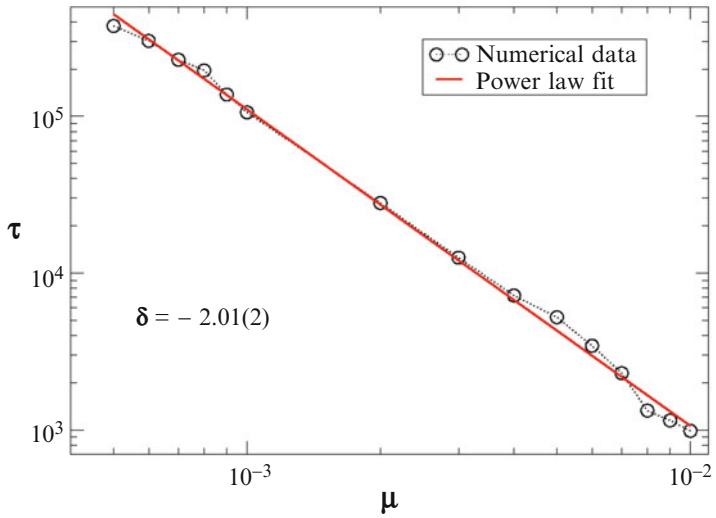


Fig. 6.9 Plot of τ vs. μ. A power law fitting furnishes $\delta = -2.01(2)$. We considered an ensemble of 5×10^3 different initial conditions along the region in the phase space where the chaotic attractor existed before the crisis. The parameters considered were $\beta = 1$ and $\varepsilon = 0.04$. The parameter α is varied as the horizontal axis with $\alpha_c = 0.93624$. Dotted line is used only as a guide to the eye

where n is the number of collisions of the particle with the moving wall and τ is a relaxation time given by

$$\tau \propto \mu^{\delta}, \tag{6.11}$$

with $\mu = \alpha - \alpha_c$, for $\alpha > \alpha_c$ where α_c denotes the critical control parameter where the boundary crisis happens and δ is an exponent that must be obtained.

The exponent δ can be obtained from different ways. A first description is by using numerical simulations. Since that the time measured in the number of collisions a particle leaves to escape the region where a chaotic attractor existed prior to the crisis depends on the initial condition, it turns out to be convenient to run an average over an ensemble of different initial conditions along the previous chaotic attractor. Figure 6.9 shows a plot of the average relaxation time τ against μ. A power law fitting to the data gives $\delta = -2.01(2)$. The parameters considered were $\varepsilon = 0.04$, $\beta = 1$, and $\alpha_c = 0.93624$. The resulting exponent obtained in Fig. 6.9 was also obtained in other crisis for different parameters in the model. For $\varepsilon = 0.033$, $\alpha_c = 0.96375$ for $m = 1$ we found $\delta = -2.11(9)$ while for $m = 2$, $\alpha_c = 0.90232$ and $\varepsilon = 0.02$ we obtained $\delta = -1.98(3)$. The numerical fittings give the two exponents close to -2 in well agreement with the results obtained in [22].

A second procedure we consider involves a description proposed by Grebogi et al. in Ref. [12]. It considers the eigenvalues of the saddle point at the parameter where the boundary crisis happens. In their investigation they also assumed the

survival probability is described by an exponential function $P(t) \propto e^{-t/\tau}$ with t denoting the time and where the average τ has an expression of the type $\tau \propto \mu^{-\gamma}$. In Grebogi's formalism (see Refs. [12, 13]), the exponent γ is obtained from the equation

$$\gamma = \frac{\ln |\beta_2|}{\ln |\beta_1 \beta_2|^2}, \tag{6.12}$$

where β_1 and β_2 are the corresponding expanding and contracting eigenvalues evaluated at the saddle point. Using their notation and considering the eigenvalues at the saddle point, we obtain an exponent $\gamma = 0.4991\ldots$, which is significantly close to $1/2$, as discussed in their paper [12]. This result however is not matching with our numerical simulations as obtained in Figs. 6.9 and 6.10. We now discuss this apparent discrepancy among the results.

When comparing the two equations describing the relaxation time, those heading the exponential, we see an ingenious difference on a minus signal that is heading the parameter μ. Because of such signal and considering Grebogi's approach [12], the expression for δ used in our simulations must be written as

$$\begin{aligned} -\delta &= \frac{\ln |\beta_2|}{\ln |\beta_1 \beta_2|^2}, \\ \delta &= -\frac{\ln |\beta_2|}{\ln |\beta_1 \beta_2|^2}, \\ &= \frac{\ln |\beta_1 \beta_2|^2}{\ln |\beta_2|}. \end{aligned} \tag{6.13}$$

This result leads us therefore to a correct estimation of the exponent with $\delta = 2.003\ldots$ obtained from the corresponding expanding and contracting eigenvalues of the saddle point. This result is in well agreement with the result obtained from numerical simulations and with the existing literature [22] and explains the apparent discrepancy of the results which indeed does not exist. In the Appendix of this chapter we present the relevant expressions obtained from the Jacobian matrix that lead to the eigenvalues of the saddle point.

As we have discussed along the chapter, the only dissipative parameter varied was α while the restitution coefficient β remained constant all the discussion. One may think what happens if the parameter β is allowed to change to. Does the boundary crisis move? Is there any significant change in the dynamics, i.e. in the phenomena itself? Indeed when dissipation is considered in both the walls, there is a combination between then to produce a type of effective restitution coefficient. As discussed in Ref. [22], the combination of the two control parameters α and β may lead to a family of boundary crisis to be observed in the parameter space. It was shown in Ref. [22] that when the parameters are varied in such a way obeying the law $\beta = A + \alpha B$ where the authors of Ref. [22] found $A = 1.927(1)$ and $B = -0.989(1)$ for the parameter $\varepsilon = 0.04$. They also considered other crisis

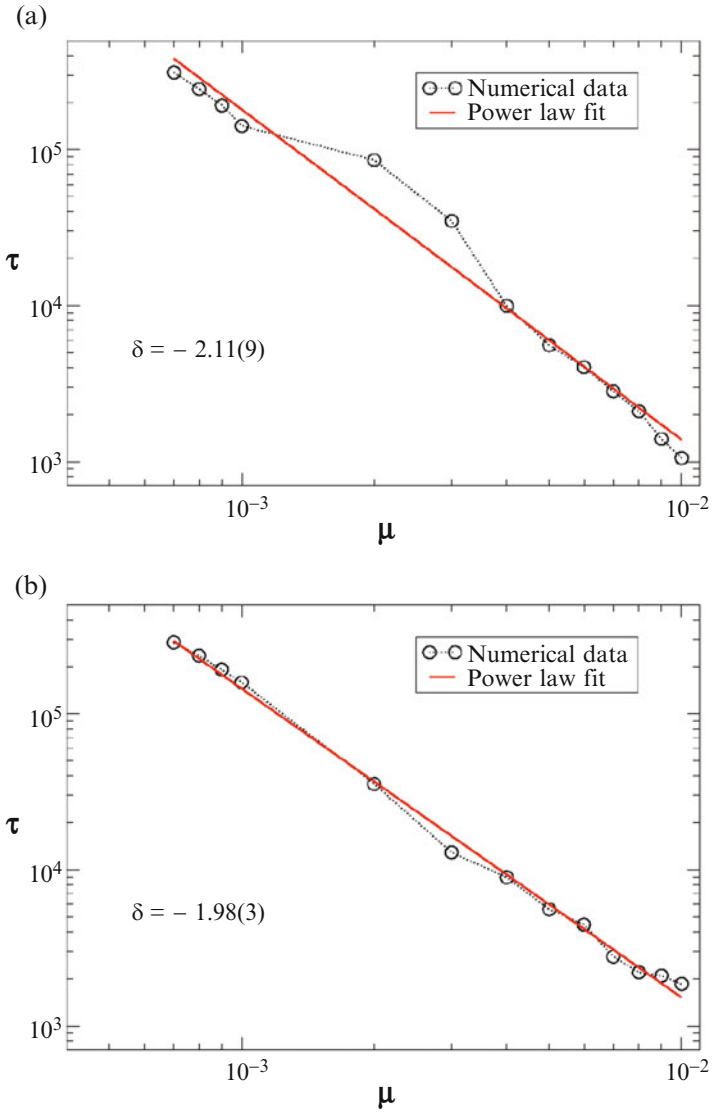


Fig. 6.10 Plot of average transient τ vs. μ for two other boundary crisis observed in the Fermi-Ulam model. The parameters used were $\beta = 1$ and: (**a**) $\varepsilon = 0.033$ and $\alpha_c = 0.96375$ considering $m = 1$ and (**b**) for $m = 2, \alpha_c = 0.90232$ and $\varepsilon = 0.02$. The numerical fittings furnish the exponent: (**a**) $\delta = -2.11(9)$ and (**b**) $\delta = -1.98(3)$. Each exponent was obtained considering an ensemble of 5×10^4 different initial conditions from a region in the phase space where the chaotic attractor likely existed before the crisis. The two exponents obtained were close to -2 in well agreement with the results reported in Ref. [22]. Dotted lines are used only as guide to the eye

happening around $\varepsilon = 0.033$, therefore finding $A = 1.956(1)$ and $B = -0.992(1)$. Therefore the variation of the parameter β retains the crisis and allows to find a family of control parameters where it is observed.

6.4 Conclusions

In this chapter we studied the chaotic transient for a boundary crisis observed in a dissipative version of a Fermi-Ulam model. The inelastic collisions with the walls led to measure contraction in the phase space hence producing attractors. We constructed the manifolds from a saddle point and show both the stable and unstable branches of the manifold tangency each other at the boundary crisis. After the crisis there are infinitely many crossings of these manifolds. Consequently the basin of attraction of the fixed point becomes porous and suitable to be visited by a likely chaotic motion. The attractor that existed prior to the boundary crisis is destroyed therefore being replaced by a chaotic transient. The survival probability a moving particle has to survive the region of the phase space where the chaotic attractor likely existed before the crisis is decaying exponentially. The relaxation time is described by a power law and that the numerical value of the exponent can be obtained from the eigenvalues of the saddle point at the critical parameter where crisis happens. For a homoclinic tangency [12] the expression for the exponent is $\gamma = \frac{\ln|\beta_2|}{\ln|\beta_1\beta_2|^2}$ leading to a numerical value of $\gamma \cong \frac{1}{2}$. Our definition however leads to the inverse of this exponent, hence our numerical findings gave $\delta = \frac{1}{\gamma} \cong 2$ and seem to be universal for this type of global bifurcation.

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Appendix

We present in this Appendix the expressions used to obtain the eigenvalues of the saddle point at the parameters leading to the boundary crisis. Since there is no fixed point for solving $G(\phi_c)$, i.e., the direct collisions, we must consider only the case of the Jacobian matrix for indirect collisions. The Jacobian matrix is written as

$$J = \begin{pmatrix} \frac{\partial V_{n+1}}{\partial V_n} & \frac{\partial V_{n+1}}{\partial \phi_n} \\ \frac{\partial \phi_{n+1}}{\partial V_n} & \frac{\partial \phi_{n+1}}{\partial \phi_n} \end{pmatrix},$$

where the coefficients of J must be evaluated at the saddle point. The terms in J are written as

$$j_{11} = \frac{\partial V_{n+1}}{\partial V_n} = \alpha\beta - (1+\beta)\varepsilon\cos(\phi)\frac{\partial \phi_{n+1}}{\partial V_n}, \tag{6.14}$$

$$j_{12} = \frac{\partial V_{n+1}}{\partial \phi_n} = -(1+\beta)\varepsilon\cos(\phi)\frac{\partial \phi_{n+1}}{\partial \phi_n}, \tag{6.15}$$

$$j_{21} = \frac{\partial \phi_{n+1}}{\partial V_n} = \frac{\partial \phi_r}{\partial V_n} + \frac{\partial \phi_l}{\partial V_n} + \frac{\partial \phi_c}{\partial V_n}, \tag{6.16}$$

$$j_{22} = \frac{\partial \phi_{n+1}}{\partial \phi_n} = 1 + \frac{\partial \phi_r}{\partial \phi_n} + \frac{\partial \phi_l}{\partial \phi_n} + \frac{\partial \phi_c}{\partial \phi_n}, \tag{6.17}$$

where the auxiliary derivatives are given by

$$\frac{\partial \phi_r}{\partial V_n} = -\frac{1-\varepsilon\cos(\phi)}{V^2}, \tag{6.18}$$

$$\frac{\partial \phi_l}{\partial V_n} = -\frac{(1-\varepsilon)}{\alpha V^2}, \tag{6.19}$$

$$\frac{\partial \phi_r}{\partial \phi_n} = \frac{\varepsilon\sin(\phi)}{V}, \tag{6.20}$$

$$\frac{\partial \phi_l}{\partial \phi_n} = 0. \tag{6.21}$$

The two terms $\frac{\partial \phi_c}{\partial V_n}$ and $\frac{\partial \phi_c}{\partial \phi_n}$ are determined by implicit derivatives and are given by

$$\frac{\partial \phi_c}{\partial V_n} = \frac{\varepsilon\sin(\phi)\left(\frac{\partial \phi_r}{\partial V_n} + \frac{\partial \phi_l}{\partial V_n}\right) - \alpha\phi_c}{\alpha V - \varepsilon\sin(\phi)}, \tag{6.22}$$

$$\frac{\partial \phi_c}{\partial \phi_n} = \frac{\varepsilon\sin(\phi)\left(1 + \frac{\partial \phi_r}{\partial \phi_n} + \frac{\partial \phi_l}{\partial \phi_n}\right)}{\alpha V - \varepsilon\sin(\phi)}, \tag{6.23}$$

where

$$\phi_c = \frac{\varepsilon - \varepsilon\cos(\phi)}{\alpha V}. \tag{6.24}$$

The eigenvalues are then given by the following expressions

$$\lambda_{1,2} = \frac{(j_{11}+j_{22}) \pm \sqrt{(j_{11}+j_{22})^2 - 4(j_{11}j_{22} - j_{12}j_{21})}}{2}. \tag{6.25}$$

To apply on the Grebogi's expressions [12] we have to normalize the eigenvalues. For the control parameters $\varepsilon = 0.04$, $\beta = 1$, and $\alpha = 0.93264$ we obtain that the expanding and contracting eigenvalues are $\beta_1 = 3.0739934591561680$ and $\beta_2 = 0.28514874278495017$, respectively.

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Chapter 7
New Trends in Chaos-Based Communications and Signal Processing



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7.1 Introduction

In the last 30 years, many possible applications of chaotic signal and systems in Telecommunication Engineering and Signal Processing have been proposed [1–6]. Although no practical standard is based on chaotic signals yet, many recently published works have shown an interest in studying the behavior of chaos-based communication systems (CBCS) in practical channels [5, 7–10].

In this chapter, we present four ongoing works in this line of research. In Sect. 7.2, a message authentication based on a chaotic fingerprint is succinctly described. The two-dimensional Hénon map [11] is one of the most employed when it comes to CBCS using chaotic synchronization [1, 12]. This way, knowing the spectral and dynamical characteristics of its orbits is fundamental. In Sect. 7.3, the spectral characteristics of the Hénon map as function of its parameters are numerically studied. Following, in Sect. 7.4, some simulation results on the chaotic

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nature of signals generated by a filtered Hénon map are accessed. Finally, in Sect. 7.5, a communication system that includes an adaptive equalizer and a switching scheme between a chaos-based modulation and a conventional one is discussed.

7.2 Chaotic Fingerprint for Physical Layer Authentication

Message authentication, which ensures that a received message comes from its acclaimed sender, is of fundamental importance for secure communication systems [13]. Traditionally, authentication methods are deployed in higher layers of the network which provide a computational secrecy to the system, meaning that algorithms running in a polynomial computer will not be able to break the secrecy in feasible time. However, exploring the stochastic nature of the wireless channel, usually seen as an undesired characteristic, is an efficient way to provide an additional security layer to the system [14]. In this context, a robust authentication method may be achieved by embedding a fingerprint in the message at the physical layer [15, 16]. The fingerprint embedding system can be deployed on top of an existing system where users employing the authentication protocol can communicate without interfering with users unaware of the authentication scheme. The fingerprints are typically generated by methods based on cryptographic hash functions [15, 16] and on the channel state information [14, 17].

In this section, it is demonstrated the viability of employing unidimensional chaotic maps for this purpose. Due to the fact that sequences generated by these maps are broadband, aperiodic, and present noise-like behavior [18, 19], they are strong candidates for the fingerprint generation process. This section follows with a brief description of the security requirements and the system overview, proceeding with the presentation of the fingerprint features and properties that tackle each of the security requirements.

7.2.1 System Overview and Security Requirements

The communication system model used to analyze the fingerprint generation method has three users sharing the same wireless medium:

- **Alice and Bob**: who are the legitimate users of the system meaning that they enforce the proposed authentication protocol. Alice sends message packets of length N with a superimposed fingerprint that depends on a secret key shared with Bob. Bob receives these message packets and based on the presence or not of a legitimate fingerprint he decides to accept or reject the messages; and
- **Eve**: who is a malicious user of the system, meaning that she is aware of the authentication protocol, but do not share a legitimate key, so she is unable to

generate valid fingerprints. Her goal is to disrupt the communication, either by impersonating a legitimate user or by keeping legitimate users from communicating.

In our model, the malicious user may do two things to disrupt the system:

- **Replay Attacks** which consists in recording a message sent along with its legitimate fingerprint and resending it in the future.
- **Impersonation Attacks** which consists in observing some legitimate messages along with their fingerprints and sending a message with a fraudulent fingerprint in an attempt to impersonate a legitimate user of the system.

7.2.2 The Authentication Protocol

The transmitter consists of a digital modulator (here, it is considered a binary phase shift keying (BPSK) modulator for the sake of simplicity), a chaotic signal generator based on a unidimensional discrete piecewise linear chaotic map, and an adder (to compose the message with the chaotic signal). We assume that the system operates in a slow block fading channel, so the fading gain h^i for the ith block is constant for a block of size N, where h^i is a Rayleigh distributed random variable. The superscript i in every variable indicates the ith block.

The system is also impaired by additive white Gaussian noise (AWGN), denoted by the vector $\boldsymbol{w}^i = \begin{bmatrix} w_1^i & w_2^i & \cdots & w_N^i \end{bmatrix}$, where w_n^i is a zero-mean Gaussian random variable with variance σ_w^2. The modulated message vector is given by $\boldsymbol{s}^i = \begin{bmatrix} s_1^i & s_2^i & \ldots & s_N^i \end{bmatrix}$, such that $s_n^i \in \{-1, +1\}$ is a bipolar signal. The signal to noise ratio (SNR) is given by $1/\sigma_w^2$.

Eve does not have a valid authentication key, so she is considered unable to recognize a valid fingerprint. So the transmitted signal for the unauthentic user simply consists of a message $\boldsymbol{x}^i = \boldsymbol{s}^i$. Thus, for this user, the signal at the receiver is given by

$$\boldsymbol{y}^i = h^i \boldsymbol{x}^i + \boldsymbol{w}^i = h^i \boldsymbol{s}^i + \boldsymbol{w}^i. \tag{7.1}$$

In a scenario with message authentication, the output of the chaotic generator acts as the fingerprint in the system and is denoted by the row vector $\boldsymbol{t}^i = \begin{bmatrix} t_1^i & t_2^i & \ldots & t_N^i \end{bmatrix}$, such that $t_n^i \in [-1, +1]$ is a limited real value. Each vector element t_{n+1}^i is a function of the previous one t_n^i, according to the chaotic map and its initial condition.

The legitimate user applies the shared key, the message, and the packet number to generate the initial condition. In order to do so the domain of the map is discretized in 2^K bins, where K is the number of bits of the key and the possible initial conditions are taken as the centers of the 2^K bins. A specific bin is determined through a bitwise exclusive or operation of the key with the concatenation of the message and the packet number i. Thus, resending the same message and

fingerprint pair at a later time would be ineffective, assigning robustness to the system against replay attacks. Due to the sensitive dependence on initial conditions (SDIC) inherent to chaotic maps [18], even two neighboring initial conditions generate an uncorrelated sequence on the long term.

In order to make the system secure to replay attacks the key goes through a bitwise exclusive or operation with the binary transmission packet number i, so every initial condition is key dependent and also message dependent, so resending the same message tag pair at a later time would be ineffective.

The message plus the fingerprint vector is given by $\boldsymbol{x}_t^i = \rho_s \boldsymbol{s} + \rho_t \boldsymbol{t}$, where ρ_s and ρ_t determine the relative energy allocation between the message and the fingerprint. In this case, the received signal is given by

$$\boldsymbol{y}^i = h^i \boldsymbol{x}_t^i + \boldsymbol{w}^i = h^i(\rho_s \boldsymbol{s}^i + \rho_t \boldsymbol{t}^i) + \boldsymbol{w}^i. \tag{7.2}$$

At the receiver, the message is first demodulated and subtracted from the received signal and then is correlated to a generated copy of the fingerprint. The decision variable for the ith block is

$$\tau^i = \frac{\boldsymbol{y}^i - h_i \rho_s \boldsymbol{s}^i}{h_i \rho_t} \cdot \boldsymbol{t}^i, \tag{7.3}$$

where the operator $\cdot$ is the inner product between two vectors. Thus, τ^i has distinct probability density function (PDF) depending whether the fingerprint is present (hypothesis $\mathcal{H}_1$) or not (hypothesis $\mathcal{H}_0$). The correspondent conditional PDFs $p_{\tau^i|\mathcal{H}_0}(\tau)$ and $p_{\tau^i|\mathcal{H}_1}(\tau)$ are obtained from

$$\tau^i|\mathcal{H}_0 = \left(\frac{1-\rho_s}{\rho_t}\right) \boldsymbol{s}^i \cdot \boldsymbol{t}^i + \frac{\boldsymbol{w}^i \cdot \boldsymbol{t}^i}{h^i \rho_t}, \tag{7.4}$$

$$\tau^i|\mathcal{H}_1 = |\boldsymbol{t}^i|^2 + \frac{\boldsymbol{w}^i \cdot \boldsymbol{t}^i}{h^i \rho_t}. \tag{7.5}$$

For the ith block, (7.4) and (7.5) correspond to the addition of a constant value to the zero-mean Gaussian random variable $v^i \triangleq \boldsymbol{w}^i \cdot \boldsymbol{t}^i/(h^i \rho_t)$. Thus, $p_{\tau^i|\mathcal{H}_0}(\tau)$ is a zero-mean Gaussian random variable (for uncorrelated vectors $\boldsymbol{s}^i$ and $\boldsymbol{t}^i$) with variance $\sigma_{v^i}^2 = N\mathbb{E}\left[(t_k^i)^2\right]\sigma_w^2/(\rho_t^2|h^i|^2)$, where $\mathbb{E}[\cdot]$ represents the expected value of a random variable. Similarly, $p_{\tau^i|\mathcal{H}_1}(\tau)$ has the same PDF, but with mean $\mathbb{E}\left[\tau^i|\mathcal{H}_1\right] = |\boldsymbol{t}^i|^2$. For more details, see [15, 20].

The main goal in an authentication system is to detect a fraudulent message (with no fingerprint present). Specifically, to prevent deciding for $\mathcal{H}_1$ when the correct hypothesis is actually $\mathcal{H}_0$ (false positive). Thus, the decision criterion compares τ^i in (7.3) with the ith block decision threshold τ_0^i, such that

$$\tau^i \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\gtrless}} \tau_0^i, \tag{7.6}$$

where τ_0^i is selected with the constraint that a safety margin α must be less than a tolerable failure probability η. In order to determine τ_0^i, we must limit the probability of false positive, $\alpha = \Pr(\tau^i > \tau_0^i | \mathcal{H}_0) < \eta$, and the attacker only obtains success if an illegitimate message ratio higher than α is accepted by the receiver. As $p_{\tau^i|\mathcal{H}_0}(\tau)$ is Gaussian distributed with zero-mean, we have that $\tau_0^i = \arg\min Q\left(\tau^i/\sigma_{v_i}\right) < \eta$ where $Q(\cdot)$ is the tail probability of the Gaussian PDF.

7.2.3 Results

In order to validate the proposed fingerprint generation method, a Monte Carlo simulation is performed using the tent map as the chaotic generator for SNR varying from 0 to 21 dB.

The metric used to evaluate the success of the protocol is the probability of rejection of a message. We simulate messages sent by Alice (where the fingerprint is present) and messages sent by Eve (no fingerprint). We consider in the simulations the block length N equals to 128, $\rho_s = 0.985$, $\rho_t = 0.173$, and the safety margin at the receiver $\eta = 0.01$. The simulation results are shown in Fig. 7.1. As can be observed, fingerprints generated by chaotic maps can be successfully used to authenticate messages in the physical layer. For the whole range of the SNR considered, the rejection probability of the messages sent by Eve is above 0.99 which according to the established criteria means that the attacker is unable to beat the authentication protocol. As Eve's messages cannot be authenticated, since she does not know the shared key, the proposed fingerprint generation method is secure to impersonation attacks. Despite that, it is worth noting that a physical layer authentication protocol is an additional security measure to the traditional higher layer authentication protocol, thus, the illegitimate messages accepted by the receiver would not be authenticated at higher layers.

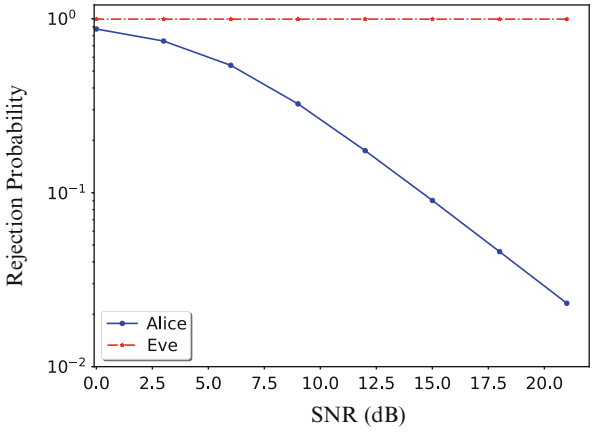


Fig. 7.1 Probability of rejection of messages sent by Alice and sent by Eve for fingerprints generated by the tent map with $\eta = 0.01$

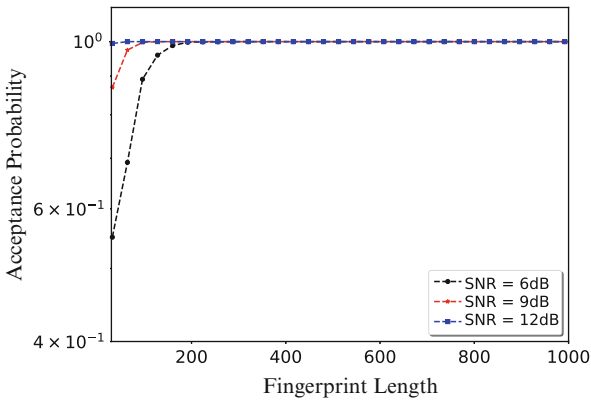


Fig. 7.2 Probability of acceptance of a message sent by Alice for three different SNR, where the fingerprint length N varies from 32 to 1000, with $\eta = 0.01$

However, by increasing the fingerprint length we can increase the rate of acceptance of legitimate messages. Figure 7.2 shows the acceptance rate of the messages transmitted by Alice when $\eta = 0.01$ for three different values of SNR. We can observe that, for a long enough fingerprint, the acceptance rate for Alice messages approaches 1 for all three SNR.

7.3 Spectral Properties of the Orbits of the Hénon Map

The study of spectral characteristics of chaotic signals is a relevant issue for practical communications. Since every real-world communication channel is bandlimited, to characterize and control the power spectral density (PSD) of the generated chaotic signals is of paramount importance [5, 21]. Recently, many works have thoroughly analyzed the PSD of signals generated by one-dimensional maps [22–26]. However, we are not aware of any systematic study of the PSD of signals generated by higher dimensional maps.

In this section, we report our initial studies involving the autocorrelation sequence (ACS) and PSD of signals generated by the two-dimensional Hénon map. We numerically analyze how they vary with the map parameters.

7.3.1 The Hénon Map

The Hénon map $f_H : \mathbb{R}^2 \to \mathbb{R}^2$ is defined as [11]

$$\boldsymbol{s}(n+1) = \boldsymbol{f}_H(\boldsymbol{s}(n)) = \begin{bmatrix} s_1(n+1) \\ s_2(n+1) \end{bmatrix} = \begin{bmatrix} s_2(n) + 1 - a s_1^2(n) \\ b s_1(n) \end{bmatrix} \tag{7.7}$$

where $\{a, b\} \in \mathbb{R}$ are parameters and $\boldsymbol{s}(n) = [s_1(n) \quad s_2(n)]^T$, with $(\cdot)^T$ indicating transposition.

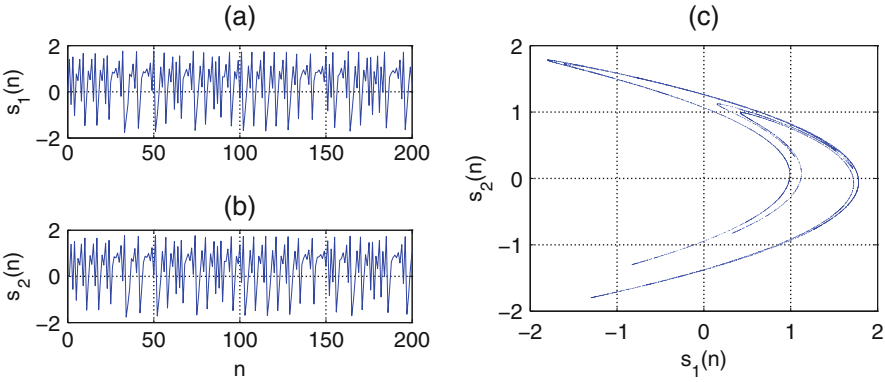


Fig. 7.3 Hénon map for $a = 1.4$ e $b = 0.3$: (**a**) and (**b**) orbit $s(n)$ with initial condition $\boldsymbol{s}(0) = [0\ 0]^T$; (**c**) attractor

Figure 7.3a, b shows $\boldsymbol{s}(n)$ with initial condition $\boldsymbol{s}(0) = [0 \quad 0]^T$ for $a = 1.4$ and $b = 0.3$. The atractor, obtained disregarding the first 10^3 iterations is presented in Fig. 7.3c. The highest Lyapunov exponent of this attractor can be numerically obtained resulting $h_H = 0.42$, confirming its chaotic nature [18, p. 201].

The bifurcation diagram of $s_1(n)$ and the highest Lyapunov exponent for $b = 0.3$ and $0 < a \leq 1.4$ are presented in Fig. 7.4. Figure 7.5 shows orbits obtained for (a) $a = 0.6$, (b) $a = 1.069$, (c) $a = 1.2$, (d) $a = 1.232$, and (e) $a = 1.4$. These values of a are signalized by dashed lines in Fig. 7.4. In cases (a) and (d) we have periodic orbits and in (b), (c), and (e) chaotic ones.

7.3.2 *Autocorrelation Sequences and Power Spectral Density*

Chaotic signals generated by a map can be considered as sample function of an ergodic stochastic process [27]. For the scalar component $s_1(n)$, the ACS $R(k)$ for an integer k is defined as

$$R(k) \triangleq \mathbb{E}\left[s_1(n)s_1(n+k)\right], \quad k \geq 0, \quad \forall n \geq 0. \tag{7.8}$$

The expected value $\mathbb{E}[\cdot]$ is taken over all initial conditions that generate chaotic signals. For negative values of k, we conveniently consider

$$R(k) \equiv R(-k), \quad k < 0. \tag{7.9}$$

The PSD $P(\omega)$ is the discrete-time Fourier transform of $R(k)$ [28]. As its integral is proportional to the signal power $R(0) = \mathbb{E}\left[s(n)^2\right]$, it represents how power is allocated in the discrete-time frequencies $0 \leq \omega < \pi$.

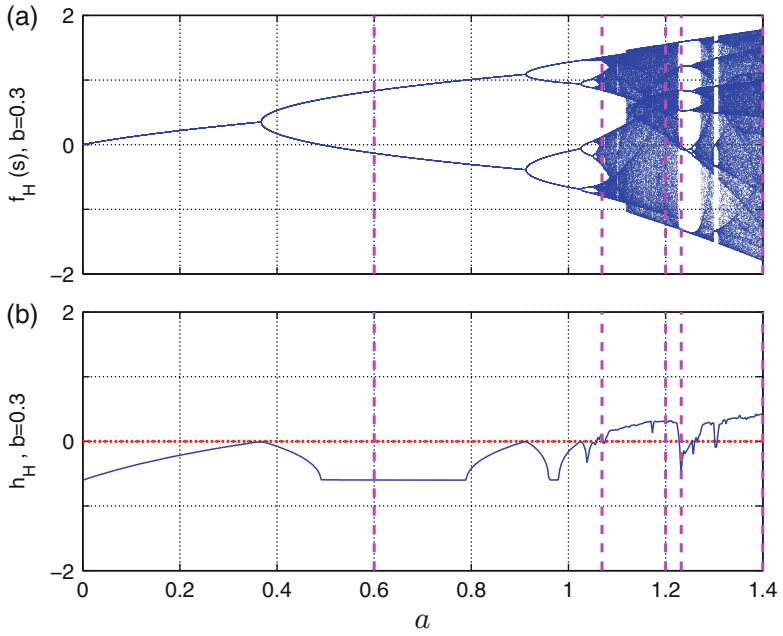


Fig. 7.4 (**a**) Bifurcation diagram for $s_1(n)$ and (**b**) highest Lyapunov exponent for the Hénon map

Figure 7.6 shows the ACS of $s_1(n)$ in (7.7) for $b = 0.3$ and the values of a considered in Fig. 7.5. Their estimated PSD using 6×10^6 samples are also shown.

In the periodic cases of (a) $a = 0.6$ (period $N = 2$) and (d) $a = 1.232$ ($N = 7$), it can be clearly seen the periodic nature of the ACS and the peaks at frequency $2\pi/N$ and its harmonics. In the chaotic cases (c) $a = 1.2$ and (e) $a = 1.4$, the ACS present a non-impulsive slowly decaying and oscillatory pattern that reflects in continuous peakless PSD. The power is concentrated predominantly in higher frequencies.

In the case (b) $a = 1.069$, we are still close to the emergence of chaos in the bifurcation diagram of Fig. 7.5. Although positive, the highest Lyapunov number is close to zero. This is reflected in a slowly decaying ACS and peaks in the PSD, specially at $2\pi/4$ and its harmonics. Therefore, we have a narrowband chaotic signal.

The general behavior of the PSD for $1.2 < a \leq 1.4$ and $b = 0.3$ is plotted in Fig. 7.7. The onset of chaos and periodic windows can be clearly noticed. It is also possible to note the power concentration in the high frequencies of the chaotic attractors.

Other simulations are presented in animated form at http://www.lcs.poli.usp.br/~marcio/NSCHenon.

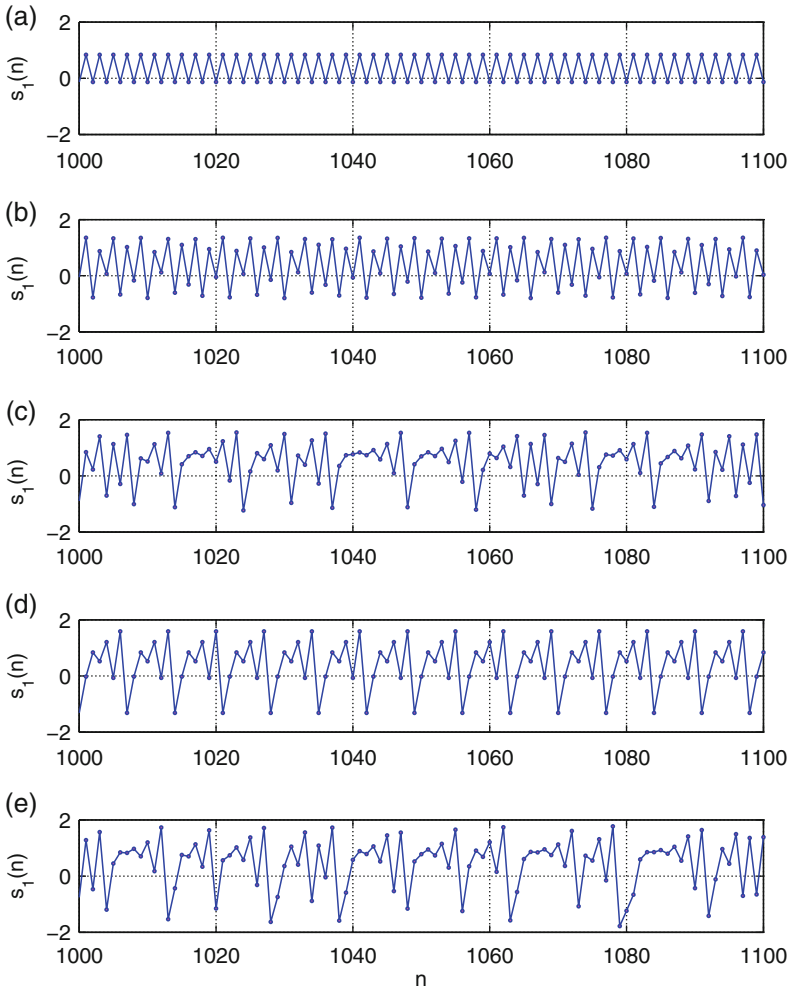


Fig. 7.5 Orbits of Hénon map for fixed $b = 0.3$ and (**a**) $a = 0.6$, (**b**) $a = 1.069$, (**c**) $a = 1.2$, (**d**) $a = 1.232$, and (**e**) $a = 1.4$

7.4 Lyapunov Exponents of a Filtered Hénon Map

One of the most interesting features of chaotic signals for communication systems is related to their SDIC. It can improve security issues involving information coding [7, 29]. This way, if the system structure or the transmitted message modifies the chaotic map, as in the system proposed in [30] and adapted to discrete-time in [12], it is necessary to determine whether the transmitted signals remain chaotic.

Chaotic signals are often broadband and the physical transmission channels are always bandlimited. Thus, the authors of [12] proposed to place linear discrete-

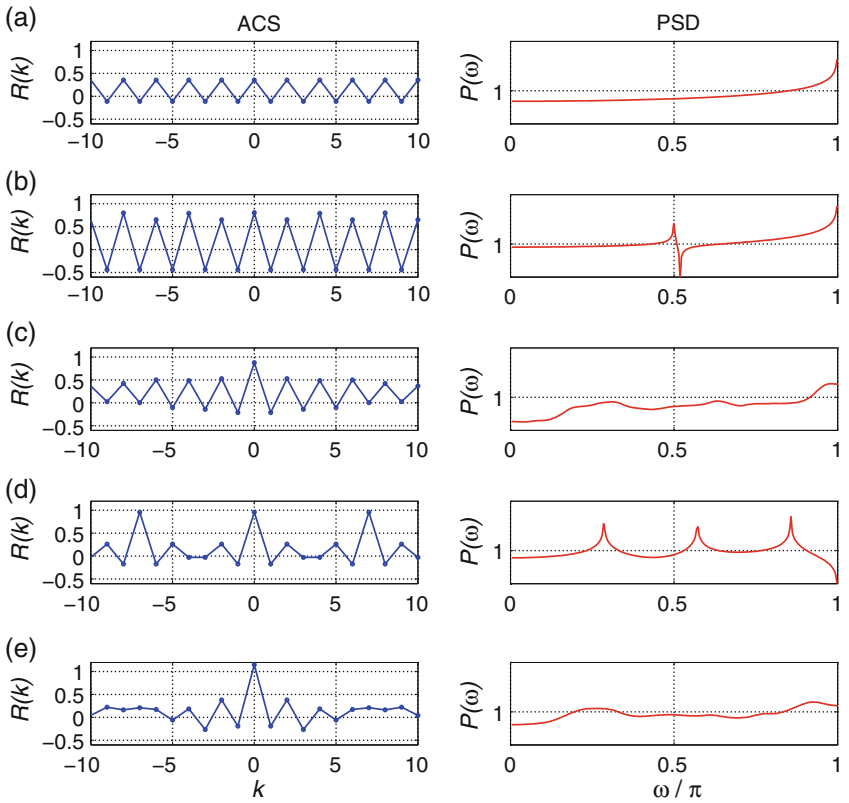


Fig. 7.6 ACS and PSD of orbits of the Hénon map for $b = 0.3$ and (**a**) $a = 0.6$, (**b**) $a = 1.069$, (**c**) $a = 1.2$, (**d**) $a = 1.232$ and (**e**) $a = 1.4$

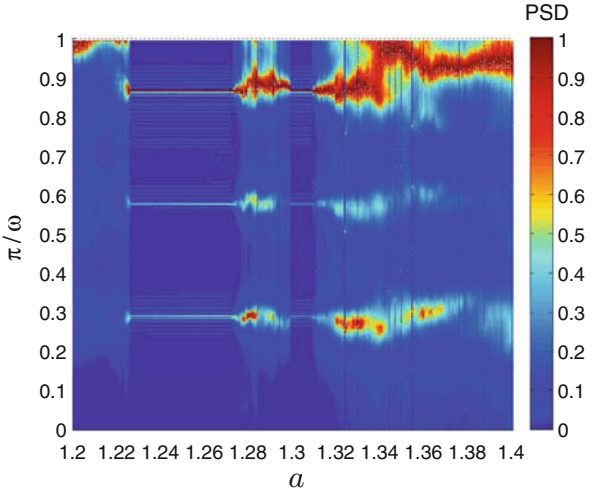


Fig. 7.7 PSD of orbits of the Hénon map for $b = 0.3$ and $1.2 < a \leq 1.4$

time lowpass filters [31] in the feedback loops in order to limit the bandwidth of the chaotic transmitted signals. The synchronization conditions for this bandlimited system have been analytically determined in [6]. However, it is not clear if the transmitted signals remain chaotic. Preliminary numerical results presented in [32] show that the insertion of a finite impulse response (FIR) filter can turn originally chaotic signals into periodic or even unbounded signals depending on the filter coefficients.

In this section, we extend the results of [32], numerically accessing the effect of the filter coefficients on the nature of the transmitted signal, considering the Hénon map as chaotic generator.

7.4.1 Hénon Map with a Linear Filter

In the system proposed in [12], the state $s_1(n)$ of (7.7) is filtered through an FIR filter with N_S coefficients before being fed back into the nonlinearity. The filtered Hénon map can then be described by

$$\boldsymbol{s}(n+1) = \begin{bmatrix} s_1(n+1) \\ s_2(n+1) \\ s_3(n+1) \end{bmatrix} = \begin{bmatrix} a - s_3^2(n) + bs_2(n) \\ s_1(n) \\ \sum_{j=0}^{N_S-1} c_j s_1(n+1-j) \end{bmatrix} \tag{7.10}$$

where $\boldsymbol{s}(n) = [\, s_1(n)\ s_2(n)\ s_3(n)\,]^T$, and c_j, $0 \leq j \leq N_S - 1$, are the filter coefficients.

In this section we numerically analyze the cases $N_S = 1$ and $N_S = 2$ postponing the analysis of filters with more coefficients for a future work.

7.4.2 Lyapunov Exponents for $N_S = 1$ and $N_S = 2$

For $N_S = 1$, (7.10) can be rewritten as

$$\boldsymbol{s}(n+1) = \begin{bmatrix} a - s_3^2(n) + bs_2(n) \\ s_1(n) \\ c_0\left(a - s_3^2(n) + bs_2(n)\right) \end{bmatrix}. \tag{7.11}$$

In particular, for $c_0 = 1$, $s_3(n) = s_1(n)$ and the dynamics is that of the original Hénon map (7.7). For the usual parameter $b = 0.3$ [11], Fig. 7.4 shows the largest Lyapunov exponent and the bifurcation diagram of $s_1(n)$ for $0 < \alpha \leq 1.4$. We can see that after a period-doubling cascading, we have a range of values of a that generate chaotic signals. In particular for the usual $a = 1.4$ we have chaos.

For $0 < c_0 \leq 1$, Fig. 7.8 shows the largest Lyapunov exponent and the bifurcation diagram of $x_1(n)$ as function of c_0 for $a = 1.4$ and $b = 0.3$. Both graphics are

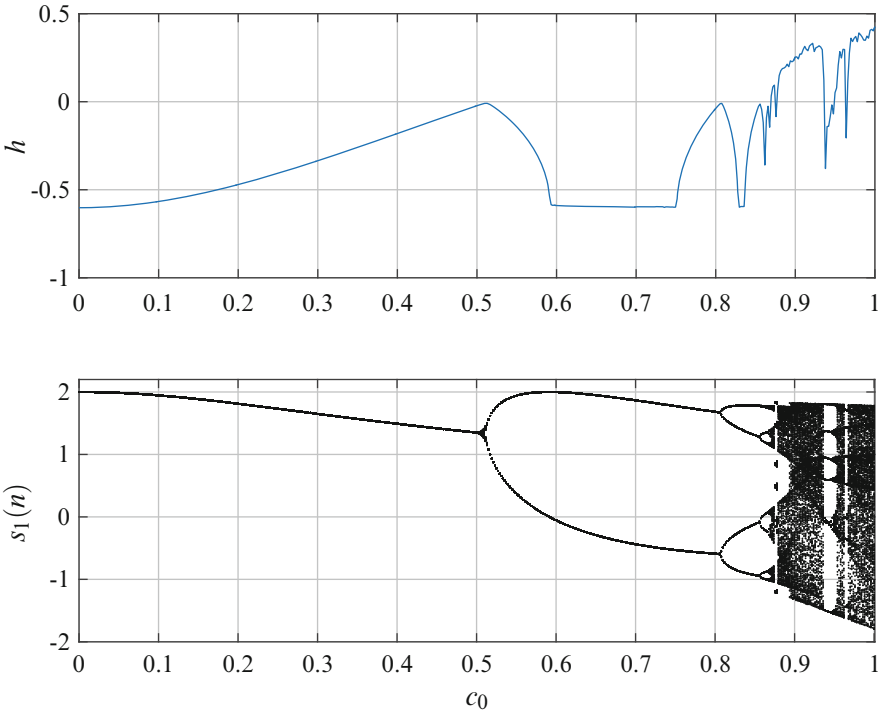


Fig. 7.8 Lyapunov exponents and bifurcation diagram for (7.11) as a function of c_0 with $a = 1.4$

distorted versions of the ones in Fig. 7.4. We clearly see that depending on the value of c_0 the chaotic condition presented for $c_0 = 1$ can vanish. It is possible to obtain chaos for $c_0 > 0.87$.

For $N_S = 2$, (7.10) can be rewritten as

$$\boldsymbol{s}(n+1) = \begin{bmatrix} a - s_3^2(n) + bs_2(n) \\ s_1(n) \\ c_0\left(a - s_3^2(n) + bs_2(n)\right) + c_1 s_1(n) \end{bmatrix}. \tag{7.12}$$

Figure 7.9 shows regions of positive and negative largest Lyapunov exponent h in the $c_0 \times c_1$ space. Blue (darker) regions represent $h < 0$ and yellow (lighter) regions represent $h > 0$. In the blank regions the orbits diverge. For low values of c_0 and c_1 chaos is lost and periodic orbits are found. It is possible to obtain chaotic signals for c_0 or c_1 sufficiently large. However, if both are simultaneously large, the orbits can diverge as is shown by the blank area.

Analyzing the area where $0.6 \leq c_1 \leq 1$, we note complicated patterns between chaotic and periodic regions. A little variation in the coefficient values can alter thoroughly the chaotic nature of the signals.

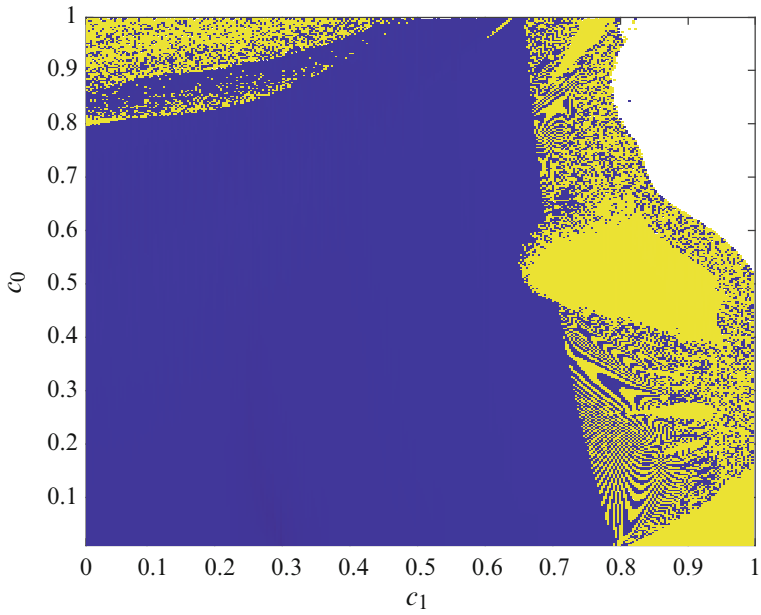


Fig. 7.9 Regions of positive and negative largest Lyapunov exponent h in the $c_0 \times c_1$ space for $\alpha = 1.4$ and $\beta = 0.3$ in (7.12). Blue (dark) regions represent $h < 0$ and yellow (light) regions represent $h > 0$. In the blank regions the orbits diverge

7.5 A Switching Scheme Between Conventional and Chaos-Based Communication Systems

A particularly interesting discrete-time CBCS was proposed in [12], based on the one by Wu and Chua [30], in which the message is fed back into the chaotic signal generator (CSG). It was shown in [12] that under some design conditions, the message is perfectly recovered in an ideal channel. However, this CBCS presents poor performance in terms of bit error rate (BER) when channel imperfections are present [33]. Besides, as ilustrated in the previous section, there is no guarantee that the transmitted signals are still chaotic.

In [34], it was proposed a CBCS using the Hénon map $\boldsymbol{f}_H(\cdot)$ in conjunction with an encoding function such that it was easy to show that the transmitted signals were chaotic. However, the performance of this system is still far from that of a conventional system without chaos.

In this section, in order to obtain better BER than before, we present two innovations:

1. a new encoding and corresponding equalization scheme for the CBCS of [34] that ensures the generation of chaotic signals for a range of parameters and presents higher immunity to intersymbol interference (ISI) and noise, thus providing lower BER for the same convergence rate;

2. inspired by many conventional protocols, like IEEE 802.11 (Wi-Fi) [35], that uses modulation and coding techniques that can adapt to the channel state, we propose an algorithm to switch between chaos-based and conventional communication.

7.5.1 Problem Formulation

Figure 7.10 shows the CBCS under consideration [34, 36]. In the scheme, which is a discrete-time lowpass equivalent for the communication system, a binary message $m(n) \in \{-1, +1\}$ is encoded by using the first component of the master state vector $\boldsymbol{x}(n)$, via a encoding function $s(n) = c\,(x_1(n), m(n))$, so that $m(n)$ can be recovered using the inverse function with respect to $m(n)$, i.e., $m(n) = c^{-1}\,(x_1(n), s(n))$. Then, the signal $s(n)$ is fed back into the CSG and transmitted through a communication channel, whose model is constituted by a transfer function $H(z)$ and AWGN. We assume an M-tap adaptive equalizer, with input regressor vector $\boldsymbol{r}(n)$ and output $\hat{s}(n) = \boldsymbol{r}^T(n)\boldsymbol{w}(n-1)$ and $\boldsymbol{w}(n-1)$ is the equalizer weight vector. The equalizer must mitigate the ISI introduced by the channel and recover the encoded signal $s(n)$ with an unavoidable delay of Δ samples.

If transmitter and receiver identically synchronize [37], i.e., if $\hat{\boldsymbol{x}}(n) \to \boldsymbol{x}(n)$, then using the output of the equalizer and the estimate of $x_1(n)$, $m(n)$ can be decoded via

$$\hat{m}(n) \triangleq c^{-1}\left(\hat{x}_1(n), \hat{s}(n)\right) \to m(n), \tag{7.13}$$

where $\hat{x}_1(n)$ is the first component of the slave state vector $\hat{\boldsymbol{x}}(n)$. Thus, the estimation error $e(n) = m(n-\Delta) - \hat{m}(n)$ can be used as an equalization criterion. Once identical synchronization between master–slave is obtained, $m(n)$ can be used to transmit information between the two systems, being $\hat{m}(n)$ the decoded binary message. We assume that there is a training sequence $\{m(n-\Delta)\}$, known in advance at the receiver. In this case, the equalizer works in the *training* (T) *mode* and updates its coefficients in a supervised manner, using the estimation error in conjunction

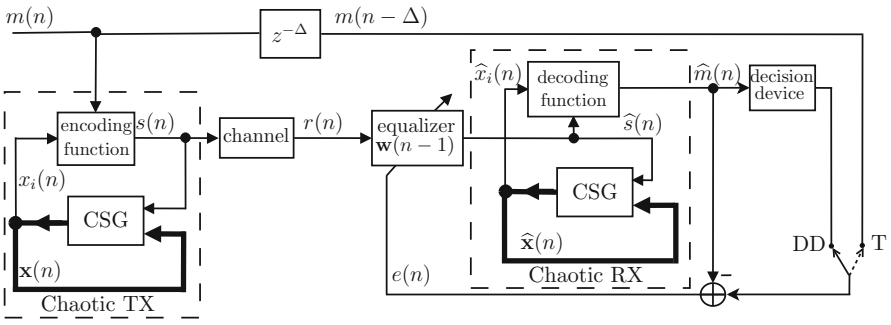


Fig. 7.10 CBCS with an equalizer

with an adaptive algorithm. If we intend to transmit information using $m(n)$, the receiver will not have access to $\{m(n-\Delta)\}$ and this sequence will be replaced by the output of the decision device [38, 39]. In this case, the equalizer works in the so-called *decision-directed* (DD) *mode*.

In this section, $\boldsymbol{f}_H(\cdot)$ of (7.7) is used in both CSGs of Fig. 7.10. Therefore, the equations governing the global dynamical system can be written as

$$\boldsymbol{x}(n+1) = \boldsymbol{A}\boldsymbol{x}(n) + \boldsymbol{b} + \boldsymbol{f}\left(s(n)\right), \tag{7.14}$$

$$\hat{\boldsymbol{x}}(n+1) = \boldsymbol{A}\hat{\boldsymbol{x}}(n) + \boldsymbol{b} + \boldsymbol{f}\left(\hat{s}(n)\right), \tag{7.15}$$

where $\boldsymbol{x}(n) \triangleq [x_1(n) \quad x_2(n)]^T$, $\hat{\boldsymbol{x}}(n) \triangleq \left[\hat{x}_1(n) \quad \hat{x}_2(n)\right]^T$,

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ b & 0 \end{bmatrix}, \boldsymbol{b} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } \boldsymbol{f}\left(s(n)\right) = \begin{bmatrix} -as^2(n) \\ 0 \end{bmatrix}. \tag{7.16}$$

In [12], it was shown that, under ideal channel conditions, i.e., when $r(n) \equiv s(n)$ and the equalizer is an identity system, identical synchronization between master and slave is obtained if all the eigenvalues of $\boldsymbol{A}$ are inside the unit circle. Since the eigenvalues of $\boldsymbol{A}$ are $\pm\sqrt{b}$, we conclude that for $|b| < 1$, master and slave identically synchronize under ideal conditions. Therefore, from (7.13), $\hat{m}(n) \to m(n)$.

7.5.2 Message Encoding

As a way to increase the parameter space where chaotic signals are generated, we propose the encoding function

$$s(n) = \eta_1 x_1(n) - \eta_2 \left[m(n) + 1\right] \text{sign}\left[\eta_1 x_1(n)\right], \tag{7.17}$$

where sign$[\cdot]$ is the signal function. The idea behind (7.17) is to decrease the disturbance on $x_1(n)$ caused by the message. In this case, if $m(n) = -1$, $s(n) = \eta_1 x_1(n)$ and if $m(n) = 1$ a constant with signal opposite to $x_1(n)$ is added to it. The corresponding decoding function is given by

$$\widehat{m}(n) = \frac{\eta_1 \widehat{x}_1(n) - \widehat{s}(n)}{\eta_2 \text{sign}[\eta_1 \widehat{x}_1(n)]} - 1. \tag{7.18}$$

By calculating the maximum Lyapunov exponent obtained in the transmitter it is possible to verify that the generated signal presents sensitive dependence on initial conditions for some values of η_1 and η_2, considering a random equiprobable binary $m(n) \in \{-1, 1\}$. Specifically, we consider $\eta_1 = 0.9$ and $\eta_2 = 0.3$, which ensures the generation of a chaotic signal and allows the equalizer to present a good performance.

7.5.3 Equalization and Switching Algorithm

To control the operation mode of the system, we introduce the parameters $\gamma_1(n)$ and $\gamma_2(n)$ in the encoding function (7.17), making

$$s(n) = \gamma_1(n)\{\eta_1 x_1(n) - \eta_2[m(n)+1]\,\mathrm{sign}[\eta_1 x_1(n)]\} + \gamma_2(n)m(n). \tag{7.19}$$

Thus, when $\gamma_1(n) = 1$ and $\gamma_2(n) = 0$, (7.19) falls back to (7.17) and the system works just like the previously presented CBCS. However, when $\gamma_1(n) = 0$ and $\gamma_2(n) = 1$, $s(n) = m(n)$ and, assuming $m(n) \in \{+1, -1\}$, a BPSK system is obtained.

To decode the message considering (7.19) and assuming $\gamma_1(n)$ and $\gamma_2(n)$ are known at the receiver,

$$\widehat{m}(n) = \frac{\gamma_1(n)\{\eta_1\widehat{x}_1(n) - \eta_2\mathrm{sign}[\eta_1\widehat{x}_1(n)]\} - \widehat{s}(n)}{\gamma_1(n)\eta_2\mathrm{sign}[\eta_1\widehat{x}_1(n)] - \gamma_2(n)}. \tag{7.20}$$

Using a stochastic gradient algorithm to adapt the equalizer, in order to minimize the instantaneous cost-function $\hat{J}(n) = e^2(n) = [m(n-\Delta) - \widehat{m}(n)]^2$, we arrive at the following update for the equalizer coefficient vector

$$\boldsymbol{w}(n) = \boldsymbol{w}(n-1) - \frac{\widetilde{\mu}\{\gamma_1(n)\eta_2\mathrm{sign}[\eta_1\widehat{x}_1(n)] - \gamma_2(n)\}}{\delta + \|\boldsymbol{r}(n)\|^2} e(n)\boldsymbol{r}(n). \tag{7.21}$$

As it can be noticed, when $\gamma_1(n) = 0$ and $\gamma_2(n) = 1$, the conventional normalized least mean squares (NLMS) algorithm [40, Chap. 7] is obtained.

When using BPSK, to maintain the mean power of $s(n)$ equal to the power of the chaotic signal, $\gamma_2(n)$ is adjusted as $\sqrt{0.415}$. To make the system practical for transmitting an actual message, besides switching between the chaotic regime and BPSK, we also consider the switching between the training (T) mode and the decision-directed (DD) mode. Thus, the system may operate in four different modes: (1) BPSK and T; (2) Chaotic and T; (3) Chaotic and DD; and (4) BPSK and DD.

The overall communication system with the switching scheme is depicted in Fig. 7.11. For each block of L samples, the switching is triggered based on thresholds applied to the estimate of the mean square error (MSE), i.e.,

$$\mathrm{MSE}(n_0) = \frac{1}{L}\sum_{k=n_0}^{n_0+L-1} e^2(k),$$

with $n_0 = 0, L, 2L, \ldots$, according to the flowchart shown in Fig. 7.12. The switching algorithm prioritizes the use of the chaotic regime whenever possible and falls back to BPSK when the communication channel condition is poor.

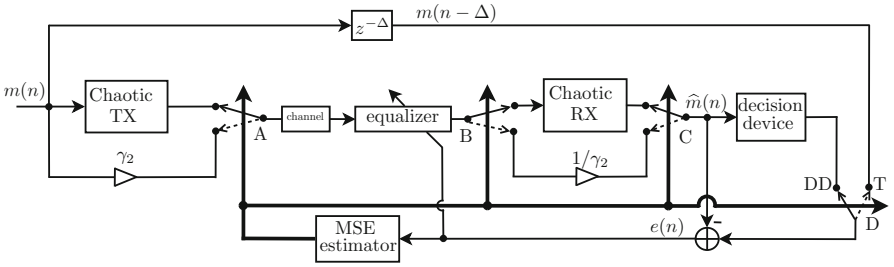


Fig. 7.11 Communication system with an equalizer and a switching scheme between conventional and chaos-based communication

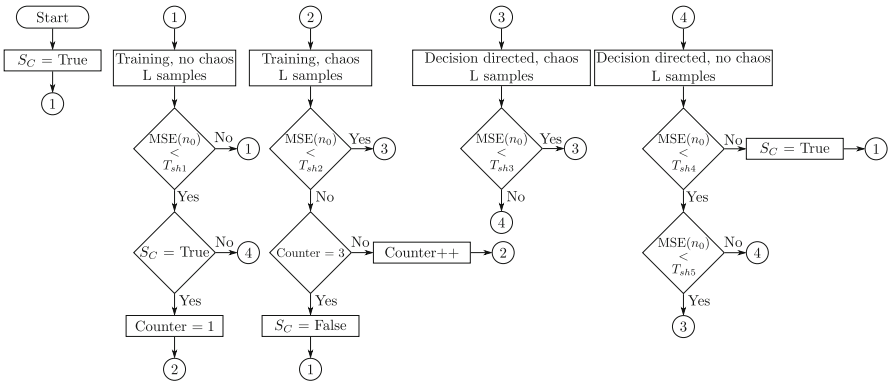


Fig. 7.12 Flowchart of the switching scheme between operation modes. $T_{sh1} \ldots T_{sh5}$ are MSE thresholds that trigger the switching between the operation modes

7.5.4 Numerical Simulation

In this simulation, we consider the communication system using the Hénon map (7.7) with parameters $a = 1.4$ and $b = 0.3$. The state vectors were initialized with $\boldsymbol{x}(0) = \mathbf{0}$ and $\hat{\boldsymbol{x}}(0) = [\,0.1 \quad -0.1\,]^T$, respectively, and we assume the transmission of a binary message $m(n) \in \{-1, \ +1\}$ with the equalizer initialized as $\boldsymbol{w}(-1) = \mathbf{0}$.

To illustrate the presented switching scheme, we first assume that the encoded sequence $s(n)$ is initially transmitted through Channel 1, with transfer function

$$H_1(z) = -0.005+0.009z^{-1}-0.024z^{-2}+0.850z^{-3}-0.218z^{-4}+0.050z^{-5}-0.016z^{-6}, \tag{7.22}$$

which is changed abruptly at $n = 250 \times 10^3$ to Channel 2 with transfer function

$$H_2(z) = -0.004 + 0.030z^{-1} - 0.104z^{-2} + 0.520z^{-3} + 0.273z^{-4} - 0.074z^{-5} + 0.020z^{-6}, \tag{7.23}$$

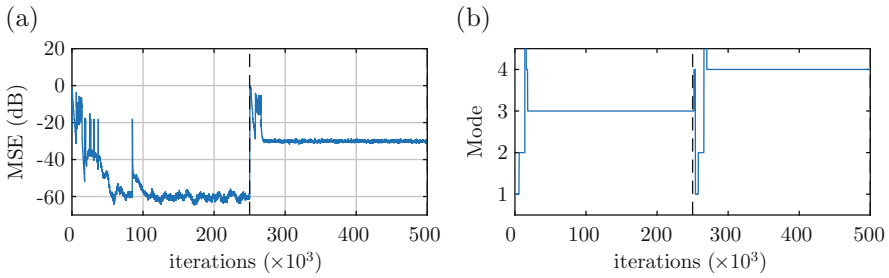


Fig. 7.13 (**a**) Squared error and (**b**) operarion mode along iterations. Communication system using the Hénon map, cNLMS$_+$ ($\widetilde{\mu}=0.01$; $\delta=10^{-2}$) and the switching scheme described in Sect. 7.5.3. Abrupt variation from Channel 1 to Channel 2 at $n = 250 \times 10^3$; $M = 12$; $\Delta = 7$

in the absence of noise [34, 41]. Figure 7.13 shows (a) the squared error, and (b) the operation mode of the communication system. The switching is performed at each $L = 2 \times 10^3$ iterations based on the flowchart shown in Fig. 7.12 with thresholds $T_{sh1} = -30\,\text{dB}$, $T_{sh2} = -40\,\text{dB}$, $T_{sh3} = -30\,\text{dB}$, $T_{sh4} = -20\,\text{dB}$, and $T_{sh5} = -35\,\text{dB}$. For the first iterations, under Channel 1, we can observe that the system switches to chaotic modulation in decision-directed mode (number 3), after a brief transient. During this transient, there are some wrong estimations of $m(n)$ but after the switching algorithm stabilizes in State 3, the message is recovered. After the abrupt variation to Channel 2 at $n = 250 \times 10^3$, the algorithm switches to BPSK modulation in decision-directed mode (State 4), after a transient period. This is due to the fact that Channel 2 inserts more ISI than Channel 1, hindering the utilization of the chaotic modulation, as we can notice by the squared error level.

7.6 Conclusions

In this chapter, we presented four recent ongoing researches involving by one hand communications and signal processing and by the other hand dynamical systems.

In Sect. 7.2, we succinctly described a fingerprint generation method using chaotic maps, as part of a physical layer authentication protocol. The results are auspicious showing that this can be a practical application of chaotic systems in signal processing.

In Sect. 7.3, we analyzed through computational simulations the ACS and PSD of orbits generated by the Hénon map. The simulations suggest that the map can be used to generate narrowband and highpass chaotic signals, what can be relevant for CBCS.

The numerical simulation presented in Sect. 7.4 shows that filtering a Hénon map can modify its chaotic regions. This way, bandlimited CBCS must be carefully projected to guarantee that the generated signals remain chaotic. In our present research we are studying the influence of filters with more coefficients.

In Sect. 7.5, we proposed an adaptive scheme that switches between a chaos-based and a conventional communication scheme. The switching is triggered based on a threshold applied to the MSE. Simulation results show that the switching and equalization algorithms can successfully recover a transmitted sequence.

Although the application of the properties of chaotic systems in engineering and applied sciences is still in its beginning, we hope that results as the ones presented here can contribute to increase this synergy.

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Chapter 8
Modeling Fireflies Synchronization



Gonzalo Marcelo Ramírez-Ávila, Jürgen Kurths, Stéphanie Depickère, and Jean-Louis Deneubourg

8.1 Introduction

Synchronization defined as the adjustment of rhythms of self-sustained oscillators due to a weak coupling is one of the most widespread complex phenomena observable both in human-made systems and in natural ones [55]. In general, synchronization is attained via a mechanism similar to a phase transition implying the emergence of structural order [44]. From a historical viewpoint, it is stated that the first observation of synchronization was done by the Dutch scientist Christian Huygens, considered the most ingenious watchmaker of all time [71] and also one of the founders of the mathematization of nature [82]. Curiously, another Dutch naturalist, Engelbert Kaempfer, described for the first time the synchronous collective behavior of large populations of fireflies, a phenomenon that he observed during his voyage to Asia (1690–1692) as stated by Buck [10]. What are the differences between their two observations? Simply, the first one

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is the detection of synchronization in a small human-made system, and the other one is the manifestation of a natural self-organized system. Being the first system composed of two pendulums continuously coupled through a shared structure consisting of one or two wooden beams in a half-timber house [6]; and the second one consisting of hundreds or thousands of fireflies seen as self-sustained-oscillators coupled by light pulses, i.e., a noncontinuous coupling. Although the results of both observations aroused curiosity and were considered as paradigmatic examples of the phenomenon of synchronization, their formal description took more than two centuries to be formulated and still now some aspects are not completely unveiled such as what determines the stability of multiple types of limit behavior [34]. For pendulums synchronization, in the last years, several works were devoted to the reproduction [51] and improvements [53] of Huygens' experimental setup, the linear stability analysis of this system, the proper explanation of the results [35], and the extensions of the synchronization phenomenon for this type of system. Thus, the obtained results showed that not only anti-phase synchronization is possible as found by Huygens but also in-phase synchronization is possible. A system of n pendulums hanging from an elastic horizontal beam, where synchronization and clustering are manifested has been modeled [25], in the same line, the extension to multiple coupled Huygens' pendulums combined with a similar system but composed of metronomes deserved the attention in the last years [34]. Concerning the synchronization of fireflies, the phenomenon was not entirely accepted before 1918, and among the statements denying the fireflies synchronous behavior we mention that of W. Craig quoted [24]:

> Dr. Edward S. Morse [50] cites a case from memory in which he saw "fireflies flashing in unison," but he gives no exact details. He quotes a paper by Mr. Blair [7] mentioning the same phenomenon; but Mr. Blair states that he never observed the synchronism himself, and he does not cite any authority who has observed it. Dr. Morse in another paper [50] quotes R. Shelford as observing a tree full of fireflies pulsating "so that at one moment the tree would be one blaze of light, whilst at another the light would be dim and uncertain." This last clause makes it appear that some fireflies were not in synchronism with the others, and thus brings in the statistical fallacy to be mentioned presently.

Similarly, P. Laurent in this letter "The supposed synchronal flashing of fireflies" [37] quoted:

> I could hardly believe my eyes, for such a thing to occur among insects is certainly contrary to all natural laws. However, I soon solved the enigma. The apparent phenomenon was caused by the twitching or sudden lowering and raising of my eyelids. The insects had nothing- whatsoever to do with it. Many times in the past twenty years I have proved that my solution was correct.

Despite the devastating above-mentioned quotations against the fireflies synchronization which were denominated as anthropomorphic explanations [18]; these shortly after were forsaken due to the increasing of careful observations and improvements in measurements and used instruments. The latter allowed the confirmation that fireflies synchronization is entirely real and has a specific functionality. The aspects mentioned above were found and developed throughout this time. Among the main features which were unveiled we emphasize: chemical [2] and

physical [5, 57, 75] mechanisms related to the fireflies light emission, genetic [4] and evolutionary [33, 45, 72] issues, the how and why fireflies synchronize in relation to the courtship and mating behavior observed in several species [42] behavior that constitutes the main functionality of synchronization [13]. The fireflies communication [40] deserved several systematic observations giving rise to establish the "language" of different species and the fact that in some species, males synchronization must be followed by the females response [83]. Among other interesting aspects, we mention the misleading courtship using some artificial light sources or LEDs to stimulate female response [48]. The tragic case of "devouring femme fatale" where a male looking for a conspecific female is deceived and devoured by a female of another species who imitates the response of the wanted partner has also been studied [43]. A catalog of communication codes in *Photinus* species [38] shows the diversity of these intercourse signals and also changes the traditional ideas that taxonomy is only possible by identifying the main morphological features, but it is also possible to identify fireflies species by knowing their courtship coding as found by Lloyd [41]. As we see, the diversity of themes and aspects that involve the synchronization of fireflies is very vast [62]. However, from mathematical modeling, there were not many attempts to formally explain the synchronization in these insects. A. Winfree proposed a model of the collective behavior of a population of oscillators that could represent electronic oscillators, secretory cells, neurons, or animals such as fireflies [80]. In the same line, other models were developed for populations of oscillators [36, 52]; in these studies, the oscillators are generic and not necessarily with the specific features of fireflies flashing. In 1990, inspired in a model of heart cells, Mirollo and Strogatz proposed a model for pulse-coupled oscillators that have the main characteristic of fireflies, i.e., the oscillator behaves in such a way that it has a charging stage until a threshold is attained and then it fires (flashes). The latter is the reason this type of oscillators are also called integrate-and-fire oscillators [47]. Thereafter, several works were devoted to study synchronization on IFOs [1]. The first qualitative model used to understand the fireflies synchronization was based upon the simple experiments exciting or inhibiting the flashing rhythm of a firefly (in particular with *Pteroptyx cribellata*). The latter allowed to formulate a resettable pacemaker model to explain the modifications on the rhythmic flash emission that eventually permits the synchronization [12]. This model facilitates the development of ideas conducting to postulate formal models from reset and threshold concepts. Precise models for fireflies were formulated, for instance, for *Pteroptyx malaccae* based on phase delay synchrony as a mechanism [28]. Another perspective for specific fireflies was constructed from some electronic devices called light-controlled oscillators (LCOs) or electronic fireflies due to their features that mimic real fireflies because they communicate through light-pulses and can synchronize [59]. The similarity between LCOs and IFOs could drive to think that the results obtained with these models are very similar but surprisingly, there are striking differences even with simplified LCOs versions [64]. It is important to remark that LCOs model can be extended to explain not only synchronization but also the response to synchronization [58] as occurring in several fireflies species, in particular, those of genus *Photinus* [38, 40].

Fireflies synchronization has multiple applications and just to mention some of them: the use of this concept to enhance the robustness of wireless communication networks [77], the better understanding of real task-execution scenarios for swarms of robots [22], the new techniques of light extraction for LEDs improvements are also inspired in fireflies luminescence [5]. The chapter is organized as follows: in Sect. 8.2, we describe the concept of entrainment and the qualitative resettable pacemaker models based on simple observations. The explanation of the first mathematical models to describe fireflies synchronization is pointed out in Sect. 8.3, including the paradigmatic IFOs, and the LCOs. The comparison of LCOs and IFOs is presented in Sect. 8.4. The generalization of LCOs model to explain the response to synchronization is exposed in Sect. 8.5. The depiction of the "solitary flash game" (SFG) is shown in Sect. 8.6. As a sort of conclusive remarks, in Sect. 8.7, we state the limitations of the models and the perspectives of the fireflies synchronization phenomenon.

8.2 Explanation of the Entrainment in Fireflies

The concept of entrainment constitutes the basis for understanding a myriad of oscillatory behaviors in living beings. Entrainment is related to the resetting of biological clocks introduced by Winfree in 1975 [81]. Among the key behaviors, circadian rhythms constitute perhaps the most known phenomena [76], in which are principally involved populations of neurons giving rise mostly to coherent oscillatory behavior and allow the adaptation to the environment [49]. In general, the entrainment of neurons is an essential aspect related to certain functionalities, and it has been widely studied, for instance, on effects of regularly spaced synaptic input [54], and phase control of neural pacemakers. Cardiac cells and their entrainment are also considered in different contexts such as the spontaneous impulse generation in the pacemaker of the heart [74], effects of regularly spaced nervous input [66], a nonlinear analysis when the cells are periodically stimulated, and the generation of cardiac dysrhythmias [32]. Fireflies constitute an excellent example of the oscillatory control strategies exhibited by most living beings: temporal coordination, prediction and preparation for repetitive events, high efficiency in energy transduction and communication, and enhanced precision of control [65]. Thus, the synchronous rhythmic behavior of the fireflies' flashes must also be tackled with the concept of entrainment. We start from the fact that male fireflies can spontaneously flash in a regular rhythm. The entrainment of the flashing rhythm of these insects is related to neural aspects as pointed out in several works (see [20]) and a saga of papers explaining in detail several noteworthy aspects associated with the control of flashing in fireflies). The set of articles firstly included topics such as the lantern as a neuroeffector organ [16]; secondly, the role of central nervous system [21]; thirdly, the peripheral excitation [17], and also aspects linked to synchronization suchlike the free run pacemaking in a synchronic *Pteroptyx* [15], or the pacemaker synchronization in *Pteroptyx cribellata* [14]. They yielded

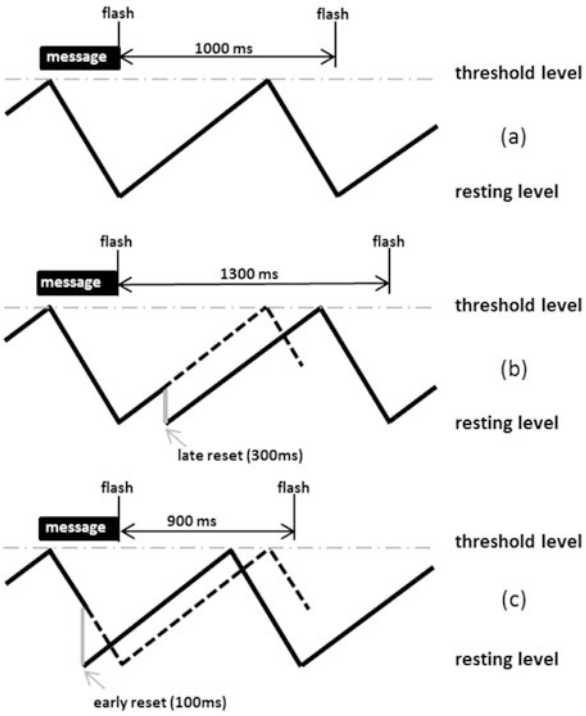


Fig. 8.1 Resettable pacemaker model suggested in [12]. (**a**) Normal or natural interflash period of 1 s. (**b**) Delayed flash that in this case appears 300 ms later. (**c**) Advance flash (early reset) cropping up 100 ms before the standard flash

the basis to propose formal models bringing out the most relevant aspects of fireflies flashing behavior. The knowledge of how entrainment works lets us see the problem of synchronization as mutual entrainment between oscillators being the reciprocal feedback decisive concerning the stability. One important aspect akin to the determination of how fireflies synchronize is related to the influence of other light sources on the rhythm of their light emission frequency. A careful observation of the influence mentioned above and the afterward simple experimental setup design allowed to realize about the resettable feature of the luminescent control of fireflies [12]. The proposed resettable pacemaker model is depicted in Fig. 8.1, where the authors considered that the instantaneous flash emitted by the oscillator in Fig. 8.1a could reset instantaneously. The signal of oscillators shown in Fig. 8.1b, c produces a delayed flash (Fig. 8.1b) or advances according to the reception of a light pulse (Fig. 8.1c).

The schematic view of fireflies' waveform permits the understanding on the entrainment of flashing fireflies. The phase resetting model introduced by Buck is shown in Fig. 8.1 and it has been applied by himself to two fireflies species, namely *Photinus pyralis* and *Pteroptyx cribellata* [11] and based on these observations, it was possible to formulate some primary models to explain how fireflies synchronize. In Sect. 8.3, we focus on three simple models characterized by the fact that they intend to reproduce the oscillatory and synchronous features of fireflies species detailed in [11]. To conclude this section, we point out that many aspects were

neglected such as temperature effects [26], chemical conditions [2, 19], or some behaviors related to competition or aggressive mimicry [43] that eventually could play a role in how synchronization is attained.

8.3 First Models: How to Explain Collective Behavior?

One of the first models arose to explain the synchronous behavior of biological oscillators is due to Winfree [80] who formulated a phase model based on relaxation oscillators whose analysis permits to solve exactly a particular situation under a mean field coupling with firing impulses [3]. Afterwards, some refinements of Winfree's model have been concocted [9] and especially, the formulation owed to Kuramoto became a paradigmatic model to study synchronization [36]. Finally, one of the few models dedicated mainly to fireflies is that formulated by Ermentrout who proposed an adaptive model for the synchronous behavior of *Pteroptyx malaccae* [27].

Before to start the analysis of IFOs and LCOs, it is imperative to mention a valuable tool to characterize the response of an oscillator to a periodic stimulus: the phase response curve (PRC). A PRC is a graphical representation of the magnitude of the phase shift produced in an oscillatory system by a pulse versus the time at which the pulse was applied. Let T_0 be the natural period of the oscillation and considering that $t = t_{F0}, 2t_{F0}, 3t_{F0}, \ldots$ are the times of successive firing events. Suppose that at $t = t_0 \in [0, T_0)$, we perturb the trajectory. Then, the new firing time is $T(t_0)$. A primary assumption is that the firing events occurring after $t = T(t_0)$ are at $t = T(t_0) + T_0, T(t_0) + 2T_0, \ldots$. Thus, the effect of the perturbation is only carried for one cycle; there is no memory of the perturbation once the event has occurred. The PRC is defined regarding the phase shift as [31]:

$$\Delta(\phi) \equiv \frac{T_0 - T(T_0\phi)}{T_0}\,, \tag{8.1}$$

where $\phi \equiv t_0/T_0 \in [0, 1)$ is the phase at which the stimulus is applied. Thus, a perturbation applied to the oscillator will generate either an advance or a delay in the phase, and it is a function of the magnitude and the timing of the perturbation. In Eq. (8.1), $\Delta(\phi)$ is positive (negative) if the perturbation acts to advance (retard) the time of the next event. We use PRCs with the aim of describing the IFOs and LCOs.

8.3.1 Integrate-and-Fire Oscillators

The IFOs have been widely used in neural systems [28] and also they constitute an important model to describe synchronous fireflies [47]. Originally, they were featured by a voltage-like state variable V_i, whose dynamics is:

$$\frac{\mathrm{d}V_i}{\mathrm{d}t} = I - \eta V_i, \qquad 0 \leq V_i \leq 1, \qquad i = 1, \ldots, N. \tag{8.2}$$

With $I > 0$ being the input, and $\eta \geq 0$ is the leakiness; more concretely, $I > 1$ ensures the existence of a PRC as shown in Fig. 8.2c. When the oscillator i reaches the threshold ($V_i = 1$), the oscillator "fires" and V_i is reset *instantaneously* to zero. The oscillators interact by a simple form of pulse coupling: when a given oscillator fires, all the other variables V_j, $j \neq i$ are increased by an amount β/N (the quotient by N is made to get reasonable behavior in the thermodynamic limit). That is,

$$\textbf{If} \quad V_i(t) = 1 \quad \Longrightarrow \quad V_j(t^+) = \min(1, V_j(t) + \beta/N), \quad \forall j \neq i. \tag{8.3}$$

Moreover, the oscillator at the state $V = 0$ (i.e., just after firing) cannot be affected by the others. The construction of an IFO's PRC starts by integrating Eq. (8.2) to find the IFO's natural period:

$$T_0 = \frac{1}{\eta} \ln \frac{I}{I - \eta}\,. \tag{8.4}$$

At $t = t_0$, due to the perturbation, the oscillator adds its voltage-like variable V by an amount β and then evaluates when the oscillator fires next. If t_0 is close enough to $t = T_0$, then the perturbation β will lift V past 1 and the oscillator will fire immediately so that $T(t_0) = t_0$. Otherwise, the fire event will be held at $t = T(t_0)$, and T is computed as follows:

$$T = t_0 + \int_{V+\beta}^{1} \frac{\mathrm{d}V}{I - \eta V}\,,$$

that finally gives

$$T = \frac{1}{\eta} \ln \frac{I - \eta\beta \mathrm{e}^{\eta t_0}}{I - \eta}\,. \tag{8.5}$$

Consequently, the PRC for the integrate-and-fire oscillator is

$$\Delta\left(\frac{t_0}{T_0}\right) = 1 - \frac{\ln\left(\frac{I-\eta\beta \mathrm{e}^{\eta t_0}}{I-\eta}\right)}{\ln\left(\frac{I}{I-\eta}\right)}\,. \tag{8.6}$$

The IFOs entrainment and some examples of PRCs are shown in Fig. 8.2.

We slightly modify this description with the aim of considering the characteristics related to fireflies, in particular to *Pteroptyx cribellata* whose oscillation period varies between 800 and 1600 ms [14]. The modified formulation considers two threshold values V^{upper} and V^{lower} not necessarily 1 and 0. When the oscillator i reaches the threshold ($V_i = 1$), the oscillator "fires" and V_i is reset *instantaneously*

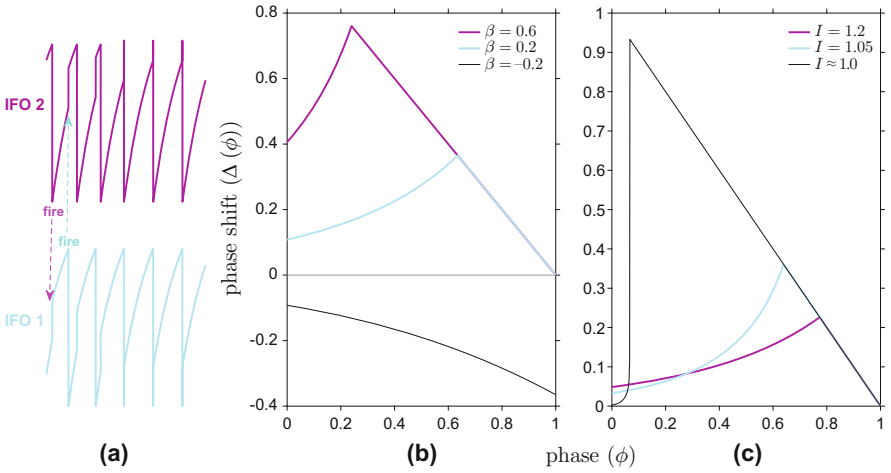


Fig. 8.2 (**a**) Entrainment and mechanism of synchronization of two coupled IFOs. PRCs for an IFO with $\eta = 1$. (**b**) Different strengths for $I = 1.25$. (**c**) Different values of I when the strength is $\beta = 0.1$

to zero. We slightly modify this description with the aim of considering the characteristics related to fireflies, in particular to *Pteroptyx cribellata* whose oscillation period varies between 800 and 1600 ms [14]. The modified formulation considers two threshold values V^{upper} and V^{lower} not necessarily 1 and 0. The mechanism when IFOs are coupled might be explained as follows: when IFO_j reaches its upper threshold, it fires, and the variables V_i of all the other IFOs are modified by adding the quantity β'_{ij} but not exceeding V^{upper}. Thus, $V_i(t^+) = \min(V^{\text{upper}}, V_i(t) + \beta_{ij})$. A general coupling scheme in which it is possible that a set of $N_f < N$ IFOs achieve their thresholds simultaneously implies that IFO_i modifies its dynamics in the form:

$$\textbf{If}\left\{V_j(t) = V^{\text{upper}}\right\} \Longrightarrow V_i(t^+) = \min\left(V^{\text{upper}}, V_i(t) + \sum_{j=1}^{N_f} \beta'_{ij}\right) \wedge \left\{V_j(t^+) = V^{\text{lower}}\right\}, \tag{8.7}$$

where $j = 1, \ldots, N_f$. The meaning of β'_{ij} is the coupling strength and represents the action of IFO_j on IFO_i; and finally, the natural period of IFO_i might be incorporated in the form:

$$\frac{\mathrm{d}V_i(t)}{\mathrm{d}t} = \frac{1}{T_{0i}}[V_{Mi} - V_i(t)], \quad V^{\text{lower}} \leq V_i(t) \leq V^{\text{upper}}, \tag{8.8}$$

with the same condition Eq. (8.7) when coupled with other IFOs.

8.3.2 Electronic Fireflies: Light-Controlled Oscillators

Originally, LCOs were conceived with the aim of mimicking fireflies synchronous behavior through a simple electronic device. An LCO functions under the primary fireflies characteristics, i.e., with the ability to emit periodic light-pulses, and receiving light pulses of other(s) LCO(s); so that, its internal flashing rhythm can be modified thus enabling synchronization. The LCOs were widely studied in different contexts such as experimental characterization [23, 59, 68, 69], noise influence [60, 63], and other aspects related to stability and synchronization [67]. The model for LCOs was built regarding their electronic components (resistors, capacitors, diodes, a source voltage characterized by V_M, and LM555 chip). The latter flip-flop constitutes the LCO's heart because this simple chip confers the oscillatory features to the LCO, establishing two well-defined thresholds: lower ($V^{\text{lower}} = V_M/3$), and upper ($V^{\text{upper}} = 2V_M/3$). The above-mentioned thresholds are associated with a binary variable $\varepsilon(t)$ that takes the values $\varepsilon(t) = 1$ or $\varepsilon(t) = 0$ during the charging stage (slow process), or discharging stage (fast process) respectively. The discharging stage changes to the charging one and vice versa when one of these thresholds is achieved. The dynamical equations describing a set of N coupled LCOs in terms of the natural charging time (T_{c0}) and the discharging or firing time (T_{d0}) are:

$$\frac{\mathrm{d}V_i(t)}{\mathrm{d}t} = \underbrace{\frac{\ln 2}{T_{c0i}}(V_{Mi} - V_i(t))\varepsilon_i(t)}_{\text{charging term}} - \underbrace{\frac{\ln 2}{T_{d0i}}V_i(t)[1-\varepsilon_i(t)]}_{\text{discharging term}} + \underbrace{\sum_{j=1}^{N}\beta_{ij}\delta_{ij}[1-\varepsilon_j(t)]}_{\text{coupling term}}, \tag{8.9}$$

where δ_{ij} indicates if LCOs i and j interact. Note that the interaction term is active only when at least one of the other LCOs is discharging. In this model, we consider symmetric coupling, such that $\beta_{ij} = \beta_{ji}$.

A simple inspection of Eq. (8.9) shows that both charging and discharging stages might be modified by the effect of the coupling with other LCO(s). The charging and the discharging times might be shortened or lengthened, respectively, when the pulsatile action due to the light of other LCOs takes place.

Similarly to the IFOs' case, we specify in Fig. 8.3 how a perturbation acts on the LCO's waveform according to the region in which it is applied.

In order to show clearly how the PRC can be obtained for an LCO, we define an arbitrary zero phase for the LCO's waveform. For convenience, we take it to be when the signal is just in the middle of the thresholds in the discharging stage so it has the voltage value $\frac{V_M}{2}$. This choice enables us to take into account the regions where there are transitions from one stage to another. These regions are particularly interesting because during a transition there can be both effects: advance or delay in the phase (Fig. 8.4).

To characterize the synchronous behavior attained by two mutually coupled LCOs with the oscillatory features of *Pteroptyx cribellata*, the Arnold tongues

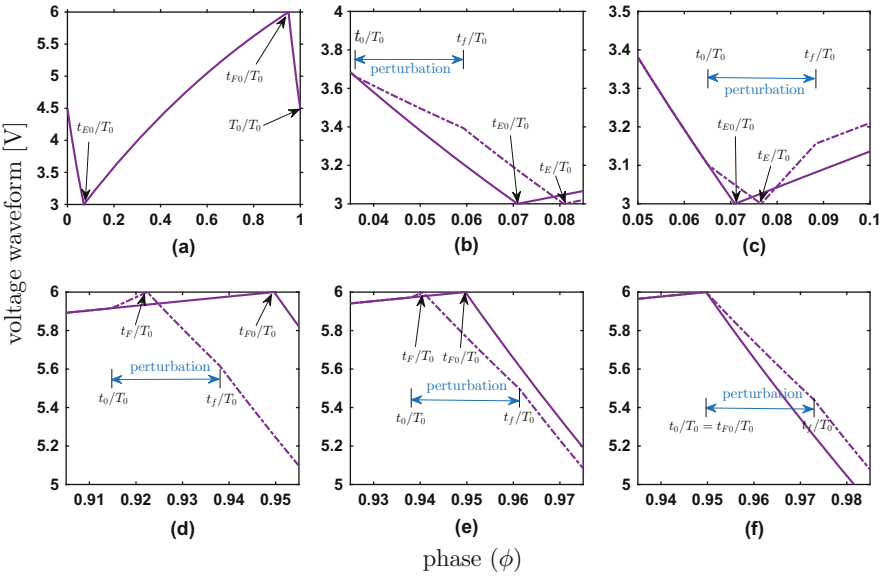


Fig. 8.3 Parameter definition to determine the LCO's PRCs. t_{E0} and t_{F0} represent the LCO's extinction time and the LCO's firing time, respectively, (without perturbation), t_0 and t_f represent the initial and the final time of the perturbation, respectively. Finally, t_E and t_F are the extinction and firing time of the perturbed LCO. (**a**) Whole unperturbed signal. (**b**) Region 1 with $t_0 = t_{E0} - 1.5\tau$. (**c**) Region 2 with $t_0 = t_{E0} - 0.25\tau$. (**d**) Region 3 with $t_0 = t_{F0} - 1.5\tau$. (**e**) Region 4 with $t_0 = t_{F0} - 0.5\tau$. (**f**) Region 5 with $t_0 = t_{F0}$. The unperturbed signal is represented with a solid line whereas the perturbed one with dash and dotted line. We have considered a coupling strength $\beta = 200$ and the duration of the pulse $\tau = 1.0$ ms

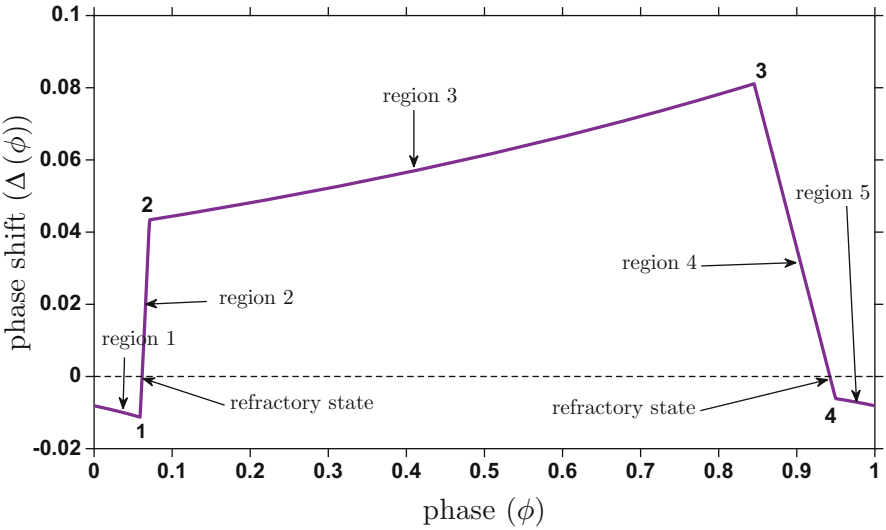


Fig. 8.4 PRC obtained from analytical solution of the LCO's differential equations. Points 1, 2, 3, and 4 are related to $t_E^{(2)}$, t_{E0}, $t_F^{(4)}$, and t_{F0}, respectively. The phases corresponding to the refractory states are also shown

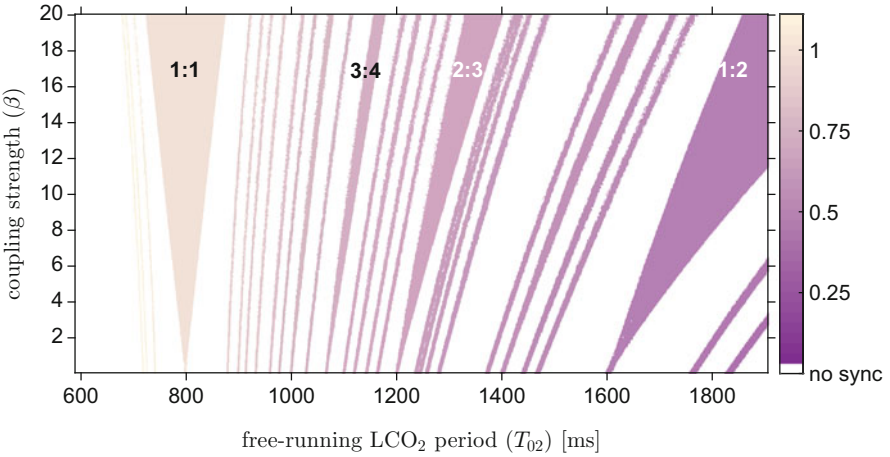


Fig. 8.5 Arnold tongues for LCOs oscillating with the features of *Pteroptyx cribellata* and considering a rapid flash. The color bar indicates the winding number or synchronization order. Some of these ratios (1:1, 2:3, 3:4, and 1:2) are explicitly shown

(synchronization regions) are shown in Fig. 8.5. These areas representing stability are obtained by considering the winding number $\rho = T_1/T_2$, where T_1 and T_2 are, respectively, the LCO_1 and LCO_2 natural oscillation periods. This result is significant because it shows in detail the phase-locking attained for two nonidentical mutually coupled LCOs which gives the possibility to have a deeper understanding of the synchronous behavior of such a system, including the possibility to control it.

Just to mention, there are other ways to model fireflies behavior. For instance, using multi-agent systems (MAS), one of their most well-known platform is NetLogo [79] where employing simple interaction rules among the agents (fireflies), and some strategies such as advanced or delayed phase, the whole population is able to synchronize. A fireflies model is available in NetLogo [78], which is based on the flashing behavior of some species: *Pteroptyx cribellata, Luciola pupilla*, and *Pteroptyx malaccae*.

8.4 From Ideal to Real: Comparing IFOs and LCOs

Despite the similarities between IFOs and LCOs, there are some features that distinguish them markedly. An important difference is that the ideal IFO considers an instantaneous resetting or flashing, but of course, in real fireflies, the flashing lasts a specified time interval that is typical for each species (see Table 8.2). In order to quantitatively establish the differences between IFOs and LCOs, we consider the same form and features for all the oscillators. Thus, analyzing populations of coupled oscillators, for IFOs, we can consider Eqs. (8.7)–(8.8) taking into account

that $V^{\text{lower}} = V_M/3$, and $V^{\text{upper}} = 2V_M/3$, i.e., the same as in the LCOs' case. On the other hand, the IFOs' and LCOs' coupling strengths are related by $\beta'_{ij} = \beta_{ij} T_{d0j}$. For LCOs, Eq. (8.9) and the threshold conditions govern their dynamics.

In what follows, we show some results evincing the differences between IFOs and LCOs. In order to characterize synchronization, we use the concept of probability of total synchronization (PTS), defined as the ratio of the number of trials in which all the oscillators achieve complete synchronization to the number of total trials. We also consider two types of coupling: (1) A *mean field* one where each one of the N oscillators is coupled to the others with a coupling strength given by $\beta_{ij} = \beta/N$, being β a constant. (2) A *distance dependent* coupling, in which the coupling strength is $\beta_{ij} \propto 1/r_{ij}^{\alpha}$, where the exponent α was found to take the value 2.11 [59].

Finally, in our analysis, we allow for situations in which identical or nonidentical oscillators might compose the ensemble. We denote the unlikeness by a difference in the period ΔT tied up with the relative variance σ_{rel} of a normal distribution linked to the possible values of the period $T_i = T_{\text{ref}} + \Delta T_i$. The quantity T_{ref} is a reference period proper to the fireflies species.

As a relevant aspect to point out the behavioral differences between IFOs and LCOs, we compute the transient to achieve total synchronization in our sets of oscillators. In order to study transients, we consider the fact that all the oscillators fire almost simultaneously as the criterion to determine whether or not a set of oscillators attains complete synchronization. When the coupling is a mean-field one, we observe in the scatter plots of Fig. 8.6 that for identical LCOs (Fig. 8.6a) and IFOs (Fig. 8.6e), mostly the whole population synchronizes. Note that circles' sizes are proportional to the PTS in percentage. The transient grows with N, but the mean values are small for both cases. On the contrary, when we consider the differences σ_{rel}, we observe for LCOs that transient escalates with N but also PTS slumps with N. When the coupling depends on the distance between oscillators, synchronization time escalates even when the oscillators are identical (Fig. 8.7a, f). The case of identical IFOs is dramatic inasmuch as populations with $N > 8$ imply noughts for the PTS. The same behavior in both kind of oscillators is observed when the differences are $\sigma_{\text{rel}} = 0.10 \times 10^{-4}$ (Fig. 8.7b, f), $\sigma_{\text{rel}} = 2.24 \times 10^{-4}$ (Fig. 8.7c, g), and $\sigma_{\text{rel}} = 7.07 \times 10^{-4}$ (Fig. 8.7d, h). It is remarkable that complete synchronization is not usual when the coupling is distance-dependent. A summary of the results is shown in Table 8.1. An exhaustive study on the differences between IFOs and LCOs can be found in [64]. As indicated above, we can affirm that despite IFOs and LCOs seem to be quite similar, the dynamical behavior might present very different behaviors. Evidently, LCOs are more realistic because they are experimentally based and validated. As a consequence, we choose LCOs as the more appropriate model for describing fireflies synchronization and as it is stated in Sect. 8.5, an extension of the LCOs might be applied not only to explain male fireflies synchronization but also the female response to synchronization what gives us more clarity concerning the functionality of fireflies synchronization.

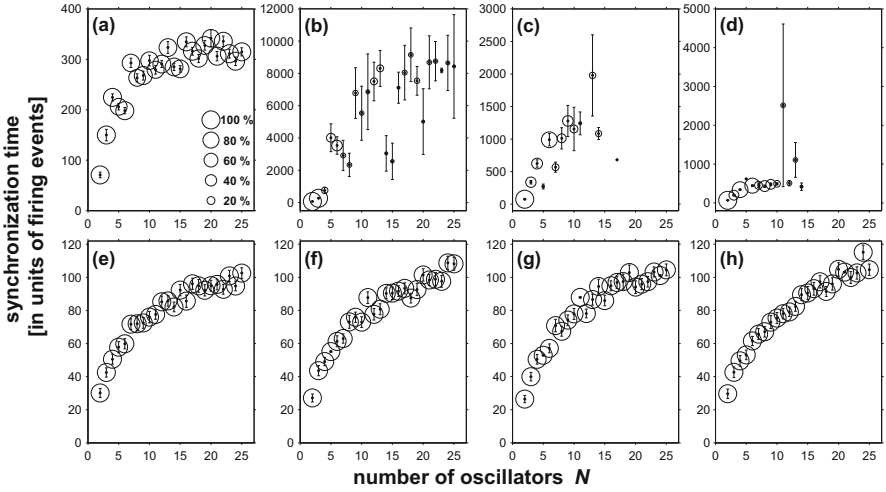


Fig. 8.6 Synchronization time as a function of N for globally coupled LCOs (top) and IFOs (bottom) being the coupling a mean-field one, when the differences σ_{rel} are: (**a**) and (**e**) 0 (identical), (**b**) and (**f**) 0.10×10^{-4}, (**c**) and (**g**) 2.24×10^{-4}, and (**d**) and (**h**) 7.07×10^{-4}. The size of the circles are proportional to PTS and the error bars represent the standard deviation

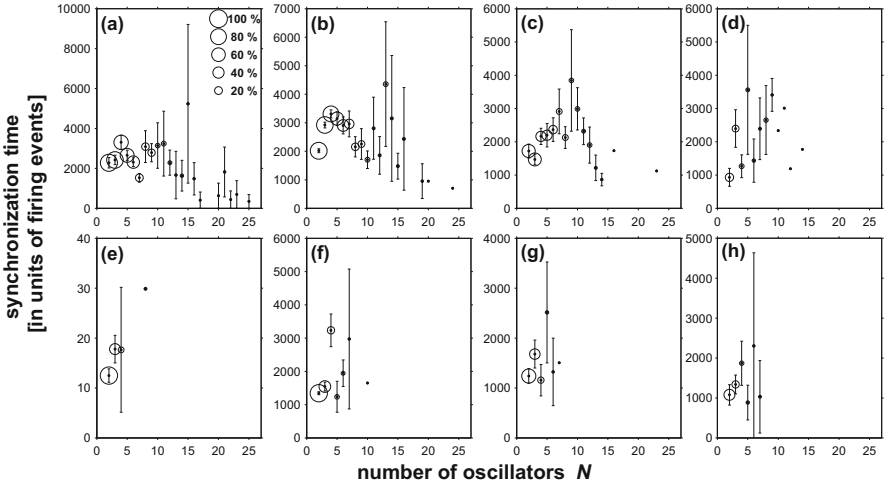


Fig. 8.7 Idem as Fig. 8.6 but for a distance-dependent coupling

Table 8.1 Comparison of different coupling configurations of LCOs and IFOs and the important changes concerning the $\langle PTS \rangle$, and synchronization time

Configurations comparison	$\langle PTS \rangle$	Sync. time
LCOs mean-field vs. LCOs distance-dependent	✓	✓
LCOs mean-field vs. IFOs mean-field	✓	✓
LCOs mean-field vs. IFOs distance-dependent	✓	✓
LCOs distance-dependent vs. IFOs mean-field	✓	✓
LCOs distance-dependent vs. IFOs distance-dependent		✓
IFOs mean-field vs. IFOs distance-dependent	✓	✓

The check mark points out the existence of an important change in the concerned variable

8.5 Beyond the Fireflies Courtship: The Response to Synchronization

Courtship is the primary cause for which fireflies synchronize. Indeed, this behavior is known from more than a century ago [46]. The mechanisms for some species, in particular for those belonging to *Photinus* genus were disemboweled, and their oscillatory features were described by Lloyd [40] and improved by Lewis and Cratsley [38].

Fireflies courtship in most species belonging to *Photinus* genus is hallmarked by the females' response to males' synchronization. A first attempt to explain the response to synchronization has been made using a modified LCOs model, in which males and females exhibit dissimilar features when they are interacting [58]. The model and the mechanisms of response to synchronization are described in [58, 61]. It is shown that not only the experimental results presented in [48] are reproduced, but it is also capable of predicting more complex and realistic situations; in particular, the behavior of other species of the genus *Photinus* as it will be described later.

As explained in [61], to model the response to synchronization, it is necessary to consider the dissimilarity between males and females in which concerns their oscillatory features. In Fig. 8.8 the terminology used in the description of a male and a female is explained. Several fireflies' species exhibit a bursting behavior for the male with n_f flashes per burst (Fig. 8.8a) followed by a quiescent or silent time T_s, a parameter that remains constant even when the oscillators are coupled. On the contrary, the female generally emits only one flash in its fast discharging process T_d which is preceded by a long-lasting charging process T_c and followed by a silent time T_s (Fig. 8.8b). We define the interburst period or the duration of a phrase T_p as the complete cycle comprising the active phase and the silent time. Consequently, the active phase takes $n_f(T_c + T_d) = Tp - Ts$. Males and females are individually considered as relaxation oscillators because they have commonly two different time scales, i.e., within each cycle, there is a slow process followed by a firing one. Each process ends at its threshold, being the lower and the upper thresholds at $V^{\text{lower}} =$

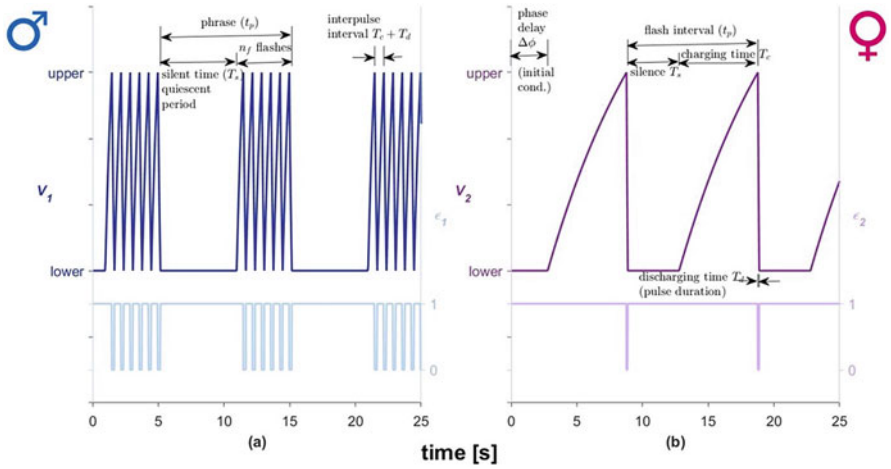


Fig. 8.8 Dynamic V and binary ε variables for the relaxation oscillators associated with (**a**) male and (**b**) female fireflies of (*Photinus carolinus*). They are characterized by the quiescent period (silent time) T_s, the active phase with n_f flashes per burst, the charging and the discharging times T_c and T_d, respectively, the interpulse interval $T_c + T_d$, the flash interval or duration of a phrase T_p, and the phase delay $\Delta\phi$ that plays the role of initial condition. In this particular case, the parameter values are, respectively, for the male (♂) and female (♀): $T_{p\male} = 10.000$ s, $n_{f\male} = 6$, $T_{c\male} = 0.500$ s, $T_{d\male} = 0.200$ s, $T_{s\male} = 5.800$ s and $\Delta\phi_{\male} = 0.603$ rad $\equiv 0.960$ s, $T_{p\female} = 10.000$ s, $n_{f\female} = 1$, $T_{c\female} = 6.000$ s, $T_{d\female} = 0.100$ s, $T_{s\female} = 3.900$ s and $\Delta\phi_{\female} = 1.750$ rad $\equiv 2.785$ s

$V_M/3 = 3$ and $V^{\text{upper}} = 2V_M/3 = 6$, respectively. We take these threshold values in connection with the experimental aspects related to the LCO, namely, the oscillator serving as the basis of the model stated in Eq. (8.2). It is also important to note that in Fig. 8.8a, the parameters values and $n_f = 6$ correspond, in biological terms, to the *Photinus carolinus* flashing pattern. It is clear that the model could be adapted to other species just in changing the relevant parameters.

The equations describing the dynamical variable V_i of each oscillator i are given by:

$$\frac{\mathrm{d}V_i(t)}{\mathrm{d}t} = \frac{\ln 2}{T_{ci}}\left(V_{Mi} - V_i(t)\right)\varepsilon_i(t) - \frac{\ln 2}{T_{di}}V_i(t)\left(1 - \varepsilon_i(t)\right), \tag{8.10a}$$

$$V_i(t) = \left(V_i(t) - V_i^{\text{lower}}\right)\varepsilon_i(t) + V_i^{\text{lower}}. \tag{8.10b}$$

As stated above, V_M is a constant that determines the lower and upper thresholds and $\varepsilon_i(t)$ is a binary variable describing the state of the ith oscillator by:

$$\begin{aligned} &\varepsilon_i(t) = 1 : \text{extinguished oscillator (charging and silent stage)} \\ &\varepsilon_i(t) = 0 : \text{fired oscillator} \qquad\quad \text{(discharging stage)}. \end{aligned}$$

The transition between the states determined by ε is described by the following relation:

$$\textbf{If} \quad V_i(t) = V_i^{\text{lower}} \quad \textbf{and} \quad \varepsilon_i(t) = 0 \quad \textbf{then} \quad \varepsilon_i(t_+) = 1; \tag{8.11a}$$

$$\textbf{If} \quad V_i(t) = V_i^{\text{upper}} \quad \textbf{and} \quad \varepsilon_i(t) = 1 \quad \textbf{then} \quad \varepsilon_i(t_+) = 0; \tag{8.11b}$$

$$\textbf{If} \quad V_i(t) = V_i^{\text{lower}} \quad \textbf{and} \quad \varepsilon_i(t) = 1 \quad \textbf{then} \quad \varepsilon_i(t_+) = 1, \tag{8.11c}$$

where t_+ in the condition given by Eq. (8.11c) is defined in the interval

$$t = [t_+ \;\; (k-1)(T_p + n_f(T_c + T_d)) + \Delta\phi] \,, \tag{8.12}$$

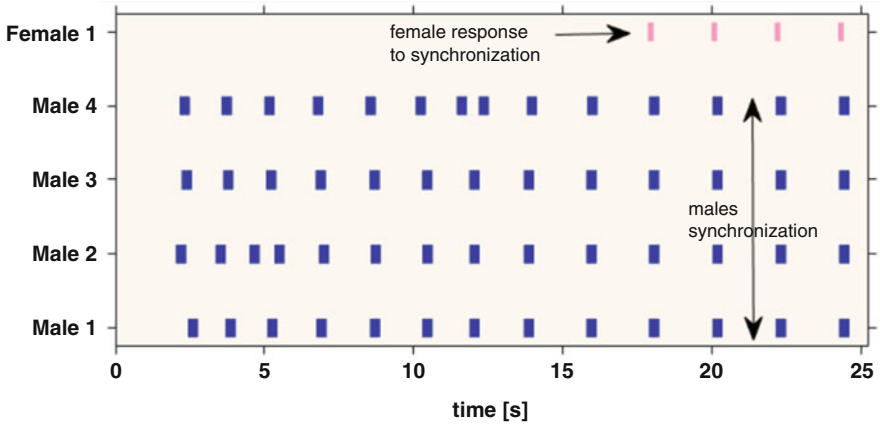
for every k flash interval or phrase, i.e., for every complete cycle comprising the active phase and the silent time. Eq. (8.12) states for the successive silent intervals.

The main feature of the considered oscillators dwells on its flashing process which allows a pulsatile coupling with other oscillators that can receive these pulses or spikes leading to a modification in their oscillatory dynamics. The dynamical equations describing a generic group of N coupled oscillators are:

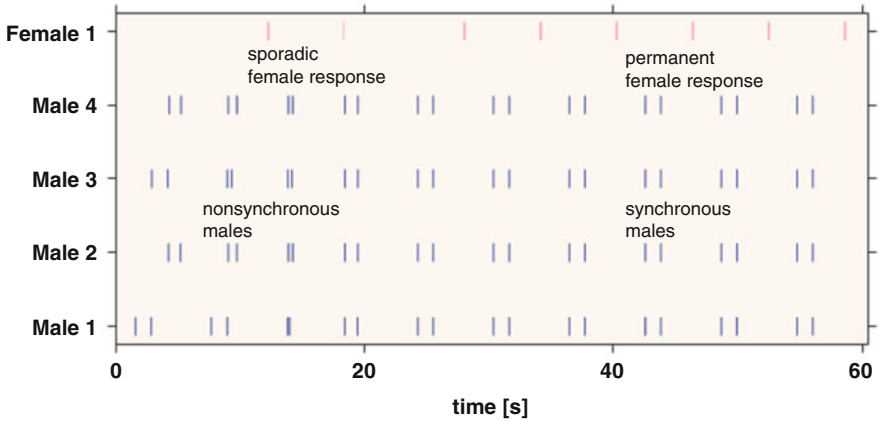
$$\frac{\mathrm{d}V_i(t)}{\mathrm{d}t} = \frac{\ln 2}{T_{c0i}}\left(V_{Mi} - V_i(t)\right)\varepsilon_i(t) - \frac{\ln 2}{T_{d0i}} V_i(t)\left(1 - \varepsilon_i(t)\right) + \theta_i \sum_{i,j=1}^{N} \beta_{ij}(1 - \varepsilon_j(t)), \tag{8.13}$$

where $i, j = 1, \ldots, N$. Conditions that are given by Eqs. (8.10b) and (8.11), which take into account the existence of a silent time, must also be followed by Eq. (8.13). The quantities T_{c0i} and T_{d0i} are the lasting time of the charge and the discharge, respectively, when there is no action on the oscillator i by other oscillators. Furthermore, we consider that oscillators are mutually coupled with a coupling strength β_{ij} that represents the pulsatile action of the oscillator j spike during its discharge upon the oscillator i. Concurrently, β_{ij} are the elements of the weighted adjacency matrix of the set. A simple inspection of Eq. (8.13) shows that both charging and discharging stages might be modified by the effect of coupling with other oscillator(s). The charging and the discharging times might be shortened or lengthened, respectively, when the pulsatile action due to the firing of other oscillator(s) takes place. The latter is determined by the value of θ that takes the value 1 for males and -1 for females. This factor is of particular importance because it determines the behavior of the oscillators when stimuli are applied to them.

Several studies have been carried out in [58] considering *Photinus carolinus* flash patterns. Here, we show in Fig. 8.9 the interaction between a set of four males and one female that according to their flashing patterns correspond to *Photinus collustrans* (Fig. 8.9a) and *Photinus greeni* (Fig. 8.9b). In the case of *Photinus collustrans*, the typical behavior of female response to synchronized males occurs at around 18 s. When considering *Photinus greeni*, it first appears a sporadic female



(a)



(b)

Fig. 8.9 Flash sequences for males (blue) and females (pink) of two different *Photinus* species: (**a**) *collustrans* ($n_{f♂} = n_{f♀} = 1$), and (**b**) *greeni* ($n_{f♂} = 2$, $n_{f♀} = 1$). In both cases, the female response to males synchronization is pointed out. We used the fireflies parameters shown in Table 8.2, coupling strength for each case $\beta_{ij} = 2.5$ and 15, and initial conditions (times for which the cycle of each oscillator starts): [1.91111, 0.1858, 0.8795, 0.5717, 8.0807] s, and [0.3848, 3.5573, 2.0924, 5.1107, 4.1023] s respectively

response when the males are not completely synchronized and eventually, at around 28 s and when the males are in synchrony, the female response becomes persistent. To obtain the above-mentioned figures, we use the oscillatory fireflies features pointed out in Table 8.2.

Table 8.2 Main oscillatory features of different *Photinus* species

Species	Gender	Flashes	Flashing (s)	Interflash (s)	Quiescence (s)
P. collustrans	♂	1	0.343	2.200	1.843
	♀	1	0.443		
P. ignitus	♂	1	0.243	4.986	4.743
	♀	1	0.314		
P. marginellus	♂	1	0.385	2.871	2.486
	♀	1	0.114		
P. pyralis	♂	1	0.729	5.657	4.928
	♀	1	0.214		
P. sabulosus	♂	1	0.129	3.786	3.657
	♀	1	0.057		
P. umbratus	♂	1	0.557	6.786	6.229
	♀	1	0.343		
P. consanguineus	♂	2	0.171	0.500	4.643
	♀	1	0.243		
P. greeni	♂	2	0.100	1.271	3.571
	♀	1	0.157		
P. consimilis	♂	8	0.086	0.271	>4.914
	♀	2	0.129	0.286	

8.6 Nice Epilogue Playing with Fireflies: The "Solitary Flash" Game

With the aim of explaining fireflies flashing behavior, a very simple algorithm has been proposed by Stewart and Strogatz [73]. Based on simple rules, it permits to capture main features of entrainment and synchronous behavior in fireflies. Originally, the rules are formulated as:

1. The game board consists of a polygon of n sides, each of them containing r boxes, i.e., $N = n \times r$ boxes on the board.
2. The first box plays the role of the flashing box, i.e., when a player (firefly) arrives at this box, it flashes.
3. Each firefly starts the game in any box (initial condition) excepting the flashing one.
4. Each firefly advances clockwise one position per time step.
5. When a firefly flashes, it remains in the flashing box one-time step, while the other fireflies go forward according to the place in the board in which they are. For instance, if the firefly is on a box of the first side, it continues to advance one position; on the other hand, if the firefly is on a box of the second side, it advances two spaces, and three if it is on a box of the third side and so on.
6. The goal of the game is that all fireflies flash synchronously and simultaneously in the shortest possible time.

The rules mentioned above might allow or not the occurrence of complete synchronization. The dynamics of the game strongly depends on the initial conditions and also on rule 5 because it determines what happens to a firefly when it approaches, arrives, or passes through the flashing box. We consider the following four complementary variants of rule 5:

(a) When a firefly is in a box located on the far side of the polygon, it could happen that it might overtake the flashing box and consequently without flashing in its cycle. This fact imposes a difficulty in attaining complete synchronization.
(b) When a firefly restarts a new cycle, it is mandatory that it arrives at the flashing box and as a result, it flashes in each cycle. This rule facilitates the achievement of complete synchronization.
(c) When two or more fireflies are nearby the flashing box, they wait until all of them are indeed in this box; at this time, all the fireflies advance one position.
(d) Finally, if we consider a similar situation as in (c) but with the modification that when the fireflies are forced to be in the flashing box, the other fireflies advance according to rule 4 and not only one position as in the precedent case.

Some frames of the game evolution reflecting the rules (a)–(d), and some other possibilities for the cardboard are shown in Fig. 8.10.

Time series for all cases (a)–(d) are presented in Fig. 8.11, where we considered four fireflies and a specified set of initial conditions $(n_{01}, n_{02}, n_{03}, n_{04}) = (4, 6, 11, 19)$. Examples of basins of attraction are depicted in Fig. 8.12 where the condition is related to the lasted time to achieve complete synchronization with collective simultaneous flashing. It is interesting to observe in Fig. 8.11a that

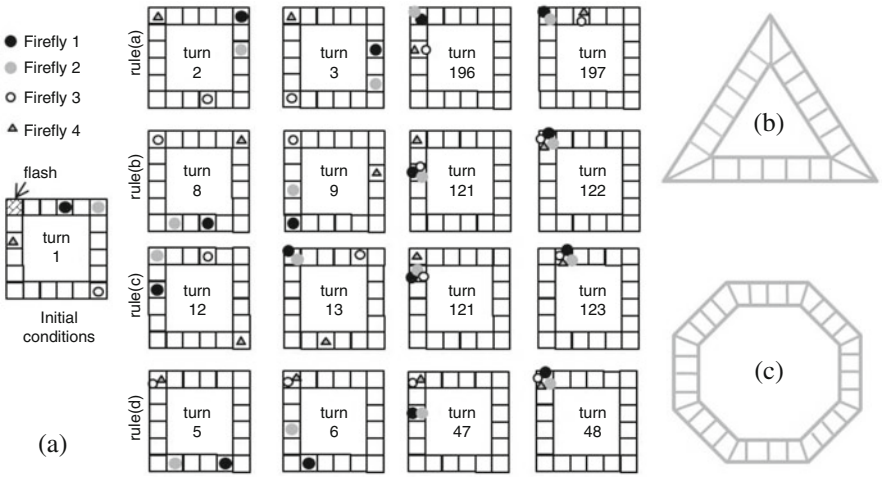


Fig. 8.10 (**a**) Frames of the game evolution for four players (fireflies) and the rules (a)–(d) considering for all the four cases the same initial conditions $(n_{01}, n_{02}, n_{03}, n_{04}) = (4, 6, 11, 19)$. Some other possibilities for the cardboard: (**b**) triangle, and (**c**) octagon

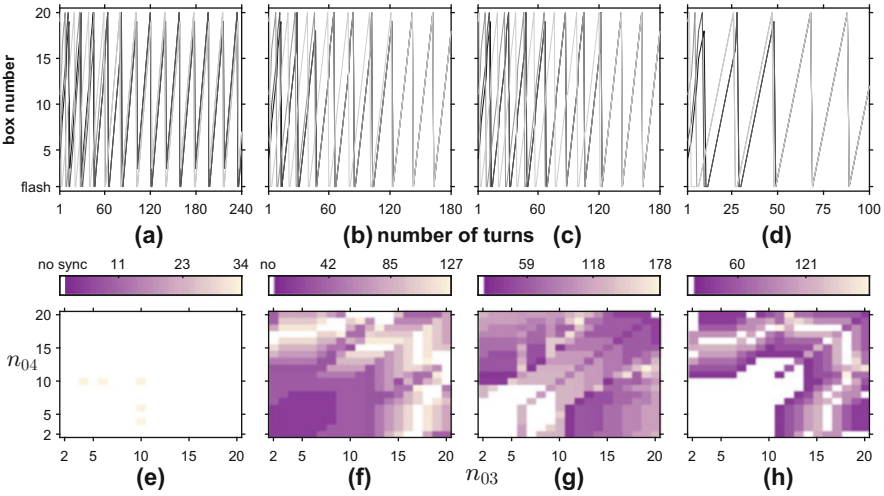


Fig. 8.11 First row: (**a**)–(**d**) Time series for the evolution of four fireflies following the rules (a)–(d) above mentioned. Second row: (**e**)–(**h**) Basins of attraction for the rules (a)–(d) considering that the initial condition for the first and second fireflies are $(n_{01}, n_{02}) = (4, 6)$. White boxes represent situations in which simultaneous, collective, and persistent flashing (in every cycle) are not achieved, i.e., there is not complete synchronization

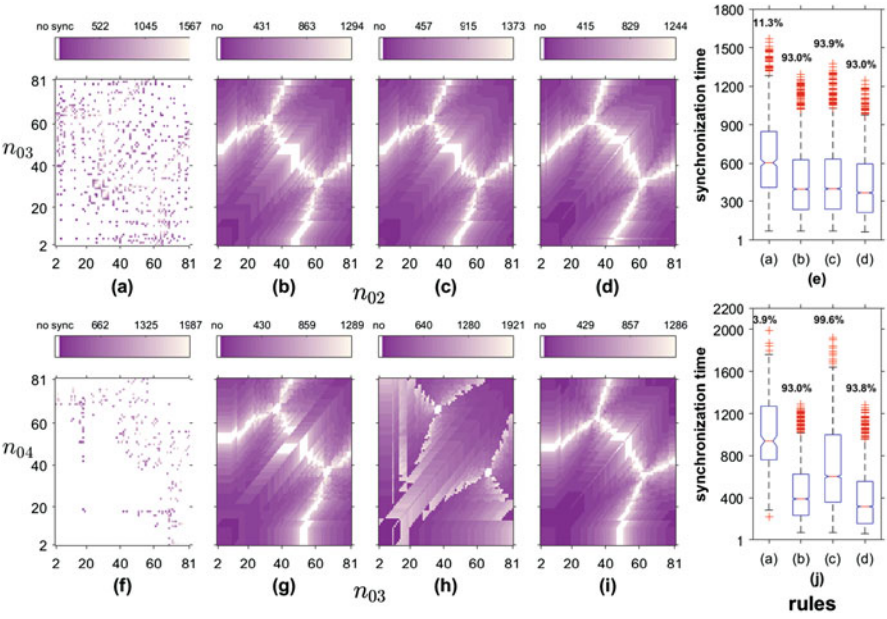


Fig. 8.12 Basins of attraction for rules (a)–(d) for (first row) three players with initial condition $(n_{01} = 5)$, and (second row) four players (fireflies) with a set of initial conditions $(n_{01}, n_{02}) = (5, 10)$, when playing with a nonagon shape board (9 sides), with 9 boxes per side. (**e**) and (**j**), box plots (median and quartiles) of the synchronization times related to the basins (**a**)–(**d**) and (**f**)–(**i**). The percentage of the synchronous events is shown in the upper part, above the whisker

the fireflies do not reach the complete synchronization condition of simultaneous flashing of all the individuals of the system. Nevertheless, considering the formal definition of synchronization, the phase difference among the fireflies remains constant from the 102nd turn; as a consequence, exhibiting synchronization but fireflies 1 and 2 do not flash in every cycle. On the contrary, Fig. 8.11b–d shows that simultaneous flashing is achieved, respectively, in 122, 142, and 48 turns (time steps). So that, in principle it indicates that in successive order the rules that facilitate synchronization are, respectively, (d), (b), (c), and (a). With a view to sustaining the recent affirmation, we obtain some other basins of attraction for three and four fireflies. Indeed, we now consider a board in nonagon form, with 9 boxes per side, and initial conditions $(n_{01}, n_{02}) = (5, 10)$; the basins of attraction for each rule are shown in Fig. 8.12a–d when the number of players is three, and in Fig. 8.12f–i when the number of players is four. The corresponding box plots displaying the medians, and the quartiles related to synchronization time as well as the percentage of complete synchronization events for each rule are shown in Fig. 8.12e and j for three and four fireflies, respectively. A visual and qualitative insight of the results set out that, as expected, rule (a) is the less favorable to attain complete synchronization. On the other hand, for three players, the results of rules (b), (c), and (d) seem to be very similar, but rule (d) is slightly more favorable to complete synchronization. Similarly, the visual inspection for four players shows us that for the rule (c), in almost all cases, complete synchronization is achieved (99.6%) but in contrast, the synchronization times are considerably longer than for rules (b) and (d).

A deeper statistical analysis corroborates in a certain way our previous assertions. Thus, for three players, the statistical analysis of both situations indicates that the median time of synchronization is the greatest for rule (a) and the shortest for rule (d). As the times of synchronization do not follow a normal distribution, a Kruskal-Wallis test was performed to look for differences in the distribution of the four rules ($\chi^2 = 436.7$, $df = 3$, $p < 2.2 \times 10^{-16}$). A Dunn's test for multiple comparisons showed that all pairwise comparisons between rules (a)–(d) are significantly different ($p < 0.001$), except between rule (b) and rule (c) ($p = 1.91 \times 10^{-1}$). The percentages of initial conditions for rules (a)–(d) leading to complete synchronization were, respectively, 11.3%, 93.0%, 93.8%, and 93.0%. Thus, we can conclude that rule (d) is characterized by a very high number of initial conditions driving to complete synchronization, with the shortest times.

For the case of four players, the median time is also the greatest for rule (a) and the shortest for rule (d) (defining t_s as the synchronization time for achieving complete synchronization, $t_s(\text{a}) > t_s(\text{c}) > t_s(\text{b}) > t_s(\text{d})$). As the Kruskal-Wallis test was significant ($\chi^2 = 2361.7, df = 3, p < 2.2 \times 10^{-16}$), a Dunn's test of multiple comparisons was performed. All the pairwise comparisons were highly significant ($p < 0.001$). The percentages of initial conditions for rules (a)–(d) leading to complete synchronization were respectively 3.9%, 93.0%, 99.6%, and 93.8%. As in the case of three players, rule (d) is characterized by a high number of initial conditions conducting to complete synchronization with the significantly shortest times.

This simple game with its four basic rules (algorithms) allows to explain complete synchronization in some species of fireflies, and it has an intrinsic richness, since these simple rules can be translated to a more technical language belonging to synchronization theory. Thus, it is possible to find a relation between the number of sides and boxes per side with the type of coupling: in the case explained above, we have an excitatory coupling between the fireflies (players). In some cases, an inhibitory coupling is also possible, as it is described in Sect. 8.5, for light-controlled oscillators (LCOs) and some species of fireflies. This type of coupling might be incorporated in the game considering that for some boxes on the game board, the player must go back a certain number of boxes according to its position. The high number of game choices (shape of the polygon, number of squares for each side of the polygon, the number of players, basic rules of the game, etc.) allows to extend the study of this model and analyze its isochronous dynamics as in [39].

8.7 What Is Missing in Better Understanding the Fireflies Synchronization and Which Are the Potentials of This Phenomenon?

Synchronization constitutes one of the central phenomena in nature, and a lot of advances have been gotten to understand how and why many systems exhibit synchronous behavior in different contexts. The case of fireflies synchronization is paradigmatic not only for the visual beauty that it represents but principally for the functionality that it typifies. Many efforts have been devoted to qualitatively describe synchrony in fireflies and also to decipher the "amorous language" that is involved with it. Mathematical models have been proposed to explain this behavior and thanks to physical-mathematical refinements, experimental work with mimicking systems, and computer simulations; nowadays it is possible to capture the main features of fireflies synchronization including the response of the female brought out by several species. Even though the models are based on rather simple ideas such as the consideration of each firefly as an oscillator and neglecting environmental influences (the weather conditions, the presence of external lights, the action of other species, the visual field of individuals, etc.), these models give the account of the main features and also have predicting power. The latter contributed to the development of applications of this phenomenon in different areas such as in the improvement of reliability and efficiency of mobile computer networks [8], in the solution of engineering problems [29], and in swarms of robots as a first approach for understanding real task-execution scenarios [22] among others. Some new impact applications are also remarkable, such as those leading to improve the efficiency of LEDs [5], or those conducting to a new method of medical diagnostics based on the way in which fireflies produce cold light [70].

Nevertheless, there are still some challenges concerning the fireflies synchronization. From a biological viewpoint, it is necessary to study more fireflies species

and whether or not they exhibit synchronous behavior. A more detailed repertory of synchronous features could also be helpful. The latter might contribute to improve the models and consequently to envisage new concepts and applications. Though some works were devoted to the study of synchronization of mobile oscillators [30, 56], it is necessary to focus the studies on fireflies through a weighted mobile network approach for populations of coupled fireflies which seem to be adequate because of the individuals' mobility, limitations on the fireflies visual field, existence of obstacles, and coupling strength dependence on the distance.

Response to synchronization deserves deeper studies and improved models allowing to characterize the females' response patterns adequately and also consider control aspects. Finally, the idea of working with mingled (both genders: males and females) and mixed (natural and artificial fireflies) groups is also a great challenge because it involves insects-robots interaction which could have profound implications at the level of primary synchronous behavior.

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Chapter 9
From Flocs to Flocks



Shannon Dee Algar, Thomas Stemler, and Michael Small

9.1 Introduction

Collective motion is a sub-class of the more general phenomena of collective behaviour. Some of the most beautiful and well-known examples of this motion are displayed by flocks of birds Fig. 9.1. A flock is, by definition, the collective noun for a group of birds—irrespective of what that group is doing. Mathematicians think slightly differently about this, typically considering a group of birds to be a flock only when it takes flight and is moving in concert, i.e. it is the action that makes it a flock, not the taxa of the individuals. In the mathematical literature we see this generalisation—bacteria can flock [26, 85], man-made colloids or drones can flock [9, 32] and non-descript dots in a numeric simulation can flock [19, 74]. While any collective noun could have been adopted, 'flock' has persisted, in part, because many of the original investigations into group dynamics were done with birds in mind [21, 62]. Flock is also a verb that invokes the right images of a group of individuals moving as if with one mind.[1] Collective motion does not, however, require a collective mind. Explanations, of how unified action occurs, that involved telepathy, military-like instruction and spread of impulse are theories of the past [18, 50]. We now understand that the sum is truly more than its parts [30]. Note that

[1]Collective nouns that perhaps don't have the same ring to them: a gaggle (of geese), murder (crows), piteousness (doves), flamboyance (flamingoes), pandemonium (parrots), murmuration (starlings), and an unkindness (raven).

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Fig. 9.1 A flock of birds

when motion is disordered we borrow the collective noun of insects and refer to the group as a swarm [72]. This is despite social insects such as ants and honey bees displaying some of the most impressive coherence and coordination. A more appropriate image is the haphazard swarms of midges whose motion is not ordered but is correlated [3].

A floc (pronounced 'flock') is an aggregate of particles that results from a process of contact and adhesion in Chemistry called flocculation. It is not a term usually connected with flocks of birds, despite likely originating from the same word, the Latin 'floccus'—a tuft, wisp of wool, cloud species [45].[2]

Flocks and flocs are complex systems, characterised by nonlinear dynamics, decentralisation and difficult to predict behaviour. Nonlinearity is known for irregularities, chaos and instabilities, but in nature tends to manifest as stability, coherence and order. These are emergent phenomena, which cannot be understood by understanding each of the components in isolation. The formation of a group and collective motion emerge from movement and decisions that likely follow a single thread of motivation on the individual's behalf. Yet, mathematical models of these processes are often treated as distinct stages: aggregation ('flockulation')—the necessary first stage of; flocking—coordinated motion of the aggregate.

Aggregation and flocking occur across many time and length scales [44, 79]—microorganelles coalesce into biological building blocks [61], which then grow, spread and self-assemble into organs that form people, who in turn crowd and travel as part of wider societies [35]. The motion of animals such as mammals, birds, fish and insects has captured the attention of scientists because the individuals and the group can be observed simultaneously [2, 20]. The ubiquity of these behaviours across species suggests that the underlying mechanisms have more to do with the interactions of the components than with the complexity of the components [11].

[2]It may now make more sense why sheep and birds (as tufts and clouds) have earned the name flock but admittedly, this is pure speculation on our part.

This has encouraged the search of general principles and spurred joint efforts across disciplines, particularly, Mathematics, Physics, Biology, Sociology and Engineering [60, 75]. The highly interdisciplinary nature of collective motion studies brings with it a range of motivations, viewpoints and tools. This chapter focusses on the approach taken by Mathematicians and Physicists, who see the birds as particles and the flock as a complex system.

Many other excellent reviews for both grouping [29, 41, 77] and flocking [10, 27, 46, 69, 75] exist. In this chapter, we aim to amalgamate the approaches taken for modelling these two aspects of group behaviour in animals with an emphasis on local rules and interactions producing emergent behaviour at the group level. We outline standard methodologies of modelling and then review some of the classic models that have helped to shape the respective fields. With reference to a new model we highlight and discuss some recommendations for future works to bridge the gap between flocs and flocks.

9.1.1 Mathematical Models

Mathematical modelling is the process of describing, characterising or predicting some aspect of the real world using the concise language of Mathematics. Models can be classified according to the method used to obtain them:

- Mechanistic models use a minimal set of assumptions about the scientific principles underlying the phenomena of interest. Assumptions are typically made using thought experiments and plausible scientific reasoning that is aligned with qualitative observations of how the system behaves. These models focus on a few characteristic features to ensure the model is mathematically tractable.
- Empirical models are based on examination of numerical data and thus are context specific. They are commonly used for analysing or describing trends, making forecasts or predictions. This requires highly realistic models with many parameters that must be estimated, increasing the model complexity and often undermining its usefulness as a mathematical tool. On the other hand, maximum entropy principles can be used to obtain models with the minimal structure consistent with observations [6, 7, 14].

Minimal models are an apparatus for detailed hypothesis testing, statistical analysis and deeper insight [19, 74]. By manipulating the inputs such as rules, parameters and initial conditions we aim to understand how these changes impact on the dynamics and resulting state of the system. Models can transcend theoretical limitations and shed new light on systems that are difficult to analyse in the laboratory or in their natural environment [34]. For example, evolutionary algorithms allow us to observe how behaviours develop over generations that would otherwise not be possible to observe due to the time scale on which they occur. With evolutionary game theory we can probe and test biological theories with rigour and see how different strategies play out [58].

The mathematical modeller's goal is not believability of the model's final behaviour. They seek an explanation of observations from their hypothetical experiments in the form of a qualitative prediction, understanding the effect of parameter values or testing broad theoretical questions [33]. With a different set of goals and tools, the biological community generally aims to be more faithful to the system they are trying to model by including additional detail, such as species specific traits, anisotropic vision, wingspan and reaction time [19, 36]. It would be naive to assume that any conclusions drawn from the model are correct in the real world. At best, the model can be valid, with plausible underlying rules, and then conclusions may be useful [42]. Conversely, convincing behaviour does not necessarily imply that the model holds a correct representation of the real world and we must be wary of realistic results that stem from unrealistic movement descriptions [41].

Because flocks are complex systems, modelling them can be challenging and many approaches have been developed. Models vary in their methodologies depending on the level of cause and effect that they aim to explore. In general, to model the underlying mechanism of movement we require: a description of the behaviour/dynamics and; a definition of the interactions—both of which may be defined in terms of either the local or the global environment. Similarly, the resulting dynamics may be analysed at the level of the individual or the group [70]. We summarise these different approaches with some examples in the classification (Table 9.1).

The analogies made with Physics in Table 9.1 are not exact for a number of reasons.

Table 9.1 Model classification scheme and examples

Cause	Effect	Physics analogy	Models
Local	Local	Classical mechanics	Typically data driven models. Small systems helping to inform and extrapolate to larger systems [5, 23, 40]
Local	Global	Kinetic theory/subset of statistical mechanics	Mainly theory driven models aimed at investigating emergence and self-organisation [19, 74]
Global	Global	Statistical physics	Rely on sufficiently large systems and universal properties and symmetries. Can complement local-global models by providing rigorous analysis [71]
Global	Local	Inverse problem	Increasingly important and fruitful with improved data collection. Animal borne tags/GPS [56, 83, 84] and computer vision [15, 16] are assisting us to infer the nature of the movement and interactions

Firstly, biological systems are far from equilibrium [7, 64, 65]. In an ideal gas, the forces are derived from the Physics, take for example the momentum transfer following a collision. There are well-established tools in Statistical Physics for analysis of equilibrium systems, however, in non-equilibrium systems such as a flock, autonomous agents have the ability to draw on energy reserves to effect movement. For the relatively short duration of flocking events an unlimited energy source can be assumed.

Secondly, the n-body problem with $n > 2$ has no closed form solution and requires numerical methods to determine trajectories at future times [49]. For small particle numbers, the micro-dynamics of each particle can be followed exactly but this becomes increasingly difficult for very large systems. In the limit, the system can be modelled by a continuum derived from a Fokker-Planck equation [63, 82] (or other) based on symmetry properties and hydrodynamic theory can then be used. This is an Eulerian (equivalently, equation based/population/continuum) description of the system. An Eulerian model is global, viewing the system as a whole with macroscopic dynamics, such as density, derived from the laws of Physics and relationships defined between observables at level of group. It is a model of the flock—but the flock must be very large. This is appropriate for systems such as bacteria but is harder to justify for larger species where the individual is the natural unit of division.

The sort of flocks typical of collective motion studies operate on a mesoscopic scale in the range of 10–1000s. A middle ground between microscopic and macroscopic descriptions is provided by kinetic theory, which uses a suitable probability distribution over the microscopic states to describe the overall state of the system. In order to capture the individual decisions and contribution a Lagrangian (equivalently agent or individual based) description is required. This bottom-up view presents a different modelling paradigm—and a very different way of looking at the world [73].[3]

Lagrangian models explicitly represent all individuals as autonomous and interdependent. This allows us to test the underlying local behavioural rules and study the consequences of self-organisation on emergent group properties such as compactness and polarisation. Other advantages include modularity, flexibility and ability to combine with other modelling techniques. Whilst previously considered impractical for very large systems, today's computing power is increasing the reach of Lagrangian models, particularly as they are well suited to parallelisation. Simulations now provide near instantaneous visual feedback, allowing for easy qualitative comparison.

Relative to other mathematical frameworks, agent based models are also fairly intuitive and thus are easily transferred to other (potentially less mathematically minded) disciplines [34]. The most rudimentary and well-known form of these models capable of emergent global behaviour are cellular automata. Agents reside

[3] A note on language: aggregation studies tend to refer to Eulerian and Lagrangian models, whilst flocking studies use continuum and agent/individual-based models.

on a grid where their state depends on that of their neighbours. Cellular automata form the basis of a number of famous results such as Schelling's model of segregation where agents' movement is dependent on small bias of neighbour colour [67] or Conway's Game of Life where the agents survival depends on the density of its neighbourhood.

9.1.2 Standard Approach

For the remainder of this chapter we focus on locally defined systems with analysis at the group level. An understanding of the governing local rules and interactions can be garnered from data at a local or global level. For example:

- locally: sensory deprived nymphs showed that stimuli from behind was the key driver of motion leading to a model of cannibalism [5, 66]; using a Newtonian force-based approach individual trajectories can be reconstructed from data collected from animal borne tags and global positioning systems (GPS) [40, 56, 83, 84].
- globally: the topological interaction network of Starlings was derived from the flock density [4]; and neighbour distributions of Surf Scoters were used to infer potentials governing individual behaviour [47].

More commonly, a number of biological ansatzes are made depending on the system being modelled.

9.1.2.1 Behavioural Rules

The dynamics of each individual can be defined by forces (either physical or artificial in nature) or dictated by a hierarchical decision making process.

Under a Newtonian framework, the position $\mathbf{x}_i$ of each of the $i = 1, 2, \ldots N$ particles evolves according to the following stochastic differential equation:

$$m\frac{\mathrm{d}^2\mathbf{x}_i}{\mathrm{d}t^2} + \mu\frac{\mathrm{d}\mathbf{x}_i}{\mathrm{d}t} = \mathbf{F}_{social} + \mathbf{F}_{external} + \mathbf{F}_{stochastic} \tag{9.1}$$

with mass, m typically set to 1 for all particles. Further simplifications reduce the dynamics to first order by assuming the mass is negligible when compared with the rescaled friction ($\mu = 1$). Friction can be either external as a drag or replaced by a relaxation function towards a desired speed [64].

External forces $\mathbf{F}_{external}$ are those that a non-active particle would equally be subjected to such as any forces associated with the medium they travel through that give rise to various sensory taxis or those of standard physical conservation laws.

Internal forces $\mathbf{F}_{internal}$ are those that a particle chooses to adhere to based on the circumstances they perceive around them. These are often referred to as effective

social forces as they tend to be a function of the state of other particles but could also result from some other complex decision making process. Social forces are defined based on common practice, empirical observation, experimental measurement or speculation. A particle may be subject to many social forces at a given time and if so are linearly additive.

A stochastic term, $\mathbf{F}_{stochastic}$ can also be introduced. The noise term is explicitly dependent on time but could also be a function the state of particle i or its neighbours. The Langevin description of the dynamics captures perturbations and random fluctuations. The origin of this randomness could lie in the particle's interpretation of the surroundings (extrinsic) or in the implementation of the movement (intrinsic).

Alternatively, particles move immediately in the direction established by a decision hierarchy (Fig. 9.2) (with stochasticity included extrinsically or intrinsically as outlined above).

Decision making algorithms have the benefit of being computationally simple and thus reasonable to assume the animal could implement it, whilst accounting for the fact that motivating factors such as danger and social interactions are influential on different scales.

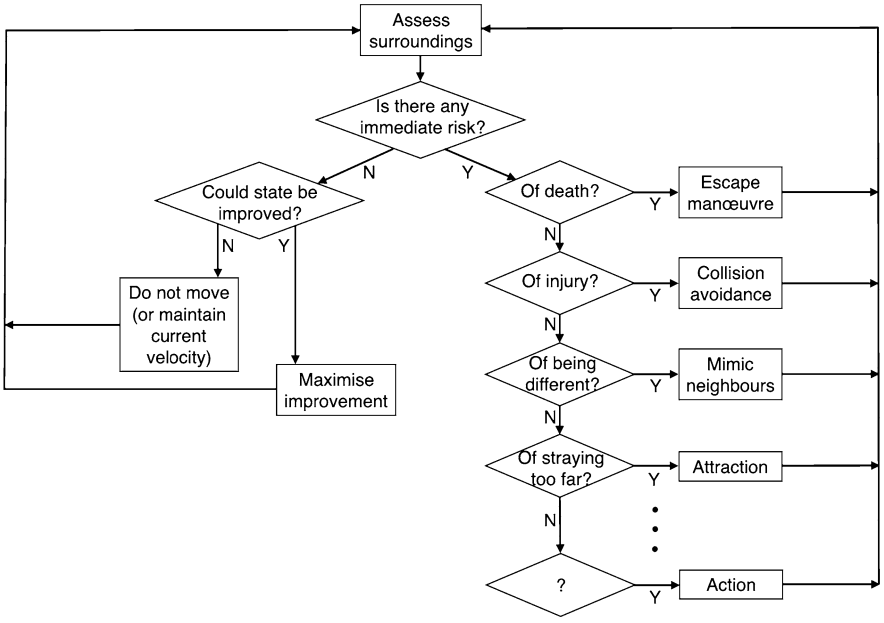


Fig. 9.2 Decision hierarchy schematic. Additional behavioural rules can be easily incorporated on top of existing structures

9.1.2.2 Interaction Network

There is no point in social forces if there is no socialising. Communication may occur as an exchange of energy or information and could be direct (active) via contact or indirect (passive) via social forces. The communication topology is rarely static as the state of each individual evolves in time. Reducing the birds in a flock to an abstraction of nodes and links generates a temporal network that can be analysed, modelled and interpreted with the tools of network science and graph theory [8].

When there exists a channel for communication between two birds they are considered to be neighbours and are linked by an edge in the network, which could be uni or bidirectional and weighted by the influence (typically assumed to be a continuous function of the distance between nodes and set to unity if all neighbours are considered equal). Mathematically, this coupling is represented by adjacency matrix A where the element A_{ij} holds the numeric weighting if the edge exists between i and j and 0 otherwise.

Neighbourhoods are usually defined using a short-range spatial metric such as the prototypical metric, topological neighbourhoods and Delaunay triangulation networks, see Fig. 9.3. Some of the other networks investigated include: infinite range or all-to-all coupling; social ties.

9.1.2.3 Simulation

With the system fully defined we want to know how it behaves. Analysis can be difficult with Lagrangian models due to their high dimensionality. The differential equations in Eq. (9.1) can rarely be solved exactly but lend themselves nicely to well-established methods of brute force numerical integration. The simplest and still the most successful approximation uses the Euler method or similarly the Euler-Maruyama method if the model includes stochastic terms. Each particles' position and velocity is updated according to these first order forward difference equations:

$$\mathbf{v}_i(t + \Delta t) = \mathbf{v}_i(t) + \mathbf{F}_i(t)\Delta t \tag{9.2}$$

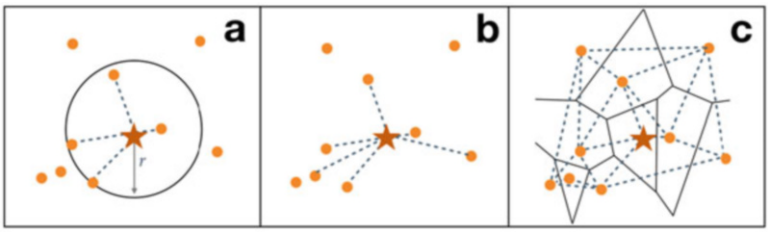


Fig. 9.3 Various common interaction networks with focal individual shown by the star. (**a**) Metric with radius r. (**b**) Topological k-NN with $k = 6$. (**c**) Delaunay triangulation, solid black line and dual Voronoi polygon tiling, dashed blue line

$$\mathbf{x}_i(t + \Delta t) = \mathbf{x}_i(t) + \mathbf{v}_i(t)\Delta t. \tag{9.3}$$

These equations are explicit first order forward difference equations because unknown future values are written only in terms of known previous or current terms (whereas backward Euler methods are implicit as they are functions of future terms and thus are much less common). For stochastic models Eq. (9.3) or (9.2) will have an additional stochastic forcing, appearing in the forward equations as a Wiener increment. Note that in the limit of large damping and no inertia we make a first order reduction with $\mathbf{v}_i = \mathbf{F}_i$ such that only Eq. (9.3) is required.

To model the system we must first set up the environment under which it will operate, including any constraints such as the boundaries and dimensionality. Indeed, many of the major differences in collective motion across species can be understood from biological constraints that should be reflected in the model. For example, ants are restricted to move on a surface, whereas birds and fish are free to move in three-dimensional space. Model and system parameters such as the time increment (frequency of decision making) and number of particles are defined and the position and velocity of each individual are initialised randomly. If forces are insufficient to ensure the group does not disperse, then periodic boundary conditions can be used to restrict particles to the surface of a torus (in two dimensions with higher dimensional generalisations) and fix the density. The particles are then released to interact and move as per the model.

Visualising the particles as dots, with small lines showing movement from the previously held position, we can observe their trajectories in time. Colour can be used to better illuminate a parameter of interest such as the individual area occupied (Fig. 9.7) or degree of polarisation (Fig. 9.4).

Depending on the model and parameters used, a spectrum of behaviours may emerge from essentially non-interacting random walkers to disordered and seemingly random swarming and the eventual onset of grouping or dynamic and rigid order.

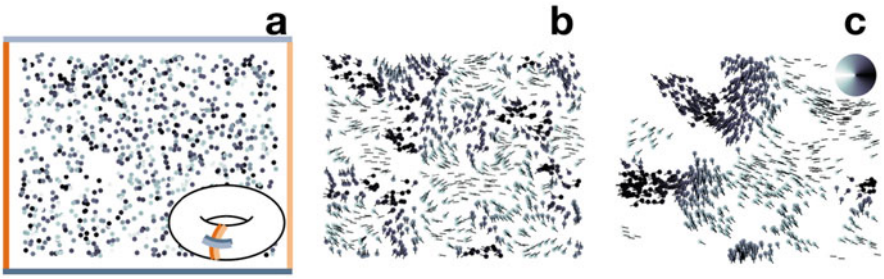


Fig. 9.4 Flocking evolution at three instances of time t with $0 < t_1 < t_2$. Particle colour indicates directional heading as per colour wheel. **a** $t = 0$. Inset: depicts periodic boundaries where colours show corresponding boundaries from the two-dimensional plane. **b** $t = t_1$. **c** $t = t_2$

9.1.2.4 Analysis

With the model now fully defined we aim to explore how it behaves under a variety of conditions, characterising the behaviour and observing its spatio-temporal evolution over time. To do this, we ask questions of the data that depend on the goals of the model and the audience its results are intended for. Aggregation models elicit questions such as: how do individual rules cause aggregations to form and persist; what is the distribution of particles/clusters; and what determines the shape and geometry of the group? On the other hand, flocking models are concerned with: the emergence of group behaviour such as polarised migration and milling; what causes transitions between behavioural regimes; and the nature of these transitions.

Through qualitative analysis an initial understanding of the behaviour and emergent group properties can be gained, which allows us to generate hypotheses regarding the form of the group, how information propagates and how the collective responds to perturbations. Scanning the parameter space informs of influential parameters and conditions required for shifts between behavioural regimes.

Quantitative analysis seeks a generalisable and statistically reliable description of observations via appropriate metrics at the group level or averaged over the individuals.

Aggregation models naturally consider the amount of aggregation, with metrics such as: convex hull; particle concentration; number of components; and network structure as natural choices. Deeper analysis may involve determining the: fractal dimension [48]; size distribution of clusters; or time to coalesce [59].

Flocking models lean heavily on methods and measures successfully used in Statistical Physics for analysis.

One of the key features of collective motion is abrupt or spontaneous transitions to (and between) these states (phases) when symmetries are broken. To quantify this an order parameter Φ is defined, typically ranging from 0 to 1. Swarming and random motion are characterised by a lack of order with $\Phi \approx 0$. Conversely, flocks display coherence—for example in the commonly observed polarised migration and rotating mills.

A natural choice for the degree of polarisation, Φ_P, is the magnitude of the normalised globally averaged velocity vector. This is effectively the motion of the flock's centre of mass:

$$\Phi_P = \frac{1}{N}\left|\Sigma_{i=1}^{N}\frac{\mathbf{v}_i}{|\mathbf{v}_i|}\right|. \tag{9.4}$$

The degree of rotation, Φ_R, about the group's centre can be measured by the angular momentum:

$$\Phi_R = \frac{1}{N}\left|\Sigma_{i=1}^{N}\frac{\mathbf{r}_{iC}\times\mathbf{v}_i}{|\mathbf{r}_{iC}\times\mathbf{v}_i|}\right| \tag{9.5}$$

where $\mathbf{r}_{iC} = \mathbf{r}_i - \frac{1}{N}\Sigma_{i=1}^{N}\mathbf{r}_i$.

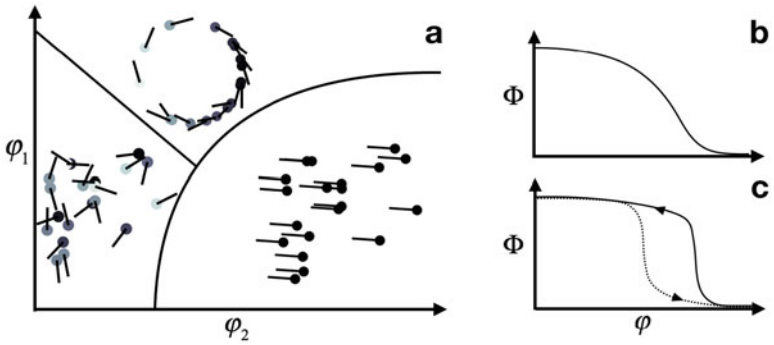


Fig. 9.5 (**a**) Phase diagram schematic. (**b**) Second order transition. (**c**) First order transition with hysteresis

Varying the relevant parameter(s), φ, a phase transition can be observed when a critical value is reached. By assessing the behaviour of Φ at criticality we can determine the nature of the transition, i.e. whether it is continuous (second order) or discontinuous (first order). Exhaustive sweeps of parameter space allow phase boundaries to be drawn. See Fig. 9.5 for a schematic example.

Phase transitions are basic feature of complex systems and have become an important area of study in collective motion as it is suggested many biological systems poise themselves at criticality to allow for near spontaneous response [53].

9.2 Floccing: Stage One

Before collective motion can occur a collective must exist. The dynamical process of many particles coming together is aggregation and the group is formed once the inter-individual distance converges to a statistical equilibrium. It goes without saying that aggregation is the most common form of group behaviour and is seen in a diverse array of systems from molecular self-assembly [79], clustering cockroaches [38] and selfish seal herds [23], to intelligent design using biomimicry [57].

There exists a large body of work on (non-active) particle aggregation (coagulation, gelation, agglomeration, flocculation and accretion), which is beyond the scope of this chapter and we point the interested reader to [39] and the references therein for a comprehensive review. For our purposes, a collective is comprised of many, similar, active particles, that are positioned with a higher density than would be expected for a random distribution. The individuals have no sense of the group as a whole and have fluid membership to and throughout the group.

The formation of these groups may be:

1. The object of selection, in which case it is the properties of the group that are emergent [22]. Some examples include:

- when grouping is the goal but not the task as in the coalescence of minimalist robots via random walk with movement ceasing when connection to another particle is made [43, 59].
- particles that move along a gradient in order to achieve a target density or neighbour distance [28].
- explicitly defined attraction to an external stimulus or cue, such as a roosting site or rendezvous site [12, 59, 80] or explicitly defined attraction to neighbours [19, 55].

  If explicit cohesion occurs without regulation for example volume exclusion, future inaction upon making contact, or short range repulsion the group will collapse. If forces are derived from an interaction potential V with corresponding force function $\mathbf{F} = -\nabla V$, there is repulsion at short range and attraction at larger inter-particle distances, decaying to zero for very large distances [24]. The corresponding Lyapunov function's minima corresponds to stable stationary states of the system and the equilibrium value (with $\mathbf{F} = 0$) determines the average nearest-neighbour distance and thus the density [27, 52].

2. A by-product of other processes and thus aggregation is itself an emergent property [22]. Generally, these are local drivers such as a desire for warmth [78] or safety (predator avoidance, confusion, dilution, anonymity, many eyes hypothesis [41]).

   Conceptually, models of these underlying processes are optimisation problems, whereby an individual moves to maximise their 'fitness'—a quantified desirable state, such as its ability to survive an attack, forage or successfully reproduce. The fitness may be fairly abstract, in the form of effort expended or a payoff. Evolutionary game theory has been used to show that there are evolutionary roots to aggregating by analysing behaviour based on individual and group payoffs, with a sharp transition between individual and group phases occurring at a critical payoff [37, 81].

Note that grouping is not always beneficial as there are some obvious costs that come with aggregation such as increased competition for mating/food, disease spread and energy disadvantages [41, 77]. Groups may also attract predators and increase their capture rate, a fact that is exploited by commercial fishing [25]. For a group to persist and its members not dissolve, the benefits must outweigh these costs. Other reasons for leaving group include transitory purpose (e.g. mating), exhaustion of the external resource attractor, maturation (locusts) or a shift in the cost/benefit relationship observed.

9.2.1 The Selfish Floc

One of the most prominent explanations for grouping is the selfish herd hypothesis, whereby each individual aims to minimise their exposure to danger by reducing

their ‘domain of danger’—the space that a predator would most likely attack from [31]. Thus, fitness is improved if the domain is reduced. From purely spatial considerations, this is a Voronoi tiling of the plane in two dimensions per Fig. 9.3 and higher order generalisations as natural extensions. Note that the maximum radius of each Voronoi polygon is bounded to mitigate infinite domains [54].

Movement towards the nearest neighbour was originally proposed as a reasonable proxy for minimising the domain [31], but it has since been shown that simple, easy to follow rules such as k-nearest neighbours tend not to result in a compact aggregation. More complex models can produce a compact aggregate but may be less realistic from the animal’s perspective. e.g. the local crowded horizon model, which requires knowledge of the state of all flock members. This is the dilemma of the selfish herd [76].

These methods do not explicitly minimise the individuals’ danger domain. A literal interpretation of the selfish herd requires that particles actively seek to improve their fitness with each movement, i.e. movement should be in the direction that best improves their situation. Current work investigates two active optimisations: local optimisation with expected gains in the subsequent time step; and a quasi-global optimisation where particles may accept short-term reductions (increases) in their fitness (domain size) for potential future reward. Simulations of the particles’ movement over time reveal rapid initial clustering followed by merging of the clusters to eventually form a tight stationary group. Due to the standard movement restrictions inhibiting movement towards already occupied space, the decisive movements tend to be made by the boundary particles with additional freedom. If these particles are also on the boundary of the entire group, then they are highly motivated and will ‘crawl’ around their cluster in an attempt to move to the inside. This exposes previously safe internal particles who will repeat the crawling process.

Any number of metrics could be used to quantify the individual and group states. Here, we choose to illustrate the individual domain area and trajectories and the average distance to the centre (Figs. 9.6 and 9.7).

Note that all Lagrangian models, where movement is not made with the aim of benefiting a neighbour, can be considered as selfish or ‘self-centred’.

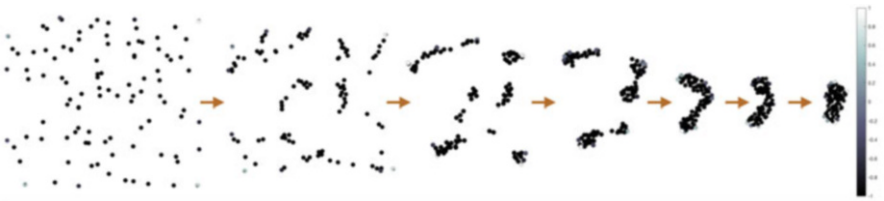


Fig. 9.6 Selfish flock evolution of group. Colour indicates normalised relative individual domain areas

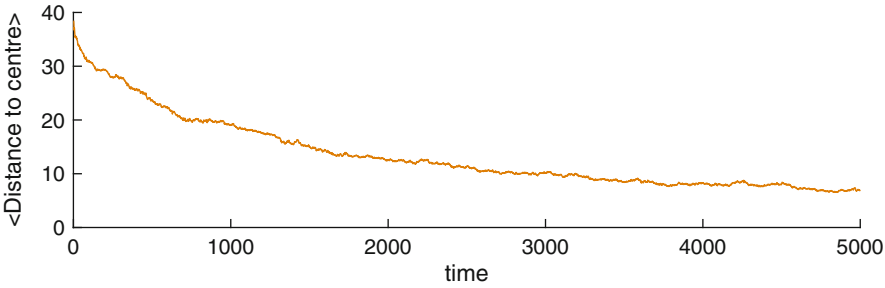


Fig. 9.7 Average distance to the centre of mass of the flock over time

9.3 Flocking: Stage Two

In 1986, Craig Reynolds, fed up with scripting the path of each individual bird in his animations, began to look at rules that would allow his bird-like objects (boids) to steer their own trajectory providing him a more efficient method with greater global controllability [62].

All boids shared three simple goals, which were prioritised as:

1. Collision avoidance—mitigate dangerous proximities.
2. Velocity matching—mimic the behaviour of other particles nearby.
3. Flock centring—do not stray too far from the group.

Surprisingly, the resulting group behaviour shared many features with real flocks. The boids were able to merge and split into sub-flocks, move in a unified direction and perform synchronised turns. The algorithm has since been used by computer animators in a number of memorable movie scenes such as the stampede scene in The Lion King.

At the time of publication, the interest that was generated mainly surrounded the fact that realistic behaviours could result from such simple rules. Less interest was shown in analysing these behaviours—until the same problem piqued the interest of Physicists. Equipped with the tools of Statistical Physics and hoping to gain deeper insight into the motion of very large flocks. Tamas Vicsek and his team stripped back the boid model to just the alignment rule. Inspired by the Heisenberg model they began to think about what it would mean for the spins in a ferromagnet to have freedom to move [60]. This was the first step towards a quantitative model and its minimal nature lent itself well to mathematical analysis.

The Vicsek model describes the movement of particles that update their orientation θ_i at each time step by assuming the average heading of their neighbours. The neighbourhood is metric with radius r and particle speed, v_0, kept constant. Formally:

$$\theta_i(t+1) = \langle\theta\rangle_r + \xi_i. \tag{9.6}$$

where ξ_i is the noise. Interpreting the motion as driven by a social force with no inertia and large damping:

$$\mathbf{F}_i(t+\Delta t) = \Sigma_{j\neq i} A_{ij} \frac{\mathbf{v}_\mathrm{j}}{|\mathbf{v}_\mathrm{j}|}. \tag{9.7}$$

In continuous time the dynamical equations are:

$$\frac{\eta}{v_0}\frac{\mathrm{d}\mathbf{v}_i}{\mathrm{d}t} = \frac{1}{v_0}\Big(A_{ij}\mathbf{v}_j\Big)^{\perp} + \boldsymbol{\xi}_i^{\perp} \tag{9.8}$$

$$\frac{\mathrm{d}\mathbf{r}_i}{\mathrm{d}t} = \mathbf{v}_i \tag{9.9}$$

where η is the friction coefficient, $\perp$ indicates a projection of a vector onto the plane orthogonal to $\mathbf{v}_i$ and $\boldsymbol{\xi}_i$ with variance $\langle\langle \boldsymbol{\xi}_i(t)\cdot\boldsymbol{\xi}_j(t')\rangle_j\rangle_{t'} = (2d)\eta T\delta_{ij}\delta(t-t')$ [13].

How noise is introduced to the system is critical to analysis of the phase transition. In the original model, stochasticity ξ_i is a scalar that is added to the angle of the velocity. It is chosen with a uniform probability from the interval $[-\xi\pi, \xi\pi]$. This results in a continuous phase transition, whereas if noise is added as a vector,$\boldsymbol{\xi}_i$, to the velocity then the phase transition is discontinuous. This was a matter of some debate [1].

The Couzin model [19] introduced rules 1 and 3 and a blind zone to the minimal Vicsek framework in an attempt to improve the biological realism, whilst maintaining simplicity. Particles determine their motion based on neighbour distribution through prioritised zones (repulsion, orientation, alignment) and then moved towards the desired direction of motion with turning speed τdt. The final state of the group is dependent on the relative spatial extent of these zones. A collective memory, or hysteresis, state was uncovered depending on whether the orientation zone was increased or decreased—changing the critical value at which the transition to/from polarisation/milling occurred.

The Reynolds, Vicsek and Couzin models were all born from very different backgrounds but share commonalities in the self-organisation and emergent group patterns from very simple, local rules. They are now central to the field of collective motion and provide a starting point for many other studies (see for example: removal of homogeneities in behavioural rules, networks or individuals [17]; purely attractive forces [68], speed variation; [51]; and inclusion of second order inertial terms; [13]).

9.3.1 The Selfish Flock and Future Directions

If the motivation behind aggregation in the first instance persists once the aggregate has formed, then the behavioural rules should reflect this.

To this end, ongoing work looks to extend the active selfish floc of the previous section to flocking. By replacing movement restrictions with repulsion and including inertial terms sustained motion via the pursuit of less risky positions within the group can occur.

Future work should continue investigations into single threads of particle motivation to continue unifying the studies of aggregation and flocking.

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Chapter 10
Network Centrality: An Introduction



Francisco Aparecido Rodrigues

10.1 Introduction

Complex systems are made up of connected elements whose interactions are nonlinear [3, 32]. Examples of complex systems include our society, the Internet, our brain, and cellular interactions [32]. The modeling of complex system behavior has attracted the attention of researchers from several areas, including Mathematics, Physics, Biology, Computer Science, and Engineering [12]. The study of these systems has several applications in Science and Technology, including the control of disease spreading [36] and adjustment of synchronization in data transmission [37]. Since the last century, researchers have verified that the structure of several complex systems can be modeled as complex networks [4, 8, 11]. These networks are characterized by a very heterogeneous organization, presenting a special set of highly connected elements, whereas the remainder of the components is low connected [5]. This ubiquitous scale-free architecture is observed in numerous systems, from food webs to collaboration networks of scientists [12], and has important implication for system dynamics. For instance, previous works have verified that the epidemic threshold for disease propagation depends on the level of network heterogeneity, being close to zero in a scale-free network [36]. Thus, scale-free networks are the ideal medium for disease transmission, explaining the quick propagation of viruses in our society, like those that cause influenza [27]. Moreover, the network organization influences the emergence of synchronization in systems like power grids and our brain [2, 41]. In this case, a set of self-sustained oscillators interacts adjusting their phases and after a given critical coupling strength, a collective

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behavior emerges. This critical coupling strength also depends on the network heterogeneity. Other dynamical processes, including cascade failures, percolation, and voter models, also depend on the network structure [6].

Since the structure of complex networks is very heterogeneous, it is expected that some nodes are more important than others in some sense. This importance can be quantified by the network centrality, as we expected that the most central nodes are the most influential ones and can propagate their information content easier than other nodes [13, 28]. However, there is no general definition of centrality and, therefore, many centrality measures have been developed [11]. In this chapter, we review the main centrality measures used to characterize complex networks structure and present their advantages and limitations. The influence of centrality on dynamical processes is also discussed in terms of epidemic spreading and synchronization. We show that central nodes are the most influential spreaders of diseases. However, the identification of these nodes through centrality measures depends on the network structure. In the case of synchronization, we show that when the natural frequency of Kuramoto oscillators is correlated with their number of connections, the system undergoes a first order phase transition [22], differently from the second order phase observed when the natural frequency is independent of network structure [41]. Applications of network centrality in some areas, including systems biology and neuroscience, are also presented here. Finally, we discuss some perspectives regarding the study of network centrality and generalizations to more intricate networks, including multilayer organization and temporal networks.

10.2 Centrality Measures

A complex network is a graph G with particularly intricate structure, made of an ordered pair of disjoint sets (V, E), where V is a set of elements called vertices (nodes) and E is a subset of ordered pairs of distinct elements of V, called edges or arcs. If the network is undirected, i.e. for every connection going from each pair i to j has a connection j to i, the links are called edges. Otherwise, directed connections are called arcs. Network edges can also have weights, e.g. indicating the strength of the interaction between two nodes. In this chapter, we consider only undirected and unweighted networks to discuss the centrality measures.

In complex networks, due to its highly heterogeneous structure, some nodes can be considered as more important than other ones. For instance, in social networks, some people like celebrities and politicians have a lot of followers and can propagate information easier than ordinary subjects. Hence, these nodes can be considered as central. However, this definition of centrality is not unique, since we can define it in terms of the load that each node receives. For example, in a street network, an urban region is central whether it presents traffic jams and is more accessed than other places. Thus, the definition of centrality is not general and depends on the application. Since we do not have a consensus about the general definition of centrality, several measures have been proposed, where each one considers specific concepts.

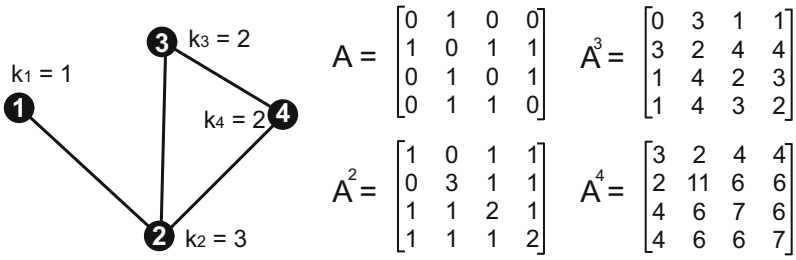


Fig. 10.1 An illustration of the concepts of degree and the power of the adjacency matrix A. The elements $(A_{ij})^n$ give the number of walks of length n between nodes i and j. The degree k_i of each node is indicated in the network

The most simple centrality measure is the degree centrality, which is defined by the number of connections attached to each node. In terms of the adjacency matrix A, which is a mathematical representation of network structure ($A_{ij} = 1$ if there is a connection between nodes i and j or $A_{ij} =$ otherwise), we can calculate the degree centrality of node i by the sum of the elements of row i of A, i.e.,

$$k_i = \sum_{j=1}^{N} A_{ij}, \tag{10.1}$$

where N is the number of nodes in the network. Figure 10.1 shows an example of a network and the degree of each node. Although the degree definition is intuitive, since it is expected that highly connected nodes are at the center of the network, it has some drawbacks. For instance, as we can see in Fig. 10.2a, the nodes with the highest degree (in black) are at the periphery of the network and, therefore, are not central. Thus, the degree centrality can be considered as a local centrality measure, as a hub (densely connected node) may not be central.

A more general definition of centrality in terms of the number of connections is given by the k-core measure [16]. The k-core is a subgraph in which all vertices have a degree at least k. This centrality measure is obtained by the k-shell decomposition, which partitions the network by iteratively removing all nodes whose degree is smaller than k. After removing these nodes, the network is re-analyzed to verify whether there are nodes with less than k connections. If such nodes are present, then they are also removed. The process is repeated until the minimum degree in the network is k. The resulting subgraph is called the k-core of the network. A node i has coreness $kc(i) = k$ whether it belongs to the k-core, but it is not in the $(k + 1)$-core. According to this measure, the most central nodes have the highest values of coreness (k-core number). Notice that high-degree nodes localized in the periphery of networks should display small values of k-core number since they are not in the main connected component when low degree nodes are removed. Therefore, only hubs at the main core of networks present the highest values of kc. The limitation of this measure lies in the fact that many nodes may be assigned to the same k-core number as we can see in Fig. 10.2b.

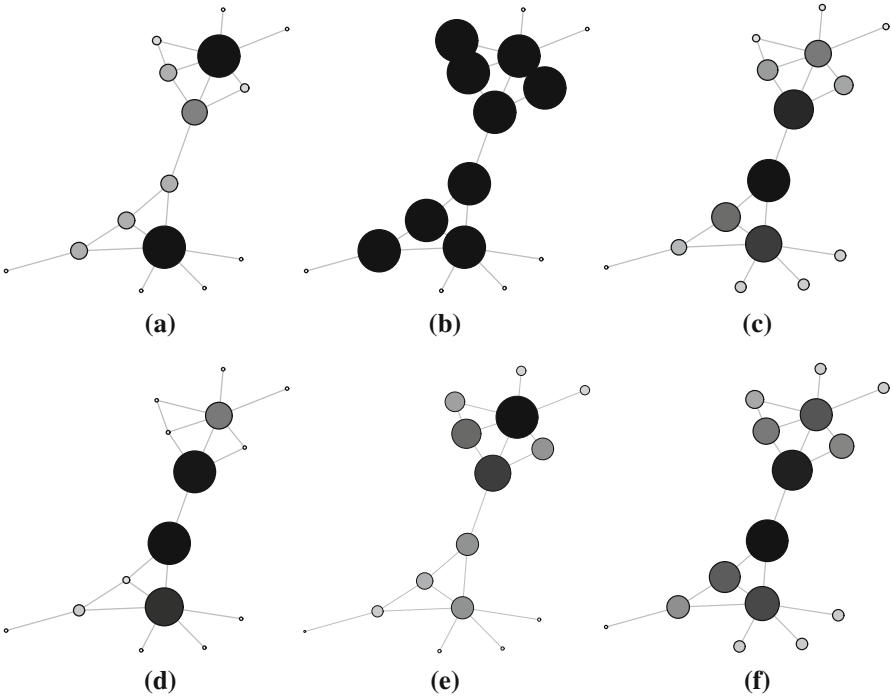


Fig. 10.2 Comparison of centrality measures: (**a**) degree centrality, (**b**) k-core, (**c**) closeness centrality, (**d**) betweenness centrality, (**e**) eigenvector centrality, and (**f**) accessibility. The size and darkness of each node are proportional to its centrality measure

Node centrality can also be defined in terms of the shortest paths. The distance between nodes i and j is given by the number of edges in the shortest path connecting them. A central node is close to all other nodes in the network in terms of this distance. This idea is enclosed in the closeness centrality measure, which is defined in terms of the average distance of each node to all others. Mathematically, the closeness centrality of i is defined as

$$C_i = \frac{N}{\sum_{j=1, j\neq i}^{N} d_{ij}}, \tag{10.2}$$

where d_{ij} is the length of the shortest path between i and j, and N is the number of nodes in the network. Closeness centrality has the advantage to be very intuitive and suitable to characterize a process in which the information travel through the shortest distances. However, similar to degree and k-core, closeness centrality also presents some limitations. It is based only on the shortest distances and, therefore, the range of variation is too narrow due to the small diameter of networks. Indeed, most complex networks present small average shortest path length, since the typical

distance increases with the logarithm of the number of nodes. The typical distance in a random network scales as $d \approx \log(N)/\log(\langle k \rangle)$, where $\langle k \rangle = \sum_i \sum_j A_{ij}/N$ is the average degree. Thus, the ratio between the largest and minimal distances is of order $\log N$, since the minimal distance is equal to one. In most real-world networks, this ratio is about six or less. Thus, we can have several nodes with the same level of centrality, although they may present different roles on information spreading. This measure is more suitable when we have spatial networks, whose distance between nodes is higher than in random networks with the same number of nodes and connections.

If we consider the flow of particles on a network, then we can define centrality in terms of the load. It is natural to think that the most central node receives the largest number of particles in a defined time interval. Assuming that these particles move following the shortest distances, the load in a node i is given by the total number of shortest paths passing through i. However, since we can have more than one shortest path between a pair of nodes a and b, it is more suitable to define the load in node i as the fraction of shortest paths connecting each pair of nodes (a, b), $a, b = 1, \ldots, N$, that includes i. Thus, mathematically [19],

$$B_i = \sum_{(a,b)} \frac{\eta(a, i, b)}{\eta(a, b)}, \tag{10.3}$$

where $\eta(a, i, b)$ is the number of shortest paths connecting vertices a and b that pass through vertex i and $\eta(a, b)$ is the total number of shortest paths between a and b. The sum is over all pairs (a, b) of distinct vertices. In this case, a central node is crossed by many paths and yields the highest value of B.

This definition of betweenness centrality considers only the shortest distances and, therefore, is not general, since information can travel long distances in a network, such as occurs on the Internet, where routers adjust the route of IP packets according to the traffic flow. Moreover, the calculus of betweenness centrality is computationally expensive, requiring $O(N^3)$ time and $O(N^2)$ space, where N is the number of nodes in the network. Even the solution proposed by Brandes [9] to calculate exact betweenness centrality, which runs in $O(NM)$, where M is the number of edges in the network, is computationally expensive for large graphs. To overcome these limitations, calculus based on random walks can be considered [34]. The betweenness centrality based on random walks is given by the expected number of visits to each node i during a random walk,

$$B_i = \sum_{a=1}^{N} \sum_{b=1}^{N} w(a, i, b) \tag{10.4}$$

where $w(a, i, b)$ is the number of times that i is visited when a random walk of length n is performed from nodes a to b. The number of steps to perform the random walk n is a parameter of this method and the solution is, therefore, approximated.

We can also define centrality in terms of the neighborhood of each node. We can assume that a node is important if it is linked to other important nodes. For instance, a person in a social network, like Twitter, can be considered important if other important people follow him/her. Thus, if this "importance" is quantified by vector x, then we can define the importance of i by

$$x_i = \frac{1}{\lambda} \sum_{k=1}^{N} A_{k,i}\, x_k, \tag{10.5}$$

where $\lambda \neq 0$ is a constant. In matrix form we have:

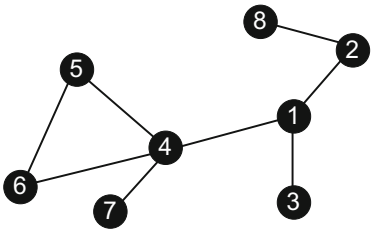
$$\lambda x = xA. \tag{10.6}$$

That is, the "importance" of node i is defined by the left-hand eigenvector of the adjacency matrix A associated with the eigenvalue λ. The entries of x are called eigenvector centrality. The eigenvalue λ is chosen as the largest eigenvalue in the absolute value of matrix A, since due to the Perron-Frobenius theorem if matrix A is irreducible then x is both unique and positive. Since λ is leading eigenvector of A, the calculation of x can be performed by the power method,

$$x^{(k)} = x^{(k-1)}A, \tag{10.7}$$

where $x^{(0)}$ is an arbitrary vector with positive entries. If we consider $x^{(0)} = \{1, 1, \ldots, 1\}$, then $x^{(1)}$ is the degree and $x^{(n)}$, $n \leq 1$, is the number of walks of length n arriving in each node (see Fig. 10.3). Indeed, the number of walks between pairs of nodes is calculated from the power of the adjacency matrix. More specifically, the number of walks of length n between every pair of nodes i and j is given by the entries of the matrix A^n. For instance, in Fig. 10.1 we can see the matrices A^2, A^3 and A^4. Therefore, the eigenvector centrality is proportional to the number of visits in each node through a random walk of infinite length. If a node is at the center of the network, then it is more accessed than the other nodes. Figure 10.3 illustrates the calculation of the eigenvector centrality through the power method. Notice that nodes with the same degree, like nodes 7, 3, and 8, have different values of eigenvector centrality. Nodes at the center of the network present the highest values of eigenvector centrality.

Eigenvector centrality also presents some limitations. Depending on the network structure, most of the weights of the eigenvector concentrate in a few nodes, like hubs. In this case, most of the nodes will present centrality close to zero and, therefore, the importance of nodes is not well quantified. For instance, in a random network with only one hub, Martin et al. [31] verified that in the localize regime, i.e., for $k > \langle k \rangle (\langle k \rangle + 1)$, where k is the degree of the hub, the hub vertex and its neighbors present the highest values of eigenvector centrality, whereas the centrality of the remaining nodes is $O(1/N)$, vanishing in the limit of large networks. After this value, the inverse participation ratio defined as



Node	x_0	x_1	x_2	x_3	x_4
1	1	3	7	15	39
2	1	2	4	9	19
3	1	1	3	7	15
4	1	4	8	23	51
5	1	2	6	14	37
6	1	2	6	14	37
7	1	1	4	8	23
8	1	1	2	4	9

Fig. 10.3 Example of eigenvector centrality calculation. The values of x are obtained by the power method (Eq. (10.7)). We do not include normalization to make the example more didactic. Notice that x_1 stores the node degree. Although nodes 7 and 8 have the same degree, their eigenvector centrality indices are very different

$$IPR = \frac{\sum_{i=1}^{N} x_i^4}{(\sum_{i=1}^{N} x_i^2)^2}, \tag{10.8}$$

where x_i is the eigenvector centrality of node i, presents a phase transition, indicating the presence of localization. The value of IPR is close to 1 if x is localized.

The limitation of the eigenvector centrality can be overcome by considering the Hashimoto non-backtracking matrix B. B is a $2M \times 2M$ non-symmetric matrix with one row and one column for each directed edge (i, j). For undirected networks, each undirected link between i and j is replaced with a pair of directed links (i, j) and (j, i). This matrix is defined for undirected networks by replacing each undirected link between i and j with a pair of directed links (i, j) and (j, i) and make B a $2M \times 2M$ non-symmetric matrix with one row and one column for each directed edge (i, j). The elements of B are defined by

$$B_{u\to v, w\to x} = \begin{cases} 1 \text{ if } v = w \text{ and } u \neq x \\ 0 \text{ otherwise.} \end{cases} \tag{10.9}$$

The entry $v_{i\to j}$ of the main eigenvector v of B gives the centrality of vertex i disregarding the contribution of node j. Thus, the centrality of node j is given by the sum of these contributions for all neighbors of j, i.e., $x_j = \sum_{i=1}^{N} A_{ij} v_{i\to j}$. Figure 10.4 illustrates the calculation of the non-backtracking matrix. In practice, the calculation of the centrality based on the non-backtracking matrix can be performed by considering the so-called Ihara (or Ihara–Bass) matrix [31]

$$B = \begin{pmatrix} 0 & D - I \\ -I & A \end{pmatrix}, \tag{10.10}$$

where I is the $N \times N$ identity matrix, D is a diagonal matrix whose element is equal to the node's degree $Dii = \sum_{j=1}^{N} A_{ij}$. The vector x of centralities is equal to the first N entries of the leading eigenvector of the $2N \times 2N$ matrix.

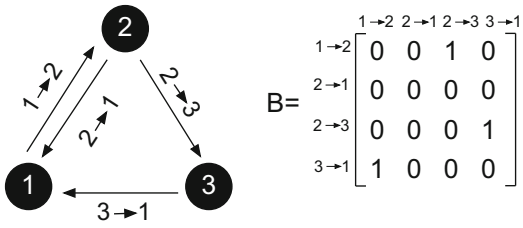


Fig. 10.4 Illustration of the non-backtracking matrix. If there is a link $i \to j$ and $j \to k$, then $B_{i\to j,j\to k} = 1$

The name non-backtracking is due to non-backtracking walks, which is a walk in which it is allowed to return to a vertex visited previously only after at least two other vertices have been visited. For instance, a sequence like $i \to j \to i$ is not allowed. The entries $(B_{uv})^n$ yield the number of non-backtracking walks of length n between the edges u and v in the original network, in the same way as powers of the adjacency matrix count the number of paths between pairs of nodes (see Fig. 10.1). These properties of B make it useful for community detection in networks by using spectral clustering [30]. In this case, when the network presents homophily, i.e., nodes in the same community are more likely connected than between communities, the eigenvalue associated with this eigenvector is separated from the bulk of the eigenvalue spectrum and communities can be easily obtained. However, when homophily is not well defined, the relevant eigenvalue dissolves into the bulk and the community identification is impossible. Thus, we have a detectability threshold for community detection in which no algorithm can label the vertices better than chance. Nevertheless, by using the non-backtracking matrix, Krzakala et al. [30] show that it is possible to detect communities above this threshold. They verified that spectral algorithms based on the non-backtracking matrix perform optimally for some popular generative models, including the stochastic block model. A complete discussion about community identification methods including those based on matrix B is presented in [18].

Although the non-backtracking matrix overcomes the localization problem, it presents some drawbacks. Since the size of B is $2M \times 2M$, the computation of its eigenvalues and eigenvectors is computationally expensive. Thus, the calculation of B is intractable for networks with millions of vertices. Moreover, non-backtracking walks must contain cycles and, therefore, subgraphs representing trees do not contain non-backtracking walks. These trees have no effect on the spectrum and can be removed from the network. However, this removal represents a loss of information about the network structure, as these subgraphs may be important for network characterization, like in community identification [18], and dynamical processes analysis, as in the case of synchronization [42]. An alternative version of the non-backtracking matrix, called flow matrix, was proposed by Newman [33] to overcome these limitations. However, the manipulation of this matrix is also expensive, since it has the same dimension as the non-backtracking matrix.

Similar to the eigenvector centrality, PageRank is another measure based on random walks [10]. In this case, the basic idea is to transform the adjacency matrix such that its elements represent the probability transition between a pair of nodes.

The network can represent a Markov chain in which each node is a state. Since this chain can have absorbing or periodic states, the PageRank algorithm transforms it to become ergodic, such that each node can be reached from every other node by following random walks. This transformation is particularly important in directed networks, since the network may present absorbing nodes, where a random walker can be trapped. As we verified for the eigenvector centrality, the spectra of adjacency matrix yield the expected number of visits in each node by following a random walk. In the case of PageRank, we have a similar idea. Thus, the PageRank is calculated by the power method,

$$\pi^T = \pi^T G, \tag{10.11}$$

where G is the Google matrix,

$$G = \kappa \left(P + \frac{ae^T}{N} \right) + \frac{(1-\kappa)}{N} ee^T. \tag{10.12}$$

The element a is the binary vector called dangling node vector (a_i is equal to one if i is a dangling node and 0 otherwise), e is a vector of ones of length N, and P is the transition probability matrix of the respective network,

$$P_{ij} = \frac{A_{ij}}{\sum_j A_{ij}}. \tag{10.13}$$

The original version of the algorithm considers $\kappa = 0.85$ [10]. The PageRank of a node i, π_i, is given by the i-th entry of the dominant eigenvector π of G, given that $\sum_i \pi_i = 1$. π_i can be understood as the probability of arriving at the node i after a large number of steps following a random walk navigation through the network.

One limitation of the PageRank measure lies in the definition of connections among all pair of nodes, which simulate the jumping after random walk navigation. This procedure selects all nodes with uniform probability, ignoring their importance. However, in real-world navigation, some web pages are more likely to be selected than other ones. For instance, pages related to news are more visited than personal websites.

Accessibility is another measure that considers random walks on networks. This measure is related to the diversity of access of individual nodes through random walks. The accessibility of the node i for a given distance h is defined by the exponential of the Shannon entropy of the transition probability matrix (Eq. (10.13)) [44], i.e.,

$$\alpha_h(i) = \exp\left(-\sum_j P_{ij}^{(h)} \log P_{ij}^{(h)} \right), \tag{10.14}$$

where $1 \leq \alpha_h(i) \leq N$. The elements of $P_{ij}^{(h)}$ yield the probability of going from i to j following a walk of length h. The maximum value of α_h corresponds to the case in which all nodes are reached with the same probability $1/N$.

This version of accessibility depends on the parameter h. However, it can be generalized by considering the matrix exponential operation [13],

$$\mathbf{P} = \frac{1}{e} \sum_{k=0}^{\infty} \frac{1}{k!} P^k = e^P, \tag{10.15}$$

where the factor $1/e$ is necessary to guarantee that $\mathbf{P}$ is a stochastic matrix [13]. Notice that the definition of $\mathbf{P}$ penalizes long walks, i.e., the shortest walks receive more weight than the longest ones. The generalized random walk accessibility is defined as

$$\alpha(i) = \exp\left(- \sum_j \mathbf{P}_{ij} \log \mathbf{P}_{ij} \right), \tag{10.16}$$

where $\mathbf{P}_{ij}$ is calculated from Eq. (10.15). This measure has been used to identify the border of networks [45]. The limitation of accessibility lies in the penalization of longer walks. Moreover, it is assumed that the random walk selects nodes with the same probability in each step, which may not occur in real-world navigation.

In Fig. 10.2 we compare the centrality measures discussed in this section. As we can see, different centrality measures may not identify the same set of central nodes. Thus, the use of centrality metrics depends on the problem and the most suitable approach is to consider more than one centrality for network characterization. For instance, one can combine a set of measures and consider multivariate statistical methods for network characterization.

Although all centrality measures present some drawbacks, they are very useful in the analysis of dynamical processes on networks. Centrality is also a fundamental property of several complex systems, like brain networks, Internet, and social networks. As follows, we discuss the importance of centrality to understanding the structure and dynamics of complex systems and some application of centrality for characterization of the structure and function of complex systems.

10.3 Centrality and Dynamical Processes in Networks

Centrality plays a fundamental role in the evolution of dynamical processes in networks, like epidemic spreading and synchronization. Due to the presence of hubs in scale-free networks, the epidemic threshold is close to zero in these structures [36]. Similarly, the critical coupling for the emergence of synchronization of Kuramoto oscillators also depends on the network heterogeneity, which decreases

with the variation of the node degree [41]. In this section, we analyze these two dynamical processes in terms of centrality measures and point out the influence of central nodes on the evolution of the dynamical process.

10.3.1 Identification of Influential Spreaders

Epidemic processes are ubiquitous in nature, society, and technology [36]. The theoretical study of infections disease aims to improve control and eradicate the pathogen agent from the population [6, 36]. Many mathematical models have been developed to understand the propagation of diseases in populations of humans and animals [27]. One particularly important model is the SIR (susceptible-infected-recovered) model, in which subjects can recover from the disease and acquire immunity. We consider this model in our analysis of centrality and epidemic outbreaks.

The importance of identification of the main propagators of disease lies in developing efficient methods for disease control. For instance, many recent works have analyzed the influence of centrality on epidemic spreading [13, 28, 38]. Here, we address this influence by the centrality measures described in Sect. 10.2. We simulate the SIR epidemic spreading by following the reactive process [21], in which at each time step, every infected node tries to transmit the disease to all of its neighbors with a fixed probability β. After that, each of these infected nodes recovers with probability μ. We consider $\mu = 1$ and $\beta = 2\lambda_c$, where $\lambda_c = \frac{1}{\Lambda_1}$ is the critical threshold of epidemic spreading and Λ_1 is the leading eigenvector of the adjacency matrix. This critical threshold depends on the network structure, as verified before in several works [36].

In Fig. 10.5, we present the relation between the average outbreak size when the disease starts in each node and its respective centrality measure. For most of the measures, the most central nodes are the most influential spreaders. However, the generalized random walk accessibility is the best metric to predict the outbreak size in the E-road network, which is a road network located mostly in Europe whose nodes represent cities, and edges are the roads. K-core and degree poorly predict the most influential spreaders. This fact is mainly due to the spatial organization of the network, which does not present hubs and, therefore, does not have a scale-free organization. Betweenness centrality also cannot predict the most influential spreaders.

A very different relationship between centrality and node outbreak size can be observed for scale-free networks. For the case of US air transportation network, where each edge represents a connection from one airport to another in 2010, we can see in Fig. 10.6 that the eigenvector centrality, degree, and k-core can predict the average outbreak size when the disease starts in each node. The accessibility can also predict, but only for high values of centrality. On the other hand, betweenness centrality is not suitable to predict the disease spreading.

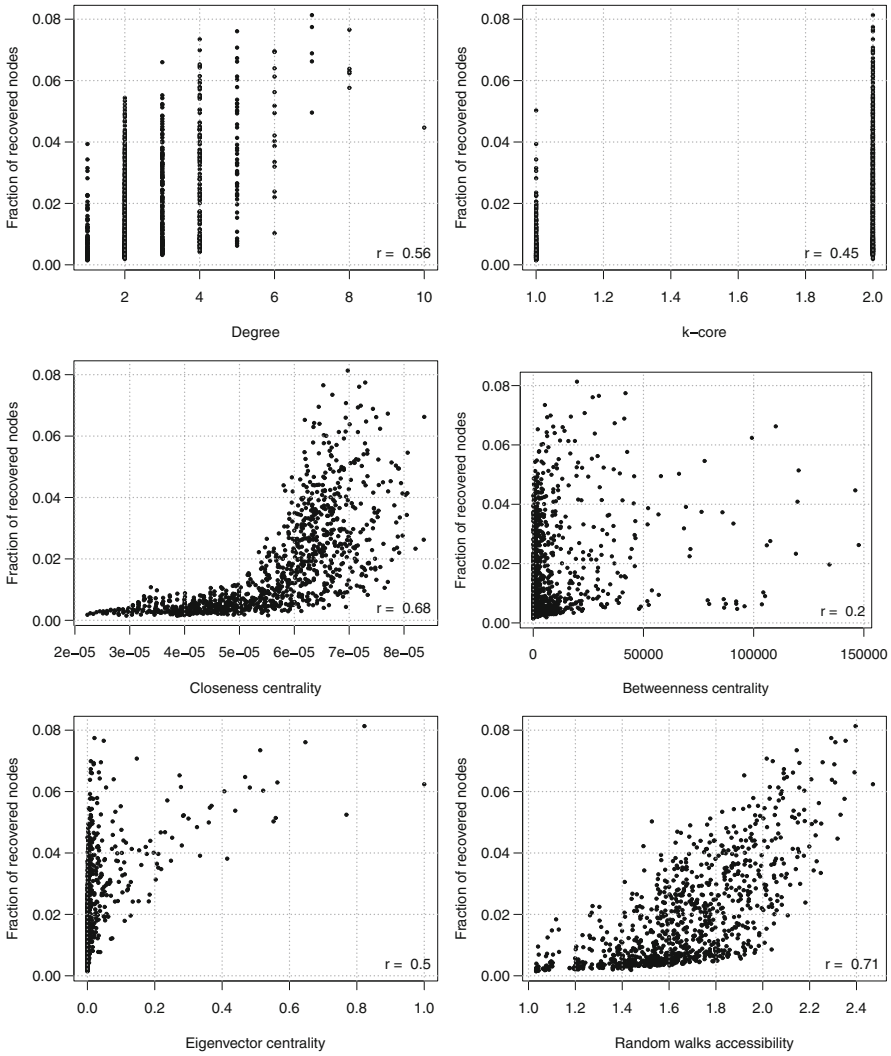


Fig. 10.5 The relation between the outbreak size and centrality measures in the E-road network. The epidemic spreading is simulated through the SIR model with discrete time, recover probability $\mu = 1$, and spreading probability $\beta = 2\lambda_c$, where $\lambda_c = 1/\lambda_{max}$ and λ_{max} is the main eigenvalue of the adjacency matrix of the network. Each point represents a city with a given centrality and spreading capacity, given by the average outbreak size. This average is calculated from 50 simulations of the SIR model starting in each node. The network has $N = 1174$ cities and $M = 1417$ roads. The Pearson correlation coefficient between the percentage of recovered individuals and centrality measures is indicated inside the plots. Only the network largest component is considered in our analysis

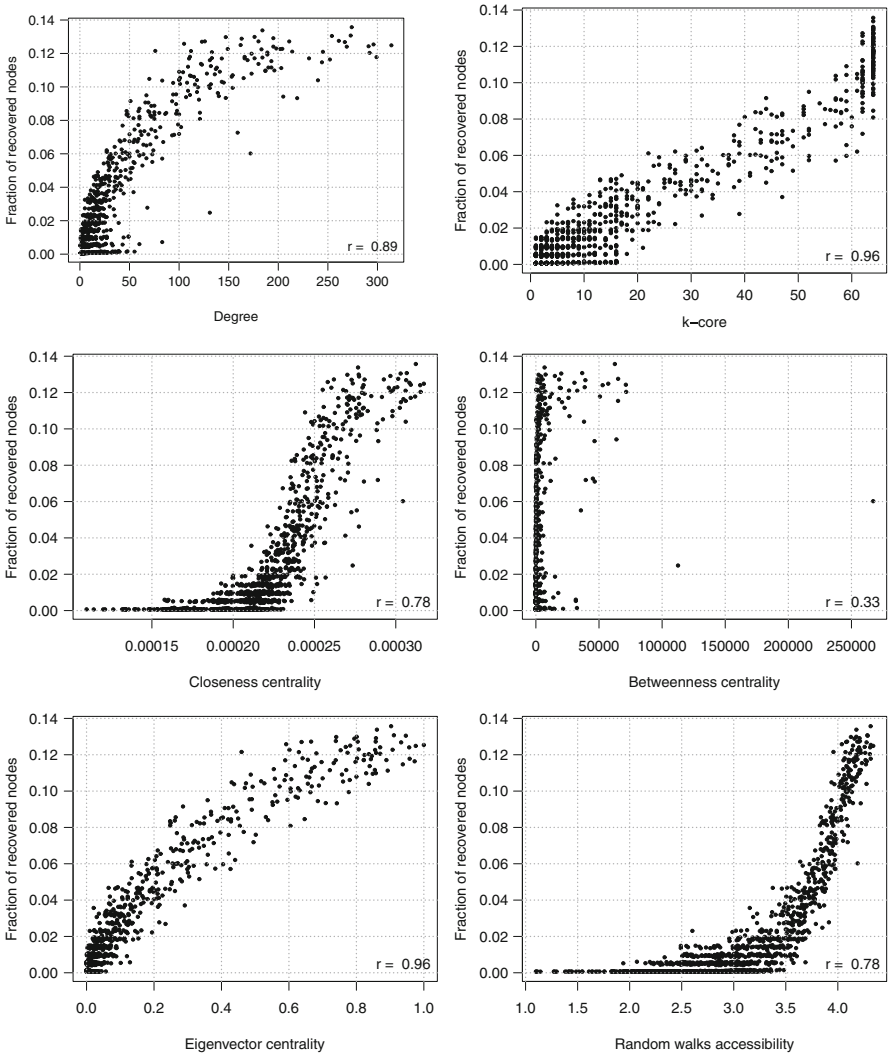


Fig. 10.6 Outbreak size and centrality measures in the US airports network. The epidemic spreading is simulated through the SIR model with discrete time and contagion probability $\beta = 2\lambda_c$, where $\lambda_c = 1/\lambda_{max}$ and λ_{max} is the main eigenvalue of the adjacency matrix A; and recover probability $\mu = 1$. Each point represents an airport with a given centrality and spreading capacity, given by the average outbreak size. This average is calculated from 50 simulations of the SIR model starting in each node. The network presents $N = 1574$ airports and $M = 28,236$ flights. The Pearson correlation coefficient between the percentage of recovered individuals and centrality measures is indicated inside the plots. Only the network largest component is considered in our analysis. We use the undirected version of the network

These two examples show that the choice of the centrality measures to predict the most influential spreaders depends on the network organization. For spatial networks, the random walks accessibility measure is the most suitable, as verified before [13]. For scale-free networks, the k-core and eigenvector centrality can determine these nodes very accurately [28]. However, due to the difficulty in the identification of the most influential spreaders for any kind of networks, this problem remains open.

10.3.2 Explosive Synchronization

Synchronization is a pervasive process present in many complex systems, such as between neurons in the central nervous system, communication networks, power grids, social interactions, animal behavior, ecosystems, and circadian rhythm [43]. These systems are made of self-sustained oscillators that interact forming a network structure. One particularly important type of synchronization is the phase synchronization, which is the process in which oscillators adjust their rhythms according to interaction with their direct neighbors [37]. A very popular model to describe the evolution of phase oscillators was introduced by Kuramoto [41]. Each self-sustained oscillator is characterized by its phase $\theta_i(t)$, $i = 1, \ldots, N$. Pairs of oscillators are connected by the sine of their phase differences. In complex networks, each oscillator i obeys an equation of motion defined as

$$\frac{d\theta_i}{dt} = \omega_i + \lambda \sum_{i=1}^{N} A_{ij} \sin(\theta_j - \theta_i), \quad i = 1, \ldots, N, \tag{10.17}$$

where λ is the coupling strength, ω_i is the natural frequency of oscillator i, and A_{ij} are the elements of the adjacency matrix A, so that $A_{ij} = 1$ when nodes i and j are connected and $A_{ij} = 0$ otherwise. The general Kuramoto model considers a random distribution of the natural frequencies and initial phases according to a specific distribution $g(\omega)$, which is independent of the network structure [2].

In most of the cases, the frequency distributions are unimodal and symmetric around a mean value ω_0 [2, 41]. It is possible to show by mean-field approximation that the critical coupling for the emergence of the synchronous state is given by [25, 40]

$$\lambda_c^{(0)} = \frac{2}{\pi g(0)} \frac{\langle k \rangle}{\langle k^2 \rangle}. \tag{10.18}$$

where $g(\omega)$ is symmetric in relation to a single local maximum ω_0 (e.g., $\omega_0 = 0$). Thus, the presence of central nodes influences the emergence of the synchronous state, since the more heterogeneous a network, the smaller the critical coupling strength. Indeed, in scale-free networks with $P(k) \sim k^{-\gamma}$, where $\gamma \leq 3$, as $N \to \infty$ the critical coupling $\lambda_c^{(0)}$ goes to zero, since the ratio $\langle k \rangle / \langle k^2 \rangle$ diverges [41].

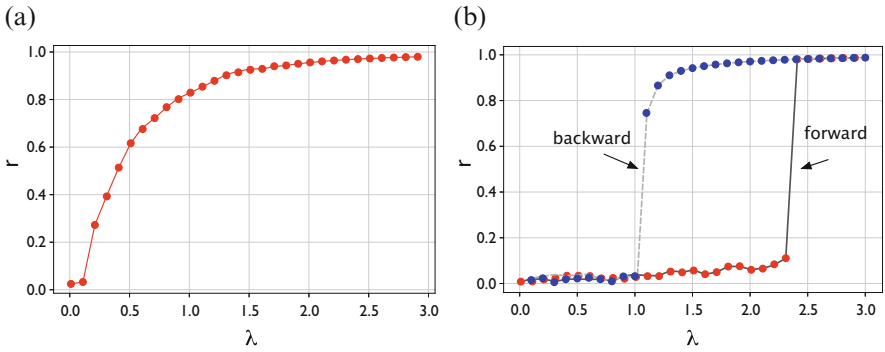


Fig. 10.7 Synchronization diagram $r(\lambda)$ for a Barabási-Albert scale-free network with $N = 500$ nodes and average degree $\langle k \rangle = 4$. (**a**) Second-order phase transition. The natural frequency of the oscillators is distributed according to a Gaussian distribution with zero mean and unit variance. (**b**) Explosive synchronization. The natural frequency is equal to the degree, $\omega_i = k_i, \forall i = 1, \ldots, N$. Notice that a hysteretic behavior is observed, indicating a discontinuous transition to synchronization. The forward and backward evolution of r are calculated adiabatically for $\lambda_0, \lambda_0 + \delta\lambda, \ldots, \lambda_0 + n\delta\lambda$, where we consider $\delta\lambda = 0.1$

The level of synchronization is quantified by the macroscopic order parameter, defined as

$$r(t)e^{i\psi(t)} = \frac{1}{N}\sum_{i=1}^{N} e^{i\theta_i(t)}, \tag{10.19}$$

where the modulus $0 \leq r(t) \leq 1$ and $\psi(t)$ is the average phase of the oscillators. When the natural frequency of each oscillator is independent of the network structure and selected at random from a unimodal and symmetric distribution $g(\omega)$, the order parameter displays a continuous (second-order) phase transition to the synchronous state. This behavior is illustrated in Fig. 10.7a. On the other hand, when the natural frequency of oscillators is correlated with the number of connection ($\omega_i = k_i$), i.e., when hubs oscillate much faster than the remainder of the nodes, a very different behavior is observed. In this case, a first order phase transition to synchronization occurs in scale-free networks [22], as shown in Fig. 10.7b. Thus, the correlation between the natural frequency and the number of connections provides a change in the system behavior. Thus, centrality plays a fundamental role on synchronization in uncorrelated networks.

Further works verified that the correlation between the natural frequency and the number of connections is not a unique condition for the emergence of explosive synchronization. The key mechanism behind explosive synchronization is the existence of sufficient large frequency mismatches between nodes and their neighbors, what occurs in non-correlated and disassortative scale-free networks [41]. However, the effect of centrality on synchronization is a key feature that deservers a deeper investigation.

10.4 Applications

Network centrality has been used to characterize complex systems in many areas, ranging from climate to biological interactions. Here we present some applications of centrality measures to characterize the structure and function of complex systems. One of the first applications of network centrality was in the prediction of protein lethality. Jeong et al. verified that the most connected proteins of the yeast *S. cerevisiae* are more likely essential than low connected proteins [26]. A similar observation was verified in genetic networks, where the P53 gene, which is a tumor-suppressor, is at the center of the network. Therefore, the disruption of p53 has severe consequences to keep cells under control [46].

In medicine, human disease genes have a tendency to encode hubs in the interactome, which may be a consequence of the evolutionary process [20]. For instance, squamous cell lung cancer genes are the most central in human protein interaction maps, sharing similar topological features as essential proteins [47]. Moreover, eigenvector and degree centrality were used to identify genes related to prostate cancer [35] with high accuracy. This relation between centrality and functions in biological networks has allowed disease gene predictions and can help to develop individualized therapies and potential cures for many diseases [20].

In cortical networks, centrality also plays a fundamental role. Centrality measures are fundamental to quantify the large-scale organization of cortical networks. For instance, by using fMRI data Achard et al. [1] verified that the human brain is dominated by a neocortical core of highly connected hubs, that have long-distance connections to other regions. These hubs are associated with network robustness since cortical networks revealed to be more resilient to a targeted attack on their hubs than a comparable scale-free network. The authors verified that regional hubs represent large-scale neocortical-thalamic circuits, which are related to human cognition and consciousness. Therefore, central nodes play a fundamental role in brain functioning. In addition to the degree, different centrality measure captures different aspects of connectivity, as verified by Zuo et al. [48]. The authors considered data from 1003 subjects and verified age-related decreases in degree centrality within precuneus and posterior cingulate regions, whereas the eigenvector centrality remains constant with age. The difference in centrality level was also verified in the case of child-onset schizophrenia. Arruda et al. [13] observed that cortical networks of schizophrenic subjects present higher values of the variance of the closeness centrality and accessibility, but small values of average k-core. By considering these measures, they verified that it is possible to perform reliable automatic diagnostics in patients with schizophrenia with a sensitivity of 90% and specificity of 74%. Among 54 network measures extracted from the cortical networks, the authors verified that only four (three related to centrality) of them differ schizophrenic and healthy subjects.

In climate data analysis, regions presenting the largest values of betweenness centrality are related to global surface ocean currents [15]. The authors verified that the nodes with the highest betweenness centrality form the backbone of the network,

that has an essential role on the oceanic surface circulation [15]. In world trade networks, central countries hold many trade partners and the richest nations have a very intense trade relationship [17]. In the worldwide air transportation network, the most connected cities are not necessarily the most central [23], which occurs because of the multi-community structure of the network. Several applications of network theory on the analysis of complex systems can be found in [12].

10.5 Perspectives

Centrality can be defined in terms of different concepts, like random walks and connectivity. The choice of the most suitable measure to characterize a complex network is application dependent, as we have verified in the case of the identification of influential spreaders. This lack of generality opens a new possibility for proposing new centrality measures. Moreover, centrality measures are little explored in the characterization of many dynamical processes, like cooperation and opinion formation. For instance, in cooperative processes, only a few works verified the importance of centrality for improving learning tasks [39].

Moreover, new kinds of networks have been studied in the last years. Many complex systems exhibit more than one type of interaction, like in social networks, where the same set of users share information in Twitter and Google+. In this case, the system's organization can be represented by a multilayer network [7, 29], in which each layer represents one type of interaction. In these networks, the definition of centrality has to consider the edges of each layer and the connections between layers. Recently, some versions of network centrality indices have been proposed for quantification of node importance [7, 14, 29]. However, there are many possibilities of research on the applications of these new metrics for characterization of complex systems, like in ecology, economy, and social networks, as well as the study of the influence of central nodes on dynamical processes in networks.

Temporal networks are also new generalizations of static complex networks [24]. In this case, the connections between nodes change with time, which affects the system behavior, as in epidemic spreading in social networks whose contacts occurs in different time intervals. Centrality measures have been adapted to these networks. However, they have mostly been generalizations of static network measures and more improvements and further analyses are needed [24]. These studies are very important for a better understanding of the evolution of complex systems and the role of centrality for dynamical processes, like epidemic spreading and synchronization in networks whose connections are not static.

Although centrality is a key concept in network theory, these generalizations and applications in many fields should help us to understand the structure, function, and dynamics of complex systems. A better understanding of the relation between centrality metrics and dynamics can provide efficient tools to control and forecast different dynamical processes.

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Chapter 11
Characterizing Flows by Complex Network Methods



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11.1 Introduction

Flows are a key concept in mathematics as well as various branches of the natural sciences (e.g., hydrology, oceanography, atmospheric sciences, astrophysics, chemistry, biology), economy (trade, monetary flows), and engineering (e.g., traffic, telecommunication). In particular, dynamical systems theory is ultimately based on the study of a system's evolution in its associated phase space, which is described by some evolution operator. For deterministic systems, the associated flow described by a set of differential or difference equations (depending on whether the system is time-continuous or time-discrete) determines uniquely the system's state at any time $t > t_0$, given its initial state at t_0 and governing equations are completely known. In turn, for stochastic systems, the evolution operator rather acts on probability density functions and thus provides probabilistic information on the development of the system's state as time proceeds.

The generic description of dynamical systems using the concept of flows finds its analogy in passive advection processes in fluids, which are described by simple

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first-order ordinary differential equations linking the position of some passive tracer particle at a given time $t > t_0$ with the imposed velocity field. The resulting *Lagrangian* perspective on the underlying flow is usually complemented by the *Eulerian* viewpoint, where the associated velocity field is analyzed based on the patterns of the vector field themselves rather than ensembles of individual trajectories advected within the latter. Indeed, for time-dependent or higher-dimensional ($d \geq 3$) fields, both aspects provide complementary information. For example, the motion of particles in a fluid can be chaotic even if the underlying velocity field is completely regular (e.g., stationary in three, or time-periodic in two dimensions), which is thus referred to as *Lagrangian chaos*.

Drawing upon the long history of graph theory, starting with Euler's solution of the classical problem of the Seven Bridges of Königsberg in 1736, statistical physicists have encountered the paradigm of complex networks as a useful concept to describe mutually interconnected structures across a variety of scientific disciplines. By now, complex network theory [1–3] has become an integral part of complex systems science, and has widely proven its relevance for modeling and understanding the structure formation and dynamics in a broad range of applications. In the course of these developments, a plethora of quantitative characteristics and algorithms for describing different aspects of network structures have been proposed [2, 4–7]. Among others, one particularly remarkable achievement is the establishment of mathematical linkages between network structure and functionality, especially in the context of synchronization phenomena [2, 8].

Besides many other fields, complex network theory has recently found its use in the context of geophysical fluid dynamics [9–12]. Here, distinct regions of a fluid domain (in the atmosphere or ocean) are represented by nodes in a geographically embedded network structure. The associated links describing the network's intrinsic connectivity pattern can represent different types of interrelationships—physical or statistical connections. On the one hand, we may consider the amount of material transported with the flow between different regions (aka nodes) within a given time window of length Δt as a proxy for the existence and strength of directed physical linkages. Practically, this relates to the probability of a passive tracer starting in one region at time t_0 to be found in another specific region at time $t = t_0 + \Delta t$, which can be estimated by studying ensembles of trajectories, hence taking a Lagrangian perspective. The most widely applied approach for establishing a network representation of this kind is thus referred to as *Lagrangian flow networks*. On the other hand, we can make use of statistical interrelationships (e.g., correlations) between the time-dependent vector field at distinct points in space, and establish network connectivity if the temporal variability patterns of the local field exhibit a sufficiently large degree of similarity [13]. Obviously, this Eulerian viewpoint is only useful if the corresponding time-dependence is sufficiently nontrivial (i.e., if the vector field is neither stationary nor time-periodic). In practical situations, the exact vector field is, however, often not known. Instead, what can be observed is another (scalar-valued or vector) field of some physical variable

that changes in space and time according to advection and diffusion, but also other dynamical processes (like chemical reactions, biological population dynamics, or others) [14, 15]. Drawing upon the similarity between spatially separated (simultaneous or mutually lagged) observations of this proxy field, one obtains a so-called *functional network* representing the underlying flow patterns together with signatures of the aforementioned physical processes. In the context of fluid dynamics, this technique relates to the Eulerian perspective on a flow field, but is itself neither strictly Eulerian nor Lagrangian, but rather presents some kind of hybrid approach. As a consequence, in contrast to Lagrangian flow networks, where an interpretation of the network structures in terms of flow patterns is relatively straightforward, the attribution of network patterns to specific flow characteristics is a non-trivial problem in the case of such (correlation-based) functional networks.

We emphasize that both, Lagrangian flow networks and functional networks have a broad range of possible applications beyond the actual field of (geophysical) fluid dynamics. Specifically, the idea of functional network representations can be used as a general tool of multivariate data analysis, thus presenting some kind of nonlinear generalization of classical dimension reduction approaches like principal component analysis [16]. The term *functional connectivity* has actually been coined in the context of neuroscience applications, where this approach has been used for identifying connections among brain regions showing similar electromagnetic activity patterns under specific tasks [17, 18]. In parallel, much work has been devoted to establishing this concept in the fields of climatology [19–22] as well as financial markets [23–25].

In turn, the Lagrangian flow network description provides a generic tool for understanding phase space structures in general dynamical systems, and can, therefore, be used independent of specific applications. Notably, this approach is based on stochastic transition matrices approximating the transfer operator of the underlying flow. Similar transition matrices play a crucial role in some specific type of complex network representations in the context of time series analysis, so-called transition networks, which are based on sequences of discrete states (in case of discrete or coarse-grained continuous state variables) or dynamical patterns (e.g., order patterns) in arbitrary uni- or multivariate time series [26–28].

This chapter provides some overview on the state of the art of both, Lagrangian flow networks and correlation-based functional networks, together with some illustrative examples drawn from relatively simple mathematical models of paradigmatic flow patterns as commonly arising in real-world geophysical flows. We start by reviewing the terminology of flows in the context of dynamical systems theory as well as fluid dynamics and beyond, followed by a brief overview of relevant concepts from complex network theory. Subsequently, we present the general ideas behind Lagrangian flow networks and correlation-based functional networks in some detail and illustrate their applications by selected insightful examples.

11.2 Terminology

11.2.1 Flows

For the sake of simplicity, let us consider a time-continuous deterministic dynamical system described by a first-order ordinary differential equation

$$\frac{d}{dt}\mathbf{x}(t) = \mathbf{F}(\mathbf{x}(t), t), \tag{11.1}$$

where $\mathbf{x}$ can be of arbitrary finite dimension. A time-discrete dynamical system (map) can be considered as a temporally coarse-grained version of Eq. (11.1) and shall thus not be further considered here.

In the context of fluid dynamics, Eq. (11.1) describes the passive advection of particles in a flow that is described by a velocity field $\mathbf{F}(\mathbf{x}(t), t) = \mathbf{v}(\mathbf{x}(t), t)$. This interpretation allows analyzing velocity fields from both, Eulerian and Lagrangian viewpoints. Specifically, an Eulerian analysis would consist of studying the spatiotemporal patterns of $\mathbf{v}(\mathbf{x}, t)$ (i.e., its local magnitude and directionality) as a whole, allowing to identify different types of behavior like regions with straight flow, rotation cells, or possible stagnation points of the flow at which the velocity modulus vanishes. However, such an analysis possibly provides only limited information on the underlying dynamical structures governing the differential behavior of the flow in different parts of the fluid domain, such as the existence and (time-dependent) location of transport barriers. In order to obtain more detailed knowledge about all aspects directly related with the (advective) transport within the flow domain, Lagrangian approaches are useful. Unlike Eulerian methods, these approaches generally consider ensembles of solutions of Eq. (11.1) with different initial conditions $\mathbf{x}(t_0)$ and evaluate their (statistical) properties in dependence on $\mathbf{x}(t_0)$ and the time window used for integrating the trajectories of the corresponding tracer particles.

More generally, the idea of complementary Eulerian and Lagrangian studies can be transferred to other vector fields. For example, consider the case of a magnetic fluid where the velocity and magnetic fields mutually depend on each other (see Fig. 11.1 for some corresponding example). The advected tracer perspective

Fig. 11.1 (continued) (**f**) $z = z_s + \pi/2$ with tracers initiated close to (**d**) $(\pi, \pi, z_s + \pi)$ (α-point, i.e., a three-dimensional saddle point of the velocity field with two-dimensional stable manifold) and (**f**) $(\pi, \pi, z_s + \pi/2)$ (β-point, i.e., a saddle point with two-dimensional unstable manifold). The combined magnetohydrodynamic equations have been simulated at kinematic Reynolds number $Re = 4.0$ and magnetic Prandtl number $Pm = 1$. Note the different subdomains shown in the three Poincaré sections. One clearly observes the periodic structures and KAM tori embedded in the layers of Lagrangian chaos in both, the velocity and magnetic field, which are to be expected since both fields are divergence-free ($\nabla \cdot \mathbf{v} = \nabla \cdot \mathbf{B} = 0$) and thus constitute three-dimensional Hamiltonian systems

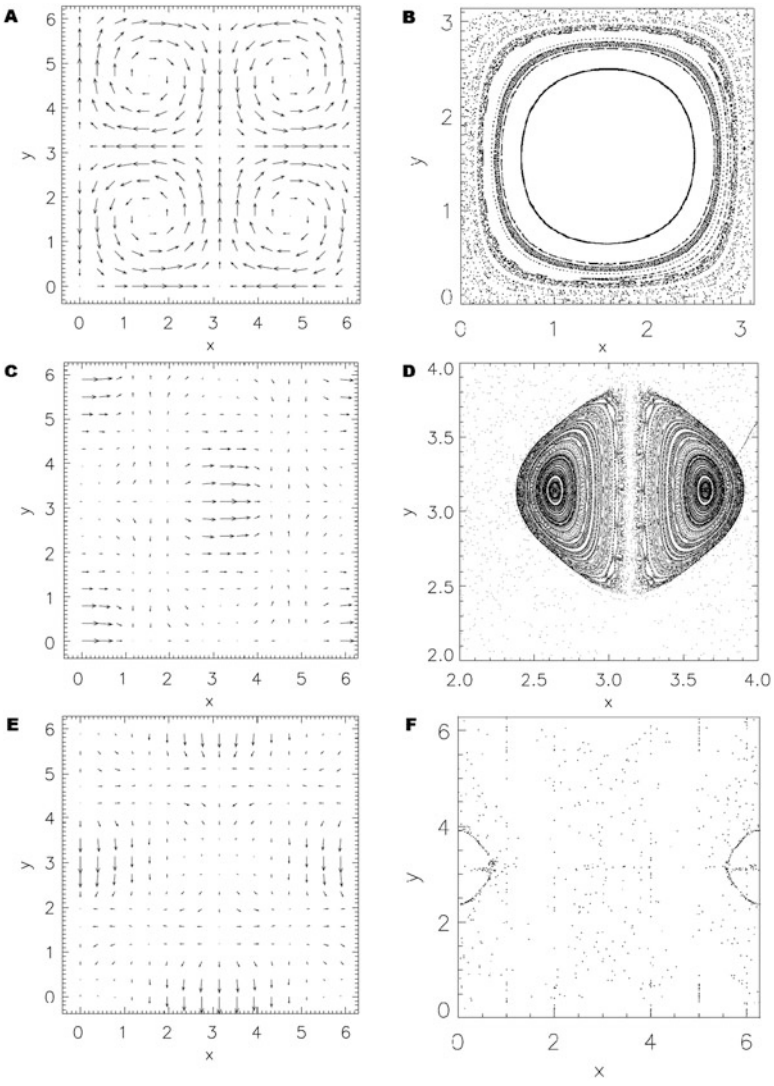


Fig. 11.1 Example of the Eulerian and Lagrangian structures embedded in the velocity and magnetic fields of a three-dimensional low-order Fourier truncation of a dynamo model driven by a Roberts flow ($\mathbf{f} = (\sin x \cos y, -\cos x \sin y, 2 \sin x \sin y)$) (for a detailed description of the system, see [29–31]). (**a**) Two-dimensional projection of the flow pattern in the $z = 0$ plane. (**b**) Poincaré section of trajectories in the velocity field at $z = 0$. (**c** and **e**) Two-dimensional projections of the magnetic field in two planes at (**c**) $z = z_s$ and (E) $z = z_s + \pi/2$, where z_s denotes the first "horizontal" layer within $z \in [0, 2\pi[$ that contains stagnation points of the flow. (**d** and **f**) Poincaré sections of trajectories of the dynamical system $\frac{d}{dt}\mathbf{x}(t) = \mathbf{B}(\mathbf{x}(t))$ at (**d**) $z = z_s + \pi$ and

describes the physical motion of some passive and non-magnetic particle in the flow with $\mathbf{x}(t)$ denoting the particle position at time t and $\mathbf{F}$ representing the velocity field $\mathbf{v}(\mathbf{x}, t)$. In turn, we may also study Eq. (11.1) with $\mathbf{F}$ representing the magnetic field $\mathbf{B}(\mathbf{x}, t)$, revealing possibly interesting properties of the latter even though $\mathbf{x}(t)$ has no direct physical interpretation in this case.

Under general conditions, $\mathbf{F} : \Omega \times \mathbb{R} \to \Omega$ describes a continuously differentiable vector field on a smooth, compact manifold $\Omega \subset \mathbb{R}^n$. If $\mathbf{F}$ does not exhibit any explicit time dependence, Eq. (11.1) describes an *autonomous* system, otherwise a *non-autonomous* or *time-dependent* one. The associated *flow map*

$$\Phi : \Omega \times \mathbb{R} \times \mathbb{R} \to \Omega;\ (\mathbf{x}_0, t_0, \Delta t) \mapsto \mathbf{x}\,(t_0 + \Delta t)$$

gives the position of a passive tracer released at point $\mathbf{x}_0$ at time t_0 after being advected for a time Δt. Here, $\mathbf{x}(\bullet) : \mathbb{R} \mapsto \Omega$ denotes the solution of Eq. (11.1) with initial condition $\mathbf{x}\,(t_0) = \mathbf{x}_0$.

11.2.2 Complex Networks

11.2.2.1 Mathematical Background

A *network* can be mathematically described as a *graph* $\mathcal{G} = (V, E)$ with V denoting a set of nodes (or vertices) and E a corresponding set of links (or edges). For the sake of simplicity, we label the elements of V with $\{1, \ldots, N\}$ so that each node of the network is uniquely identified by one index value (i.e., the cardinality of the node set reads $\#V = N$) and $E \subseteq V \times V$. In common language, we speak of a *complex network* if the associated network topology (see below) is non-trivial, i.e., does neither exhibit the structural characteristics of a regular graph nor those of a completely random network.

The most convenient way to describe the connectivity structure of $\mathcal{G}$ is the so-called *adjacency matrix* $\mathbf{A}$, which is a binary matrix with entries $A_{ij} = 1$ if $(i, j) \in E$ and $A_{ij} = 0$ otherwise. In case of an undirected network, $\mathbf{A}$ is symmetric, i.e., $\mathbf{A}^T = \mathbf{A}$. Otherwise, there exist some $i, j \in V$ for which we have a link from node i to node j, but not vice versa.

For some applications to be discussed in this chapter, it is furthermore relevant to also allow for additional weights associated with the individual links. For example, in the case of Lagrangian flow networks, a corresponding link weight naturally represents the fraction of a flow (or probability of a passive tracer) starting from a specific fluid region to end in some other. In this case, we can extend $\mathbf{A}$ to a weighted adjacency matrix $\mathbf{W}$. Commonly, a corresponding weighted network can be transformed into an unweighted one by thresholding, i.e., by setting $A_{ij} = \Theta(W_{ij} - W_{ij}^*)$, where W_{ij}^* describes a characteristic weight above which a link between nodes i and j is considered to be relevant for the overall network structure, and $\Theta(\bullet)$ is the Heaviside function.

11.2.2.2 Topological Network Characteristics

Topological network characteristics describe different aspects of the network structure by taking only information from the associated adjacency matrix into account. They can be grouped according to different classification schemes. On the one hand, we may distinguish local (per-node or per-link) network properties from mesoscale features (characterizing subgraphs) and global characteristics (describing the entire network structure with single characteristic numbers). On the other hand, one can differentiate between properties taking into account only local information (associated with a single node, i.e., contained in individual rows or columns of $\mathbf{A}$ or $\mathbf{W}$), mesoscale information (considering the connectivity structure of a node i's *neighbors* $\mathcal{N}_i = \{j \in V : A_{ij} = 1\}$ as well) or global adjacency information (on connectivity properties between all node pairs in the network). The latter group of characteristics commonly makes use of the concept of paths, i.e., mutually exclusive ordered sequences of edges that need to be traversed to get from a certain node i to another node j. Most importantly, shortest paths (i.e., paths of minimum length l_{ij}) are often used to define topological network characteristics. In the following, we will provide a brief overview of some selected examples of topological network measures which have found particularly wide use among various fields of application. For a more detailed account on these as well as other measures, we refer to the corresponding literature [1, 2, 4–6].

The simplest network properties are the (global) *link density* $\rho = \sum_{i,j=1}^{N} A_{ij}/N^2$, measuring the fraction of possible edges that actually exist in the network (i.e., $\rho = \#E/(\#V)^2$) and the (local) node *degree* $k_i = \sum_{j=1}^{N} A_{ij}$. Regarding the latter, in the case of directed networks, one needs to distinguish between out-degree $k_i^{out} = \sum_{j=1}^{N} A_{ij}$ and in-degree $k_i^{in} = \sum_{j=1}^{N} A_{ji}$. Moreover, a generalization to weighted networks is provided by the node strength $s_i = \sum_{j=1}^{N} W_{ij}$, which can be further refined for directed networks in full analogy to the node degree. All aforementioned measures capture exclusively local adjacency information by definition. Another generalization of the node degree is the so-called eigenvector centrality e_i, which is defined as the coefficient of the first eigenvector (corresponding to the largest eigenvalue of $\mathbf{A}$) associated with a given node i.

The most common examples of network characteristics that also take connectivity information from the neighborhood of nodes into account are clustering measures. Specifically, the *local clustering coefficient* $\mathcal{C}_i$ gives the probability that two randomly selected neighbors $j, k \in \mathcal{N}_i$ of a node i are mutually connected themselves. Taking the arithmetic mean over all nodes i yields the *global clustering coefficient* $\mathcal{C} = \sum_{i=1}^{N} \mathcal{C}_i/N$. Since the value of this measure can be severely affected by the probability distribution of the node degree (i.e., low and high degree nodes contribute equally strong to the sum), a common alternative is provided by the *network transitivity*, which gives the fraction of connected node triplets in the network that exhibit all three possible links [32]. While the formulation of the aforementioned three measures is relatively straightforward for undirected and unweighted networks, directed networks exhibit different possible clustering

coefficients distinguishing between the different directionality patterns of the three considered links [33]. The latter is closely related to the concept of network motifs [34], i.e., connected subgraphs of a certain size (here, 3-node subgraphs).

Finally, we mention some examples from the large class of path-based network characteristics. At the local scale, two corresponding per-node measures are *closeness* c_i and *local efficiency* $\mathscr{E}_i$, which are defined as the inverse arithmetic and harmonic means of the shortest path lengths l_{ij} between a given node i and all other nodes j of the network, respectively [35, 36]. In case of networks consisting of disconnected components, c_i would take a value of zero since the shortest path length between two nodes from two separate components is infinite. In order to account for this, one commonly sets $l_{ij} = N - 1$ in such case. A similar approach is taken in case of the corresponding global network measure, *average path length* $\mathscr{L} = \sum_{i \neq j} l_{ij}/N(N-1)$, while the global counterpart of $\mathscr{E}_i$, the *global efficiency* $\mathscr{E}$ is not affected by this complication. In case of directed networks, distinguishing measures based on incoming and outgoing links, respectively, is straightforward, while the generalization to weighted networks may require the transformation of link weights into link-specific distances, which allows for different functional descriptions.

Beyond measuring the shortest path lengths with respect to individual nodes, it can be useful to also quantify the share of specific nodes or links on the shortest paths between all pairs of nodes in the networks. This property, which can often be understood as highlighting bottlenecks for flow processes on a given network structure, is described by the so-called (node or edge) *betweenness*. Similarly to the degree, eigenvector centrality and closeness, betweenness provides a measure related to the centrality of nodes with respect to the overall network architecture.

Since in many applications, long distances on a network structure are not of much relevance, recent work has suggested the use of truncated path-based measures like *cutoff closeness* or *cutoff betweenness*, the definition of which involves only node pairs that exhibit a shortest path length below a prescribed maximum value [37–39]. The study of such truncated measures has recently been found particularly useful in the context of Lagrangian flow networks [40].

11.2.2.3 Spatial Network Characteristics

In many cases, flow networks exhibit a natural embedding in geographical space or, in the case of dynamical systems, some more general metric (phase) space. In such cases, topological network properties which characterize the connectivity structure may miss important network characteristics that originate from the spatial embedding of the graph [41]. In turn, spatial network measures taking geometric properties into account may provide useful complementary information. One classical example for such a property is the spatial link length (or link distance) distribution $p(d)$, where d_{ij} describes the metric distance between two nodes $i, j \in V$ that are connected by a corresponding link (i.e., $A_{ij} = 1$). Beyond such metrically scaled characteristics, the orientation of edges associated with a single

node or globally is another potentially relevant characteristic. Examples for spatial network measures characterizing such properties include the orientation (or trend) entropy used in the context of street network analysis [42–44] and the local and global edge anisotropy [45].

It is important to recognize that spatial embedding of a network may crucially affect some of the topological characteristics, especially in cases where the probability of two nodes to be connected by a link depends on their spatial distance (e.g., in functional brain or climate networks) [46, 47]. In such cases, for example the clustering properties may be elevated in comparison with values expected if the former dependency is missing [48]. As a consequence, in order to differentiate inherent topological properties from such induced by the spatial structure of the graph, one promising strategy is employing surrogate networks that respect certain spatial network characteristics while destroying the topological properties by some spatially constrained rewiring of edges [49].

11.3 Lagrangian Flow Networks

In a fluid dynamic context, Lagrangian flow networks are based on defining connectivity among different parts of a fluid according to the amount of material transported between them [10–12, 50–52]. In the case of general dynamical systems, nodes represent regions of phase space, and the links describe the system's evolution between subsequent instances of time [50–53]. Previous work on fluid flows has shown that it is possible to identify key quasi-stationary structures embedded in the flow, so-called *almost-invariant* or *coherent sets*, from the associated transfer operator by making use of spectral methods. This *set-oriented* approach has already found many applications [52–61]. Practically, the transfer operator is approximated by some transition matrix (describing the probabilities of fluid transport between pairs of regions within a predefined time step Δt), which can be interpreted as the weighted adjacency matrix of a Lagrangian flow network. This identification allows drawing upon concepts from complex network theory like community detection methods [9, 10, 62] or path-finding algorithms [11, 12] to obtain further information on the structural organization of the flow.

In the following, we will summarize the main concepts underlying the construction of Lagrangian flow networks and illustrate them by some simple examples.

11.3.1 Mathematical Background

Numerical studies of flows commonly rely on a spatial discretization of the (compact) flow domain Ω into a (sufficiently large) number N of spatially connected yet mutually disjoint boxes B_i, $i = 1, \ldots, N$. This partition can be obtained by using either boxes of equal or adaptive size, the latter being particularly relevant in the

context of set-oriented approximations of invariant structures within the flow [63]. Taking the time interval $[t_0, t_0 + \Delta t]$ as a basis for further analysis, the proportion of mass transported from B_i to B_j is given by

$$P_{ij}\left(t_0, \Delta t\right) = \frac{\mathrm{vol}\left(B_i \cap \Phi\left(B_j, t_0 + \Delta t, -\Delta t\right)\right)}{\mathrm{vol}\left(B_i\right)}, \tag{11.2}$$

where $\mathrm{vol}\left(X\right)$ is the Lebesque measure of a subset $X \subseteq \Omega \subset \mathbb{R}^n$. The thus defined matrix $\mathbf{P} = \left(P_{ij}\right)$ is referred to as the *transition matrix* $\mathbf{P}\left(t_0, \Delta t\right)$ of the flow. In a dynamical systems context, it provides a spatial discretization of the associated *transfer operator* or *Perron-Frobenius operator*, a construction known as *Ulam's method* [64, 65]. Obviously, the detailed properties and approximation quality of Eq. (11.2) depend on the choice of the partition, with finer partitions commonly improving the accuracy of the approximation, but dramatically increasing the associated computational efforts. A detailed discussion on the numerical estimation and the resulting Markov properties of $\mathbf{P}$ can be found in [40, 66, 67].

In Lagrangian flow networks, the transition matrix (Eq. (11.2)) is interpreted as a weighted adjacency matrix, and each box B_i of the considered spatial partition is identified with a node i of the network. Consequently, a weighted link between two nodes i and j exists if and only if material is transported between the associated boxes B_i and B_j during a time step Δt. An unweighted yet directed network representation of the flow is obtained by setting

$$A_{ij}\left(t_0, \Delta t\right) := \Theta(P_{ij}\left(t_0, \Delta t\right)). \tag{11.3}$$

11.3.2 Examples

11.3.2.1 Meandering Flow Model

Real-world examples for meandering flows include important ocean currents like the Gulf stream and the Kuroshio. One of the most widely studied models of such meandering flows [68–71] is based on the stream function

$$\Psi\left(x, y, t\right) := 1 - \tanh\left[\frac{y - B\left(t\right)\cos\left(k\left(x - ct\right)\right)}{\sqrt{1 + k^2 B^2\left(t\right)\sin^2\left(k\left(x - ct\right)\right)}}\right] \tag{11.4}$$

with

$$B\left(t\right) = B_0 + \varepsilon\cos\left(\omega t + \theta\right). \tag{11.5}$$

The explicit equations of motion in x and y-direction can be easily derived as the solutions of

$$\dot{x} = \frac{\partial\psi}{\partial y}, \;\; \dot{y} = -\frac{\partial\psi}{\partial x}, \tag{11.6}$$

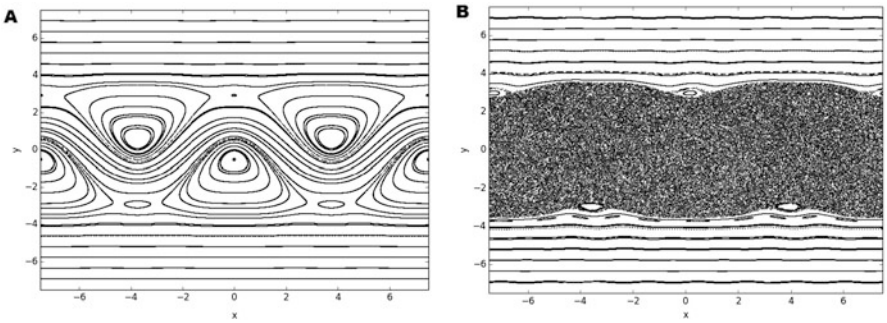


Fig. 11.2 Stroboscopic map of the meandering flow model (Eq. (11.4)) with (**a**) $\varepsilon = 0$ and (**b**) $\varepsilon = 0.3$. 80 particles were initialized on four different vertical axes with $x = -3.25, 0, 3.25, 7.5$ and advected for 2000 time steps of length $\Delta t = 62.5$ corresponding to the time periodicity of the flow

Here, we study the set of model parameters $k = 2\pi/L_x$, $L_x = 7.5$, $B_0 = 1.2$, $c = 0.12$ and $\theta = \pi/2$, Moreover, we consider the cases $\varepsilon = 0$ and $\varepsilon = 0.3$. The choice of $\omega = 4kc \approx 0.4$ ensures the flow's periodicity in time with a period of $L_x/c = 62.5$.

Figure 11.2a shows a stroboscopic map of the meandering flow with $\varepsilon = 0$. The main meander is clearly discernible from the regular vortices that form "north" and "south" of it. Additional small tori are visible at every turning point of the meandering wave. If we set $\varepsilon = 0.3$ and thereby consider the meander with varying amplitude, the vortices merge with the main meander and form a broad band of chaotic dynamics. Only some small tori remain with their locations as in the undisturbed setting, see Fig. 11.2b.

For studying this model in terms of the properties of the resulting Lagrangian flow network representation, we partition the domain $[-7.5, 7.5] \times [-7.5, 7.5]$ into 200×200 boxes.

For the case $\varepsilon = 0$, in- and out-degree patterns capture the main features of the meandering flow (see Fig. 11.3a). In particular, the central meander region is delimited by ridges in the degree field. Vortex-like structures, so-called rotation cells, are visible in both, in- and out-degree, "north" and "south" of the meander with an opposite direction of rotation. More detailed numerical analysis (not shown) reveals that the resulting structures of in- and out-degrees closely match those of the time-backward and time-forward finite-time Lyapunov exponents [72], respectively, as expected from previous studies [10, 40]. Specifically, since ridges in the FTLE field have been shown to be transport barriers [73], we expect the same to hold for ridges in the degree field. Since the regions north and south of the central meander are separated by pronounced ridges in the degree field we expect only minimal mixing between them [68].

Figure 11.3b–d demonstrates the capability of the concept of cutoff closeness to resolve different structures when using different cutoff values. Note that the cutoff-1 out-closeness is analytically related with the out-degree via a simple,

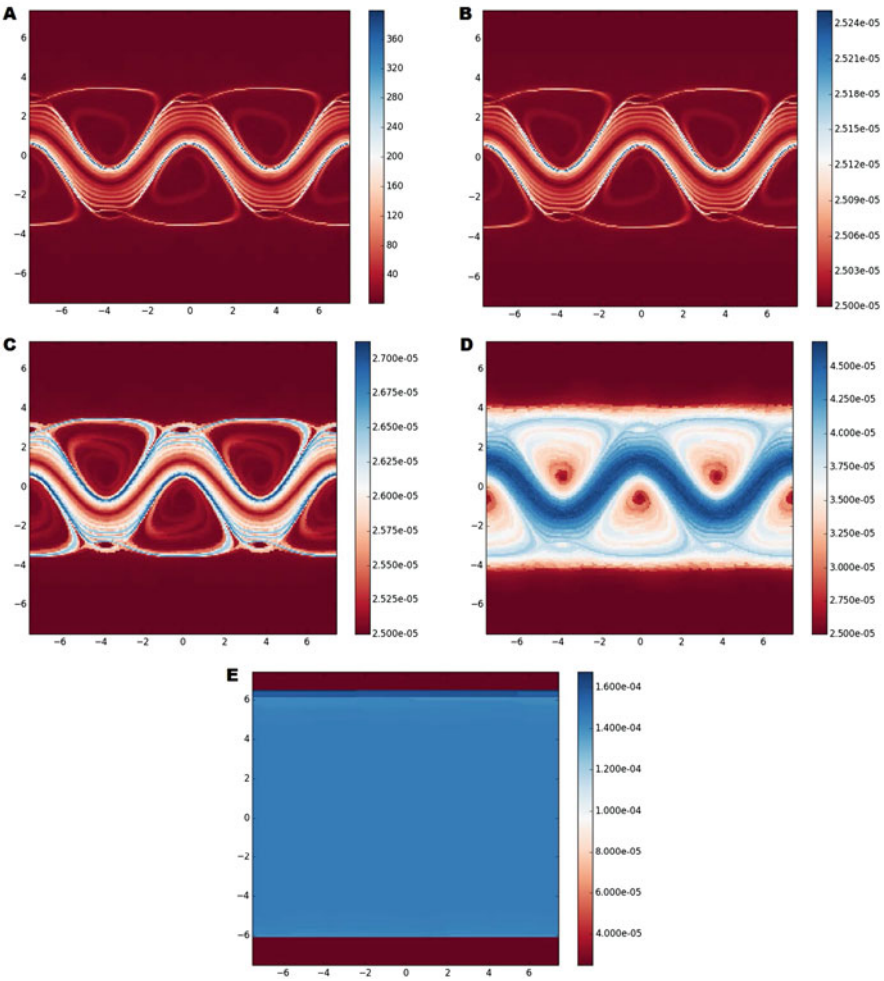


Fig. 11.3 Spatial patterns (color-coded) of (**a**) out-degree, cutoff out-closeness with cutoffs (**b**) 1, (**c**) 3 and (**d**) 20, and (**e**) normal out-closeness for the meandering flow (Eq. (11.4)) with $\varepsilon = 0$

strictly monotonic transform [40]. In general, for low cutoff values the meander and vortex boundaries are relatively well expressed. With increasing cutoff, the overall highest values of cutoff closeness migrate to the central meander of the flow and then spread over the whole flow domain. In case of standard closeness (i.e., considering a cutoff at N, Fig. 11.3e), all nodes with degree larger than 1 take the same value (and so do nodes with degree 1). A clear separation between the central, meandering region and the almost static "far field" is visible. However, this separation results exclusively from discretization effects, since far from the central meander, the velocity modulus is so low that particles practically don't leave their

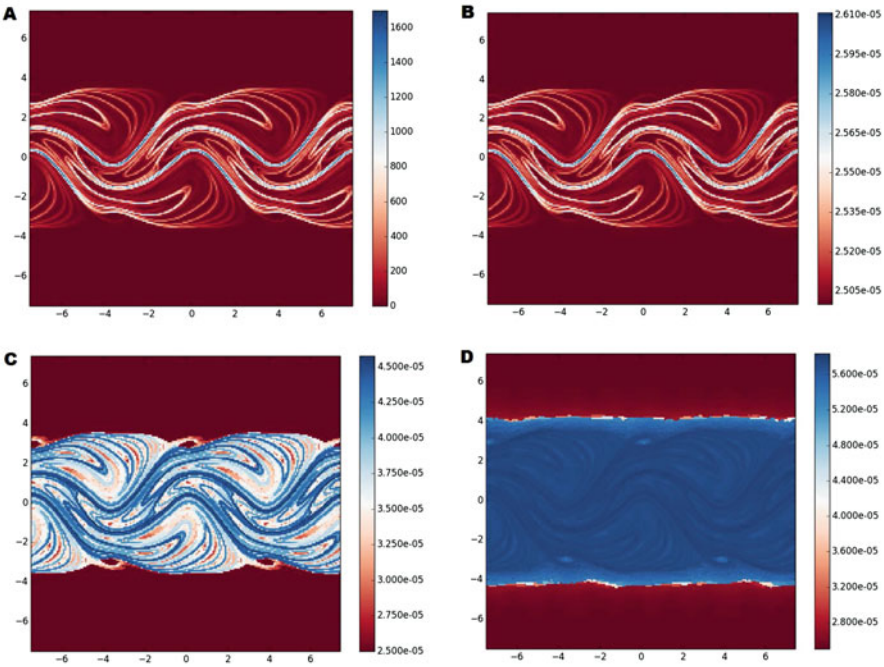


Fig. 11.4 Same as in Fig. 11.3a–d for $\varepsilon = 0.3$

initial boxes anymore within the considered time step Δt. In a similar spirit, the homogeneity of the structures exhibited by the cutoff closeness in the corresponding regions of the flow may be explained as well.

If we turn on the periodic modulation of the flow field ($\varepsilon = 0.3$), the resulting out-degree field has a far more complex filamental structure (Fig. 11.4a). This is in accordance with earlier findings [68] reporting that the main dynamics in the meander is chaotic for this configuration of ε and ω. North-South mixing is still weak, but more significant than for $\varepsilon = 0$. Comparing the closeness with cutoff for the meandering flow with and without periodic excitation, we indeed find that the mixing happens a lot faster in case of a periodic modulation (Fig. 11.4b–d). For $\varepsilon = 0$, at a cutoff of 20 only the central region of the meander exhibits high cutoff out-closeness values, whereas the rotation cells show low values. In turn, for $\varepsilon = 0.3$ the whole jet including the rotation cells displays high values. It is interesting to note that the periodic modulation hardly affects the dynamics around the KAM tori located at the wave crests. In both configurations of ε, these tori remain visible as local minima of the cutoff closeness field (except for very high cutoffs, when all closeness values are practically the same).

Betweenness is one of the network properties that has not yet been studied extensively in the context of Lagrangian flow networks. When calculating its cutoff version, we find that this measure highlights the invariant tori also observed in

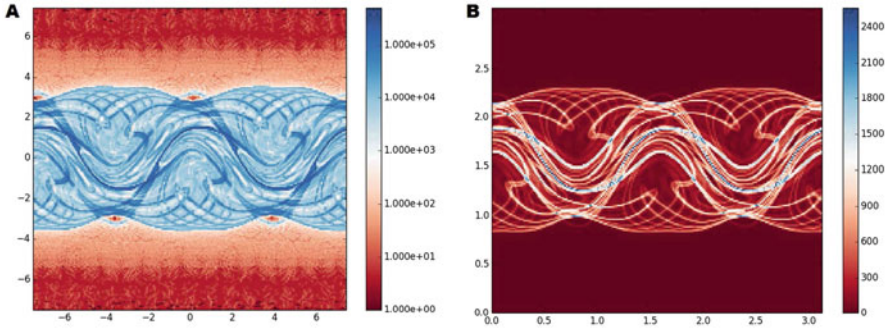


Fig. 11.5 Spatial patterns (color-coded) of (**a**) cutoff-3 betweenness and (**b**) total degree for the meandering flow field (Eq. (11.4)) with periodic modulation ($\varepsilon = 0.3$)

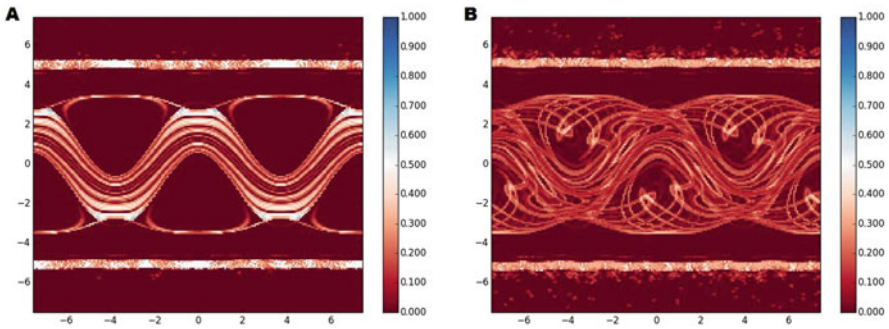


Fig. 11.6 Spatial patterns (color-coded) of local clustering coefficients (taken for a non-directional version of the Lagrangian flow network) for the meandering flow field (Eq. (11.4)) (**a**) without and (**b**) with periodic modulation ($\varepsilon = 0.3$)

the closeness field as pronounced, almost global minima (Fig. 11.5a). Notably, this property is visible at both, low and high cutoff values. Beyond this remarkable observation, cutoff betweenness exhibits a filamental structure, which resembles closely the pattern of the total degree (i.e., the sum of in- and out-degree, see Fig. 11.5b).

Finally, we also show here the patterns of local clustering coefficients for the two cases without and with periodic modulation (Fig. 11.6). As expected, in both cases, some substructure of the main meander is recovered as a dominating feature. In addition, some "horizontal" lines are characterized by elevated clustering coefficients. Given the definition of this measure, it is most likely that these structures mainly arise due to the specific discretization in space and time employed in our numerical calculations and are not stable if the spatial partition and/or the time step of our analysis are varied. Note that clustering coefficients are particularly suited to identify time-periodic structures with periods corresponding to three times the considered time step [74].

11.3.2.2 Driven Vortex Flow

Another omnipresent pattern in many geophysical flows are vortices (e.g., cyclones, tornadoes, or ocean gyres). A generic model describing a chain of periodically driven vortices [75–78] can be represented by the stream function [78]

$$\psi(x, y, t) = A(1 + \varepsilon \sin \pi t) \sin y + B \sin 2x \sin y, \tag{11.7}$$

where the first term generates a shear flow and the second term a chain of driven vortices. ε again measures the strength of the temporal modulation that is responsible for the emergence of chaotic particle motion in the flow. Here, we consider the parameter values $A = 8.35$, $B = -2.55$, and $\varepsilon = 0.2$ as in [40, 78]. In this case, a thin layer of Lagrangian chaos surrounds one large vortex that is embedded in a laminar shear flow.

The degree and closeness characteristics of the Lagrangian flow networks derived from this model have been extensively studied in [40]. Here, we only show the patterns of cutoff betweenness (Fig. 11.7), which have not been discussed in the original work. For low cutoffs, the measure mainly highlights the invariant manifolds of the embedded saddle point as ridges of the cutoff betweenness field, as well as the regions of laminar flow by low betweenness values. At high cutoffs, we observe a pronounced local minimum in the center of the vortex region and further local minima in the centers of the KAM tori. The system's primary hyperbolic orbit (corresponding to the system's saddle point at $\varepsilon = 0$) coincides with a pronounced local maximum of the betweenness field. When relieving the cutoff, the betweenness patterns retain some of these large-scale features while becoming far more noisy at smaller scales, which is most likely due to discretization effects and the huge number of paths considered.

11.3.3 Methodological Alternatives and Applications

Using network measures for studying Lagrangian patterns in geophysical flows has been proposed first in [9, 10]. A notable alternative yet closely related approach was recently introduced by Hadjighasem et al. [79], who studied ensembles of trajectories with different initial conditions and employed a weighted network representation based upon the mutual distances between individual trajectories to obtain a meaningful clustering of phase space. This technique was then used for detecting coherent vortices embedded in the studied flow pattern. In a similar spirit, it has been recently proposed to generate network patterns from Lagrangian trajectories encoding whether or not two trajectories encounter the vicinity of each other at least once during their evolution [80].

In general, Lagrangian flow networks present a probabilistic approach to Lagrangian dynamics. One main field of application of such approaches is identifying transport barriers associated with (almost) invariant structures embedded

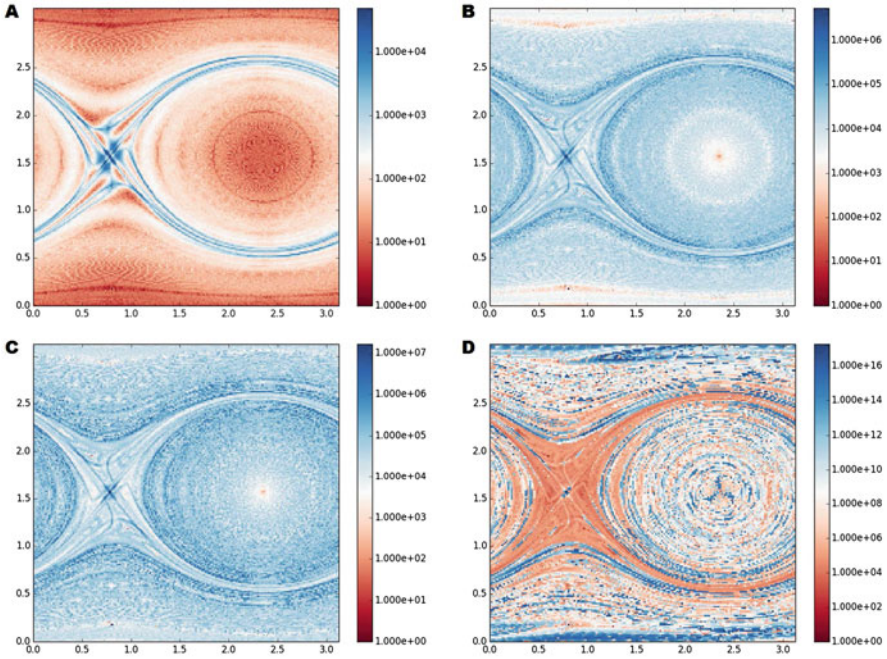


Fig. 11.7 Spatial patterns (color-coded) of cutoff betweenness with cutoff (**a**) 2, (**b**) 10, (**c**) 20 and (**d**) without cutoff for the driven vortex model (Eq. (11.7))

in the flow [57, 64, 81]. Interpreting the transition matrix as a weighted adjacency matrix allows revisiting the results of a large number of previous studies from a complex network perspective. For example, the structure of eigenvectors of the transition matrix allows the detection of almost-invariant sets in time-dependent flows [64], which has been applied to characterizing the approximate locations of the five so-called *garbage patches* in the global ocean surface flow [82]. The same patches were also detected by van Sebille et al. [83] based upon a transition matrix estimated from surface drifter data. Subsequent developments have led to novel measures of nonlinear stretching in terms of families of so-called *finite-time entropies* [10, 84]. A perturbation-theoretic framework for analyzing the efficiency of absorption and flow modification strategies in the context of removing contaminants from flows has been recently presented by Fujiwara et al. [85].

More explicit network interpretations of the transition matrix of flow systems have resulted in a couple of atmospheric and oceanic applications. Among others, Ser-Giacomi et al. established a relationship between the node degree and nonlinear stretching [10] and used the concept of betweenness to obtain maximum flow pathways [11]. Rossi et al. [9] employed a community detection algorithm for complex networks to identify distinct hydrodynamic provinces in the Mediterranean Sea from the associated Lagrangian flow network. Further applications by the same group of authors addressed the detection of key flow structures in the atmosphere and ocean as well as their linkage with marine biology.

11.4 Correlation-Based Functional Networks

Energy and heat transport in the atmosphere and ocean are among the most important drivers of spatiotemporal climate variability. In order to identify the corresponding key structures, functional climate networks have been introduced that represent the statistical similarity between the temporal fluctuations of one or more climate variables observed at different spatial locations (including meteorological stations [86], grid points in climate model outputs or reanalysis datasets [19], or even locations of paleoclimate archives [87]). In most recent studies on climate networks, scalar variables like air temperature, geopotential height, or precipitation have been investigated, while the explicit study of vector fields (especially wind fields) has not yet been systematically addressed [88]. However, most of the identified network patterns have been ultimately related with the atmospheric or ocean circulation, drawing upon general insights from fluid dynamics that the spatiotemporal variability structures of scalar observables might provide a window into analyzing the underlying (unknown) flow pattern.

11.4.1 Mathematical Background

In a series of papers [14, 15, 45], Molkenthin et al. demonstrated how to obtain analytical expressions for the correlation structures of a scalar field originating from a discretization of the advection-diffusion equation. For an incompressible fluid ($\nabla \cdot \mathbf{v} = 0$), a generalized version of this equation (including external forcing, dissipation and noise) can be formulated as

$$\frac{\partial T(\mathbf{x},t)}{\partial t} = \kappa \Delta T(\mathbf{x},t) - \mathbf{v}(\mathbf{x},t) \cdot \nabla T(\mathbf{x},t) - bT(\mathbf{x},t) + F(\mathbf{x}) + \sqrt{D}\xi(\mathbf{x},t), \quad (11.8)$$

where T is the considered scalar field (e.g., temperature or concentration). The five terms on the right-hand side of this equation correspond to (1) diffusion (with a diffusion coefficient κ), (2) advection within the flow field $\mathbf{v}(\mathbf{x},t)$, (3) dissipation (i.e., damping of fluctuations at a rate b), (4) deterministic forcing, and (5) stochastic fluctuations (commonly represented by Gaussian white noise with zero mean, unit variance and zero correlations between values differing in spatial coordinate or time). For the sake of simplicity, any possible back-reaction of $T(\mathbf{x},t)$ on the velocity field $\mathbf{v}(\mathbf{x},t)$ is neglected.

Given proper discretization steps (for the sake of simplicity, we will set $\Delta t = 1$ in the following) and boundary conditions (see [45] for details), for a *stationary* flow field without dissipation and forcing, the dynamics of the vector $\mathbf{T} = (T(\mathbf{x}_1), \ldots, T(\mathbf{x}_N))$ representing the discretized scalar field can be expressed as

$$\mathbf{T}(t+1) = \mathbf{P}\mathbf{T}(t) + s\boldsymbol{\varepsilon}(t), \quad (11.9)$$

where $\boldsymbol{\varepsilon}$ is a vector of independent Gaussian random variables with zero mean and unit variance, s some parameter depending on the stochastic diffusion constant D and the spatial and temporal discretization steps, and $\mathbf{P}$ a one-step transition matrix approximating the time-evolution operator of the advection-diffusion process that is determined by $\mathbf{v}(\mathbf{x})$. Specifically, $\mathbf{P}$ is conceptually related with the transition matrix in the case of Lagrangian flow networks. Iterating the previous equation for subsequent time steps, we can replace the recursive expression (11.9) for $\mathbf{T}$ by an explicit one as

$$\mathbf{T}(t+1) = s \sum_{k=0}^{t} \mathbf{P}^{t-k} \boldsymbol{\varepsilon}(k), \tag{11.10}$$

where the advective dynamics is fully encoded in the matrix elements of $\mathbf{P}$.

For a general *time-dependent* flow, the transition matrix $\mathbf{P}$ becomes explicitly time-dependent as well. Moreover, the effects of additional forcing and dissipation can be integrated with $\mathbf{P}$ [15], replacing the matrix product in the last equation by a general *propagator matrix*

$$\mathbf{G}_{tt'} = e^{-(t+1-t')b\Delta t} \prod_{k=t'}^{t} \mathbf{P}(k) \, \forall \, t \geq t' \tag{11.11}$$

and finally leading to the following expression for the discretized scalar field:

$$\mathbf{T}(t+1) = \mathbf{G}_{t0}\mathbf{T}(0) + \Delta t \sum_{k=0}^{t} \mathbf{G}_{t,t+1-k}\mathbf{F} + s \sum_{k=0}^{t} \mathbf{G}_{t,t+1-k}\boldsymbol{\varepsilon}(t-k), \tag{11.12}$$

where $\mathbf{F}$ describes the spatially discretized forcing term.

Making use of the above representations of $\mathbf{T}$, the associated spatial covariance matrix can be written as [15]

$$\mathbf{R}_T(t) = s^2 \sum_{k=0}^{t-1} \mathbf{G}_{t-1,t-k}\mathbf{G}_{t-1,t-k}^T, \tag{11.13}$$

which in the case of zero forcing, no dissipation and a stationary velocity field reduces to [45]

$$\mathbf{R}_T(t) = s^2 \sum_{t'=0}^{t} \sum_{k=0}^{t'-1} (\mathbf{P}\mathbf{P}^T)^{t'-1-k}. \tag{11.14}$$

Normalization by the corresponding diagonal elements yields the spatial lag-zero correlation matrix $\mathbf{C}_T(t)$ of the scalar field $T(\mathbf{x}, t)$, which can be used to generate a functional network representation.

11.4.2 Example

For illustrative purposes, we utilize here a two-dimensional stationary meandering flow model similar to that studied above in the context of Lagrangian flow networks, but with a slightly different mathematical structure. Specifically, we consider the stream function [69]

$$\Psi(x, y) = 1 - \tanh\left[\frac{y - \sin 2x}{\cos(\arctan(\cos 2x))}\right] \tag{11.15}$$

which has already been studied in [45]. The effect of dissipation is neglected in the following.

In addition to the unperturbed flow pattern, we consider three cases of perturbations:

1. *Secluded region*: We define a region of the flow that is partly disconnected from the rest of the fluid domain. In geophysical fluid dynamics, this setting might approximate large differences in altitude or other geographical boundaries. Practically, the transition matrix in this setting is obtained by subdividing the fluid domain into regions, estimating $\mathbf{P}$ from the data, and then multiplying those matrix elements with a small factor $f \ll 1$ that correspond to flows between nodes situated in different regions.
2. *Externally forced region*: Another form of perturbation, which has already been studied in [45], is employing a localized forcing. In case of $T(\mathbf{x})$ representing a temperature field, this could, for example, describe some external heating of a confined spatial region, or just a region with a heat capacity differing from that in the rest of the fluid domain. In this case, we can simply add some constant to the diagonal elements of $\mathbf{P}$, i.e., consider a perturbed evolution matrix $\mathbf{P}' = \mathbf{P} + \mathbf{H}$ with $\mathbf{H}$ being a diagonal matrix with $h_{ii} > 0$ in the forced region and $h_{ii} = 0$ otherwise.
3. *Interacting flows*: In order to model heat exchange between two separate flows (e.g., ocean and atmospheric flow), we may utilize a generalization of the theoretical treatment described above, starting from two coupled recursive equations for the two individual scalar fields and associated respective flows. If we consider the corresponding effect to be unidirectional, the effect of the secondary flow field can be understood as an external perturbation to the advection-diffusion dynamics in the primary one, which allows for an analytical estimation of the correlation matrix. For illustrative processes, we assume here a fully-local influence of the secondary scalar field on the primary one with a fixed rate and a linear (Hagen-Poisseuille like) flow pattern in y direction representing the secondary flow.

For all four considered situations, an illustration of the generated flow pattern is provided in Fig. 11.8. Here, we study the flow pattern on a regular spatial grid consisting of 40×40 nodes. The summations used in the analytical evaluation of

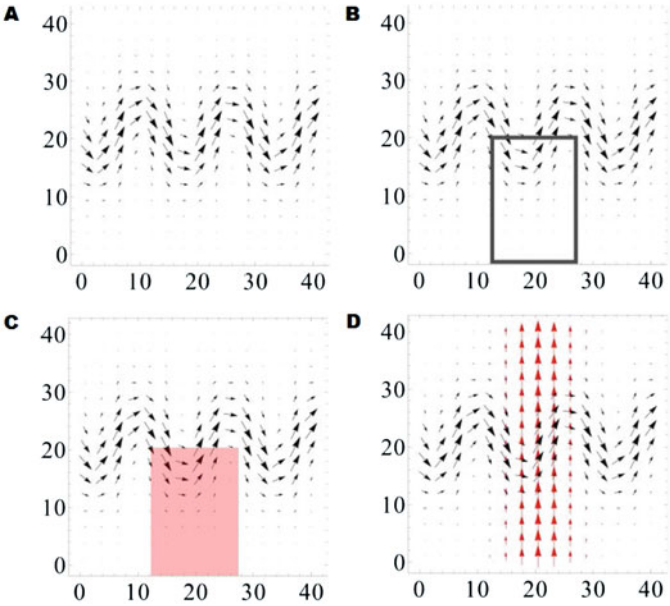


Fig. 11.8 Schematic visualization of the meandering flow pattern (**a**) without perturbation, (**b**) with secluded region (black line), (**c**) with externally forced region (red rectangle), and (**d**) with interacting flow (red arrows)

the correlation matrices are restricted to 20 terms for reasons of computational efficiency. It has been checked (not shown) that this truncation does not affect the resulting functional network representations markedly.

Since lag-zero correlations are a symmetric quantity, the functional network representations based upon this similarity measure are symmetric and, hence, undirected. In what follows, we will consider fully connected weighted networks instead of their unweighted (threshold-based) counterparts, where the weighted adjacency matrix is defined as $W_{ij} = \left|C_{ij}\right|$.

Figure 11.9 shows the resulting patterns of node strength s_i. In agreement with [14], the unperturbed advection-diffusion system exhibits the largest node strengths in regions of particularly high absolute velocity. This observation can be explained by the flow transporting random temperature fluctuations across vast parts of the fluid medium, which is exactly what the correlation structure describes. Specifically, fast parts of the flow can propagate these fluctuations over large spatial distances and, thus, among many other nodes of the flow network. The aforementioned effect is preserved in case of a secluded region. However, in the subdomain where links to the rest of the fluid have been cut, the node strength is slightly reduced. Moreover, a similar reduction applies to the fast flowing regions along the cutting line. In turn, an external forcing (heating) in a confined region elevates the correlation among the affected nodes (common superimposed trend) and thus leads to an elevated node

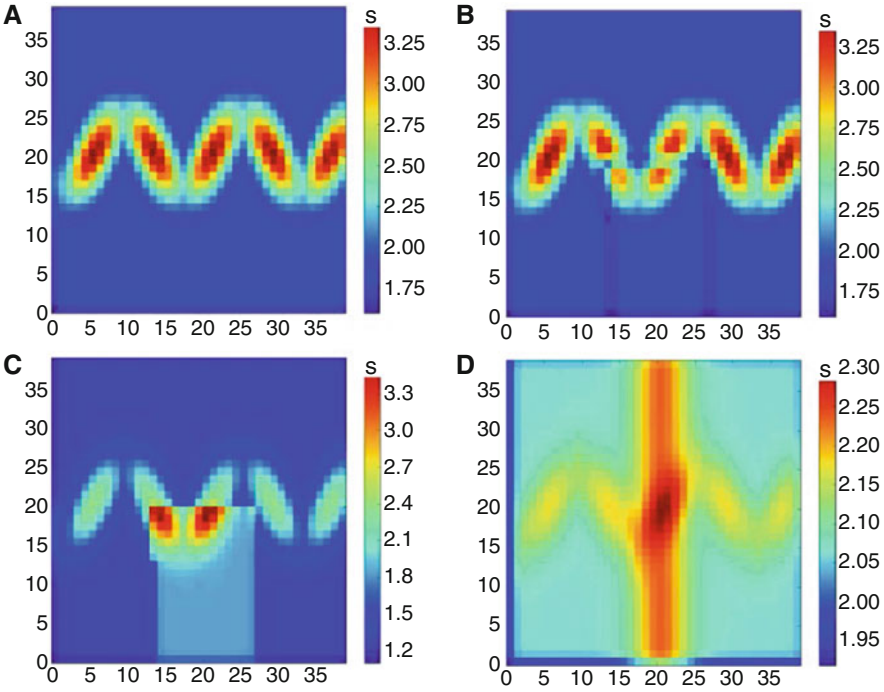


Fig. 11.9 Patterns of node strength for the weighted functional network representation of the meandering flow (**a**) without perturbation, (**b**) with secluded region, (**c**) with externally forced region, and (**d**) with interacting secondary flow

strength. A similar effect is also found if additional heat is induced into the system by a superimposed secondary flow. Taken together, all three types of perturbations have marked effects on the network structure. However, when solely analyzing topological characteristics like node strength, it appears challenging to diagnose the type of perturbation [45].

Among the spatial network characteristics, it has been suggested [45] that the local edge angle anisotropy may provide a useful complement of the classical topological network measures like node strength. For a two-dimensional flow, we define a reference axis (e.g., the $y = 0$ axis in the present example) and an edge angle between two nodes representing the spatial positions $\mathbf{x}_i = (x_i, y_i)$ and $\mathbf{x}_j = (x_j, y_j)$ as $\phi_{ij} = \arctan((y_j - y_i)/(x_j - x_i))$. Then, we can express the edge angle anisotropy of a weighted functional network as the mean resultant length associated with the normalized vector sum, i.e.,

$$R_i = \frac{1}{s_i} \left| \sum_{j=1}^{N} W_{ij} e^{2\pi i \phi_{ij}} \right| . \tag{11.16}$$

R_i thus measures how well aligned the edges associated with a given node i are in (two-dimensional) Euclidean space. Very low values close to 0 would indicate an isotropic connectivity, while high values close to 1 correspond to an almost parallel orientation of the local links. Thus, we might deduce information on possible underlying mechanisms, i.e., distinguish advection as a spatially directed process from more isotropic diffusion.

For our meandering flow model, the patterns obtained without and with perturbations are shown in Fig. 11.10. For the unperturbed system, the highest anisotropy is found in regions with fast and almost parallel fluid motion, however, exhibiting a perpendicular offset with respect to the position of the highest velocity modulus. In turn, the connectivity pattern is least anisotropic at the turning points of the flow. This general pattern also persists in the presence of perturbations, but is altered in different ways. In case of some secluded flow region, the anisotropy is either reduced or elevated along the boundaries of that region, depending on their relative orientation with respect to the main flow direction. A similar observation can be made in case of the externally forced region, with the exception that here, the distorted boundaries show both, reduced and elevated anisotropy values at their inner and outer parts, respectively. Together with the node strength, the edge angle anisotropy can therefore be used to discriminate between stronger connectivity due to fast flow and external forcing, respectively. Patterns that are common to s_i and R_i

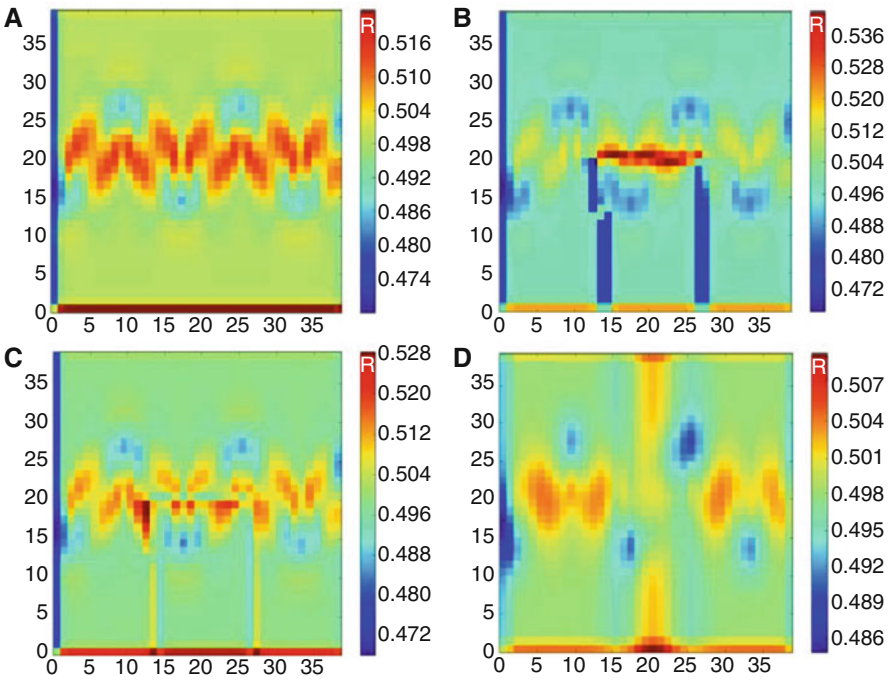


Fig. 11.10 As in Fig. 11.9 for the weighted edge angle anisotropy R_i

(here, larger values in straight and lower ones in bent regions of the flow) most likely originate from the underlying (and possibly unknown) background flow, whereas an elevated node strength without markedly altered anisotropy rather points to some perturbed region. Finally, interaction between the advection-diffusion system and a secondary straight flow perpendicular to the average direction of the meandering velocity field leads to a blurring of the original anisotropy pattern and the emergence of a superposed pattern of high anisotropy tracing the fast-flow regions of the secondary velocity field. However, in those regions where the fast-flow segments of both fields overlap, anisotropy is reduced, since links point into different directions according to the two main flow directions.

In order to summarize our findings, Fig. 11.11 shows estimates of the two-dimensional joint probability density functions of node strength and edge angle anisotropy for all four cases. In the unperturbed case, the highest probability is found for $R_i \approx 0.5$ and $s_i \approx 1.9$, corresponding to the large group of nodes located outside the fast-flow regime, i.e., in a region dominated by diffusive processes. In turn, nodes within the core part of the meandering flow have higher anisotropy in the straight, and lower anisotropy in the curved flow segments. In the case with some secluded region, we observe additional nodes with high anisotropy values (above $R_i \approx 0.51$) originating from the cutting of non-aligned links at the boundary of the secluded region located in the core flow zone. In case of an externally perturbed region, we find two peaks in the probability profile at $R_i \approx 0.5$ and $s_i \approx 1.6$ as well as $s_i \approx 2.0$, respectively, differing in node strength but not in anisotropy. These two peaks correspond to the slow-flowing "heated" region (higher strength) and the fast-flowing "non-heated" region (lower strength), respectively. Hence, plotting both characteristics against each other can actually help differentiating between these two regimes. Finally, in the case of interacting flows, the overall range of values taken by both measures is markedly reduced, exhibiting the highest probability around $R_i \approx 0.5$ and $s_i \approx 2.0$. Notably, the anisotropy (strength) associated with this peak is slightly reduced (elevated) in comparison with the unperturbed case, which can be explained by the additional directionality induced by the secondary flow.

11.4.3 Applications

Despite their intrinsic limitations, correlation-based functional network representations of key climate variables have recently provided important new insights into key processes associated with atmospheric and ocean dynamics. Among other achievements, such network approaches have been utilized for identifying statistical precursors of abrupt transitions or tipping points in terms of characteristic changes in degree, assortativity, clustering coefficient or degree-distribution kurtosis of the networks when approaching a bifurcation point [89–91]. In a similar spirit, percolation in such networks can provide indications of approaching abrupt transitions [92]. Changing connectivity patterns in global surface air temperatures have been used for studying the teleconnectivity structure of large-scale climate phenomena

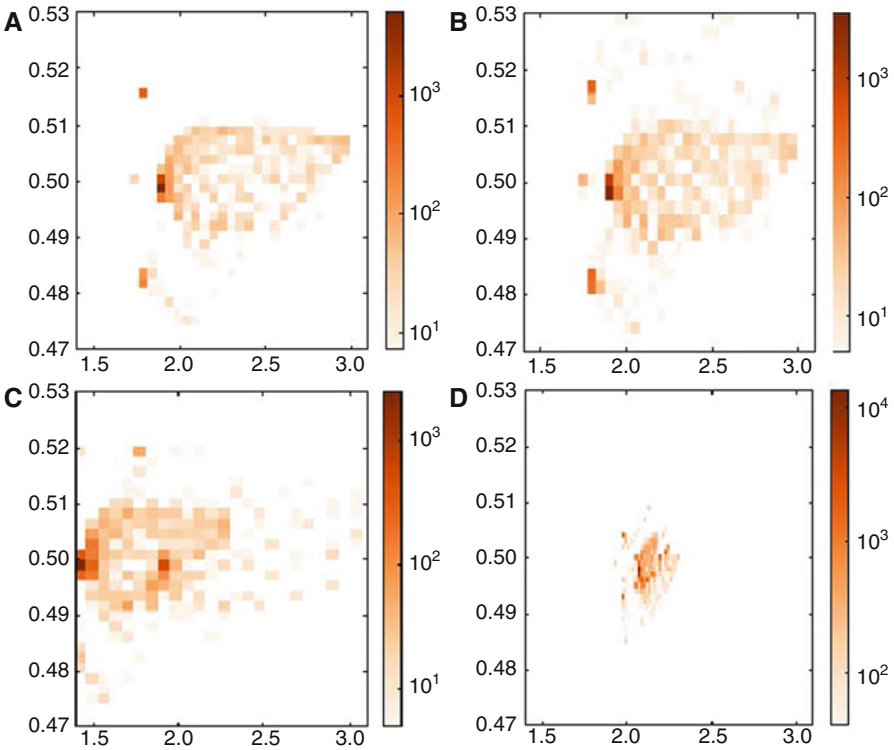


Fig. 11.11 As in Fig. 11.9 for the two-dimensional probability density functions of node strength s_i (horizontal axis) and edge angle anisotropy R_i (vertical axis)

like the El Niño Southern Oscillation (ENSO) [20, 93–95], classifying the different phases of this phenomenon [96, 97], or even predict the emergence of El Niño based on changes in the network properties [98, 99]. By replacing correlations by other similarity measures like event synchronization [100], key patterns of extreme precipitation and their relationships with the emergence of heavy flooding events have been studied for different regions of the world [101–105].

In a more explicit fluid dynamics context, the functional network approach can be used to obtain a complementary description of mixing regimes in time-dependent flows. Notably, mixing (characterized by the intersecting trajectories of particles that are advected by the flow) has been extensively studied using the Lagrangian flow network approach (see [40] and the references therein). However, the interrelation between mixing and the evolution of some scalar field evolving according to an underlying time-dependent non-periodic velocity field has been less studied so far and needs to be addressed in future work. Among others, one particular research avenue has been opened by recent work on anomalous diffusion in certain types of time-dependent flows [106] and provides an additional possible field of application of both, Lagrangian and correlation-based functional flow networks.

11.5 Conclusions

Complex network methods provide a novel approach to understanding the structural organization of flows across a variety of scientific disciplines. In this chapter, we have summarized the present state of the art in this active and exciting new field of research, together with some illustrative examples highlighting which kinds of information can be provided by different types of *flow networks*. In particular, correlation-based functional networks and Lagrangian flow networks constitute two important classes of network representations of dynamical systems that have already proven their potentials in a variety of applications in both, dynamical systems theory and the Earth sciences. Since the underlying developments have been obtained relatively recently, especially the Lagrangian flow network approach is still in its infancy, and major breakthroughs regarding both, the conceptual understanding and application of this concept are still to be expected. However, also the more mature functional network method still offers manifold avenues for future research. In particular, the attribution of network structures to underlying flow properties is far from being fully understood. Moreover, there might be various other ways of employing network theory in the context of flow system analysis, so that there is a reason to believe that the stage of methodological development reached so far will not be the end of the corresponding endeavors.

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